



US012410442B2

(12) **United States Patent**
Burns et al.

(10) **Patent No.: US 12,410,442 B2**

(45) **Date of Patent: Sep. 9, 2025**

(54) **COMPOSITIONS AND METHODS FOR
 TREATING SENSORINEURAL HEARING
 LOSS USING OTOFERLIN DUAL VECTOR
 SYSTEMS**

USPC 435/320.1
 See application file for complete search history.

(71) Applicants: **Decibel Therapeutics, Inc.**, Boston,
 MA (US); **Regeneron
 Pharmaceuticals, Inc.**, Tarrytown, NY
 (US)

(72) Inventors: **Joseph Burns**, Newton, MA (US);
Kathryn Ellis, Arlington, MA (US);
Adam Palermo, Somerville, MA (US);
Martin Schwander, Auburndale, MA
 (US); **Jonathon Whitton**, Cambridge,
 MA (US); **Leah Sabin**, Goldens Bridge,
 NY (US); **Christos Kyratsous**,
 Irvington, NY (US); **Meghan
 Drummond Samuelson**, Katonah, NY
 (US)

(73) Assignees: **Regeneron Pharmaceuticals, Inc.**,
 Tarrytown, NY (US); **Decibel
 Therapeutics, Inc.**, Boston, MA (US)

(*) Notice: Subject to any disclaimer, the term of this
 patent is extended or adjusted under 35
 U.S.C. 154(b) by 976 days.

(21) Appl. No.: **17/395,711**

(22) Filed: **Aug. 6, 2021**

(65) **Prior Publication Data**

US 2021/0395781 A1 Dec. 23, 2021

Related U.S. Application Data

(63) Continuation of application No.
 PCT/US2020/017257, filed on Feb. 7, 2020.

(60) Provisional application No. 62/965,770, filed on Jan.
 24, 2020, provisional application No. 62/928,279,
 filed on Oct. 30, 2019, provisional application No.
 62/802,890, filed on Feb. 8, 2019.

(51) **Int. Cl.**
C12N 15/80 (2006.01)
A61K 48/00 (2006.01)
C07K 14/705 (2006.01)
C12N 15/86 (2006.01)

(52) **U.S. Cl.**
 CPC **C12N 15/86** (2013.01); **A61K 48/005**
 (2013.01); **C07K 14/705** (2013.01); **C12N**
2800/40 (2013.01); **C12N 2830/008** (2013.01);
C12N 2840/44 (2013.01)

(58) **Field of Classification Search**
 CPC **C12N 15/86**; **C12N 2800/40**; **C12N**
2830/008; **C12N 15/52**; **C12N 15/902**;
C12N 2750/14144; **C12N 2830/50**; **C12N**
2800/50; **A61K 35/761**; **A61K 31/711**;
C07K 14/4716; **C07K 14/50**; **C12Y**
301/03001; **C12Y 402/01011**; **C12Y**
207/10001

(56) References Cited

U.S. PATENT DOCUMENTS

6,436,392 B1	8/2002	Engelhardt et al.
6,544,786 B1	4/2003	Xiao et al.
6,808,922 B1	10/2004	Bebbington et al.
6,897,045 B2	5/2005	Engelhardt et al.
7,803,622 B2	9/2010	Engelhardt et al.
8,236,557 B2	8/2012	Dongsheng et al.
8,298,818 B2	10/2012	Boye et al.
10,214,572 B2	2/2019	Boye et al.
11,325,956 B2	5/2022	Boye et al.
11,525,139 B2	12/2022	Simons et al.
11,660,353 B2	5/2023	Burns et al.
11,781,145 B2	10/2023	Simons et al.
11,807,867 B2	11/2023	Simons et al.
12,188,041 B2	1/2025	Dyka et al.
12,233,136 B2	2/2025	Burns et al.
2003/0219741 A1	11/2003	Isogai et al.
2004/0072154 A1	4/2004	Morris et al.
2007/0161031 A1	7/2007	Trinklein et al.
2007/0161110 A1	7/2007	Iida et al.
2008/0249052 A1	10/2008	Duan et al.
2010/0003218 A1	1/2010	Duan et al.
2010/0266551 A1	10/2010	Richard et al.
2012/0003190 A1	1/2012	Yamoah et al.

(Continued)

FOREIGN PATENT DOCUMENTS

WO	WO-01/25465 A1	4/2001
WO	WO-2001/070972 A2	9/2001

(Continued)

OTHER PUBLICATIONS

Liang et al., "Characterization of the Human and Mouse Unconventional Myosin XV Genes Responsible for Hereditary Deafness DFNB3 and Shaker 2," *Genomics*. 61(3):243-258 (1999).

(Continued)

Primary Examiner — Robert B Mondesi

Assistant Examiner — Mohammad Y Meah

(74) *Attorney, Agent, or Firm* — Clark & Elbing LLP

(57) ABSTRACT

The disclosure features compositions and methods for the treatment of sensorineural hearing loss and auditory neuropathy, particularly forms of the disease that are associated with mutations in otoferlin (OTOF), by way of OTOF gene therapy. The disclosure provides a variety of compositions that include a first nucleic acid vector that contains a polynucleotide encoding an N-terminal portion of an OTOF protein and a second nucleic acid vector that contains a polynucleotide encoding a C-terminal portion of an OTOF protein. These vectors can be used to increase the expression of OTOF in a subject, such as a human subject suffering from sensorineural hearing loss.

19 Claims, 25 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2012/0087862	A1	4/2012	Hood et al.
2013/0210895	A1	8/2013	Boye et al.
2014/0249208	A1	9/2014	Bancel et al.
2014/0256802	A1	9/2014	Boye et al.
2015/0065562	A1	3/2015	Yazicioglu et al.
2015/0209406	A1	7/2015	Chen
2016/0022836	A1	1/2016	Banfi et al.
2016/0076054	A1	3/2016	Auricchio et al.
2018/0015172	A1	1/2018	Muzyczka et al.
2018/0055908	A1	3/2018	Petit et al.
2018/0327779	A1	11/2018	Colella et al.
2019/0002916	A1	1/2019	Kalatzis et al.
2019/0153050	A1	5/2019	Boye et al.
2019/0185864	A1	6/2019	Simons et al.
2019/0309326	A1	10/2019	Maclaren et al.
2020/0155705	A1	5/2020	Burns et al.
2020/0157573	A1	5/2020	Boye et al.
2021/0130421	A1	5/2021	Boye et al.
2021/0236654	A1	8/2021	Burns et al.
2021/0388045	A1	12/2021	Burns et al.
2021/0395778	A1	12/2021	Dyka et al.
2022/0064671	A1	3/2022	Maranga et al.
2022/0265865	A1	8/2022	Burns et al.
2023/0149565	A1	5/2023	Boye et al.
2024/0011039	A1	1/2024	Simons et al.
2024/0131186	A1	4/2024	Burns et al.
2024/0148905	A1	5/2024	Palermo et al.
2024/0309399	A1	9/2024	Hu et al.

FOREIGN PATENT DOCUMENTS

WO	WO-2008/088895	A2	7/2008
WO	WO-2009/100438	A2	8/2009
WO	WO-2013/075008	A1	5/2013
WO	WO-2013/158879	A1	10/2013
WO	WO-2014/140051	A1	9/2014
WO	WO-2014/170480	A1	10/2014
WO	WO-2014/193716	A2	12/2014
WO	WO-2016/131981	A1	8/2016
WO	WO-2016/139321	A1	9/2016
WO	WO-2017/049252	A1	3/2017
WO	WO-2017/100791	A1	6/2017
WO	WO-2017/216560	A1	12/2017
WO	WO-2018/039375	A1	3/2018
WO	WO-2018/145111	A1	8/2018
WO	WO-2018/162748	A1	9/2018
WO	WO-2018/204734	A1	11/2018
WO	WO-2019/162396	A1	8/2019
WO	WO-2019/165292	A1	8/2019
WO	WO-2019/183641	A1	9/2019
WO	WO-2020/093018	A1	5/2020
WO	WO-2020/097372	A1	5/2020
WO	WO-2020/148458	A1	7/2020
WO	WO-2021/087296	A1	5/2021
WO	WO-2024/173835	A2	8/2024

OTHER PUBLICATIONS

Yuhe, Liu, "Preparation of adeno-associated virus vector and its application in cochlea transgenic research," *Chinese Journal of Otolaryngology*. 4(4):343-347 (2006) (6 pages).

Hirsch et al., "Delivering Transgenic DNA Exceeding the Carrying Capacity of AAV Vectors," available in PMC Aug. 3, 2016, published in final edited form as: *Methods Mol Biol.* 13382:21-39 (2016) (20 pages).

Majewski and Ott, "GT Repeats Are Associated with Recombination on Human Chromosome 22," *Genome Res.* 10(8): 1108-1144 (Aug. 2000) (7 pages).

Lostal et al., "Full-Length Dystrophin Reconstitution with Adeno-Associated Viral Vectors," *Human Gene Ther.* 25(6): 552-562 (Jun. 2014) (11 pages).

Gao et al., "The Dystrophin Complex: structure, function and implications for therapy," available in PMC Jul. 1, 2016, published in final edited form as: *Compr Physiol.* 5(3): 1223-1239 (Jul. 2015) (33 pages).

Dyka et al., "Dual adeno-associated virus vectors result in efficient in vitro and in vivo expression of an oversized gene, MYO7A," *Human Gene Ther Methods.* 25(2): 166-77 (Apr. 2014) (12 pages).

Pryadkina et al., "A comparison of AAV strategies distinguishes overlapping vectors for efficient systemic delivery of the 6.2 kb Dysferlin coding sequence," *Mol Ther Methods Clin Dev.* 2: 15009 (Mar. 2015) (12 pages).

Geleoc et al., "Sound strategies for hearing restoration," available in PMC Aug. 29, 2014, published in final edited form as: *Science.* 344(6184):1241062 (May 2014) (20 pages).

Al-Moyed et al., "A dual-AAV approach restores fast exocytosis and partially rescues auditory function in deaf otoferlin knock-out mice," *EMBO Mol Med.* 11(1):e9396 (2019) (13 pages).

GenBank Accession No. JN953192.1, "Mus musculus targeted KO-first, conditional ready, lacZ-tagged mutant allele Myo15:tmla(EUCOMM)Wtsi; transgenic," retrieved from <<https://www.ncbi.nlm.nih.gov/nucleotide/JN953192>>, dated Nov. 5, 2011 (11 pages).

Caberlotto et al., "Usher type 1G protein sans is a critical component of the tip-link complex, a structure controlling actin polymerization in stereocilia," *Proc Natl Acad Sci U S A.* 108(14):5825-30 (2011) (14 pages).

Schlabach et al., "Synthetic design of strong promoters," *Proc Natl Acad Sci U S A.* 107(6):2538-43 (2010).

International Search Report and Written Opinion for International Application No. PCT/US2019/029366, mailed Sep. 10, 2019 (17 pages).

Boëda et al., "A specific promoter of the sensory cells of the inner ear defined by transgenesis," *Hum Mol Genet.* 10(15):1581-1589 (2001).

Belyantseva et al., "Myosin XVa localizes to the tips of inner ear sensory cell stereocilia and is essential for staircase formation of the hair bundle," *Proc Natl Acad Sci U S A.* 100(24):13958-63 (2003).

GenBank Accession No. JN957158.1, "Mus musculus targeted non-conditional, lacZ-tagged mutant allele Myo15:tmla(EUCOMM)Wtsi; transgenic," retrieved from <<https://www.ncbi.nlm.nih.gov/nucleotide/JN957158.1>>, dated Nov. 5, 2011 (12 pages).

International Search Report and Written Opinion for International Patent Application No. PCT/US2020/017292, mailed Jun. 26, 2020 (18 pages).

Akil et al., "Dual AAV gene therapy restores hearing in a mouse model for human genetic Deafness," *International Symposium on Inner Ear Therapies (ISIET)*, Marrakech, Morocco. 21 (2017) (Abstract only).

Akil et al., "Dual AAV-mediated gene therapy restores hearing in a DFNB9 mouse model," *Proc Natl Acad Sci U S A.* 116(10):4496-4501 (2019).

Al-Moyed et al., "A dual AAV viral vector approach partially restores exocytosis and rescues hearing in deaf otoferlin knock-out mice," *ARO Abstracts.* 41:76 (2018) (Abstract only).

Alemi, "Progress Report: AOS Research Grant: Restoration of Hearing in the Otoferlin Knockout Mouse using Viral Gene Therapy," 145th Annual Meeting of the American Otological Society, Inc, Apr. 21-22, San Diego, California. 68 (2012) (Abstract only).

Choi et al., "Identities and frequencies of mutations of the otoferlin gene (OTOF) causing DFNB9 deafness in Pakistan," available in PMC Oct. 1, 2012, published in final edited form as: *Clin Genet.* 75(3):237-243 (2009) (10 pages).

Duan et al., "Expanding AAV Packaging Capacity with Trans-splicing or Overlapping Vectors: A Quantitative Comparison," *Mol Ther.* 4(4):383-91 (2001).

McClements et al., "Adeno-associated Virus (AAV) Dual Vector Strategies for Gene Therapy Encoding Large Transgenes," *Yale J Biol Med.* 90(4):611-623 (2017).

Trapani et al., "Effective delivery of large genes to the retina by dual AAV vectors," *EMBO Mol Med.* 6(2):194-211 (2014).

Trapani et al., "Improved dual AAV vectors with reduced expression of truncated proteins are safe and effective in the retina of a mouse model of Stargardt disease," *Hum Mol Genet.* 24(23):6811-25 (2015).

(56)

References Cited

OTHER PUBLICATIONS

- Yasunaga et al., "OTOF Encodes Multiple Long and Short Isoforms: Genetic Evidence That the Long Ones Underlie Recessive Deafness DFNB9," *Am J Hum Genet.* 67(3):591-600 (2000).
- International Search Report and Written Opinion for International Application No. PCT/US2020/017257, mailed Apr. 29, 2020 (22 pages).
- Xu et al., "Trans-Splicing Adeno-Associated Viral Vector-Mediated Gene Therapy Is Limited by the Accumulation of Spliced mRNA but Not by Dual Vector Coinfection Efficiency," available in PMC Jun. 19, 2008, published in final edited form as: *Hum Gene Ther.* 15(9):896-905 (2004) (17 pages).
- Ghosh et al., "A Hybrid Vector System Expands Adeno-associated Viral Vector Packaging Capacity in a Transgene-independent Manner," *The American Society of Gene Therapy.* 16(1):124-130 (2008).
- Ghosh et al., "Efficient Transgene Reconstitution with Hybrid Dual AAV Vectors Carrying the Minimized Bridging Sequences," *Hum Gene Ther.* 22(1):77-83 (2011).
- International Search Report and Written Opinion for International Patent Application No. PCT/US2020/058265, mailed Feb. 8, 2021 (15 pages).
- Michalski et al., "Genetics of auditory mechano-electrical transduction," *Pflügers Arch.* 467(1):49-72 (2015).
- Skarnes et al., "A conditional knockout resource for the genome-wide study of mouse gene function," available in PMC Feb. 14, 2013, published in final edited form as: *Nature.* 474(7351):337-342 (2011) (18 pages).
- Boye et al., "Transduction and Tropism of an Abbreviated Form of CMV-Chicken B-Actin Promoter (CBA) With AAV in Mouse Retina," *ARVO Annual Meeting Abstract May 2006*, published in: *Investigative Ophthalmology & Visual Science.* 47: 852 (2006) (2 pages) (Abstract only).
- Lovell, "Mouse DNA sequence from clone RP23-135F6 on chromosome 11," *European Nucleotide Archive, EMBL-EBI.* (2012) (15 pages).
- Higashimoto et al., "The woodchuck hepatitis virus post-transcriptional regulatory element reduces readthrough transcription from retroviral vectors," *Gene Ther.* 14(17):1298-304 (2007).
- Wang, Aihui, Dissertation: "Molecular Cloning of an Unconventional Myosin MYO15 and the Identification of Mutations of MYO15 Responsible for Human Nonsyndromic Deafness DFNB3," Doctor of Philosophy, Graduate Program in Genetics, Michigan State University (1999) (140 pages).
- Corns et al., "Mechanotransduction is required for establishing and maintaining mature inner hair cells and regulating efferent innervation," *Nat Commun.* 9(1):4015 (Oct. 2018) (15 pages).
- International Search Report and Written Opinion for PCT/US2022/017058, dated Jun. 14, 2022 (14 pages).
- Yoshimura et al., "Enhanced viral-mediated cochlear gene delivery in adult mice by combining canal fenestration with round window membrane inoculation," *Scientific Reports.* 8:2980 (with supplemental material) (Feb. 2018) (14 pages).
- Pangrsic et al., "Otoferlin: a multi-C₂ domain protein essential for hearing," *Trends in Neurosciences.* 35(11): 671-680 (2012) (10 pages).
- Holt et al., "Split otoferlin reunited," *EMBO Molecular Medicine.* 11(1):e9995 (Jan. 2019) (3 pages).
- Suzuki et al., "Cochlear gene therapy with ancestral AAV in adult mice: complete transduction of inner hair cells without cochlear dysfunction," *Scientific Reports.* 7(1):45524 (Apr. 2017) (11 pages).
- Tertrais et al., "Viral Transfer of Mini-Otoferlins Partially Restores the Fast Component of Exocytosis and Uncovers Ultrafast Endocytosis in Auditory Hair Cells of Otoferlin Knock-Out Mice," *J. Neurosci.* 39(18):3394-3411 (May 2019) (18 pages).
- Petr-Silva et al., "Novel Properties of Tyrosine-mutant AAV2 Vectors in the Mouse Retina," *Molecular Therapy.* 19(2):293-301 (Feb. 2011) (9 pages).
- American Academy of Audiology, "Children's Hospital of Philadelphia Performs First Gene Therapy Procedure to Treat Genetic Hearing Loss in United States," <<https://www.audiology.org/childrens-hospital-of-philadelphia-performs-first-gene-therapy-procedure-to-treat-genetic-hearing-loss-in-united-states/>>, dated Jan. 26, 2024 (2 pages).
- Yoshimura et al., "Targeted Allele Suppression Prevents Progressive Hearing Loss in the Mature Murine Model of Human *TMC1* Deafness," *Molecular Therapy.* 27(3):681-690 (with supplemental material) (Mar. 2019) (17 pages).
- Akil et al., "Surgical Method for Virally Mediated Gene Delivery to the Mouse Inner Ear through the Round Window Membrane," *Journal of Visualized Experiments.* 97(1):e52187 (Mar. 2015) (7 pages).
- Liu et al., "Specific and Efficient Transduction of Cochlear Inner Hair Cells with Recombinant Adeno-associated Virus Type 3 Vector," *Molecular Therapy.* 12(4):725-733 (Oct. 2005) (9 pages).
- "Basics of sound, the Ear, and Hearing," *Hearing Loss: Determining Eligibility for Social Security Benefits.* Edited by Robert A. Dobie and Susan Van Hemel, 42-68 (2004) (61 pages).
- Akil et al., "AAV-Mediated Gene Delivery to the Inner Ear," *Adeno-Associated Virus Vectors: Design and Delivery.* Methods in Molecular Biology. Edited by Michael J. Castle, 271-282 (2019) [published online on Jan. 1, 2019] (16 pages).
- Langouet-Astrie et al., "Characterization of intravitreally delivered capsid mutant AAV2-Cre vector to induce tissue-specific mutations in murine retinal ganglion cells," *Experimental Eye Research.* 151(1):61-67 (Jul. 2016) (7 pages).
- Li et al., "A novel bispecific molecule delivered by recombinant AAV2 suppresses ocular inflammation and choroidal neovascularization," *J. Cell. Mol. Med.* 21(8):1555-1571 (Aug. 2017) (17 pages).
- Lopes-Pacheco et al., "Self-complementary and tyrosine-mutant rAAV vectors enhance transduction in cystic fibrosis bronchial epithelial cells," *Experimental Cell Research.* 372:99-107 (Sep. 2018) (9 pages).
- Petr-Silva et al., "High-efficiency Transduction of the Mouse Retina by Tyrosine-mutant AAV Serotype Vectors," *Molecular Therapy.* 17(3):463-471 (Mar. 2009) (9 pages).
- Kilpatrick et al., "Adeno-associated virus-mediated gene delivery into the scala media of the normal and deafened adult mouse ear," *Gene Therapy.* 18(6):569-578 (Jan. 2011) (10 pages).
- Tao et al., "Delivery of Adeno-Associated Virus Vectors in Adult Mammalian Inner-Ear Cell Subtypes Without Auditory Dysfunction," *Human Gene Therapy.* 29(4):492-506 (Nov. 2017) (15 pages).
- Zhang et al., "Cochlear Gene Therapy for Sensorineural Hearing Loss: Current Status and Major Remaining Hurdles for Translational Success," *Front. Mol. Neurosci.* 11(221):1-15 (Jun. 2018) (15 pages).
- Roux et al., "Otoferlin, Defective in a Human Deafness Form, Is Essential for Exocytosis at the Auditory Ribbon Synapse," *Cell.* 127(2):277-289 (Oct. 2006) (13 pages).
- "Genetic Hearing Loss With No. Associated Abnormalities," *Hereditary Hearing Loss and Its Syndromes, Third Edition.* Helga V. Toriello and Shelley D. Smith. 164-165 (2013) (4 pages).
- Ahmed et al., "Emerging Gene Therapies for Genetic Hearing Loss," *JARO.* 18(5):649-670 (Aug. 2017) (22 pages).
- Zhang et al., "Temperature sensitive auditory neuropathy," *Hearing Research.* 335(1):53-63 (Jan. 2016) (11 pages).
- Hamosh et al., "OTOFERLIN; OTOF," OMIM. (Apr. 2015) (8 pages) retrieved via The Wayback Machine on Jul. 29, 2015, URL: <<https://web.archive.org/web/20150729163826/http://omim.org/entry/603681>>.
- International Search Report and Written Opinion for International Patent Application No. PCT/US2024/016218, mailed Jul. 12, 2024 (19 pages).
- Kim et al., "Direct isolation and identification of promoters in the human genome," *Genome Res.* 15(6):830-9 (Jun. 2005) (11 pages).
- U.S. Appl. No. 17/290,082, Dyka et al.
- U.S. Appl. No. 16/952,016, Boye et al.
- U.S. Appl. No. 17/916,308, Boye et al.
- GenPept Accession NP_001274418.1, dated Apr. 23, 2017, (4 pages) retrieved from <https://www.ncbi.nlm.nih.gov/protein/566559996?sa=46&safkey=73202094>.
- McClements et al., "A fragmented adeno-associated viral dual vector strategy for treatment of diseases caused by mutations in

(56)

References Cited**OTHER PUBLICATIONS**

large genes leads to expression of hybrid transcripts," *J Genet Syndr Gene Ther.* 7(5):311 (Nov. 14, 2016) (16 pages).

Avraham, "What's hot about otoferlin," *Embo J.* 35(23):2502-4 (Dec. 1, 2016).

NCBI Reference Sequence: NM_001632.5, "Homo sapiens alkaline phosphatase, placental (ALPP), mRNA" (Apr. 4, 2024) (5 pages).

NCBI Reference Sequence: NP_000251.3, "unconventional myosin-VIIa isoform 1 [Homo sapiens]" (Dec. 11, 2024) (5 pages).

GenBank: U39226.1, "Human myosin VIIa (USH1B) mRNA, complete cds" (Jul. 11, 1996) (4 pages).

Akil et al., "Restoration of Hearing in the VGLUT3 Knockout Mouse Using Virally Mediated Gene Therapy," *Neuron.* 75:283-293 (2012).

Al-Hussaini et al., "Mature retinal pigment epithelium cells are retained in the cell cycle and proliferate in vivo," *Mol Vis.* 14:1784-91 (2008).

Allocca et al., "Serotype-dependent packaging of large genes in adeno-associated viral vectors results in effective gene delivery in mice," *J Clin Invest.* 118(5):1955-64 (May 2008) (11 pages).

Chen et al., "Molecular cloning and domain structure of human myosin-VIIa, the gene product defective in Usher syndrome 1B," *Genomics* 36(3):440-8 (Sep. 15, 1996).

Daya et al., "Gene therapy using adeno-associated virus vectors," *Clin Microbiol Rev.* 21(4):583-93 (Oct. 2008).

Dong et al., "Characterization of genome integrity for oversized recombinant AAV vector," *Mol Ther.* 18(1):87-92 (Jan. 2010).

Duan et al., "Circular intermediates of recombinant adeno-associated virus have defined structural characteristics responsible for long-term episomal persistence in muscle tissue," *J Virol.* 72(11):8568-77 (Nov. 1998).

Hashimoto et al., "Lentiviral gene replacement therapy of retinas in a mouse model for Usher syndrome type 1B," *Gene Ther.* 14(7):584-94 (Apr. 2007) (21 pages).

Jacobson et al., "Usher syndromes due to MYO7A, PCDH15, USH2A or GPR98 mutations share retinal disease mechanism," *Hum Mol Genet.* 17(15):2405-15 (Aug. 1, 2008).

Lai et al., "Evidence for the failure of adeno-associated virus serotype 5 to package a viral genome $\geq$ 8.2 kb," *Mol Ther.* 18(1): 75-9 (Jan. 2010).

Li et al., "High-efficiency transduction of fibroblasts and mesenchymal stem cells by tyrosine-mutant AAV2 vectors for their potential use in cellular therapy," *Hum Gene Ther.* 21(11):1527-43 (Nov. 2010) (18 pages).

Lopes et al., "Retinal gene therapy with a large MYO7A cDNA using adeno-associated virus," *Gene Ther.* 20(8):824-33 (Aug. 2013) (21 pages).

Weil et al., "Human myosin VIIA responsible for the Usher 1B syndrome: a predicted membrane-associated motor protein expressed in developing sensory epithelia," *Proc Natl Acad Sci USA.* 93(8):3232-7 (Apr. 16, 1996).

Wu et al., "Effect of genome size on AAV vector packaging," *Mol Ther.* 18(1):80-6 (Jan. 2010).

Yan et al., "Inverted terminal repeat sequences are important for intermolecular recombination and circularization of adeno-associated virus genomes," *J Virol.* 79(1):364-79 (Jan. 2005).

Freni et al., "Cochlear Implant Surgery: Endomeatal Approach versus Posterior Tympanotomy," *Int. J. Environ. Res. Public Health* 17:4187 (Jun. 2020) (9 pages).

"Types of CFTR Mutations," Cystic Fibrosis Foundation. <<https://www.cff.org/research-clinical-trials/types-cftr-mutations#:~:text=>>, accessed Mar. 1, 2025 (9 pages).

Barnes et al., "Remarkable Rigidity of the Single α -Helical Domain of Myosin-VI As Revealed by NMR Spectroscopy," *J Am Chem Soc.* 141(22):9004-9017 (Jun. 2019).

Laine et al., "Cell cycle regulation in the inner ear sensory epithelia: role of cyclin D1 and cyclin-dependent kinase inhibitors," *Dev Biol.* 337(1):134-46 (Jan. 2010).

Orthwein et al., "A mechanism for the suppression of homologous recombination in G1 cells," *Nature.* 528(7582):422-6 (Dec. 2015); retraction in: *Nature.* 638(8051):844 (Feb. 2025) (35 pages).

Regalado et al., "Some deaf children in China can hear after gene therapy treatment," *MIT Technology Review.* <<https://www.technologyreview.com/2023/10/27/1082551/gene-treatment-deaf-children-hearing-china/>>, published Oct. 27, 2023 (7 pages).

Zhang et al., "Single amino acid change alters specificity of the multi-allelic wheat stem rust resistance locus SR9," *Nat Commun.* 14(1):7354 (Nov. 2023) (12 pages).

FIG. 1A

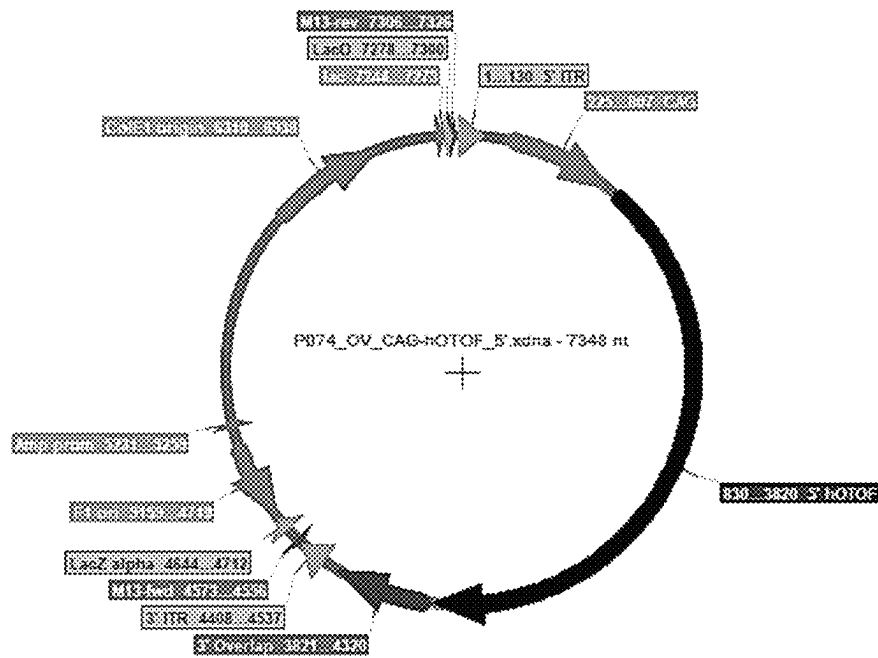


FIG. 1B

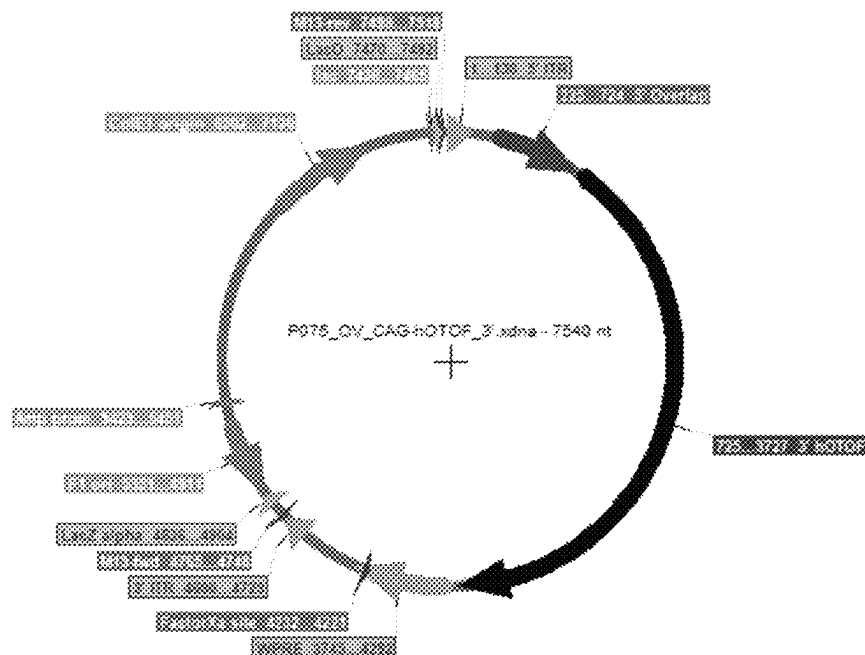


FIG. 2A

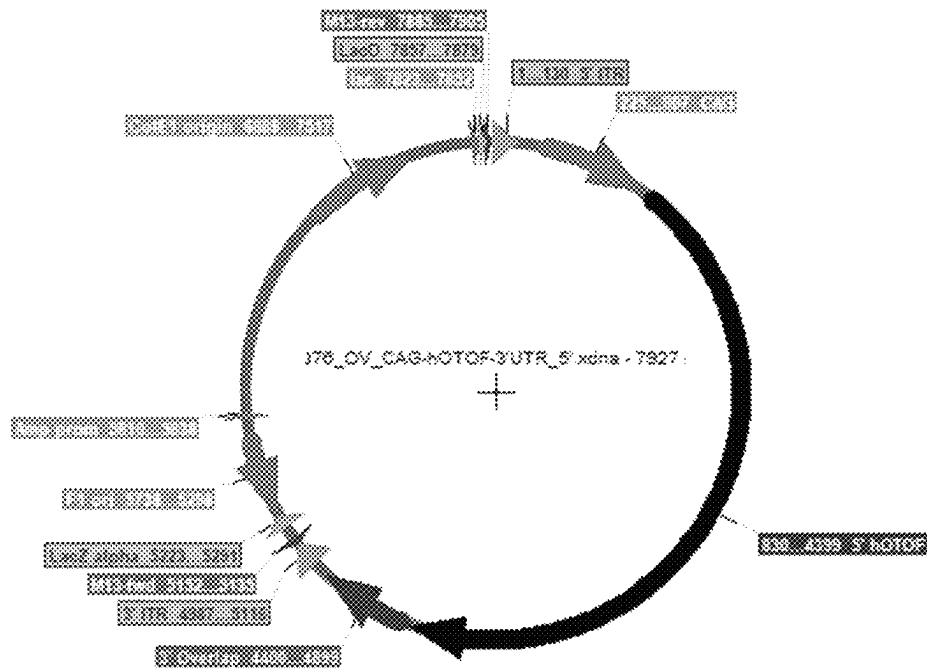
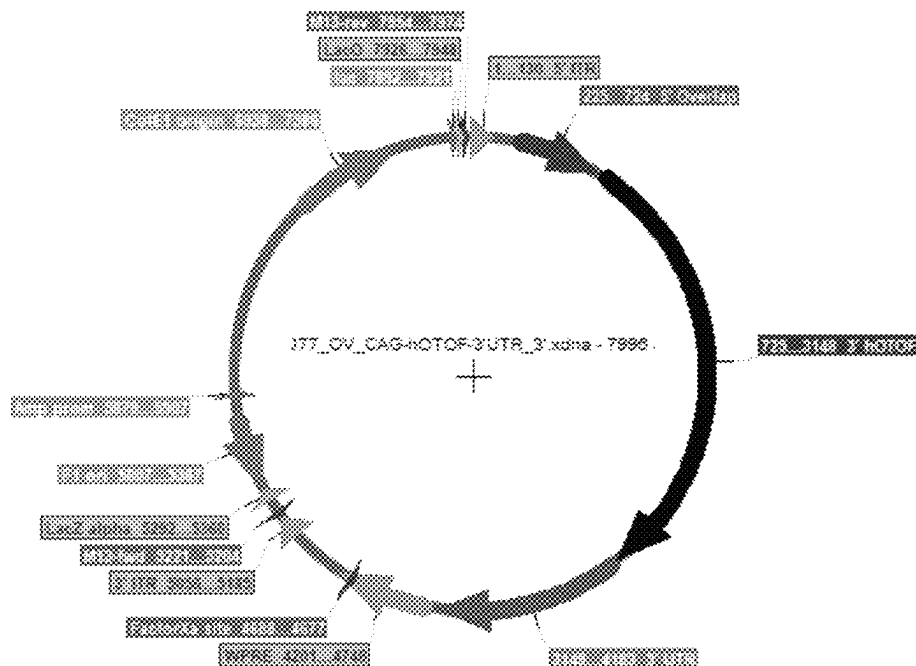


FIG. 2B



[illegible]

FIG. 4A

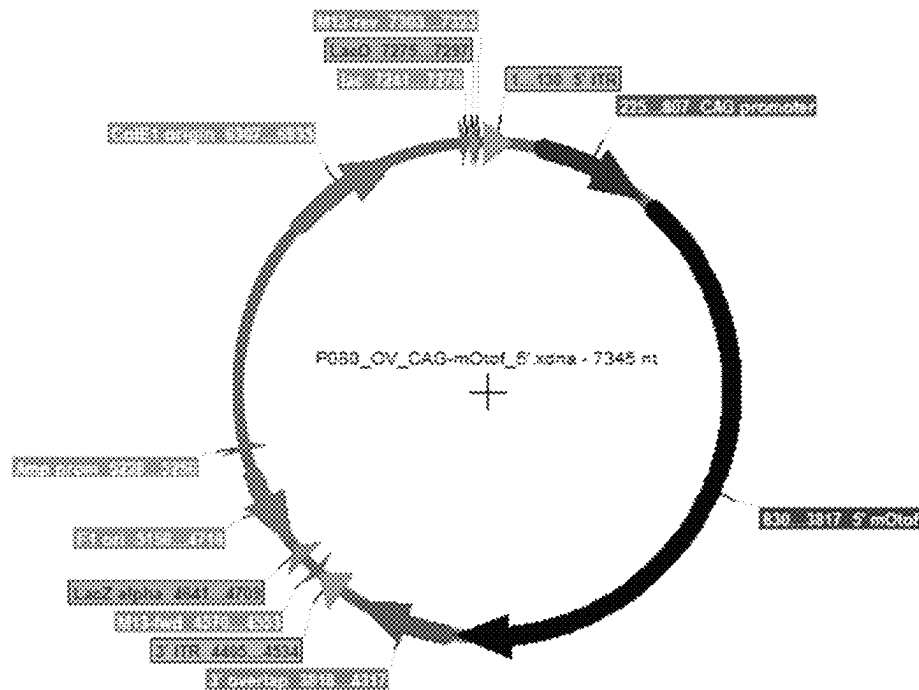


FIG. 4B

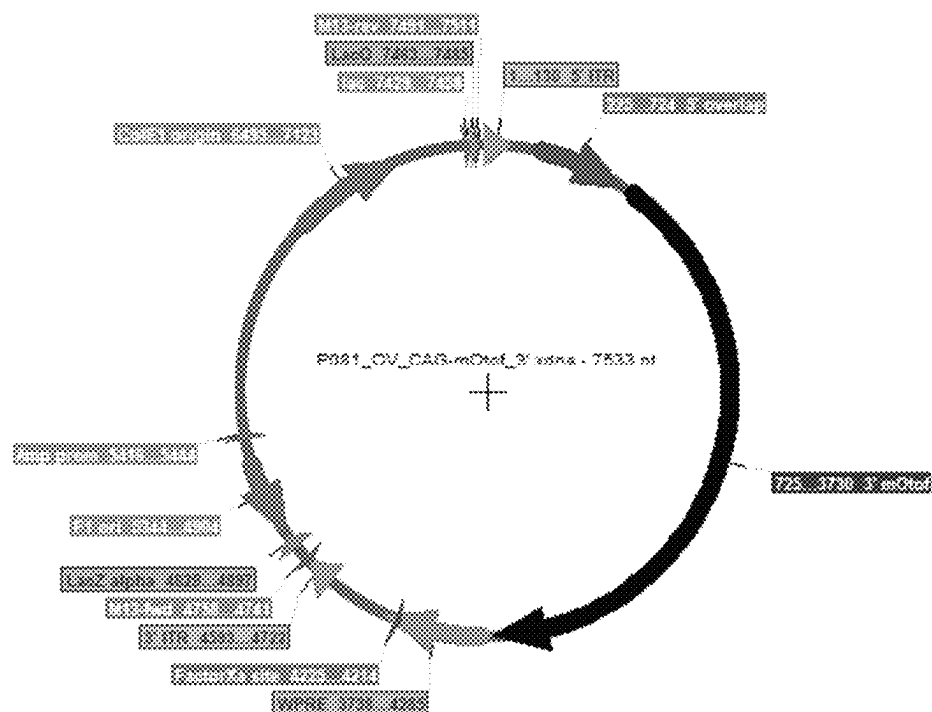


FIG. 6A

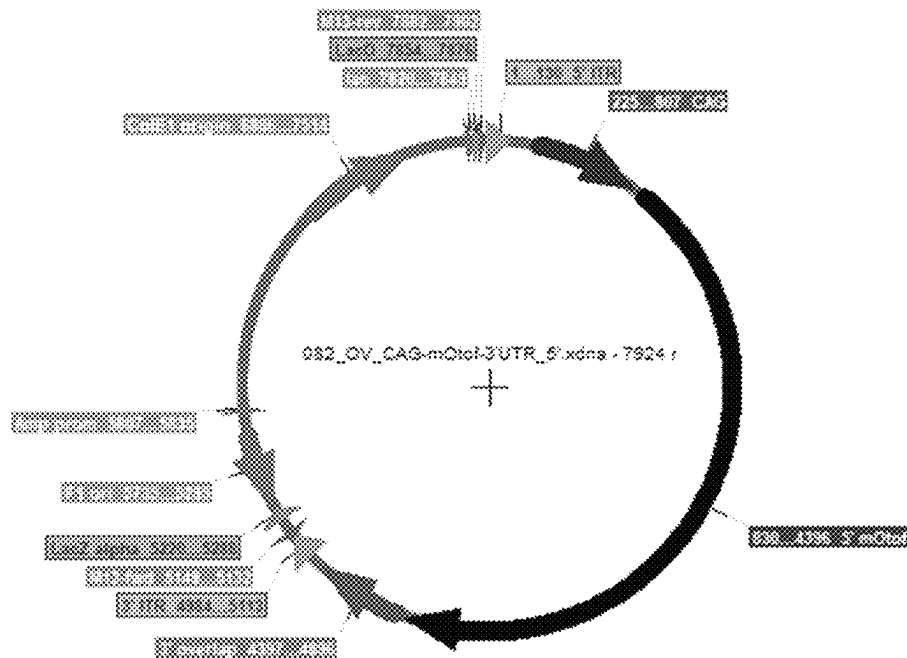


FIG. 6B

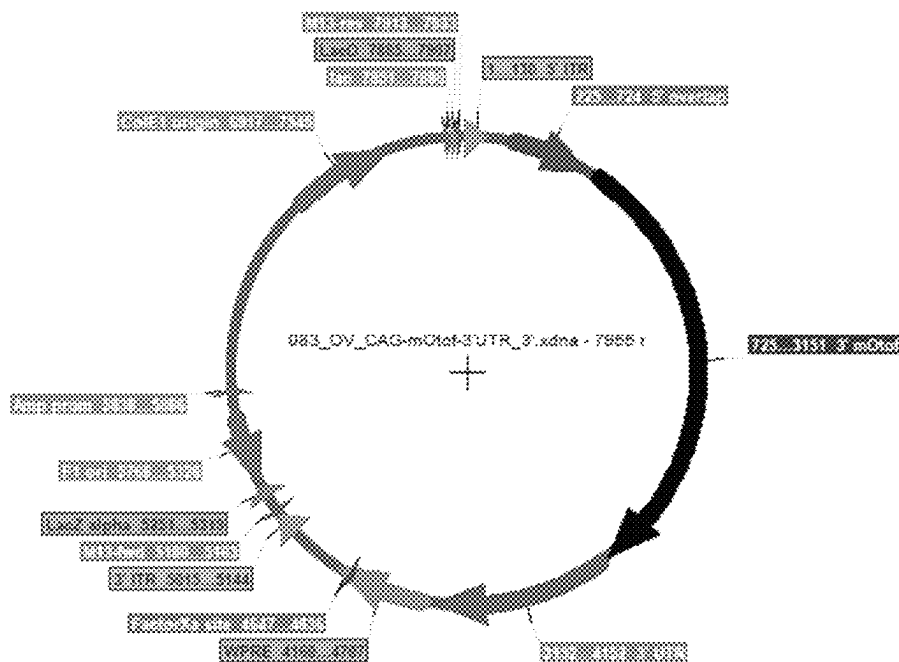


FIG. 8A

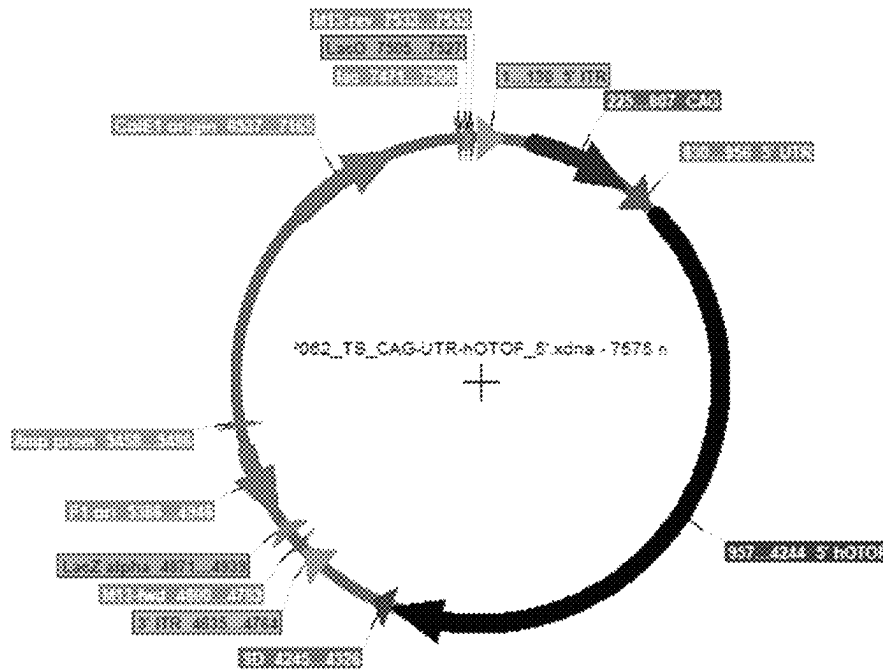


FIG. 8B

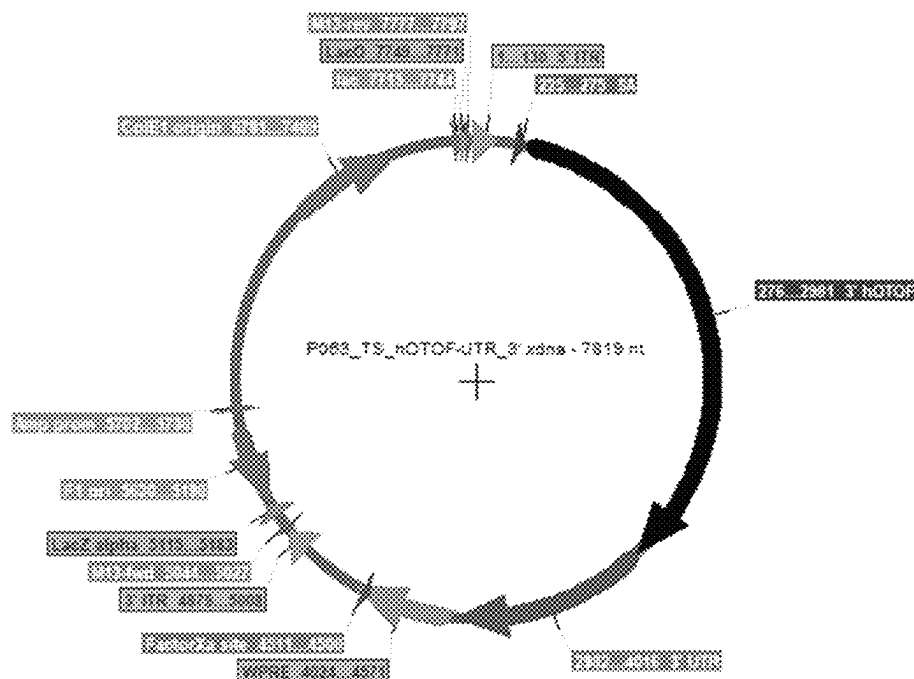


FIG. 9A

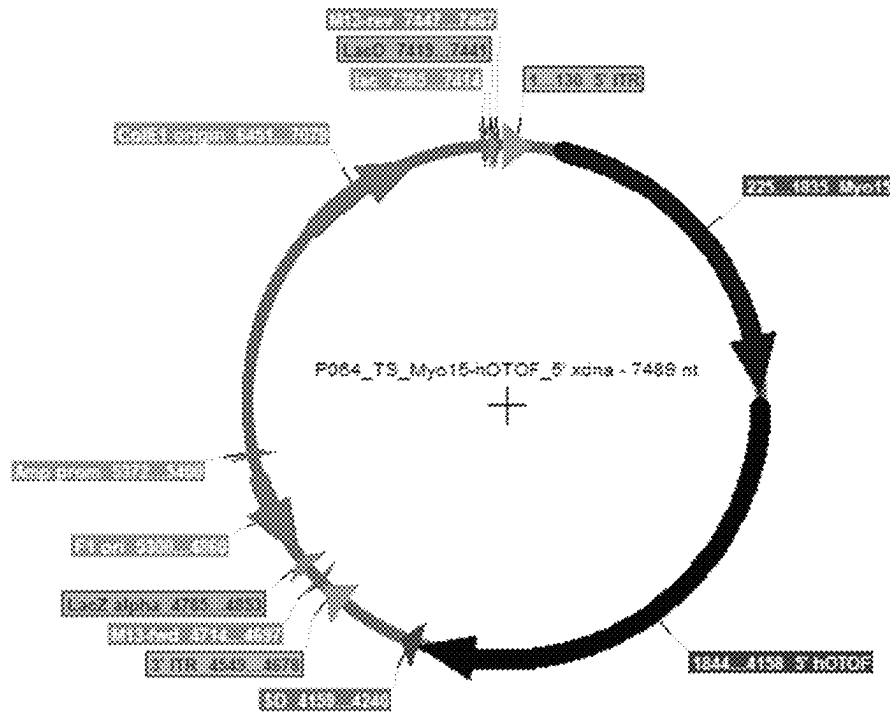


FIG. 10A

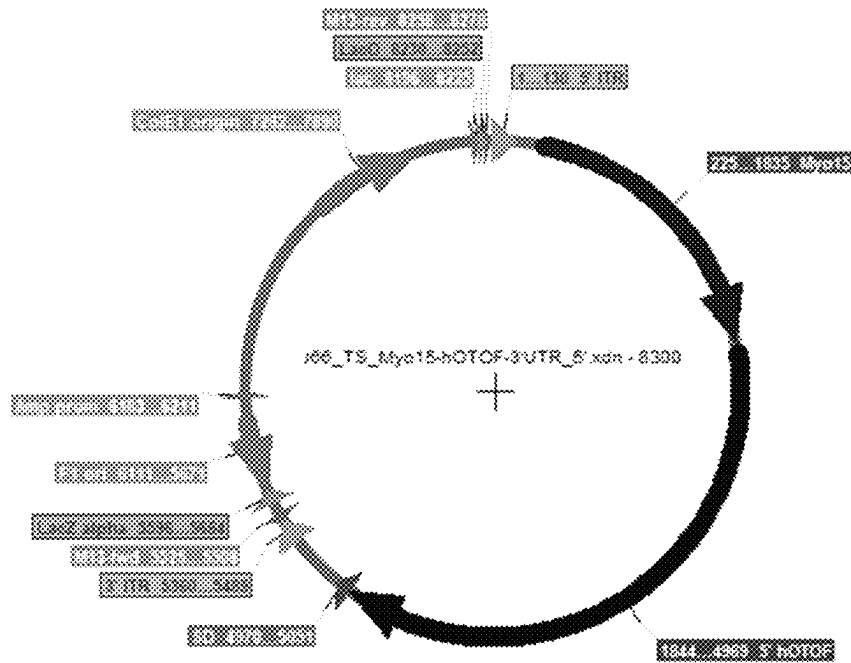


FIG. 10B

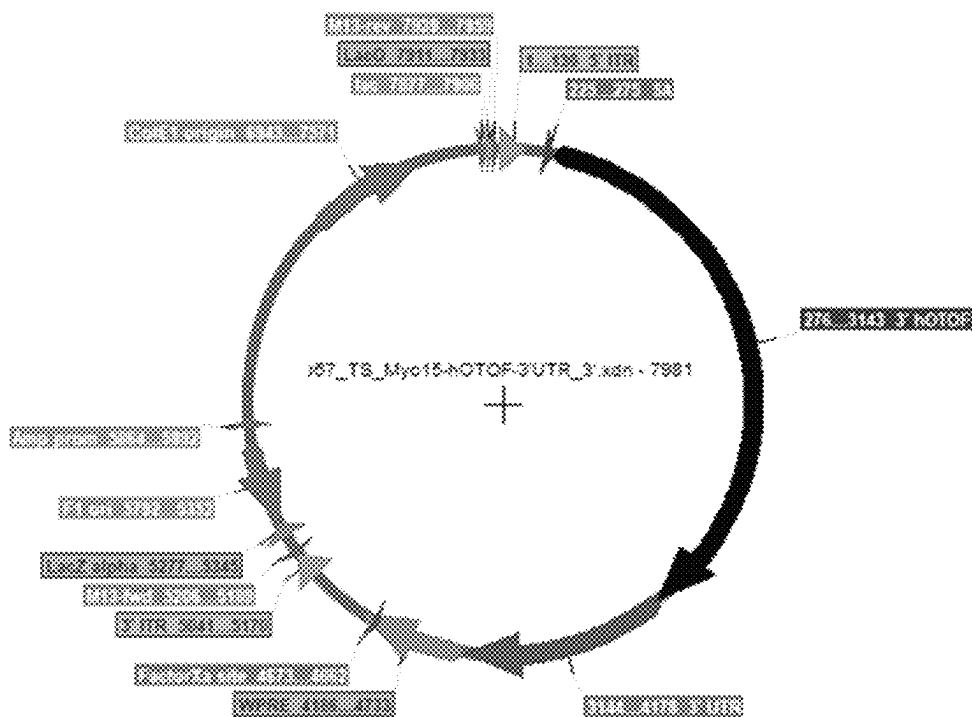


FIG. 11A

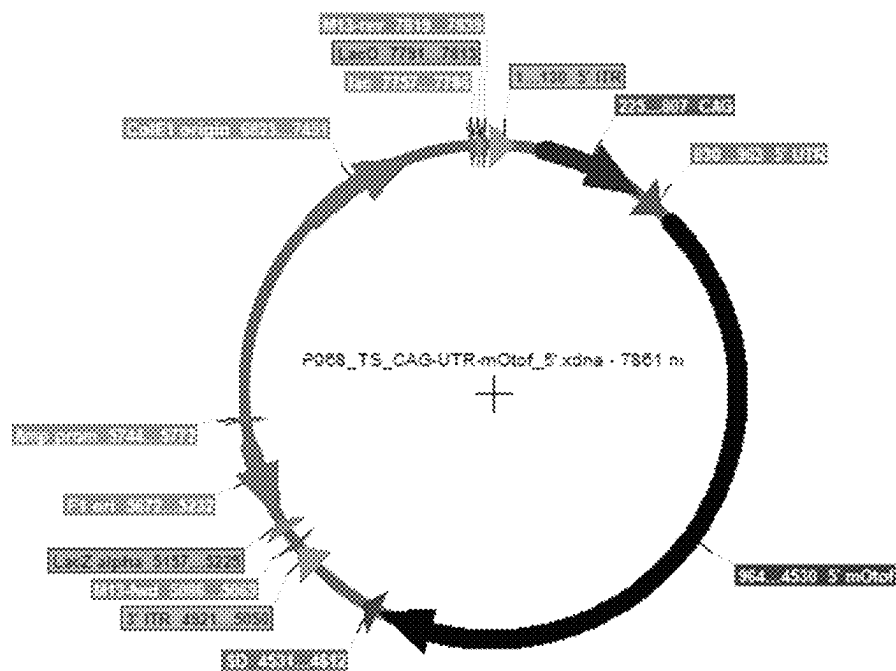


FIG. 11B

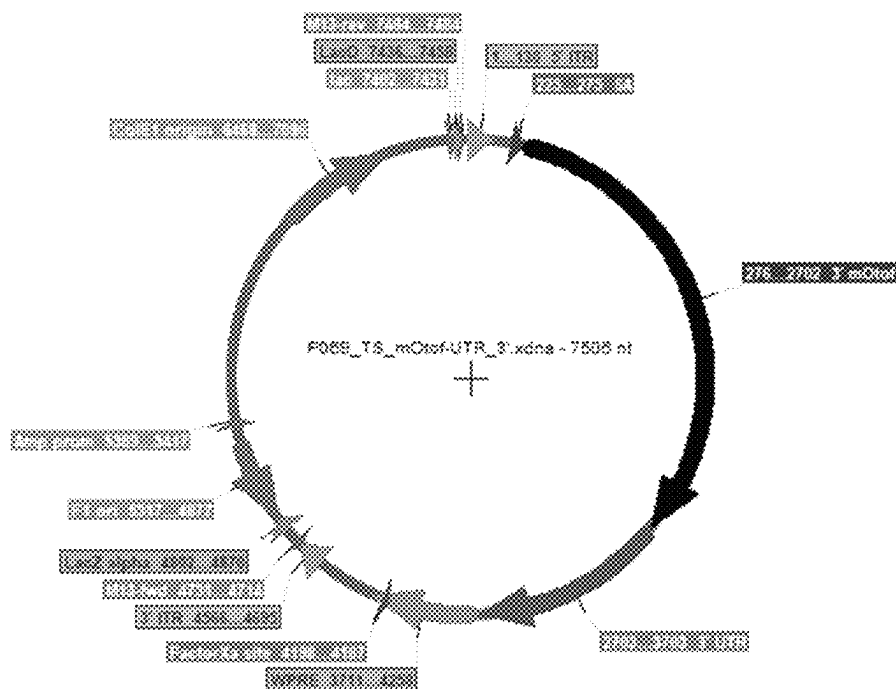


FIG. 12A

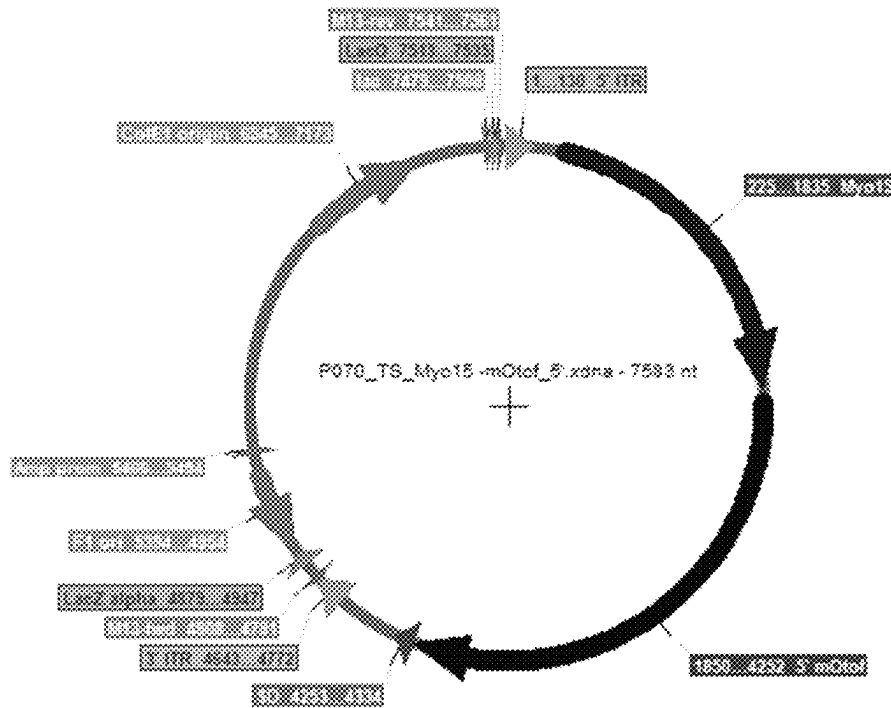


FIG. 13A

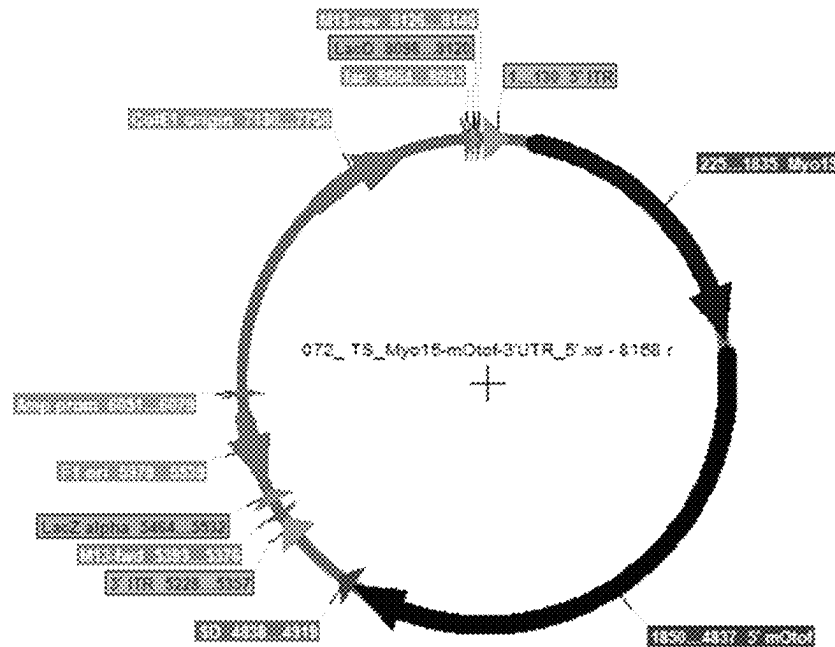


FIG. 13B

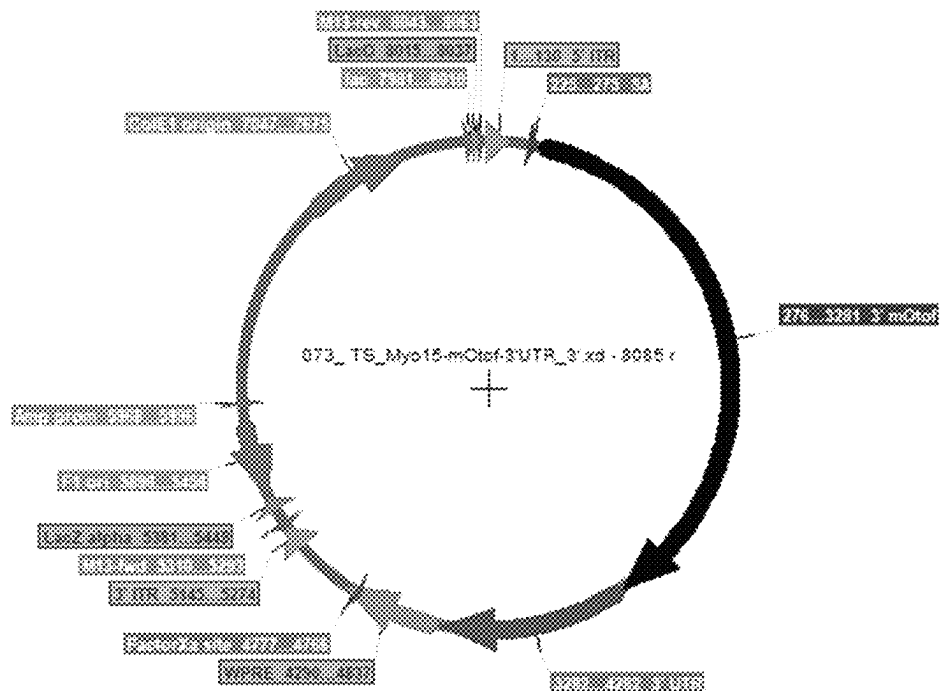


FIG. 14A

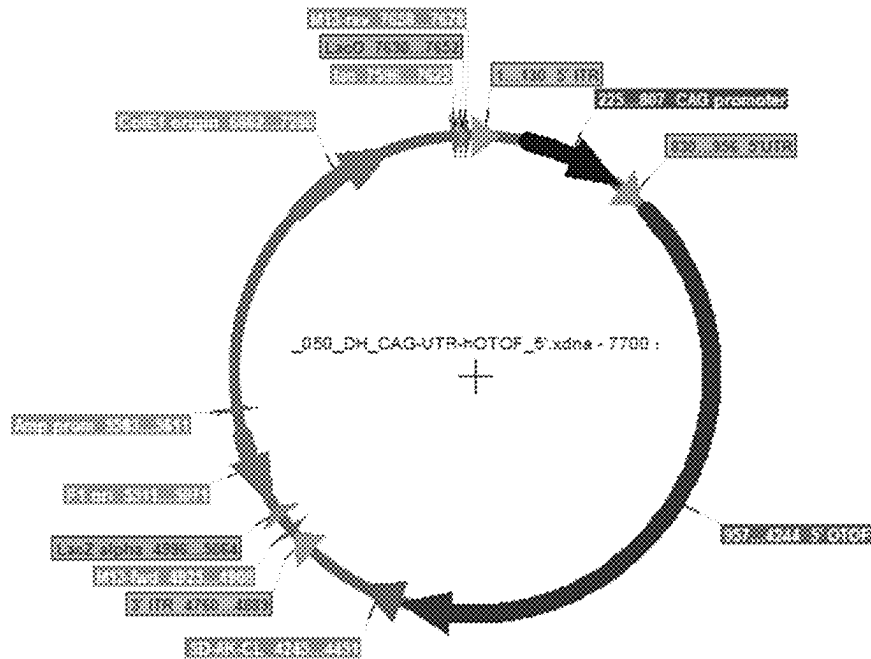


FIG. 15A

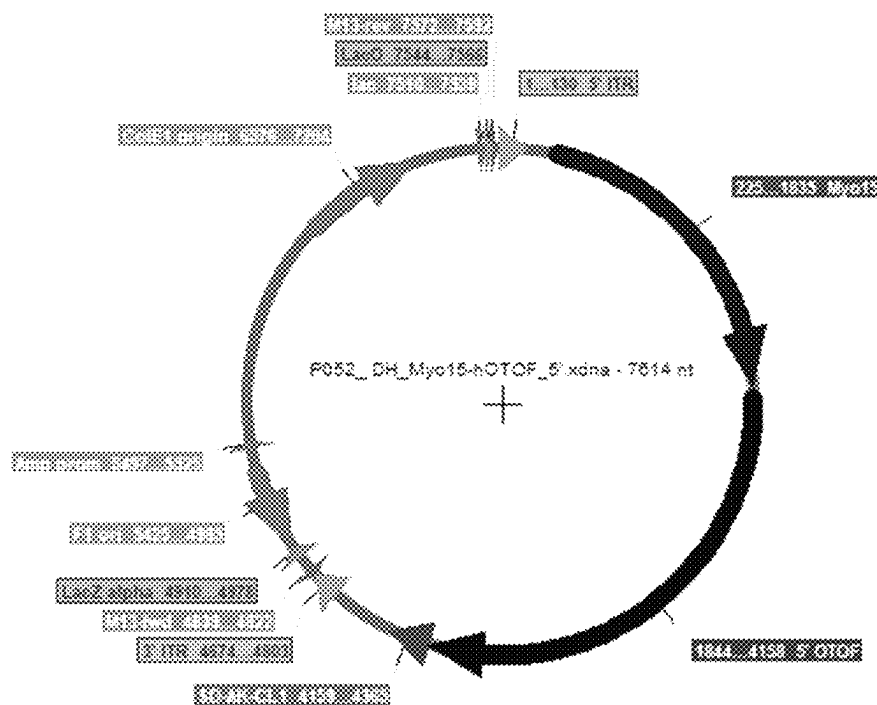


FIG. 15B

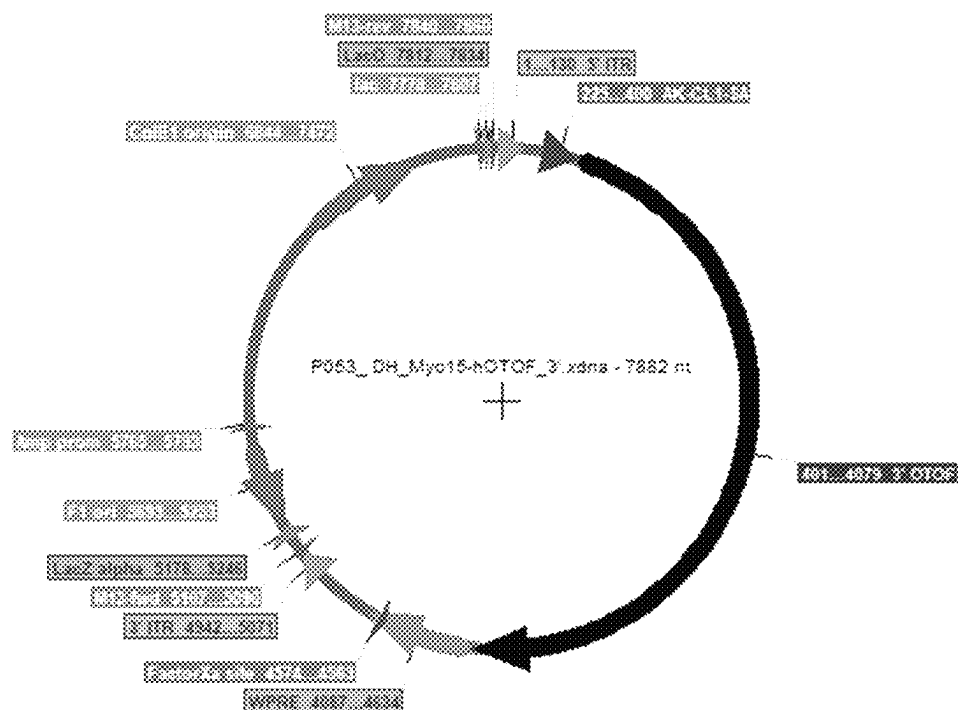


FIG. 16A

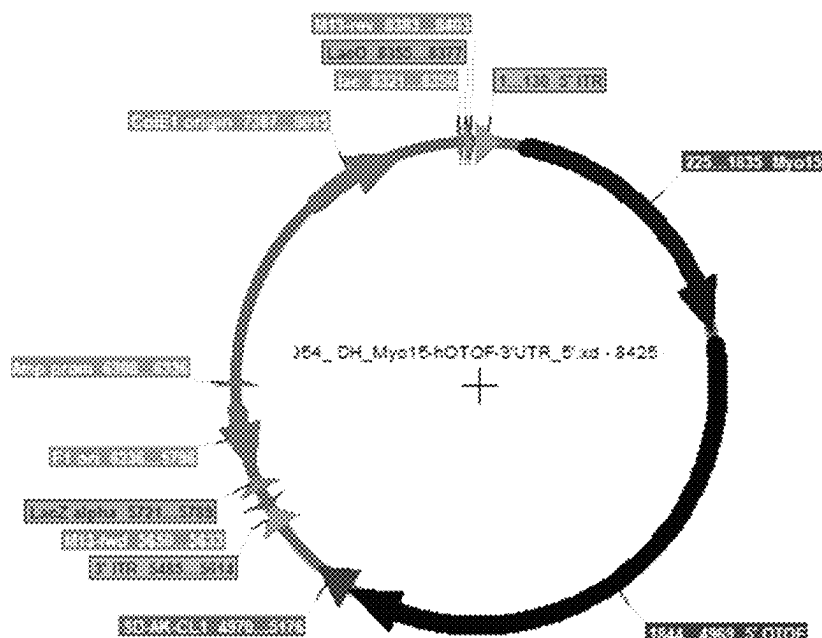


FIG. 17A

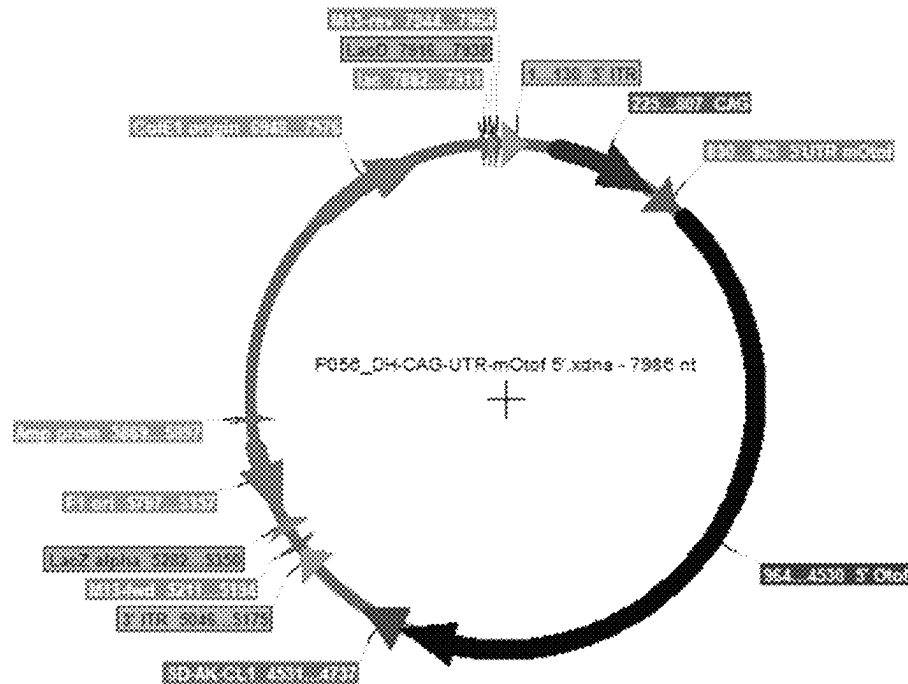


FIG. 17B

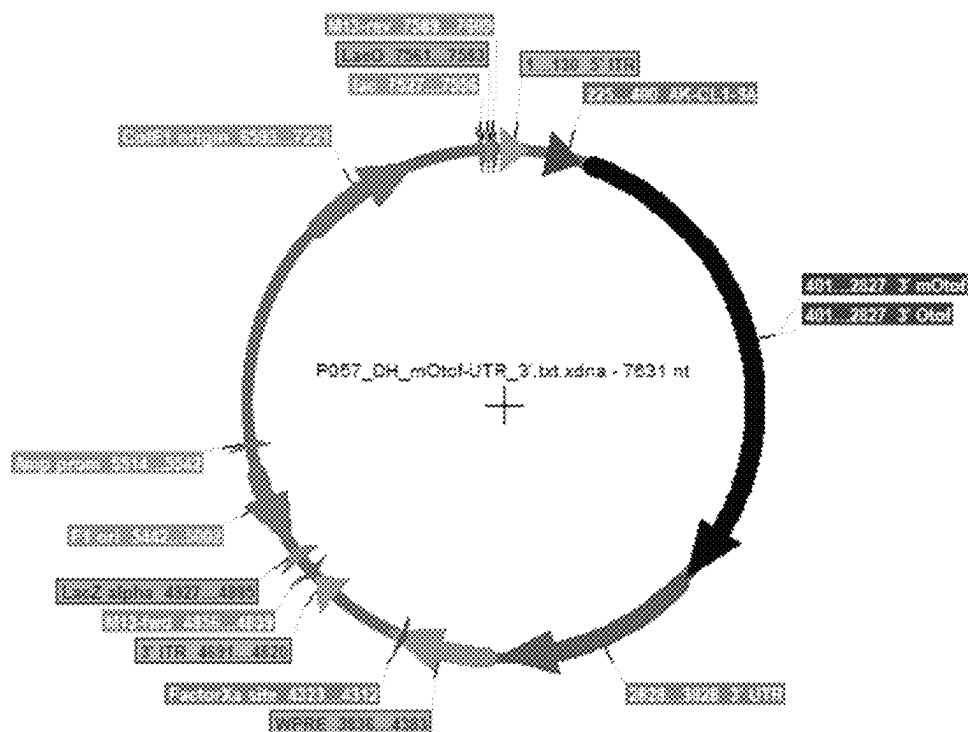


FIG. 19A

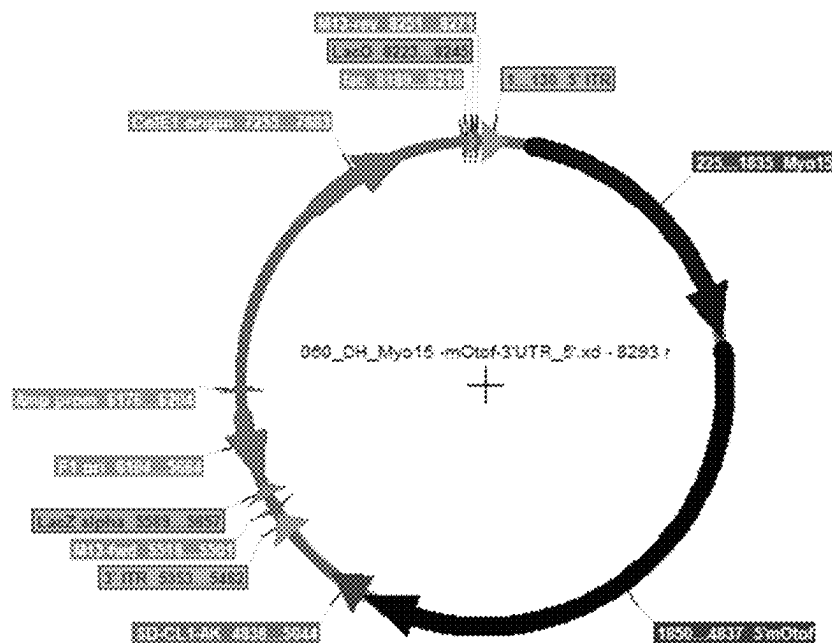


FIG. 19B

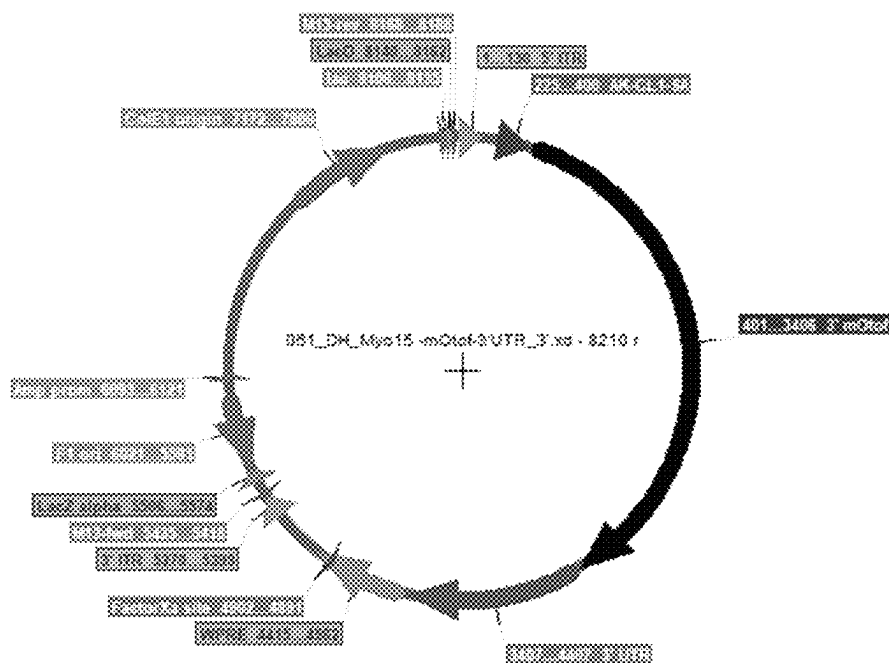
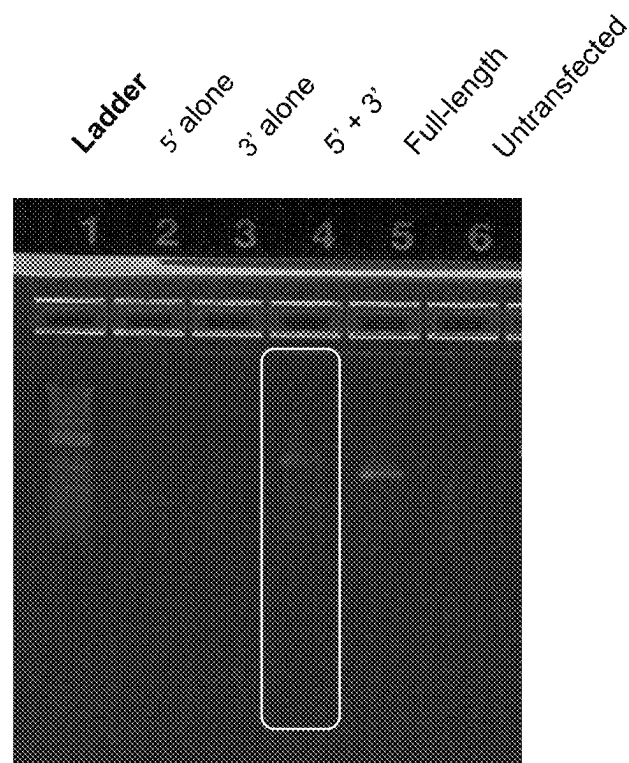


FIG. 20A

Dual hybrid vector



Dual hybrid: Myo15-hOTOF

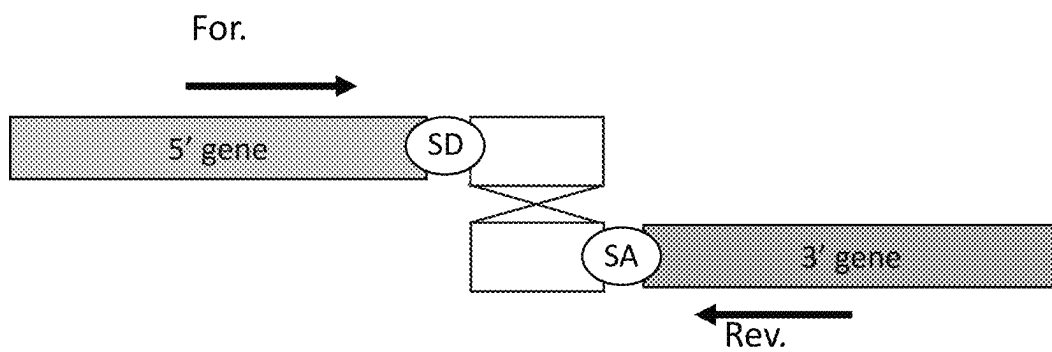


FIG. 20B

Overlapping vector

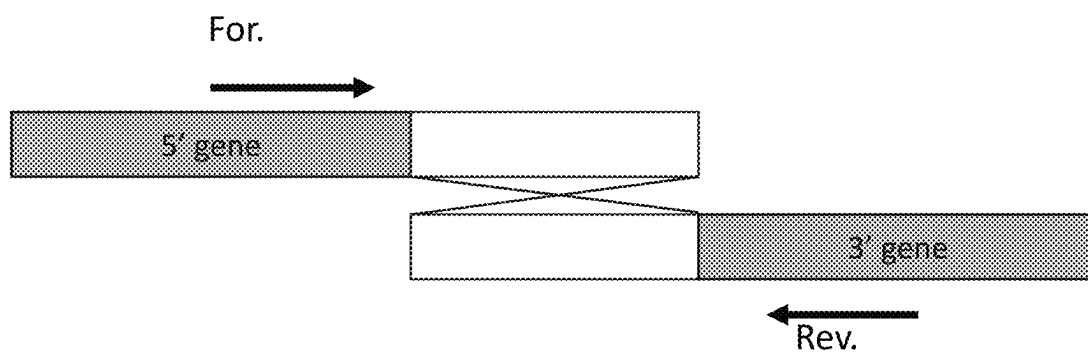
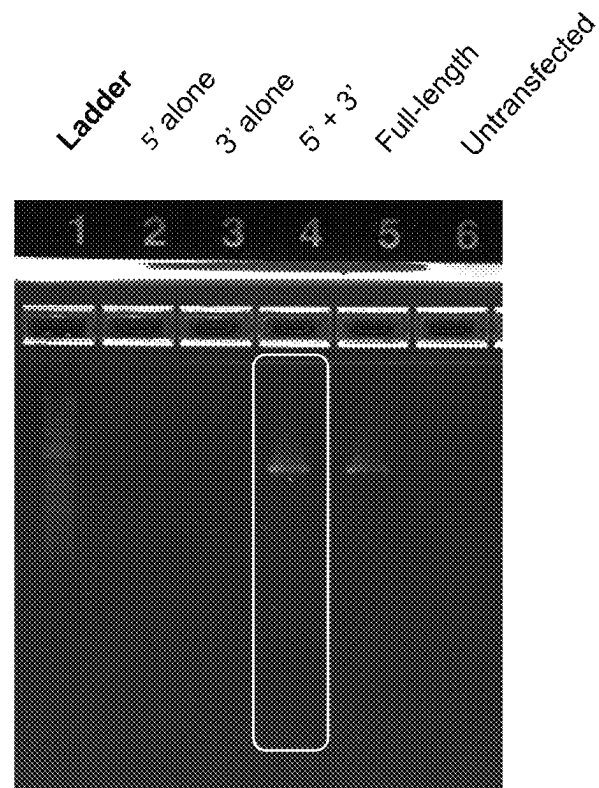


FIG. 21A

Dual hybrid vector

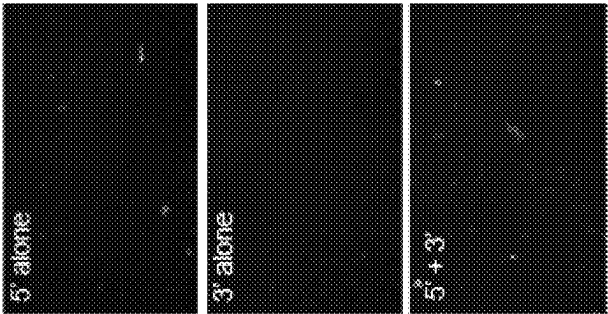


FIG. 21B

Trans-splicing vector

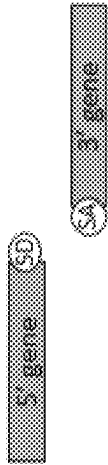
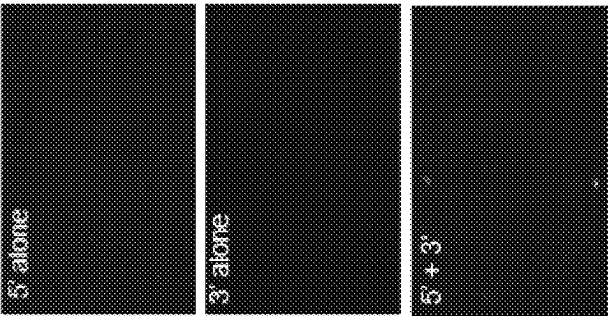


FIG. 21C

Overlapping vector

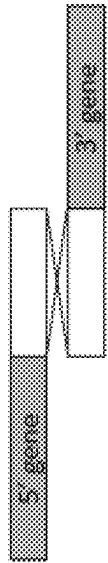
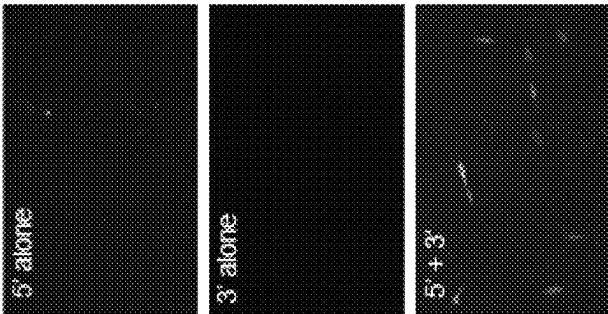


FIG. 22A

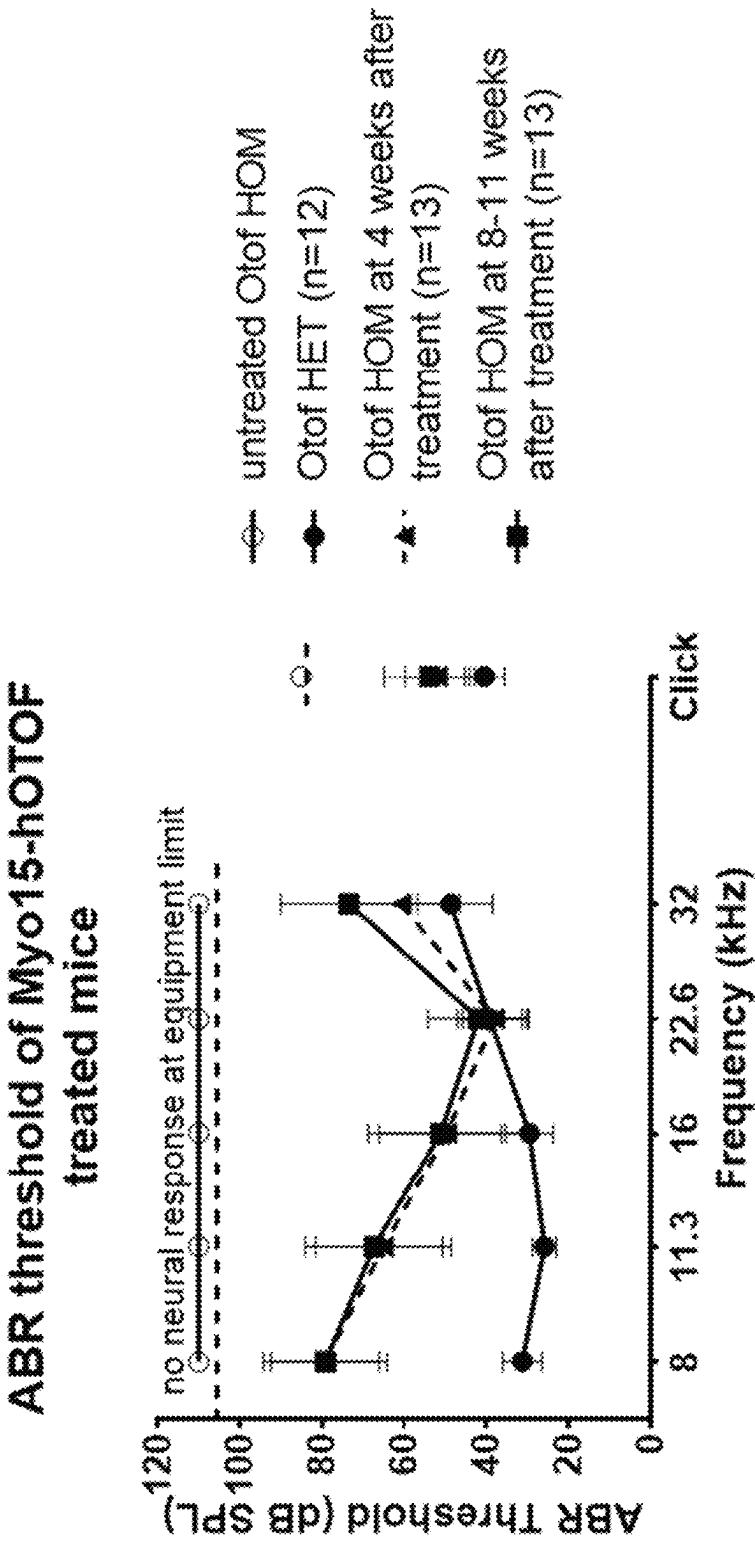


FIG. 22B

ABR threshold of smCBA-hOTOF
treated mice

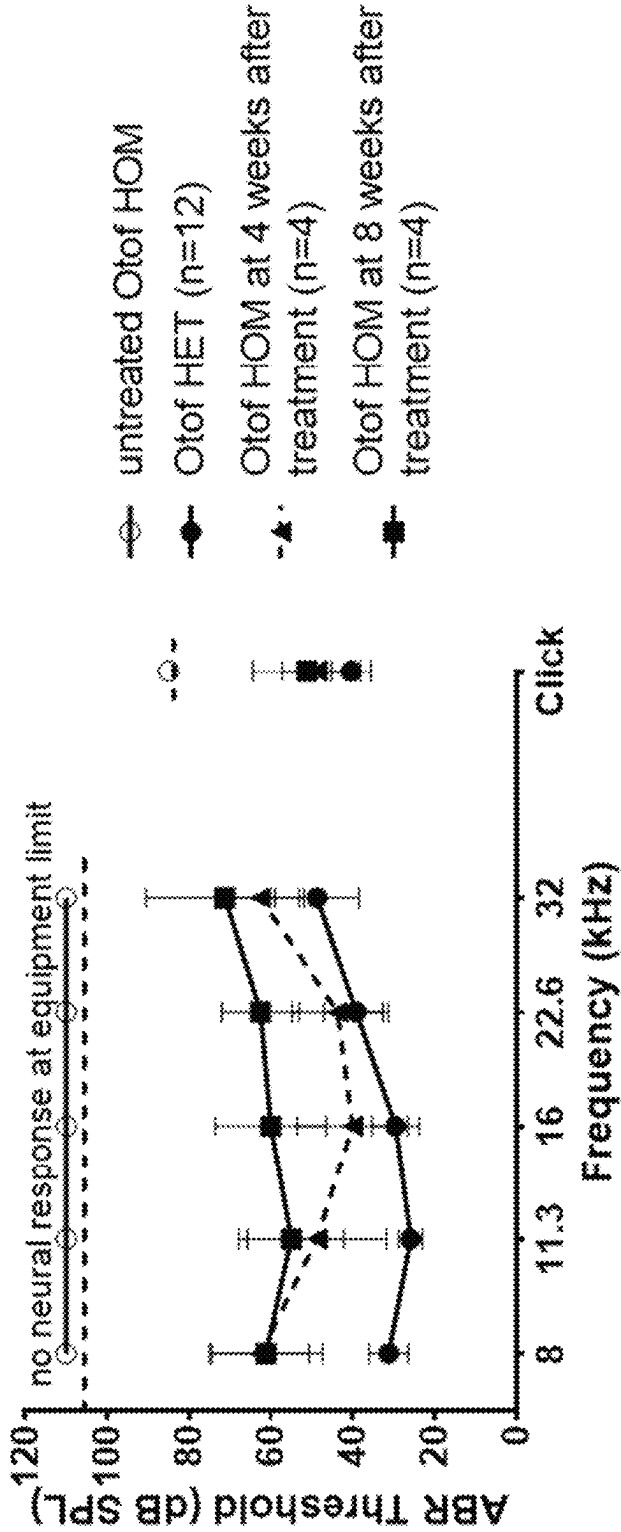
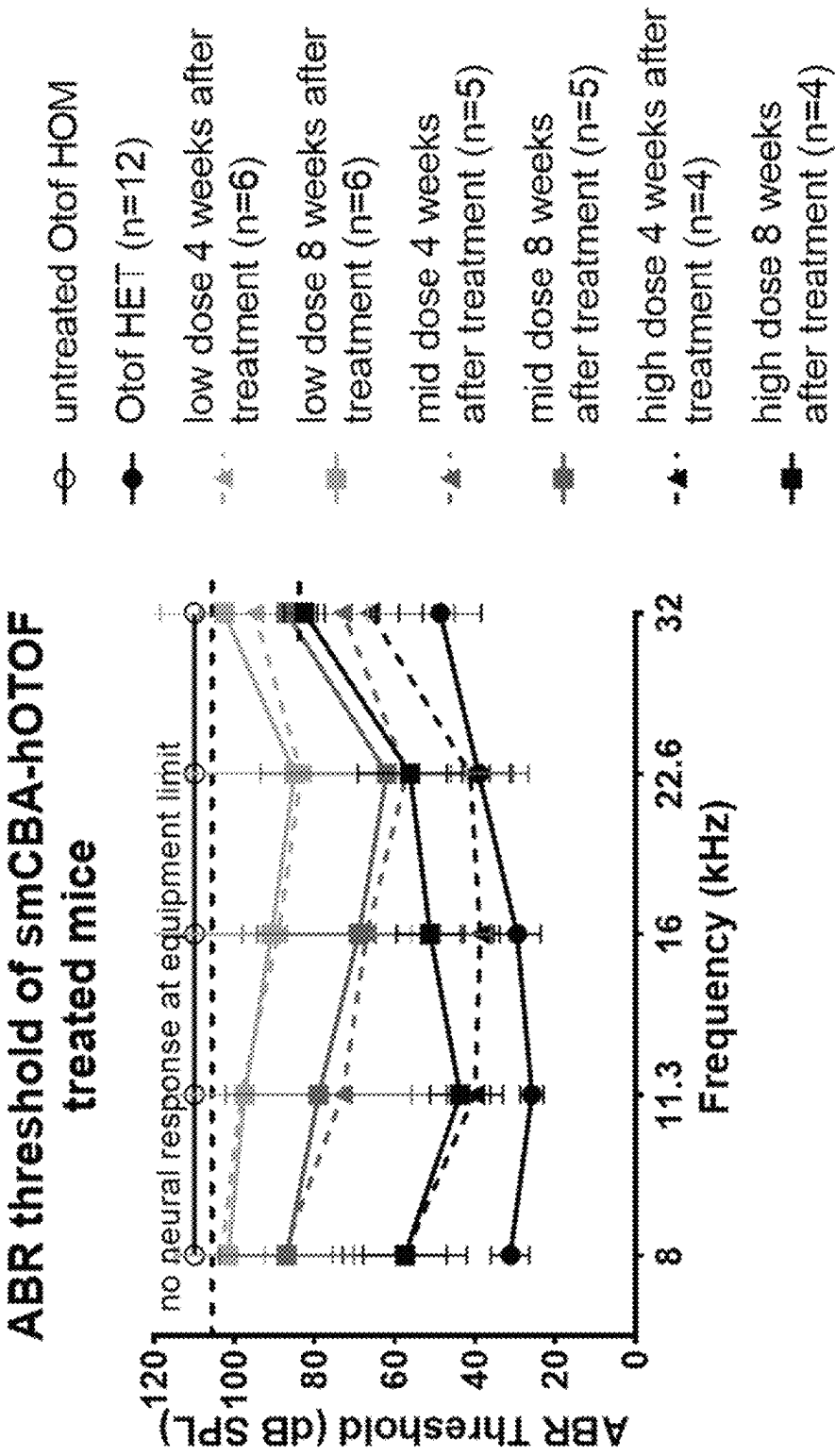


FIG. 22C



1

COMPOSITIONS AND METHODS FOR TREATING SENSORINEURAL HEARING LOSS USING OTOFERLIN DUAL VECTOR SYSTEMS

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created Aug. 4, 2021, is named 51471-003005_Sequence_Listing_08_04_21_ST25, and is 246,168 bytes in size.

FIELD OF THE INVENTION

Described herein are compositions and methods for the treatment of sensorineural hearing loss and auditory neuropathy, particularly forms of the disease that are associated with mutations in otoferlin (OTOF), by way of OTOF gene therapy. The disclosure provides dual vector systems that include a first nucleic acid vector that contains a polynucleotide encoding an N-terminal portion of an OTOF protein and a second nucleic acid vector that contains a polynucleotide encoding a C-terminal portion of an OTOF protein. These vectors can be used to increase the expression of or provide wild-type OTOF to a subject, such as a human subject suffering from sensorineural hearing loss.

BACKGROUND

Sensorineural hearing loss is a type of hearing loss caused by defects in the cells of the inner ear or the neural pathways that project from the inner ear to the brain. Although sensorineural hearing loss is often acquired, and can be caused by noise, infections, head trauma, ototoxic drugs, or aging, there are also congenital forms of sensorineural hearing loss associated with autosomal recessive mutations. One such form of autosomal recessive sensorineural hearing loss is associated with mutation of the otoferlin (OTOF) gene, which is implicated in prelingual nonsyndromic hearing loss. In recent years, efforts to treat hearing loss have increasingly focused on gene therapy as a possible solution; however, OTOF is too large to allow for treatment using standard gene therapy approaches. There is a need for new therapeutics to treat OTOF-related sensorineural hearing loss.

SUMMARY OF THE INVENTION

The present invention provides compositions and methods for treating sensorineural hearing loss or auditory neuropathy in a subject, such as a human subject. The compositions and methods of the disclosure pertain to dual vector systems for the delivery of a polynucleotide encoding an otoferlin (OTOF) protein to a subject having or at risk of developing sensorineural hearing loss or auditory neuropathy (e.g., a subject with a mutation in OTOF). For example, using the compositions and methods described herein, a first nucleic acid vector and a second nucleic acid vector that each encode a portion of a functional OTOF protein may be delivered to a subject by way of viral gene therapy. The compositions and methods described herein may also be used to increase expression of a WT OTOF protein in a cochlear hair cell (e.g., an inner hair cell).

In a first aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter

2

selected from the group consisting of a myosin 15 (Myo15) promoter, a vesicular glutamate transporter 3 (VGLUT3) promoter, and a fibroblast growth factor 8 (FGF8) promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an otoferlin (OTOF) protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombinant nucleic acid that encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a myosin 15 (Myo15) promoter, a vesicular glutamate transporter 3 (VGLUT3) promoter, and a fibroblast growth factor 8 (FGF8) promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an otoferlin (OTOF) protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombinant nucleic acid that encodes a full-length OTOF protein.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a cytomegalovirus (CMV) promoter, a truncated CMV-chicken β -actin promoter (smCBA promoter), a Myo15 promoter, a Myosin 7A (Myo7A) promoter, a Myosin 6 (Myo6) promoter, a POU Class 4 Homeobox 3 (POU4F3) promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombinant nucleic acid that encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a cytomegalovirus (CMV) promoter, a smCBA promoter, a Myo15 promoter, a Myosin 7A (Myo7A) promoter, a Myosin 6 (Myo6) promoter, a POU Class 4 Homeobox 3 (POU4F3) promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein; and a second nucleic acid vector

3

containing a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombined nucleic acid that encodes a full-length OTOF protein.

In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors are adeno-associated virus (AAV) vectors. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombined nucleic acid that encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombined nucleic acid that encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4,

4

AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombined nucleic acid that encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein; and a second nucleic acid vector containing a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein and a polyadenylation (poly(A)) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide partially overlaps with the second coding polynucleotide, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the two first and second nucleic acid vectors undergo homologous recombination to form a recombined nucleic acid that encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In some embodiments of any of the foregoing aspects, the first and second coding polynucleotides that encode the OTOF protein (e.g., the human OTOF protein) do not contain introns.

In some embodiments of any of the foregoing aspects, the OTOF protein is a mammalian OTOF protein.

In some embodiments of any of the foregoing aspects, the OTOF protein is a murine OTOF protein. In some embodiments of any of the foregoing aspects, the murine OTOF protein has at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity) to the sequence of SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, or SEQ ID NO: 9. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 6.

In some embodiments of any of the foregoing aspects, the OTOF protein is a human OTOF protein. In some embodiments of any of the foregoing aspects, the human OTOF protein has at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity) to the sequence of SEQ

ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, or SEQ ID NO: 5. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 1. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 2. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 3. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 4. In some embodiments of any of the foregoing aspects, the OTOF protein comprises or consists of the sequence of SEQ ID NO: 5.

In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV1. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV6. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80L65. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is DJ/9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is 7m8. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2quad(Y-F). In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is PHP.B. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV8. In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors have the same serotype (e.g., both the first and second nucleic acid vector are AAV vectors having an AAV1 serotype or an AAV9 serotype). In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors have different serotypes (e.g., the first nucleic acid vector is an AAV having an AAV1 serotype, and the second nucleic acid vector is an AAV having an AAV9 serotype).

In some embodiments of any of the foregoing aspects, each of the first and second coding polynucleotides encode about half of the OTOF protein sequence.

In some embodiments of any of the foregoing aspects, wherein the first coding polynucleotide overlaps with the second coding polynucleotide by at least 1 kilobase (kb).

In some embodiments of any of the foregoing aspects, the region of overlap between the first and second coding polynucleotides is centered at an OTOF exon boundary. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes an N-terminal portion of the OTOF protein containing the OTOF N-terminus to 500 kb 3' of the exon boundary at the center of the overlap region; and the second coding polynucleotide encodes a C-terminal portion of the OTOF protein containing 500 kb 5' of the exon boundary at the center of the overlap region to the OTOF C-terminus.

In some embodiments of any of the foregoing aspects, the exon boundary at the center of the overlap region is not within a portion of the first coding polynucleotide or second coding polynucleotide that encodes a C2 domain.

In some embodiments of any of the foregoing aspects, the promoter is a Myo15 promoter.

In some embodiments of any of the foregoing aspects, the promoter is a long promoter (e.g., a promoter that is longer than 1 kb, e.g., 1.1 kb, 1.25 kb, 1.5 kb, 1.75 kb, 2 kb, 2.5 kb,

3 kb or longer). In some embodiments, the long promoter is a Myo15 promoter that is longer than 1 kb (e.g., a Myo15 promoter comprising or consisting of a sequence with at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the sequence of SEQ ID NO: 36).

In some embodiments of any of the foregoing aspects, the exon boundary is selected such that the first coding polynucleotide encodes the entire C2C domain and the second coding polynucleotide encodes the entire C2D domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide contains exons 1-21 of a polynucleotide encoding the OTOF protein and 500 kb 3' of the exon 21/22 boundary; and the second coding polynucleotide contains 500 kb 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first nucleic acid vector and the second nucleic acid vector do not contain OTOF untranslated regions (UTRs).

In some embodiments of any of the foregoing aspects, the promoter is a short promoter (e.g., a promoter that is 1 kb or shorter, e.g., approximately 1 kb, 950 bp, 900 bp, 850 bp, 800 bp, 750 bp, 700 bp, 650 bp, 600 bp, 550 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp or shorter). In some embodiments, the short promoter is a CAG promoter. In some embodiments, the short promoter is a CMV promoter. In some embodiments, the short promoter is a smCBA promoter. In some embodiments, the short promoter is a Myo15 promoter that is 1 kb or shorter (e.g., a Myo15 promoter having a sequence with at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to any one of SEQ ID NOs: 38, 39, or 49-60).

In some embodiments of any of the foregoing aspects, the exon boundary is within a portion of the first coding polynucleotide and the second coding polynucleotide that encodes the C2D domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide contains exons 1-24 of a polynucleotide encoding the OTOF protein and 500 kb 3' of the exon 24/25 boundary; and the second coding polynucleotide contains 500 kb 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors include OTOF UTRs (e.g., full-length 3' and 5' UTRs).

In some embodiments of any of the foregoing aspects, the exon boundary is selected such that the first coding polynucleotide encodes the entire C2D domain and the second coding polynucleotide encodes the entire C2E domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide contains exons 1-28 of a polynucleotide encoding the OTOF protein and 500 kb 3' of the exon 28/29 boundary; and the second coding polynucleotide contains 500 kb 5' of the exon 28/29 boundary and exons 29-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the second nucleic acid vector contains a full-length OTOF 3' UTR.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a Myo15 promoter, a VGLUT3 promoter, and an FGF8 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein

positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a Myo15 promoter, a VGLUT3 promoter, and an FGF8 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a Myo15 promoter, a VGLUT3 promoter, and an FGF8 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a Myo15 promoter, a VGLUT3 promoter, and an FGF8 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide

that encodes an N-terminal portion of a human OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors are adeno-associated virus (AAV) vectors. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG

promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, and a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide; and a second nucleic acid vector containing a splice acceptor signal sequence, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter,

an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, and wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein.

In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors are AAV vectors. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein

the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter selected from the group consisting of a CAG promoter, a CMV promoter, a smCBA promoter, a Myo15 promoter, a Myo7A promoter, a Myo6 promoter, a POU4F3 promoter, an OTOF promoter, an FGF8 promoter, and a VGLUT3 promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a composition comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of a human OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In another aspect, the invention provides a dual vector system comprising: a first nucleic acid vector containing a promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of a human OTOF protein, a splice donor signal sequence positioned at the 3' end of the first coding polynucleotide, and a recombinogenic region positioned 3' of the splice donor signal sequence; and a second nucleic acid vector containing a second recombinogenic region, a splice acceptor signal sequence positioned 3' of the second recombinogenic region, a second coding polynucleotide that encodes a C-terminal portion of a human

OTOF protein positioned at the 3' end of the splice acceptor signal sequence, and a poly(A) sequence positioned at the 3' end of the second coding polynucleotide; wherein the first coding polynucleotide and the second coding polynucleotide do not overlap, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and wherein the first and second nucleic acid vectors are AAV vectors having a serotype selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S.

In some embodiments of any of the foregoing aspects, the first and second coding polynucleotides that encode the OTOF protein (e.g., the human OTOF protein) do not contain introns.

In some embodiments of any of the foregoing aspects, the OTOF protein is a mammalian OTOF protein.

In some embodiments of any of the foregoing aspects, the OTOF protein is a murine OTOF protein. In some embodiments of any of the foregoing aspects, the murine OTOF protein has at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity) to the sequence of SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, or SEQ ID NO: 9. In some embodiments of any of the foregoing aspects, the murine OTOF protein comprises or consists of the sequence of SEQ ID NO: 6.

In some embodiments of any of the foregoing aspects, the OTOF protein is a human OTOF protein. In some embodiments of any of the foregoing aspects, the human OTOF protein has at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity) to the sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, or SEQ ID NO: 5. In some embodiments of any of the foregoing aspects, the human OTOF protein comprises or consists of the sequence of SEQ ID NO: 1. In some embodiments of any of the foregoing aspects, the human OTOF protein comprises or consists of the sequence of SEQ ID NO: 2. In some embodiments of any of the foregoing aspects, the human OTOF protein comprises or consists of the sequence of SEQ ID NO: 3. In some embodiments of any of the foregoing aspects, the human OTOF protein comprises or consists of the sequence of SEQ ID NO: 4. In some embodiments of any of the foregoing aspects, the human OTOF protein comprises or consists of the sequence of SEQ ID NO: 5.

In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV1. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV6. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80L65. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is DJ/9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is 7m8. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2quad(Y-F). In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is PHP.B. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV8. In some embodiments

of any of the foregoing aspects, the first and second nucleic acid vectors have the same serotype (e.g., both the first and second nucleic acid vector are AAV vectors having an AAV1 serotype or an AAV9 serotype). In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors have different serotypes (e.g., the first nucleic acid vector is an AAV having an AAV1 serotype, and the second nucleic acid vector is an AAV having an AAV9 serotype).

In some embodiments, of any of the foregoing aspects, the first and second recombinogenic regions are the same. In some embodiments, of any of the foregoing aspects, the recombinogenic region is an AP gene fragment or an F1 phage AK gene. In some embodiments of any of the foregoing aspects, the recombinogenic region is an F1 phage AK gene. In some embodiments of any of the foregoing aspects, the F1 phage AK gene comprises or consists of the sequence of SEQ ID NO: 19. In some embodiments of any of the foregoing aspects, the recombinogenic region is an AP gene fragment. In some embodiments of any of the foregoing aspects, the AP gene fragment comprises or consists of the sequence of any one of SEQ ID NOs: 62-67.

In some embodiments of any of the foregoing aspects, the first nucleic acid vector further includes a degradation signal sequence positioned 3' of the recombinogenic region; and the second nucleic acid vector further includes a degradation signal sequence positioned between the recombinogenic region and the splice acceptor signal sequence. In some embodiments of any of the foregoing aspects, the degradation signal sequence comprises or consists of the sequence of SEQ ID NO: 22.

In some embodiments of any of the foregoing aspects, each of the first and second coding polynucleotides encode about half of the OTOF protein sequence.

In some embodiments of any of the foregoing aspects, the division between the first and second coding polynucleotides is at an OTOF exon boundary.

In some embodiments of any of the foregoing aspects, the OTOF exon boundary is not within a portion of the first coding polynucleotide or second coding polynucleotide that encodes a C2 domain.

In some embodiments of any of the foregoing aspects, the promoter is a short promoter (e.g., a promoter that is 1 kb or shorter, e.g., approximately 1 kb, 950 bp, 900 bp, 850 bp, 800 bp, 750 bp, 700 bp, 650 bp, 600 bp, 550 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp or shorter). In some embodiments, the short promoter is a CAG promoter. In some embodiments, the short promoter is a CMV promoter. In some embodiments, the short promoter is a smCBA promoter. In some embodiments, the short promoter is a Myo15 promoter that is 1 kb or shorter (e.g., a Myo15 promoter having a sequence with at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to any one of SEQ ID NOs: 38, 39, or 49-60).

In some embodiments of any of the foregoing aspects, the exon boundary is selected such that the first coding polynucleotide encodes the entire C2D domain and the second coding polynucleotide encodes the entire C2E domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-26 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 27-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-28 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 29-48 of a polynucleotide encoding the OTOF protein. In

some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors contain OTOF UTRs (e.g., full length 3' and 5' UTRs).

In some embodiments of any of the foregoing aspects, the promoter is a Myo15 promoter.

In some embodiments of any of the foregoing aspects, the promoter is a long promoter (e.g., a promoter that is longer than 1 kb, e.g., 1.1 kb, 1.25 kb, 1.5 kb, 1.75 kb, 2 kb, 2.5 kb, 3 kb or longer). In some embodiments, the long promoter is a Myo15 promoter that is longer than 1 kb (e.g., a Myo15 promoter comprising or consisting of a sequence with at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the sequence of SEQ ID NO: 36).

In some embodiments of any of the foregoing aspects, the OTOF exon boundary is not within a portion of the first coding polynucleotide or second coding polynucleotide that encodes a C2 domain.

In some embodiments of any of the foregoing aspects, the exon boundary is selected such that the first coding polynucleotide encodes the entire C2C domain and the second coding polynucleotide encodes the entire C2D domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-19 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 20-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-20 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 21-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first nucleic acid vector and the second nucleic acid vector do not contain OTOF UTRs.

In some embodiments of any of the foregoing aspects, the exon boundary is within a portion of the first coding polynucleotide and the second coding polynucleotide that encodes the C2D domain. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-25 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 26-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the first coding polynucleotide encodes exons 1-24 of a polynucleotide encoding the OTOF protein and the second coding polynucleotide encodes exons 25-48 of a polynucleotide encoding the OTOF protein. In some embodiments of any of the foregoing aspects, the second nucleic acid vector contains a full-length OTOF 3' UTR.

In some embodiments of any of the foregoing aspects, the splice donor sequence comprises or consists of the sequence of SEQ ID NO: 20. In some embodiments of any of the foregoing aspects, the splice donor sequence comprises or consists of the sequence of SEQ ID NO: 68.

In some embodiments of any of the foregoing aspects, the splice acceptor sequence comprises or consists of the sequence of SEQ ID NO: 21. In some embodiments of any of the foregoing aspects, the splice acceptor sequence comprises or consists of the sequence of SEQ ID NO: 69.

In some embodiments, the first and second nucleic acid vectors contain inverted terminal repeats (ITRs). In some embodiments of any of the foregoing aspects, the ITRs are AAV2 ITRs.

In some embodiments of any of the foregoing aspects, the poly(A) sequence is a bovine growth hormone (bGH) poly(A) signal sequence.

15

In some embodiments of any of the foregoing aspects, the second nucleic acid vector contains a Woodchuck Hepatitis Virus Posttranscriptional Regulatory Element (WPRE). In some embodiments of any of the foregoing aspects, the WPRE comprises or consists of the sequence of SEQ ID NO: 23. In some embodiments of any of the foregoing aspects, the WPRE comprises or consists of the sequence of SEQ ID NO: 61.

In some embodiments of any of the foregoing aspects, the composition contains a pharmaceutically acceptable excipient.

In another aspect, the invention provides a kit containing a composition of the invention.

In another aspect, the invention provides a method of increasing OTOF expression in a subject in need thereof by administering to the subject a therapeutically effective amount of a composition of the invention.

In another aspect, the invention provides a method of treating a subject having or at risk of developing sensorineural hearing loss by administering to the subject a therapeutically effective amount of a composition of the invention.

In another aspect, the invention provides a method of treating a subject having or at risk of developing auditory neuropathy by administering to the subject a therapeutically effective amount of a composition of the invention.

In another aspect, the invention provides a method of increasing OTOF expression in a subject in need thereof by administering to the subject a therapeutically effective amount of a pair of nucleic acid vectors listed in Table 4.

In another aspect, the invention provides a method of treating a subject having or at risk of developing sensorineural hearing loss by administering to the subject a therapeutically effective amount of a pair of nucleic acid vectors listed in Table 4.

In another aspect, the invention provides a method of treating a subject having or at risk of developing auditory neuropathy by administering to the subject a therapeutically effective amount of a pair of nucleic acid vectors listed in Table 4.

In some embodiments of any of the foregoing aspects, the subject has a mutation in OTOF.

In some embodiments of any of the foregoing aspects, the subject has been identified as having a mutation in OTOF.

In some embodiments of any of the foregoing aspects, the method further includes the step of identifying the subject as having a mutation in OTOF prior to administering the composition or the pair of nucleic acid vectors.

In some embodiments of any of the foregoing aspects, the method further includes the step of evaluating the hearing of the subject prior to administering the composition or the pair of nucleic acid vectors.

In some embodiments of any of the foregoing aspects, the composition or the pair of nucleic acid vectors is administered locally to the ear. In some embodiments of any of the foregoing aspects, the nucleic acid vectors are administered concurrently. In some embodiments of any of the foregoing aspects, the nucleic acid vectors are administered sequentially.

In some embodiments of any of the foregoing aspects, the method increases OTOF expression in a cochlear hair cell. In some embodiments of any of the foregoing aspects, the cochlear hair cell is an inner hair cell.

In some embodiments of any of the foregoing aspects, the subject is a mammal. In some embodiments of any of the foregoing aspects, the subject is a human.

In some embodiments of any of the foregoing aspects, the method further includes the step of evaluating the hearing of

16

the subject after administering the composition or the pair of nucleic acid vectors. In some embodiments of any of the foregoing aspects, the method further includes the step of evaluating OTOF expression after administering the composition or the pair of nucleic acid vectors.

In some embodiments of any of the foregoing aspects, the composition or the pair of nucleic acid vectors increases OTOF expression in a cell (e.g., a cochlear hair cell), improves hearing (e.g., as assessed by standard tests, such as audiometry, auditory brainstem response (ABR), electrocochleography (ECOG), and otoacoustic emissions), prevents or reduces hearing loss, delays the development of hearing loss, slows the progression of hearing loss, improves speech discrimination, or improves hair cell function.

In some embodiments of any of the foregoing aspects, the composition or the pair of nucleic acid vectors is administered in an amount sufficient to increase OTOF expression in a cochlear hair cell, prevent or reduce hearing loss, delay the development of hearing loss, slow the progression of hearing loss, improve hearing (e.g., as assessed by standard tests, such as audiometry, ABR, ECOG, and otoacoustic emissions), improve speech discrimination, or improve hair cell function.

In another aspect, the invention provides a method of increasing OTOF expression in a cell by introducing a composition of the invention into the cell.

In another aspect, the invention provides a method of increasing OTOF expression in a cell by introducing a pair of nucleic acid vectors listed in Table 4 into the cell.

In some embodiments of any of the foregoing aspects, the cell is a cochlear hair cell. In some embodiments of any of the foregoing aspects, the cell is an inner hair cell.

In some embodiments of any of the foregoing aspects, the cell is a mammalian cell. In some embodiments of any of the foregoing aspects, the cell is a human cell.

In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors has at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity) to the sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, or SEQ ID NO: 5. In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors comprises or consists of the sequence of SEQ ID NO: 1. In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors comprises or consists of the sequence of SEQ ID NO: 2. In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors comprises or consists of the sequence of SEQ ID NO: 3. In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors comprises or consists of the sequence of SEQ ID NO: 4. In some embodiments of any of the foregoing aspects, the OTOF protein encoded by the pair of nucleic acid vectors comprises or consists of the sequence of SEQ ID NO: 5.

In some embodiments of any of the foregoing aspects, the nucleic acid vectors are AAV vectors. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is selected from the group consisting of AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, or 7m8. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV1. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is

AAV9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV6. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is Anc80L65. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is DJ/9. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is 7m8. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV2quad(Y-F). In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is PHP.B. In some embodiments of any of the foregoing aspects, the serotype of the AAV vectors is AAV8. In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors have the same serotype (e.g., both the first and second nucleic acid vector are AAV vectors having an AAV1 serotype or an AAV9 serotype). In some embodiments of any of the foregoing aspects, the first and second nucleic acid vectors have different serotypes (e.g., the first nucleic acid vector is an AAV having an AAV1 serotype, and the second nucleic acid vector is an AAV having an AAV9 serotype).

In some embodiments of any of the foregoing aspects, the vectors contain AAV2 ITRs.

In some embodiments of any of the foregoing aspects, the second nucleic acid vector in the pair of nucleic acid vectors contains a WPRE. In some embodiments of any of the foregoing aspects, the WPRE comprises or consists of the sequence of SEQ ID NO: 23. In some embodiments of any of the foregoing aspects, the WPRE comprises or consists of the sequence of SEQ ID NO: 61.

In some embodiments of any of the foregoing aspects, the nucleic acid vectors are overlapping dual vectors.

In some embodiments of any of the foregoing aspects, the nucleic acid vectors are trans-splicing dual vectors.

In some embodiments of any of the foregoing aspects, the nucleic acid vectors are dual hybrid vectors.

In some embodiments of any of the foregoing aspects, the recombinogenic region in the dual hybrid vectors is an AP gene fragment or an F1 phage AK gene. In some embodiments of any of the foregoing aspects, the F1 phage AK gene comprises or consists of the sequence of SEQ ID NO: 19. In some embodiments of any of the foregoing aspects, the AP gene fragment comprises or consists of the sequence of any one of SEQ ID NOs: 62-67. In some embodiments of any of the foregoing aspects, the first nucleic acid vector in the pair of nucleic acid vectors further contains a degradation signal sequence positioned 3' of the recombinogenic region; and the second nucleic acid vector in the pair of nucleic acid vectors further contains a degradation signal sequence positioned between the recombinogenic region and the splice acceptor sequence. In some embodiments of any of the foregoing aspects, the degradation signal sequence comprises or consists of the sequence of SEQ ID NO: 22.

In some embodiments of any of the foregoing aspects, the splice donor sequence in the first nucleic acid vector comprises or consists of the sequence of SEQ ID NO: 20. In some embodiments of any of the foregoing aspects, the splice donor sequence comprises or consists of the sequence of SEQ ID NO: 68.

In some embodiments of any of the foregoing aspects, the splice acceptor sequence in the second nucleic acid vector comprises or consists of the sequence of SEQ ID NO: 21. In some embodiments of any of the foregoing aspects, the

splice acceptor sequence comprises or consists of the sequence of SEQ ID NO: 69.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a first region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 24 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 26 and/or SEQ ID NO: 27, operably linked to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 25 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 31 and/or SEQ ID NO: 32, optionally containing a linker including one to one hundred nucleotides (e.g., 1-5, 1-10, 1-15, 1-20, 1-25, 1-30, 1-35, 1-40, 1-45, 1-50, 1-60, 1-70, 1-80, 1-90, 10-20, 10-30, 10-40, 10-50, 10-60, 10-70, 10-80, 10-90, 10-100, 20-30, 20-40, 20-50, 20-60, 20-70, 20-80, 20-90, or 20-100 nucleotides) between the first region and the second region. In some embodiments, the first region comprises or consists of the sequence of SEQ ID NO: 24. In some embodiments, the second region comprises or consists of the sequence of SEQ ID NO: 25.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 36. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 38. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 39. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 53. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 54. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 59. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 60.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a first region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 25 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 31 and/or SEQ ID NO: 32, operably linked to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 24 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 26 and/or SEQ ID NO: 27, optionally containing a linker including one to one hundred nucleotides (e.g., 1-5, 1-10, 1-15, 1-20, 1-25, 1-30, 1-35, 1-40, 1-45, 1-50, 1-60, 1-70, 1-80, 1-90, 10-20, 10-30, 10-40, 10-50, 10-60, 10-70, 10-80, 10-90, 10-100, 20-30, 20-40, 20-50, 20-60, 20-70, 20-80, 20-90, or 20-100 nucleotides) between the first region and the second region. In some embodiments, the first region comprises or consists of the sequence of SEQ ID NO: 25. In some embodiments, the second region comprises or consists of the sequence of SEQ ID NO: 24.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 37. In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 58.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 24 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 26 and/or SEQ ID NO: 27. In some embodiments, the region comprises or consists of the sequence of SEQ ID NO: 24.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 25 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 31 and/or SEQ ID NO: 32. In some embodiments, the region comprises or consists of the sequence of SEQ ID NO: 25.

In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 26. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 27. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 26 and the sequence of SEQ ID NO: 27. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 28. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 29. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 30. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 24 contains the sequence of SEQ ID NO: 50.

In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 31. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 32. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 33. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 34. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 35. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 25 contains the sequence of SEQ ID NO: 55.

In some embodiments of any of the foregoing aspects, the Myo15 promoter has at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence of any one of SEQ ID NOs: 50-58. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 50. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 51. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 52. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 53. In some embodiments, the Myo15 pro-

motor comprises or consists of the sequence of SEQ ID NO: 54. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 55. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 56. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 57. In some embodiments, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 58.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a first region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 40 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 42, joined (e.g., operably linked) to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 41 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 43 and/or SEQ ID NO: 44, optionally containing a linker including one to four hundred nucleotides (e.g., 1-5, 1-10, 1-15, 1-20, 1-25, 1-30, 1-35, 1-40, 1-45, 1-50, 1-60, 1-70, 1-80, 1-90, 1-100, 1-125, 1-150, 1-175, 1-200, 1-225, 1-250, 1-275, 1-300, 1-325, 1-350, 1-375, 1-400, 10-20, 10-30, 10-40, 10-50, 10-60, 10-70, 10-80, 10-90, 10-100, 20-30, 20-40, 20-50, 20-60, 20-70, 20-80, 20-90, 20-100, 30-100, 40-100, 50-100, 50-150, 50-200, 50-250, 50-300, 50-350, 50-400, 100-150, 100-200, 100-250, 100-300, 100-350, 100-400, 150-200, 150-250, 150-300, 150-350, 150-400, 200-250, 200-300, 200-350, 200-400, 250-300, 250-350, 250-400, 300-400, or 350-400 nucleotides) between the first region and the second region. In some embodiments, the first region comprises or consists of the sequence of SEQ ID NO: 40. In some embodiments, the second region comprises or consists of the sequence of SEQ ID NO: 41.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 48.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of the sequence of SEQ ID NO: 49.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a first region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 41 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 43 and/or SEQ ID NO: 44, joined (e.g., operably linked) to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 40 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 42, optionally containing a linker including one to four hundred nucleotides (e.g., 1-5, 1-10, 1-15, 1-20, 1-25, 1-30, 1-35, 1-40, 1-45, 1-50, 1-60, 1-70, 1-80, 1-90, 1-100, 1-125, 1-150, 1-175, 1-200, 1-225, 1-250, 1-275, 1-300, 1-325, 1-350, 1-375, 1-400, 10-20, 10-30, 10-40, 10-50, 10-60, 10-70, 10-80, 10-90, 10-100, 20-30, 20-40, 20-50, 20-60, 20-70, 20-80, 20-90, 20-100, 30-100, 40-100, 50-100, 50-150, 50-200, 50-250, 50-300, 50-350, 50-400, 100-150, 100-200, 100-250, 100-300, 100-350, 100-400, 150-200, 150-250, 150-300, 150-350, 150-400, 200-250, 200-300, 200-350, 200-400, 250-300, 250-350, 250-400, 300-400, or 350-400 nucleotides) between the first region and the second region. In some embodiments, the first region comprises or consists

21

of the sequence of SEQ ID NO: 41. In some embodiments, the second region comprises or consists of the sequence of SEQ ID NO: 40.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 40 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 42. In some embodiments, the region comprises or consists of the sequence of SEQ ID NO: 40.

In some embodiments of any of the foregoing aspects, the Myo15 promoter comprises or consists of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to SEQ ID NO: 41 or a functional portion or derivative thereof including the sequence of SEQ ID NO: 43 and/or SEQ ID NO: 44. In some embodiments, the region comprises or consists of the sequence of SEQ ID NO: 41.

In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 40 contains the sequence of SEQ ID NO: 42.

In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 43. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 44. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 44. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 43 and the sequence of SEQ ID NO: 44. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 45. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 46. In some embodiments of any of the foregoing aspects, the functional portion of SEQ ID NO: 41 contains the sequence of SEQ ID NO: 47.

In some embodiments of any of the foregoing aspects, the Myo15 promoter induces transgene expression when operably linked to a transgene and introduced into a hair cell.

Definitions

As used herein, the term “about” refers to a value that is within 10% above or below the value being described.

As used herein, “administration” refers to providing or giving a subject a therapeutic agent (e.g., a composition containing a first nucleic acid vector containing a polynucleotide that encodes an N-terminal portion of an otoferlin protein and a second nucleic acid vector containing a polynucleotide that encodes a C-terminal portion of an otoferlin protein), by any effective route. Exemplary routes of administration are described herein below.

As used herein, the term “cell type” refers to a group of cells sharing a phenotype that is statistically separable based on gene expression data. For instance, cells of a common cell type may share similar structural and/or functional characteristics, such as similar gene activation patterns and antigen presentation profiles. Cells of a common cell type may include those that are isolated from a common tissue (e.g., epithelial tissue, neural tissue, connective tissue, or muscle tissue) and/or those that are isolated from a common organ, tissue system, blood vessel, or other structure and/or region in an organism.

As used herein, the term “cochlear hair cell” refers to group of specialized cells in the inner ear that are involved in sensing sound. There are two types of cochlear hair cells:

22

inner hair cells and outer hair cells. Damage to cochlear hair cells and genetic mutations that disrupt cochlear hair cell function are implicated in hearing loss and deafness.

As used herein, the terms “conservative mutation,” “conservative substitution,” and “conservative amino acid substitution” refer to a substitution of one or more amino acids for one or more different amino acids that exhibit similar physicochemical properties, such as polarity, electrostatic charge, and steric volume. These properties are summarized for each of the twenty naturally-occurring amino acids in table 1 below.

TABLE 1

Representative physicochemical properties of naturally-occurring amino acids					
Amino Acid	3 Letter Code	1 Letter Code	Side-chain Polarity	Electrostatic character at physiological pH (7.4)	Steric Volume†
Alanine	Ala	A	nonpolar	neutral	small
Arginine	Arg	R	polar	cationic	large
Asparagine	Asn	N	polar	neutral	intermediate
Aspartic acid	Asp	D	polar	anionic	intermediate
Cysteine	Cys	C	nonpolar	neutral	intermediate
Glutamic acid	Glu	E	polar	anionic	intermediate
Glutamine	Gln	Q	polar	neutral	intermediate
Glycine	Gly	G	nonpolar	neutral	small
Histidine	His	H	polar	Both neutral and cationic forms in equilibrium at pH 7.4	large
Isoleucine	Ile	I	nonpolar	neutral	large
Leucine	Leu	L	nonpolar	neutral	large
Lysine	Lys	K	polar	cationic	large
Methionine	Met	M	nonpolar	neutral	large
Phenylalanine	Phe	F	nonpolar	neutral	large
Proline	Pro	P	non-polar	neutral	intermediate
Serine	Ser	S	polar	neutral	small
Threonine	Thr	T	polar	neutral	intermediate
Tryptophan	Trp	W	nonpolar	neutral	bulky
Tyrosine	Tyr	Y	polar	neutral	large
Valine	Val	V	nonpolar	neutral	intermediate

†based on volume in Å³: 50-100 is small, 100-150 is intermediate, 150-200 is large, and >200 is bulky

From this table it is appreciated that the conservative amino acid families include (i) G, A, V, L and I; (ii) D and E; (iii) C, S and T; (iv) H, K and R; (v) N and Q; and (vi) F, Y and W. A conservative mutation or substitution is therefore one that substitutes one amino acid for a member of the same amino acid family (e.g., a substitution of Ser for Thr or Lys for Arg).

As used herein, the term “degradation signal sequence” refers to a sequence (e.g., a nucleotide sequence that can be translated into an amino acid sequence) that mediates the degradation of a polypeptide in which it is contained. Degradation signal sequences can be included in the nucleic acid vectors of the invention to reduce or prevent the expression of portions of otoferlin proteins that have not undergone recombination and/or splicing. An exemplary degradation signal sequence for use in the invention is GCCTGCAAGAACTGGTTCAGCAGCCTGAGC-CACCTTCGTGATCCACCTG (SEQ ID NO: 22).

As used herein, the terms “effective amount,” “therapeutically effective amount,” and a “sufficient amount” of a composition, vector construct, or viral vector described herein refer to a quantity sufficient to, when administered to the subject in need thereof, including a mammal, for

example a human, effect beneficial or desired results, including clinical results, and, as such, an “effective amount” or synonym thereto depends upon the context in which it is being applied. For example, in the context of treating sensorineural hearing loss, it is an amount of the composition, vector construct, or viral vector sufficient to achieve a treatment response as compared to the response obtained without administration of the composition, vector construct, or viral vector. The amount of a given composition described herein that will correspond to such an amount will vary depending upon various factors, such as the given agent, the pharmaceutical formulation, the route of administration, the type of disease or disorder, the identity of the subject (e.g. age, sex, weight) or host being treated, and the like, but can nevertheless be routinely determined by one skilled in the art. Also, as used herein, a “therapeutically effective amount” of a composition, vector construct, or viral vector of the present disclosure is an amount which results in a beneficial or desired result in a subject as compared to a control. Note that when a combination of active ingredients is administered, the effective amount of the combination may or may not include amounts of each ingredient that would have been effective if administered individually. As defined herein, a therapeutically effective amount of a composition, vector construct, viral vector or cell of the present disclosure may be readily determined by one of ordinary skill by routine methods known in the art.

Dosage regime may be adjusted to provide the optimum therapeutic response.

As used herein, the term “endogenous” describes a molecule (e.g., a polypeptide, nucleic acid, or cofactor) that is found naturally in a particular organism (e.g., a human) or in a particular location within an organism (e.g., an organ, a tissue, or a cell, such as a human cell, e.g., a human cochlear hair cell).

As used herein, the term “express” refers to one or more of the following events: (1) production of an RNA template from a DNA sequence (e.g., by transcription); (2) processing of an RNA transcript (e.g., by splicing, editing, 5' cap formation, and/or 3' end processing); (3) translation of an RNA into a polypeptide or protein; and (4) post-translational modification of a polypeptide or protein.

As used herein, the term “exogenous” describes a molecule (e.g., a polypeptide, nucleic acid, or cofactor) that is not found naturally in a particular organism (e.g., a human) or in a particular location within an organism (e.g., an organ, a tissue, or a cell, such as a human cell, e.g., a human cochlear hair cell). Exogenous materials include those that are provided from an external source to an organism or to cultured matter extracted there from.

As used herein, the term “hair cell-specific expression” refers to production of an RNA transcript or polypeptide primarily within hair cells (e.g., cochlear hair cells) as compared to other cell types of the inner ear (e.g., spiral ganglion neurons, glia, or other inner ear cell types). Hair cell-specific expression of a transgene can be confirmed by comparing transgene expression (e.g., RNA or protein expression) between various cell types of the inner ear (e.g., hair cells vs. non-hair cells) using any standard technique (e.g., quantitative RT PCR, immunohistochemistry, Western Blot analysis, or measurement of the fluorescence of a reporter (e.g., GFP) operably linked to a promoter). A hair cell-specific promoter induces expression (e.g., RNA or protein expression) of a transgene to which it is operably linked that is at least 50% greater (e.g., 50%, 75%, 100%, 125%, 150%, 175%, 200% greater or more) in hair cells (e.g., cochlear hair cells) compared to at least 3 (e.g., 3, 4,

5, 6, 7, 8, 9, 10, or more) of the following inner ear cell types: Border cells, inner phalangeal cells, inner pillar cells, outer pillar cells, first row Deiter cells, second row Deiter cells, third row Deiter cells, Hensen's cells, Claudius cells, inner sulcus cells, outer sulcus cells, spiral prominence cells, root cells, interdental cells, basal cells of the stria vascularis, intermediate cells of the stria vascularis, marginal cells of the stria vascularis, spiral ganglion neurons, Schwann cells.

As used herein, the terms “increasing” and “decreasing” refer to modulating resulting in, respectively, greater or lesser amounts, of function, expression, or activity of a metric relative to a reference. For example, subsequent to administration of a composition in a method described herein, the amount of a marker of a metric (e.g., OTOF expression) as described herein may be increased or decreased in a subject by at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or 98% or more relative to the amount of the marker prior to administration. Generally, the metric is measured subsequent to administration at a time that the administration has had the recited effect, e.g., at least one week, one month, 3 months, or 6 months, after a treatment regimen has begun.

As used herein, the term “intron” refers to a region within the coding region of a gene, the nucleotide sequence of which is not translated into the amino acid sequence of the corresponding protein. The term intron also refers to the corresponding region of the RNA transcribed from a gene. Introns are transcribed into pre-mRNA, but are removed during processing, and are not included in the mature mRNA.

As used herein, “locally” or “local administration” means administration at a particular site of the body intended for a local effect and not a systemic effect. Examples of local administration are epicutaneous, inhalational, intra-articular, intrathecal, intravaginal, intravitreal, intrauterine, intra-lesional administration, lymph node administration, intratumoral administration, administration to the inner ear, and administration to a mucous membrane of the subject, wherein the administration is intended to have a local and not a systemic effect.

As used herein, the term “operably linked” refers to a first molecule that can be joined to a second molecule, wherein the molecules are so arranged that the first molecule affects the function of the second molecule. The term “operably linked” includes the juxtaposition of two or more components (e.g., a promoter and another sequence element) such that both components function normally and allow for the possibility that at least one of the components can mediate a function that is exerted upon at least one of the other components. The two molecules may or may not be part of a single contiguous molecule and may or may not be adjacent. For example, a promoter is operably linked to a transcribable polynucleotide molecule if the promoter modulates transcription of the transcribable polynucleotide molecule of interest in a cell. In additional embodiments, two portions of a transcription regulatory element are operably linked to one another if they are joined such that the transcription-activating functionality of one portion is not adversely affected by the presence of the other portion. Two transcription regulatory elements may be operably linked to one another by way of a linker nucleic acid (e.g., an intervening non-coding nucleic acid) or may be operably linked to one another with no intervening nucleotides present.

As used herein, the terms “otoferrin” and “OTOF” refer to the gene associated with nonsyndromic recessive deafness

DNFB9. The terms “otoferlin” and “OTOF” also refer to variants of wild-type OTOF protein and nucleic acids encoding the same, such as variant proteins having at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.9% identity, or more) to the amino acid sequence of a wild-type OTOF protein (e.g., any one of SEQ ID NOs: 1-5) or polynucleotides having at least 85% sequence identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.9% identity, or more) to the nucleic acid sequence of a wild-type OTOF gene, provided that the OTOF analog encoded retains the therapeutic function of wild-type OTOF. As used herein, OTOF may refer to the protein localized to inner hair cells or to the gene encoding this protein, depending upon the context, as will be appreciated by one of skill in the art.

As used herein, the term “plasmid” refers to a to an extrachromosomal circular double stranded DNA molecule into which additional DNA segments may be ligated. A plasmid is a type of vector, a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. Certain plasmids are capable of autonomous replication in a host cell into which they are introduced (e.g., bacterial plasmids having a bacterial origin of replication and episomal mammalian plasmids). Other vectors (e.g., non-episomal mammalian vectors) can be integrated into the genome of a host cell upon introduction into the host cell, and thereby are replicated along with the host genome. Certain plasmids are capable of directing the expression of genes to which they are operably linked.

As used herein, the terms “nucleic acid” and “polynucleotide,” used interchangeably herein, refer to a polymeric form of nucleosides in any length. Typically, a polynucleotide is composed of nucleosides that are naturally found in DNA or RNA (e.g., adenosine, thymidine, guanosine, cytidine, uridine, deoxyadenosine, deoxythymidine, deoxyguanosine, and deoxycytidine) joined by phosphodiester bonds.

However, the term encompasses molecules containing nucleosides or nucleoside analogs containing chemically or biologically modified bases, modified backbones, etc., whether or not found in naturally occurring nucleic acids, and such molecules may be preferred for certain applications. Where this application refers to a polynucleotide it is understood that both DNA, RNA, and in each case both single- and double-stranded forms (and complements of each single-stranded molecule) are provided. “Polynucleotide sequence” as used herein can refer to the polynucleotide material itself and/or to the sequence information (i.e., the succession of letters used as abbreviations for bases) that biochemically characterizes a specific nucleic acid. A polynucleotide sequence presented herein is presented in a 5 to 3 direction unless otherwise indicated.

As used herein, the terms “complementarity” or “complementary” of nucleic acids means that a nucleotide sequence in one strand of nucleic acid, due to orientation of its nucleobase groups, forms hydrogen bonds with another sequence on an opposing nucleic acid strand. The complementary bases in DNA are typically A with T and C with G. In RNA, they are typically C with G and U with A. Complementarity can be perfect or substantial/sufficient. Perfect complementarity between two nucleic acids means that the two nucleic acids can form a duplex in which every base in the duplex is bonded to a complementary base by Watson-Crick pairing. “Substantial” or “sufficient” complementary means that a sequence in one strand is not completely and/or perfectly complementary to a sequence in an

opposing strand, but that sufficient bonding occurs between bases on the two strands to form a stable hybrid complex in set of hybridization conditions (e.g., salt concentration and temperature). Such conditions can be predicted by using the sequences and standard mathematical calculations to predict the T_m (melting temperature) of hybridized strands, or by empirical determination of T_m by using routine methods. T_m includes the temperature at which a population of hybridization complexes formed between two nucleic acid strands are 50% denatured (i.e., a population of double-stranded nucleic acid molecules becomes half dissociated into single strands). At a temperature below the T_m , formation of a hybridization complex is favored, whereas at a temperature above the T_m , melting or separation of the strands in the hybridization complex is favored. T_m may be estimated for a nucleic acid having a known G+C content in an aqueous 1 M NaCl solution by using, e.g., $T_m = 81.5 + 0.41(\% \text{ G+C})$, although other known T_m computations take into account nucleic acid structural characteristics.

As used herein, the term “promoter” refers to a recognition site on DNA that is bound by an RNA polymerase. The polymerase drives transcription of the transgene. Exemplary promoters suitable for use with the compositions and methods described herein include ubiquitous promoters (e.g., the CAG promoter, cytomegalovirus (CMV) promoter, and a truncated form of the chimeric CMV-chicken β -actin promoter (CBA), in which the hybrid chicken β -actin/rabbit β -globin intron is greatly shortened to produce a smaller version of the promoter called smCBA), cochlear hair cell-specific promoters (e.g., the Myosin 15 (Myo15) promoter, the Myosin 7A (Myo7A) promoter, the Myosin 6 (Myo6) promoter, the POU Class 4 Homeobox 3 (POU4F3) promoter, and inner hair cell-specific promoters (e.g., the Fibroblast growth factor 8 (FGF8) promoter, the vesicular glutamate transporter 3 (VGLUT3) promoter, and the OTOF promoter).

“Percent (%) sequence identity” with respect to a reference polynucleotide or polypeptide sequence is defined as the percentage of nucleic acids or amino acids in a candidate sequence that are identical to the nucleic acids or amino acids in the reference polynucleotide or polypeptide sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent nucleic acid or amino acid sequence identity can be achieved in various ways that are within the capabilities of one of skill in the art, for example, using publicly available computer software such as BLAST, BLAST-2, or Megalign software. Those skilled in the art can determine appropriate parameters for aligning sequences, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. For example, percent sequence identity values may be generated using the sequence comparison computer program BLAST. As an illustration, the percent sequence identity of a given nucleic acid or amino acid sequence, A, to, with, or against a given nucleic acid or amino acid sequence, B, (which can alternatively be phrased as a given nucleic acid or amino acid sequence, A that has a certain percent sequence identity to, with, or against a given nucleic acid or amino acid sequence, B) is calculated as follows:

$$100 \text{ multiplied by (the fraction } X/Y \text{)}$$

where X is the number of nucleotides or amino acids scored as identical matches by a sequence alignment program (e.g., BLAST) in that program’s alignment of A and B, and where Y is the total number of nucleic acids in B. It will be

appreciated that where the length of nucleic acid or amino acid sequence A is not equal to the length of nucleic acid or amino acid sequence B, the percent sequence identity of A to B will not equal the percent sequence identity of B to A.

The term “derivative” as used herein refers to a nucleic acid, peptide, or protein or a variant or analog thereof comprising one or more mutations and/or chemical modifications as compared to a corresponding full-length wild-type nucleic acid, peptide, or protein. Non-limiting examples of chemical modifications involving nucleic acids include, for example, modifications to the base moiety, sugar moiety, phosphate moiety, phosphate-sugar backbone, or a combination thereof.

As used herein, the term “pharmaceutical composition” refers to a mixture containing a therapeutic agent, optionally in combination with one or more pharmaceutically acceptable excipients, diluents, and/or carriers, to be administered to a subject, such as a mammal, e.g., a human, in order to prevent, treat or control a particular disease or condition affecting or that may affect the subject.

As used herein, the term “pharmaceutically acceptable” refers to those compounds, materials, compositions and/or dosage forms, which are suitable for contact with the tissues of a subject, such as a mammal (e.g., a human) without excessive toxicity, irritation, allergic response and other problem complications commensurate with a reasonable benefit/risk ratio. Preferably, the term “pharmaceutically acceptable” means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in mammals, and more particularly in humans.

As used herein, the term “recombinogenic region” refers to a region of homology that mediates recombination between two different sequences.

As used herein, the term “regulatory sequence” includes promoters, enhancers and other expression control elements (e.g., polyadenylation signals) that control the transcription or translation of the polynucleotides that encode OTOF. Such regulatory sequences are described, for example, in Goeddel, *Gene Expression Technology: Methods in Enzymology* 185 (Academic Press, San Diego, Calif., 1990); incorporated herein by reference.

As used herein, the term “sample” refers to a specimen (e.g., blood, blood component (e.g., serum or plasma), urine, saliva, amniotic fluid, cerebrospinal fluid, tissue (e.g., placental or dermal), pancreatic fluid, chorionic villus sample, and cells) isolated from a subject.

As used herein, the term “transfection” refers to any of a wide variety of techniques commonly used for the introduction of exogenous DNA into a prokaryotic or eukaryotic host cell, e.g., electroporation, lipofection, calcium-phosphate precipitation, DEAE-dextran transfection, Nucleofection, squeeze-poration, sonoporation, optical transfection, Magnetofection, impalefection and the like.

As used herein, the terms “subject” and “patient” refer to an animal (e.g., a mammal, such as a human), veterinary animals (e.g., cats, dogs, cows, horses, sheep, pigs, etc.) and experimental animal models of diseases (e.g., mice, rats). A subject to be treated according to the methods described herein may be one who has been diagnosed with hearing loss (e.g., hearing loss associated with a mutation in OTOF), or one at risk of developing these conditions. Diagnosis may be performed by any method or technique known in the art. One skilled in the art will understand that a subject to be treated according to the present disclosure may have been subjected to standard tests or may have been identified,

without examination, as one at risk due to the presence of one or more risk factors associated with the disease or condition.

As used herein, the terms “transduction” and “transduce” refer to a method of introducing a vector construct or a part thereof into a cell. Wherein the vector construct is contained in a viral vector such as for example an AAV vector, transduction refers to viral infection of the cell and subsequent transfer and integration of the vector construct or part thereof into the cell genome.

As used herein, “treatment” and “treating” of a state, disorder or condition can include: (1) preventing, delaying, or reducing the incidence and/or likelihood of the appearance of at least one clinical or sub-clinical symptom of the state, disorder or condition developing in a subject that may be afflicted with or predisposed to the state, disorder or condition, but does not yet experience or display clinical or subclinical symptoms of the state, disorder or condition; or (2) inhibiting the state, disorder or condition, i.e., arresting, reducing or delaying the development of the disease or a relapse thereof or at least one clinical or sub-clinical symptom thereof; or (3) relieving the disease, i.e., causing regression of the state, disorder or condition or at least one of its clinical or sub-clinical symptoms. The benefit to a subject to be treated is either statistically significant or at least perceptible to the patient or to the physician.

As used herein, the term “vector” includes a nucleic acid vector, e.g., a DNA vector, such as a plasmid, an RNA vector, virus or other suitable replicon (e.g., viral vector). A variety of vectors have been developed for the delivery of polynucleotides encoding exogenous proteins into a prokaryotic or eukaryotic cell. Examples of such expression vectors are disclosed in, e.g., WO94/11026; incorporated herein by reference as it pertains to vectors suitable for the expression of a gene of interest. Expression vectors suitable for use with the compositions and methods described herein contain a polynucleotide sequence as well as, e.g., additional sequence elements used for the expression of proteins and/or the integration of these polynucleotide sequences into the genome of a mammalian cell. Certain vectors that can be used for the expression of OTOF as described herein include vectors that contain regulatory sequences, such as promoter and enhancer regions, which direct gene transcription. Other useful vectors for expression of OTOF contain polynucleotide sequences that enhance the rate of translation of these genes or improve the stability or nuclear export of the mRNA that results from gene transcription. These sequence elements include, e.g., 5' and 3' untranslated regions and a polyadenylation signal site in order to direct efficient transcription of the gene carried on the expression vector. The expression vectors suitable for use with the compositions and methods described herein may also contain a polynucleotide encoding a marker for selection of cells that contain such a vector. Examples of a suitable marker include genes that encode resistance to antibiotics, such as ampicillin, chloramphenicol, kanamycin, or nourseothricin.

As used herein, the term “wild-type” refers to a genotype with the highest frequency for a particular gene in a given organism.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a CAG promoter operably linked to exons 1-24 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1 and the 500 kb immediately 3' of the exon

24/25 boundary (FIG. 1A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23) (FIG. 1B).

FIGS. 2A and 2B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a CAG promoter operably linked to exons 1-28 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1 and the 500 kb immediately 3' of the exon 28/29 boundary (FIG. 2A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 28/29 boundary and exons 29-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 2B).

FIGS. 3A and 3B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-21 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1 and the 500 kb immediately 3' of the exon 21/22 boundary (FIG. 3A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23) (FIG. 3B).

FIGS. 4A and 4B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a CAG promoter operably linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6 and the 500 kb immediately 3' of the exon 24/25 boundary (FIG. 4A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23) (FIG. 4B).

FIGS. 5A and 5B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, a CMV enhancer, and a CMV promoter operably linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6 and the 500 kb immediately 3' of the exon 24/25 boundary (FIG. 5A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23) (FIG. 5B).

FIGS. 6A and 6B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a CAG promoter operably linked to exons 1-28 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6 and the 500 kb immediately 3' of the exon 28/29 boundary (FIG. 6A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 28/29 boundary and exons 29-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 6B).

FIGS. 7A and 7B are maps of the 5' and 3' vectors in an overlapping dual vector system. The 5' vector contains AAV2 ITRs, and a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-21 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6 and the 500 kb immediately 3' of the exon 21/22 boundary (FIG. 7A). The 3' vector contains AAV2 ITRs, the 500 kb immediately 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23) (FIG. 7B).

FIGS. 8A and 8B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a CAG promoter operably linked to the full-length human OTOF 5' UTR and exons 1-26 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a splice donor sequence (SD) (FIG. 8A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 27-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 8B).

FIGS. 9A and 9B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a splice donor sequence (SD) (FIG. 9A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 20-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23) (FIG. 9B).

FIGS. 10A and 10B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-25 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a splice donor sequence (SD) (FIG. 10A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 26-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 10B).

FIGS. 11A and 11B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a CAG promoter operably linked to the full-length mouse OTOF 5' UTR and exons 1-28 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a splice donor sequence (SD) (FIG. 11A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 29-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 11B).

FIGS. 12A and 12B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-20 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a splice donor sequence (SD) (FIG. 12A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 21-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23) (FIG. 12B).

FIGS. 13A and 13B are maps of the 5' and 3' vectors in a trans-splicing dual vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a splice donor sequence (SD) (FIG. 13A). The 3' vector contains AAV2 ITRs, a splice acceptor sequence (SA), exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 13B).

FIGS. 14A and 14B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a CAG promoter operably linked to the full-length human OTOF 5' UTR and exons 1-26 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 14A). The 3'

31

vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 27-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 14B).

FIGS. 15A and 15B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 15A). The 3' vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 20-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23) (FIG. 15B).

FIGS. 16A and 16B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-25 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 16A). The 3' vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 26-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 16B).

FIGS. 17A and 17B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a CAG promoter operably linked to the full-length mouse OTOF 5' UTR and exons 1-28 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 17A). The 3' vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 29-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 17B).

FIGS. 18A and 18B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-20 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 18A). The 3' vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 21-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23) (FIG. 18B).

FIGS. 19A and 19B are maps of the 5' and 3' vectors in a dual hybrid vector system. The 5' vector contains AAV2 ITRs, a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1) (FIG. 19A). The 3' vector contains AAV2 ITRs, a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23) (FIG. 19B).

32

FIGS. 20A and 20B are images demonstrating that polynucleotides encoded by dual hybrid vectors (FIG. 20A) and overlapping vectors (FIG. 20B) undergo recombination in cell culture (top). Beneath each image is a schematic depicting the type of recombination that occurs in each dual vector system (bottom). Inner ear-derived House Ear Institute-Organ of Corti 1 (HEI-OC1) cells were transfected with plasmids containing either the 5' half of OTOF alone, the 3' half of OTOF alone, or both the 5' and 3' halves together. For the dual hybrid vector experiment, the 5' vector contained a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1); and the 3' vector contained a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 20-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23). For the overlapping dual vector experiment, the 5' vector contained a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-21 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1 and the 500 kb immediately 3' of the exon 21/22 boundary; and the 3' vector contained the 500 kb immediately 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, and a WPRE sequence (SEQ ID NO: 23). Full-length otoferlin was transfected as a positive control and untransfected cells were used as a negative control. Genomic DNA was extracted from cells using a standard column from a gDNA isolation kit. PCR primers were designed to anneal outside of the region of splice sites or overlap and PCR was performed using standard molecular biology techniques. Amplicons were visualized using gel electrophoresis. The white box in the gel image indicates the lane in which 5' and 3' halves were transfected together and where a roughly 1 kb amplicon is seen, indicating recombination.

FIGS. 21A-21C are images demonstrating that polynucleotides encoded by dual hybrid vectors (FIG. 21A), trans-splicing vectors (FIG. 21B), and overlapping vectors (FIG. 21C) undergo recombination in cell culture and produce OTOF protein (top). Beneath each image is a schematic depicting the type of recombination that occurs in each dual vector system (bottom). Inner ear derived HEI-OC1 cells were transfected with plasmids containing either the 5' half of otoferlin alone, the 3' half of otoferlin alone, or both 5' and 3' halves together. For the dual hybrid vector experiment, the 5' vector contained a CAG promoter operably linked to the full-length human OTOF 5' UTR and exons 1-26 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, a splice donor sequence (SD), a recombinogenic region (AK), and a degradation signal sequence (CL1); and the 3' vector contained a recombinogenic region (AK), a degradation signal sequence (CL1), a splice acceptor sequence (SA), exons 27-48 of a polynucleotide encoding the human OTOF protein of SEQ ID NO: 1, the full-length human OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23). For the trans-splicing dual vector experiment, the 5' vector contained a Myo15 promoter (SEQ ID NO: 36) operably linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a splice donor sequence (SD); and the 3' vector contained a splice acceptor sequence (SA), exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, the full-length mouse OTOF 3' UTR, and a WPRE sequence (SEQ ID NO: 23). For the overlapping dual vector experiment, the 5' vector contained a CMV promoter operably

linked to exons 1-24 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6 and the 500 kb immediately 3' of the exon 24/25 boundary; and the 3' vector contained the 500 kb immediately 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding the mouse OTOF protein of SEQ ID NO: 6, and a WPRE sequence (SEQ ID NO: 23). Cells were fixed in 4% PFA 48 hours post transfection and processed for immunohistochemistry. An antibody specific to OTOF protein labels some cells in the 5' alone transfected cells, no cells in the 3' alone transfected cells, and more cells than both previous conditions combined in the 5'+3' transfected cells.

FIGS. 22A-22C are a series of graphs showing electrophysiological signatures of hearing function in mice treated with viral vectors expressing OTOF via dual hybrid vector systems. Homozygous OTOF-Q828X mice (a mouse model of human OTOF mutation p.Gln828Ter) were treated with an AAV1-Myo15 (SEQ ID NO: 38)-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector system by injection through the round window membrane and auditory brainstem response (ABR) thresholds were used to assess hearing function (FIG. 22A). Untreated animals (untreated Otof HOM) showed no detectable recovery in hearing function, whereas treated animals exhibited robust recovery, which was consistent from four weeks post-treatment (Otof HOM at 4 weeks after treatment) to eight weeks post-treatment (Otof HOM at 8-11 weeks after treatment). ABR thresholds from heterozygous mice (Otof HET) were also tested.

In another set of experiments, Homozygous OTOF-Q828X mice were treated with an AAV1-truncated chimeric CMV-chicken β -actin (smCBA)-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector system by injection through the round window membrane and ABR thresholds were used to assess hearing function as described above (FIG. 22B). Untreated animals showed no detectable recovery in hearing function, while treated animals exhibited a robust recovery at four weeks post-treatment (Otof HOM at 4 weeks after treatment). When the same mice were evaluated at eight weeks post-treatment (Otof HOM at 8 weeks after treatment), ABR thresholds increased, suggesting less durable recovery with the smCBA promoter. ABR thresholds from heterozygous mice were also tested.

In yet another set of experiments, homozygous OTOF-Q828X mice were treated with an AAV1-smCBA-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector system by injection through the round window membrane at low, medium (mid), and high doses and ABR thresholds were used to assess hearing function at four weeks and eight weeks post-treatment (FIG. 22C). A dose-dependent recovery in ABR thresholds was observed at both timepoints. When comparing the eight weeks versus the four weeks timepoints, recovery was steady for the low and mid doses, but decreased for the high dose animals. ABR thresholds for HET animals were also tested.

DETAILED DESCRIPTION

Described herein are compositions and methods for the treatment of sensorineural hearing loss or auditory neuropathy in a subject (such as a mammalian subject, for instance, a human) by administering a first nucleic acid vector containing a promoter and a polynucleotide encoding an N-terminal portion of an otoferlin (OTOF) protein (e.g., a wild-type (WT) OTOF protein) and a second nucleic acid vector containing a polynucleotide encoding a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence. When introduced into a mammalian cell, such as

a cochlear hair cell, the polynucleotides encoded by the two nucleic acid vectors can combine to form a nucleic acid molecule that encodes the full-length OTOF protein. The compositions and methods described herein can, therefore, be used to induce or increase expression of WT OTOF in cochlear hair cells of a subject who has an OTOF deficiency (e.g., low OTOF expression or an OTOF mutation that impairs OTOF expression or function).

Otoferlin

OTOF is a 230 kDa membrane protein that contains at least six C2 domains implicated in calcium, phospholipid, and protein binding. It is encoded by a gene that contains 48 exons, and the full-length protein is made up of 1,997 amino acids. OTOF is located at ribbon synapses in inner hair cells, where it is believed to function as a calcium sensor in synaptic vesicle fusion, triggering the fusion of neurotransmitter-containing vesicles with the plasma membrane. It has also been implicated in vesicle replenishment and clathrin-mediated endocytosis, and has been shown to interact with Myosin VI, Rab8b, SNARE proteins, calcium channel Cav1.3, Ergic2, and AP-2. The mechanism by which OTOF mediates exocytosis and the physiological significance of its interactions with its binding partners remain to be determined.

Otoferlin-Associated Hearing Loss

OTOF was first identified by a study investigating the genetics of a non-syndromic form of deafness, autosomal recessive deafness-9 (DFNB9). Mutations in OTOF have since been found to cause sensorineural hearing loss in patients throughout the world, with many patients carrying OTOF mutations having auditory neuropathy, a disorder in which the inner ear detects sound, but is unable to properly transmit sound from the ear to the brain. These patients have an abnormal auditory brainstem response (ABR) and impaired speech discrimination with initially normal otoacoustic emissions. Patients carrying homozygous or compound heterozygous mutations often develop hearing loss in early childhood, and the severity of hearing impairment has been found to vary with the location and type of mutation in OTOF.

The compositions and methods described herein can be used to treat sensorineural hearing loss or auditory neuropathy by administering a first nucleic acid vector containing a polynucleotide encoding an N-terminal portion of an OTOF protein and a second nucleic acid vector containing a polynucleotide encoding a C-terminal portion of an OTOF protein. The full-length OTOF coding sequence is too large to include in the type of vector that is commonly used for gene therapy (e.g., an adeno-associated virus (AAV) vector, which is thought to have a packaging limit of 5 kb). The compositions and methods described herein overcome this problem by dividing the OTOF coding sequence between two different nucleic acid vectors that can recombine in a cell to reconstitute the full-length OTOF sequence. These compositions and methods can be used to treat subjects having one or more mutations in the OTOF gene, e.g., an OTOF mutation that reduces OTOF expression, reduces OTOF function, or is associated with hearing loss. When the first and second nucleic acid vectors are administered in a composition, the polynucleotides encoding the N-terminal and C-terminal portions of OTOF can combine within a cell (e.g., a human cell, e.g., a cochlear hair cell) to form a single nucleic acid molecule that contains the full-length OTOF coding sequence (e.g., through homologous recombination and/or splicing).

The nucleic acid vectors used in the compositions and methods described herein include nucleic acid sequences

35

that encode wild-type OTOF, or a variant thereof, such as a nucleic acid sequences that, when combined, encode a protein having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97, 98%, 99, or more, sequence identity) to the amino acid sequence of wild-type human or mouse OTOF. The polynucleotides used in the nucleic acid vectors described herein encode an N-terminal portion and a C-terminal portion of an OTOF amino acid sequence in Table 2 below (e.g., two portions that, when combined, encode a full-length OTOF amino acid sequence listed in Table 2, e.g., any one of SEQ ID NOs: 1-5).

According to the methods described herein, a subject can be administered a composition containing a first nucleic acid vector and a second nucleic acid vector that encode an

36

N-terminal and C-terminal portion, respectively, of a nucleic acid sequence encoding the amino acid sequence of any one of SEQ ID NOs: 1-5, or a nucleic acid sequence encoding an amino acid sequence having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the amino acid sequence of any one of SEQ ID NOs: 1-5, or a nucleic acid sequence encoding an amino acid sequence that contains one or more conservative amino acid substitutions relative to any one of SEQ ID NOs: 1-5 (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more conservative amino acid substitutions), provided that the OTOF analog encoded retains the therapeutic function of wild-type OTOF (e.g., the ability to regulate exocytosis at ribbon synapses).

TABLE 2

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
1	OTOF-201 protein (NP_919224.1), human otoferlin isoform a, 1997 aa	MALLIHLKTVSELGRGDRIAKVTRFGQSFYSRVLENCEDVADFDE TFRWPVASSIDRNEMLEIQVFNYSKVFSNKLIGTFRMVLQKVVEES HVEVTDTLIDNNNAIKTSLCCEVRYQATDGTGVSWDGDFLGDES LQEEKDSQETDGLLPGSRPSSRPPEKESFRRAGRSVFSAMKLGK NRSHKEEPQRDPDEPAVLEMEDLDHLAIRLGDGLDPDSVSLASVTAL TTNVSNKRSPKPDIKMEPSAGRPMDYQVSITVIEARQLVGLNMDPVV CVEVGDDKKYTSMKESTNCPYNYEFVFDHVSVDVDFDKIKISVI HSKNLLRSGTLVGSFKMDVGTVYSQPEHQPHHKWAILSDPDDISS GLKGYVKCDVAVVGKGDNIKTPHKANETDEDDIEGNLLLPPEGVPPE RQWARFVYKIVRAEGLPRMNTSLMANVKKAFIGENKDLVDPPYQV FFAGQKGTSTVQKSSYEPLWNEQVVFDTLFPPLCKRMKVQIRDS KVNDVAIGTHFIDLRKISNDGDKGFLPTLGPAAVNMVYGSTRTNYTL DEHQDLNEGLGEGVSFRARLLGLAVEIVDTSNPELTSSTEVQVEQ ATPISSESCAGKMEFFLFGAFLEASMDRRNGDKPITFEVTIGNYGN EVDGLSRPQRPRPRKEPGEDEEVDLIQNASDDEAGDAGDLASVSS TPPMRPQVTDNRNYFHLPLYLERKPCIYIKSWWPDQRRRLYNANIMD HIADKLEEGLDNIQEMIKTEKSYPERRLRGVLEELSCGCCRFSLAD KDQGHSSRTRLDRELRKSCMRELENMGGQARMLRAQVKRHTVRD KLRLCQNFQKRLFLADEPQHSIPDIFIWMSNNKRVAYARVPSKD LLFSIVEEETGKDCAKVKTFLKLPGKRGFSAGWTVQAKVELYLW LGLSKQRKEFLCGLPCGFQEVKAAQGLGLHAFPPVSLVYTKKQAF QLRAHMYQARSLFAADSSGLSDPPARVFFINQSQCTEVLNETLCT WDQMLVFDNLELYGEAHELDDPPPIVIEIYDQDSMGKADFMGRFT AKPLVKMADEAYCPRFPQLEYQIYRGNATAGDLLAAPELLQIG PAGKADLPPINGPVDVDRGPIMPVPMGIRPVLISKYRVEVLFWGLRD LKRVLNAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFVEV DLPENELLHPPPLNIRVDCRAFGRYTLVGSHAVSSLRRFIYRPPDRS APSWNTTVRLRRCRVLCNGSSSHSTGEVVTMEPEVPIKKLET MVKLDATSEAVVKVDVAEEKEKKKKKGTAEPEEEEPDESMLD WWSKYFASIDTMKBQLRQQEPGSGIDLEEKEVDNTEGLKSGMKGK EKARAAKEKKKKTQSSGSGQSEAEKPKKIDELKVYPKELESE FDNFEDWLHTFNLRLRGKTGDDGSTEERIVGRFKGSLCVYKVPL PEDVSREAGDYSTYGMFQGISNDPINVLRVYVVRATDLHPADIN GKADPYIAIRLGKTDIRDKENYISKQLNPVFGKSFIDEASFPMESMLT VAVYDWDLVGTDDLIGETKIDLENRFYSKHRATCGIAQTYSYTHGYNI WRDPMKPSQILTRLCKDGKVDGPHFGPPGRVKVANRVFTGPSEIE DENGQRKPTDEHVALLALRHWDIPRAGCRLVPEHVETRLPNPD KPGIEQGRLELWVDMFPMMPAGTPLDISPRKPKKYELRVIIWNT DEVVLEDDDDFFTGEKSSDIFVRGWLKGQEDKQDQTDVHYHSLTGE GNFNWRYLFPFDYLAEEKIVISKESMFSWDETEYKIPARLTQIWI DADHFSADDFLGAIELDLNRFPRGAKTAKQCTMEMATGEVDVPLV SIFKQKRVKGVWPLLRANENDEFELTGKVEAELHLLTAEEAEKNPV GLARNEPDLEKPNRPDTSFIWFLNPLKSARYFLWHTYRWLLKLL LLLLLLLLLLALFLYSVPGYLVKKILGA
2	OTOF-202 protein (NP_004793.2), human otoferlin isoform b, 1230 aa	MIKTEKSYPERRLRGVLEELSCGCCRFSLADKDQGHSSRTRLDRE RLKSCMRELENMGGQARMLRAQVKRHTVRDKLRLCQNFQKRLRF LADEPQHSIPDIFIWMSNNKRVAYARVPSKDLLFSIVEEETGKDC KVKTLFLKLPGKRGFSAGWTVQAKVELYLWGLSKQRKEFLCGL PCGFQEVKAAQGLGLHAFPPVSLVYTKKQAFQLRAHMYQARSLFA ADSSGLSDPPARVFFINQSQCTEVLNETLCTWDQMLVFDNLELYG EAHELDDPPPIVIEIYDQDSMGKADFMGRFTAKPLVKMADEAYC RFPQLEYQIYRGNATAGDLLAAPELLQIGPAGKADLPPINGPVDV DRGPIMPVPMGIRPVLISKYRVEVLFWGLRD LKRVLNAQVDRPRVD IECAGKGVQSSLIHNYKKNPNFNTLVKWFVEVDLPENELLHPPPLNIRV VDCRAFGRYTLVGSHAVSSLRRFIYRPPDRSAPSWNTTGEVVTME PEVPIKKLETMVKLDATSEAVVKVDVAEEKEKKKKKGTAEPEE

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		EEEPDESMLDWWSKYFASIDTMKEQLRQQEPSGIDLEEKEEVDNT EGLKGSMDGKEKARAAKEEKKKTQSSSGSGGSEAEKKPKIDE LKVYPKELESEFDNFEDWLHTFNLLRGKTGDDDEDGSTEERIVGRF KGSLCVYKVPLEDPVSREAGYDSTYGMFQGIPSNDPINVLVRVYV RATDLHPADINGKADPYIAIRLGKTDIRDKENYISKQLNPVFGKSFDI EASFPMESMLTVAVYDWDLVGTDDLIGETKIDLENRFYSKHRATCG IAQTYSTHGYNIWRDPMKPSQILTRLCKDGKVDGPHFGPPGRVKVA NRVFTGSPSEIEDENGQRKPTDEHVALALRHWEDIPRAGCRLVPEH VETRPLLNPDKPGIEQGRLELWVDMFPMMPAGTPLDISPRKPKK YELRVI IWNTDEVVLEDDFFTGEKSSDIFVRGWLKGQEDKQDTD VHYHSLTGEENFNWRYLFPFDYLAEEKIVISKESMFSWDETEYKI PARLTLQIWDADHFSADDFLGAI ELDLNRFPAGAKTAKQCTMEMAT GEVDVPLVSI FKQKRVKGWVPLARNENDEFELTGKVEAELHLLTA EAEKNPVGLARNEDPLEKPNRPDTSFIWFLNPLKSARYFLWHTY RWLLKLLLLLLLLLLLALFLYSVPGYLVKKILGA
3	OTOF-203 protein (NP_919304.1), human otoferlin isoform d, 1230 aa	MIKTEKSYPERRLRGVLEELS CGCCRFSLADKDGHSSTRLDRE RLKSCMRELENMGQARMLRAQVKRHTVRDKLRLCQNFLQKLR LADEPQHSIPDIFIWMSNNKRVAYARVPSKDLLFSIVEETGKDCA KVKTFLKLPGRGFGSAGWTVAQVELYLWLGLSKQRKEFLCGL PCGFQEVKAAQGLGLHAFPPVSLVYTKQAFQLRAHMYQARSLFA ADSSGLSDPFARVFFINQSQCTEVLNETLCPDQMLVFDNLELYG EAHEL RDDPPIIVIEIYDQDSMGKADFMGRTFAPKPLVKMADEAYC PPRFPQLEYQIYRGNATAGDLLAAFELLQIGPAGKADLPINGPVDV DRGPI MPVPMGIRPVLSKYRVEVLFWGLRDLKRVNLAQVDRPRVDI ECAGKGVQSSLIHNYKKNPNFNTLVKWFVLDLPELHLLPLNIRV VDCRAFGRYTLVGSHAVSSLRRFIYRPPDRSAPS WNTTGEVVV TMEPEVPIKKLETMVKL DATSEAVVKVDVAEEKEKKKKKGTAE EPEEEEPDESMLDWWSKYFASIDTMKEQLRQQEPSGIDLEEKEEVDNT EGLKGSMDGKEKARAAKEEKKKTQSSSGSGGSEAEKKPKIDE LKVYPKELESEFDNFEDWLHTFNLLRGKTGDDDEDGSTEERIVGRF KGSLCVYKVPLEDPVSREAGYDSTYGMFQGIPSNDPINVLVRVYV RATDLHPADINGKADPYIAIRLGKTDIRDKENYISKQLNPVFGKSFDI EASFPMESMLTVAVYDWDLVGTDDLIGETKIDLENRFYSKHRATCG IAQTYSTHGYNIWRDPMKPSQILTRLCKDGKVDGPHFGPPGRVKVA NRVFTGSPSEIEDENGQRKPTDEHVALALRHWEDIPRAGCRLVPEH VETRPLLNPDKPGIEQGRLELWVDMFPMMPAGTPLDISPRKPKK YELRVI IWNTDEVVLEDDFFTGEKSSDIFVRGWLKGQEDKQDTD VHYHSLTGEENFNWRYLFPFDYLAEEKIVISKESMFSWDETEYKI PARLTLQIWDADHFSADDFLGAI ELDLNRFPAGAKTAKQCTMEMAT GEVDVPLVSI FKQKRVKGWVPLARNENDEFELTGKVEAELHLLTA EAEKNPVGLARNEDPLEKPNRPDTSFIWFLNPLKSARYFLWHTY RWLLKLLLLLLLLLLLALFLYSVPGYLVKKILGA
4	OTOF-208 protein (NP_919303.1), human otoferlin isoform c, 1307 aa	MMTDTQDGPSESSQIMRSLTPLINREEAFGEAGEAGLWPSITHTPD SQEEGLNDIQEMI KTEKSYPERRLRGVLEELS CGCCRFSLADKDG GHSSSTRLDREERLKS CMRELENMGQARMLRAQVKRHTVRDKLR LCQNFLQKLRFLADEPQHSIPDIFIWMSNNKRVAYARVPSKDLLFS IVEETGKDCAKVKTFLKLPGRGFGSAGWTVAQVELYLWLGLS KQRKEFLCGLPCGFQEVKAAQGLGLHAFPPVSLVYTKQAFQLRA HMYQARSLFAADSSGLSDPFARVFFINQSQCTEVLNETLCPDQ MLVFDNLELYGEAHEL RDDPPIIVIEIYDQDSMGKADFMGRTFAPKPL VKMADEAYCPRFPQLEYQIYRGNATAGDLLAAFELLQIGPAGK ADLPPI NGPVDVDRGPI MPVPMGIRPVLSKYRVEVLFWGLRDLKRV NLAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFVLDLPE NELLHPLNIRVDCRAFGRYTLVGSHAVSSLRRFIYRPPDRSAPS WNTTVRLRLRCRVLCNGSSSHSTGEVVVTEPEVPIKKLETMVK LDATSEAVVKVDVAEEKEKKKKKGTAEPEEEEPDESMLDWWS KYFASIDTMKEQLRQQEPSGIDLEEKEEVDNT EGLKGSMDGKEKAR AAKEEKKKTQSSSGSGGSEAEKKPKIDELKVYPKELESEFDNF EDWLHTFNLLRGKTGDDDEDGSTEERIVGRF KGSLCVYKVPLEDP VSREAGYDSTYGMFQGI PSNDPINVLVRVYVVRATDLHPADINGKA DPYIAIRLGKTDIRDKENYISKQLNPVFGKSFDI EASFPMESMLTVAV YDWDLVGTDDLIGETKIDLENRFYSKHRATCGIAQTYSTHGYNIWR DPMKPSQILTRLCKDGKVDGPHFGPPGRVKVANRVFTGSPSEIED ENGQRKPTDEHVALALRHWEDIPRAGCRLVPEHVETRPLLNPDK GIEQGRLELWVDMFPMMPAGTPLDISPRKPKYELRVI IWNTDE VVLEDDFFTGEKSSDIFVRGWLKGQEDKQDTDVHYHSLTGEEN FNWRYLFPFDYLAEEKIVISKESMFSWDETEYKI PARLTLQIWDAD HFSADDFLGAI ELDLNRFPAGAKTAKQCTMEMATGEVDVPLVSI FKQKRVKGWVPLARNENDEFELTGKVEAELHLLTA EAEKNPVGL ARNEPDPLEKPNRPDTSFIWFLNPLKSARYFLWHTYRWLLKLL LLLLLLALFLYSVPGYLVKKILGA

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
5	OTOF-205 protein (NP_001274418.1), human otoferlin isoform e, 1997 aa	MALLIHLKTVSELRGRGDRIAKVTFRGQSFYSRVLENCEVDADFDE TFRWPVASSIDRNEMLEIQVFNYSKVFSNKLI GTFRMVLQKVVEES HVEVTDTLIDNNAI IKTSLCEVEVRYQATDGTVGSWDDGDFLGDES LQEEEEKDSQETDGLLPGRSPSSRPPEKSFRRAGRSVFSAMKLGK NRSHKEEPQRDEPAVLEMEDLDHLAIRLGDGLDPDSVSLASVTAL TTNVSNKRSKPDIKMEPSAGRPMQVQVITVIEARQLVGLNMDPVV CVEVGDDKKYTSMKESTNCPYYNEYFVDFHVSFDPVMFDKI IKISVI HSKNLLRSGTLVGSFKMDVGTVYSQPEHQFHHKWAILSDPDDISS GLKGYVKCDVAVVGKGDNIKTPH KANETDEDDIEGNLLLP EGVPPPE RQWARFYVKIYRAEGLPRMNTSLMANVKKAFI GENKDLVDPYVQV FFAGQKGKTSVQKSSYEPLWNEQVVFDTLFPPLCKRMKVQIRDS KVNDVAIGTHFIDL RKISNDGDKGFLPTLGP AWVNMVYSTRNYTLL DEHQDLNEGLGEGVSFRARLLGLAVEIVDTSNPELTSSTEVQVEQ ATPISESCAGKMEFFLFGAPLEASMDRRNGDKPI TFEVTIGNYGN EVDGLSRPQRPRPRKEPGDEEVDLIQNASDDEAGDAGDLASVSS TPPMRPQVTD RNYFHLPYLERKPCIYIKSWWPDQRRRLYNANIMD HIADKLEEGLNDIQEMI KTEKSYPERRLRGVLEELSCGCCRFSLAD KDQGHSSRTRLDRELRKSCMRELENMGQQA RMLRAQVKRHTVRD KLRLCQNFLOKLRLFLADEPQHSIPDIFWMMSSNNKRIAYARVP SKD LLFSIVEEETGKDCAKVKTFLKLPGKRGFSGAGWTVQAKVELYLW LGLSKQRKEFLCGLPCGFQEVKAAQGLGLHAFPPVSLVYTKQAF QLRAHMYQARSLFAADSSGLSDPFARVFFINQSQCTEVLNETLCPT WDQMLVFDNLELYGEAHEL RDDPPII VI E IYDQDSMGKADFMGRTF AKPLVKMADEAYCPRFPQLEYYQIYRGNATAGDLLAA FELLQIG PAGKADLPPI NGPVDVDRGPI MPVPMGIRPVLSKYRVEVLFWGLRD LKRVLNAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFV DLPENELLHPPNLNIRVVD CRAFTYTLVGSHAVSSLRRFIYRPPDRS APSWNTTVRLRRRCVLCNGGSSSHSTGEVVVTEPEVP IKKLET MVKLDATSEAVVKVDVAEEEEKKKKKKGTAEEPEEEEPDESMLD WWSKYFASIDTMKEQLRQQEP SGIDLEEKEVDNTEGLKGSMKGK EKARAAKEEKKKTQSSGSGQGS EAPKPKPIDELKVYPKELESE FDNFEDWLHTFNLRLGKTGDD EGGSTEERIVGRFKGSLCVYKVP PEDVSR EAGYDSTYGMFQGISNDP INVLRVYVVRATDLHPADIN GKADPYIAIRL GKTDIRDKENYISKQLNPVFGKSF D IEASFPMESMLT VAVYDWDLVGTDDLIGETKIDLENRFYSKHRATCGIAQTYSTHGYNI WRDPMKPSQILTRLCKDGKVDGPHFGPPGRVKVANRVFTGPSEIE DENGQRKPTDEHVALALRHWEDIPRAGCRLVPEHVETRPLLNPD KPGIEQGRLELWVDMFPMMPAPGTPLDISPRKPKYELRVI I WNT DEVVLEDDDFTEKSSDIFVRGWLKGQEDKQD TDVHYHSLTGE GNFNWRYLFPFDYLA AEEKIVISKESMFSWDETEYKIPARLTQIW DADHFSADDFLGAIELDLNRFPRGAKTAKQCTMEMATGEVDVPLV SIFKQKRVKGWNP LLARNENDEFELTGKVEALHLLTAEAEKNPV GLARNEPDLEKPNRPDTAFVWFLNPLKSIKYLICTRYKWLIIKIVLAL LGLMLGLFLYSLPGYMKVKKLLGA
6	mOTOF-201_1 protein (NP_114081.2), mouse otoferlin isoform 2, 1997 aa	MALIVHLKTVSELRGKGDRIAKVTFRGQSFYSRVLENCEGVADFDE TFRWPVASSIDRNEVLEIQIFNYSKVFSNKLI GTFMVLQKVVEENR VEVTDTLMDDSNAIIKTSLSMEVRYQATDGTVGWDDGDFLGDES LQEEKDSQETDGLLPGRSPSTRISGEKSFRRAGRSVFSAMKLGKTR SHKEEPQRQDEPAVLEMEDLDHLAQLGDGLDPDSVSLASVTALTS NVSNKRSKPDIKMEPSAGRPMQVQVITVIEARQLVGLNMDPVVCV EVGDDKKYTSMKESTNCPYYNEYFVDFHVSFDPVMFDKI IKISVIHS KNLLRSGTLVGSFKMDVGTVYSQPEHQFHHKWAILSDPDDISAGLK GYVKCDVAVVGKGDNIKTPH KANETDEDDIEGNLLLP EGVPPPERQ WARFYVKIYRAEGLPRMNTSLMANVKKAFI GENKDLVDPYVQVFFA GQKGKTSVQKSSYEPLWNEQVVFDTLFPPLCKRMKVQIRDSKVN DVAIGTHFIDL RKISNDGDKGFLPTLGP AWVNMVYSTRNYTLLDEH QDLNEGLGEGVSFRARLLMLGLAVEILDTSNPELTSSTEVQVEQATP VSECTGRMEFFLFGAPLEASMDRKNNGDKPI TFEVTIGNYGNV DGMSRPLRPRPRKEPGDEEVDLIQNSSDDEGDEAGDLASVSSTP PMRPQITDRNYFHLPYLERKPCIYIKSWWPDQRRRLYNANIMDHIA DKLEEGLNVDQEMI KTEKSYPERRLRGVLEELSCGCHRFSLSDKD QGRSSRTRLDRELRKSCMRELES MGQQA KSLRAQVKRHTVRDKL RSCQNFLOKLRLFLADEPQHSIPDVFIMMSSNNKRIAYARVP SKDLL FSIVEEELGKDCAKVKTFLKLPGKRGFSGAGWTVQAKLELYLWLG LSKQRKDFLCGLPCGFEEVKAAQGLGLHSFPPISLVYTKQAFQLR AHMYQARSLFAADSSGLSDPFARVFFINQSQCTEVLNETLCPTWD QMLVFDNLELYGEAHEL RDDPPII VI E IYDQDSMGKADFMGRTF LVKMADEAYCPRFPQLEYYQIYRGSATAGDLLAA FELLQIGPSG KADLPPI NGPVDMDRGP IMPVPMGIRPVLSKYRVEVLFWGLRDLKR VNLAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFV ENELLHPPNLNIRVVD CRAFTYTLVGSHAVSSLRRFIYRPPDRSAP NWNTTVRLRGCHRLRNGGSSSRPTGEVVVSMEPEEPVKKLET VKLDATSDAVVKVDVAEDEKERRKKKKKGPS EEP EEEEPDESMLD WWSKYFASIDTMKEQLRQHETSGTDLEEKEEMESAEGLKGP MKS

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		KEKSRAAKEEKKKKNQSPGPGQGSEAPEKKKAKIDELKVYPKELES EFDSFEDWLHTFNLRLRGKTGDEDEGSTEEERIVGRFKGSLCVYKV PLPEDVSRAGYDPTYGMFQGISNDPINVLVRIYVVRATDLHPADI NGKADPYIAIKLGKTDIRDKENYISKQLNPVFGKSFIDIEASFPMESML TVAVYDWDLVGTDDLIGETKIDLENRFYSKHRATCGIAQYTSIHGYN IWRDPMKPSQILTRLCKEGKVDGPHFGPHGRVVRVANRVFTGPSEIE DENGQRKPTDEHVALSALRHWEDIPRVGCRVLVPEHVETRPLLNPD KPGIEQGRLELWVDMFPMMPAPGTPLDISPRKPKYELRVIVWNT DEVVLEDDDDFTGEKSSDIFVRGWLKGQEDKQDQTDVHYHSLTGE GNFNWRYLFPFDYLAEEKIVMSKESMFSWDETEYKIPARLTQLI WDADHFSADDFLGAIELDLNRFPRGAKTAKQCTMEMATGEVDVPL VSIFKQKRVKGWWPLLARNENDEFELTGKVEAELHLLTAAEEAKNP VGLARNEPDPLEKPNRPDTSFIWFLNPLKSARYFLWHTYRWLLLK FLLFLLLLLFALFLYSLPGYLAKKILGA
7	mOTOF-201_2 protein (NP_001273350.1), mouse otoferlin isoform 3, 1977 aa	MALIVHLKTVSELRGKGDRIAKVTFRGQSFYSRVLENCEGVADFDE TFRWPVASSIDRNEVLEIQIFNYSKVFSNKLIGTCFVQLQKVVLENR VEVTDTLMDDSNAIIKTSLSMEVRYQATDGTGVPWDDGDFLGDESL QEEKDSQETDGLLPGSRPSTRISGEKSPRRAGRSVFSAMKLGKTR SHKEEPQRQDEPAVLEMEDLDHLAIQLGDLDPDSVSLASVTALTS NVSNKRKSPDIKMEPSAGRPMDYQVSIITVIEARQLVGLNMDPVVVCV EVGDDKKYTSMKESTNCPYYNEYFVDFHVSFDPVDFDKIIKISVIHS KNLLRSGTLVGSFKMDVGTVYSQPEHQFHHKMAILSDPDDISAGLK GYVKCDVAVVGKGDNIKTPHKAETDEDDIEGNLLLPPEGVPPERQ WARFVYVKIYRAEGLPRMNTSLMANVKKAFIENKDLVDPYVQVFFA GQKGKTSVQKSSYEPLWNEQVVFDTLFPPLCKRMKVQIRSDSKVN DVAIGTHFIDLRKISNDGDKGFLPTLGPWVNMVYGSTRNITLLDEH QDLNEGLGEGVSFRARLMLGLAVEILDTSNPELTSSTEVQVEQATP VSECTGRMEEFFLFGAFLEASMDRKNKDKPITFEVTIGNYGNV DGMSRPLRPRPRKEPGDEEVDLIQNSDDDEAGDLASVSSTP PMRPQITDRNYFHLPLYLERKPCIYIKSWWPDQRRRLYNANIMDHIA DKLEEGLNVDQEMIKTEKSYPERRLRGVLEELSCGCHRFSLSDKD QGRSSRTRLDRELRLKSCMRELESMGQQAQSLRAQVKRHTVRDKL RSCQNFLQKLRLADEPQHSIPDVFIWMSNNKRIAYARVPSKDLL FSIVEEELGKDCAKVKTFLFKLPGKRGFGSAGWTVQAKLELYLWLG LSKQRKDFLCGLPCGFEEVKAQGLGLHSFPPISLVYTKQAFQLR AHMYQARSLFAADSSGLSDPFARVFFINQSQCTEVNLTLCPTWD QMLVFDNLELYGEAHELDDPPIVIEIYDQDSMGKADFMGRTPAKP LVKMADEAYCPRFPQLEYYQIYRGSATAGDLLAAPELLQIGPSG KADLPPINGPVDMDRGPIMPVPVGIRPVLSKYRVEVLFWGLRDLKR VNLAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFVVDLP ENELLHPLNIRVVD CRAFTGRYTLVGSHAVSSLRRFIYRPPDRSAP NWNTTGEVVVSMEPEEPVKLETVMVKLDATSDAVVKVDVAEDEKE RKKKKKKGPSEPEEEEEPEDESMLDWSKYFASIDTMKEQLRQHET SGTDL EEKEEMESAEGLKGPMSKEKSRAAKEEKKKKNQSPGPG QGSEAPEKKKAKIDELKVYPKELESEFDSFEDWLHTFNLRLRGKTGD DEDGSTEEERIVGRFKGSLCVYKVLPLPEDVSRAGYDPTYGMFQGI PSNDPINVLVRIYVVRATDLHPADINGKADPYIAIKLGKTDIRDKENYI SKQLNPVFGKSFIDIEASFPMESMLTVAVYDWDLVGTDDLIGETKIDL ENRFYSKHRATCGIAQYTSIHGYNIWRDPMKPSQILTRLCKEGKVD GPHFGPHGRVVRVANRVFTGPSEI EDENGQRKPTDEHVALSALRHW EDIPRVGCRVLVPEHVETRPLLNPDKPGIEQGRLELWVDMFPMMP APGTPLDISPRKPKYELRVIVWNTDEVVLEDDDDFTGEKSSDIFVR GWLKGQEDKQDQTDVHYHSLTGEGNFNWRYLFPFDYLAEEKIV MSKESMFSWDETEYKIPARLTQLIWDADHFSADDFLGAIELDLNRF PRGAKTAKQCTMEMATGEVDVPLVSIFKQKRVKGWWPLLARNEN DEFELTGKVEAELHLLTAAEEAKNPVGLARNEPDPLEKPNRPDTSFI WFLNPLKSARYFLWHTYRWLLLKFLLLFLLLLLFALFLYSLPGYLAKK ILGA
8	mOTOF-202_1 protein (NP_001093865.1), mouse otoferlin isoform 1, 1992 aa	MALIVHLKTVSELRGKGDRIAKVTFRGQSFYSRVLENCEGVADFDE TFRWPVASSIDRNEVLEIQIFNYSKVFSNKLIGTCFVQLQKVVLENR VEVTDTLMDDSNAIIKTSLSMEVRYQATDGTGVPWDDGDFLGDESL QEEKDSQETDGLLPGSRPSTRISGEKSPRSKGREKTGGGRDGEHK AGRSVFSAMKLGKTRSHKEEPQRQDEPAVLEMEDLDHLAIQLGDL LDPDSVSLASVTALTSNVSNKRKSPDIKMEPSAGRPMDYQVSIITVIE ARQLVGLNMDPVVVCVEVGDDKKYTSMKESTNCPYYNEYFVDFHVS SPDVMFDKIIKISVIHSKNLLRSGTLVGSFKMDVGTVYSQPEHQFHH KMAILSDPDDISAGLKGYVKCDVAVVGKGDNIKTPHKAETDEDDIE GNLLLPPEGVPPERQWARFVYVKIYRAEGLPRMNTSLMANVKKAFIGE NKDLVDPYVQVFFAGQKGKTSVQKSSYEPLWNEQVVFDTLFPPLC KRMKVQIRSDSKVNDVAIGTHFIDLRKISNDGDKGFLPTLGPWVNM MYGSTRNITLLDEHQDLNEGLGEGVSFRARLMLGLAVEILDTSNPE LTSSTEVQVEQATPVSECTGRMEEFFLFGAFLEASMDRKNKDKP ITFEVTIGNYGNVVDGMSRPLRPRPRKEPGDEEVDLIQNSDDDEG

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		DEAGDLASVSSTPPMRPQITDRNYFHLPLYLERKPCIIYKSWWPDQR RRLYNANIMDHIADKLEEGLNVDQEMIKTEKSYPERRLRGVLEELSC GCHRFSLSDKDQGRSSRTRLDRERLKSCMRELESMGQQAASLR AQVVRHTVRDKLRSCQNFLQKLRLFLADEPQHSIPDVFIIWMSNNK RIAYARVPSKDLLFSIVEEELGKDCAKVKTFLKLPGRGFGSAGWT VQAKLELYLWLGLSKQRKDFLCGLPCGFEEVKAAQGLGLHSFPPIS LVYTKKQAFQLRAHMYQARSIFAADSSGSLDPPFARVFFINQSQCTE VLNETLCPTWQMLVFDNLELYGEAHELDDPPIIVIEIYDQDSMGK ADFMGRRTFAKPLVKMADEAYCPRFPQLEYQIYRGSATAGDLLA AFELLQIGPSGKADLPPINGPVDMDRGPIMPVPVGIKRVLSKYRVEV LFWGLRDLKRVNLAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNT LVKWFVVDLPENELLHPLNIRVVDCAFGRYTLVGSHAVSSLRRFI YRPPDRSAPNWNNTTGEVVVSMEPEEPVKKLETMVKLDATSDAVVK VDVAEDEKERKKKKKGPSSEPEEEEPDESMLDWSKYFASIDTM KEQLRQHETSGTDLEEKEEMESAEGKGPMSKEKSRAAKEEKKK KNQSPGPGQGSEAPEKKKAKIDELKVYPKELESEFDSFEDWLHTF NLLRGKTGDDGSTEERIVGRFKGSLCVYKVPDPEDVSREAGY DPTYGMFQGISNDPINVLRIVVVRATDLHPADINGKADPYIAIKLG KTDIRDKENYISKQLNPVFGKSFIDEASFPMESMLTVAVYDWDLVG TDDLIGETKIDLENRFYSKHRATCGIAQTYSIHGYNIRWDPMPKPSQIL TRLCKEGKVDGPHFGPHGRVVRANRVFTGPSEIEDENGQRKPTDE HVALSALRHWEDI PRVGCRLVPEHVETRPLLNPDKPGIEQGRLELW VDMFPMMPAPGTPLDISPRKPKYELRVIVWNTDEVVLEDDDDFFT GEKSSDIFVRGWLKGQEDKQDQTDVHYHSLTGEGNFNWRYLFPF DYLAABEEKIVMSKKESMFSDTEYKIPARLTLQIWDADHFSADDF LGAI ELDLNRFPAGAKTAKQCTMEMATGEVDVPLVSIKQKRVKGV WPLLARNENDEFELTGKVEAELHLLTAAEAENPVGLARNEPDPLE KPNRPDTAFVWFLNPLKSIKYLICTRYKWLIIKIVLALLGLLMLALFLY SLPGYMKVKKLLGA
9	mOTOF-202_2 protein (NP_001300696.1), mouse otoferlin isoform 4, 1977 aa	MALIVHLKTVSELRGKGDRIAKVTFRGQSFYSRVLENCEGVADFDE TFRWPVASSIDRNEVLEIQIFNYSKVFSNKLIGTCFVQLQKVVEENR VEVTDTLMDDSNAIIKTSLSMEVRYQATDGTGVPWDDGDFLGDSESL QEEKDSQETDGLLPGSRPSTRISGEKSPRRAGRSVFSAMKLGKTR SHKEEPQRQDEPAVLEMEDLDHLAIQLGDGLDPSVSLASVTAITS NVSNKRSPDKIMKPSAGRPMQYQVSIITVIEARQLVGLNMDPVVVCV EVGDDKKYTSMKESTNCPYYNEYFVDFHVSFDPVDFDKIKISVIHS KNLLRSGTLVGSFKMDVGTVYSQPEHQFHHKWAILSDPDDISAGLK GYVKCDVAVVGKGDNIKTPHKANETDEDDIEGNLLLPEGVPPERQ WARFVYVKIYRAEGLPRMNTSLMANVKKAFIENKDLVDPYVQVFFA GQKGKTSVQKSSYEPLWNEQVVFDTLFPPLCKRMKVQIRDSKVN DVAIGTHFIDLKISNDGDKGFLPTLGPWVWVNGSTRNYYTLLEDEH QDLNEGLGEGVSFRARLMLGLAVEILDTSNPELTSSTEVQVEQATP VSECTGRMEFFLFGAFLEASMDIRKNGDKPITFEVTIGNYGNV DGMSRPLRPRPRKEPGDEEVDLIQNSSDDEGDEAGDLASVSSTP PMRPQITDRNYFHLPLYLERKPCIIYKSWWPDQRRRLYNANIMDHI ADKLEEGLNVDQEMIKTEKSYPERRLRGVLEELSCGCHRFSLSDKD QGRSSRTRLDRERLKSCMRELESMGQQAASLRAQVVRHTVRDKL RSCQNFLQKLRLFLADEPQHSIPDVFIIWMSNNKRIAYARVPSKDLL FSIVEEELGKDCAKVKTFLKLPGRGFGSAGWTQAKLELYLWLGL LSKQRKDFLCGLPCGFEEVKAAQGLGLHSFPPISLVYTKKQAFQLR AHMYQARSIFAADSSGSLDPPFARVFFINQSQCTEVLNETLCPTWD QMLVFDNLELYGEAHELDDPPIIVIEIYDQDSMGKADFMGRRTFAK PLVKMADEAYCPRFPQLEYQIYRGSATAGDLLAFAPELLQIGPSG KADLPPINGPVDMDRGPIMPVPVGIKRVLSKYRVEVLFWGLRDLK RVNLAQVDRPRVDIECAGKGVQSSLIHNYKKNPNFNTLVKWFVVDLP ENELLHPLNIRVVDCAFGRYTLVGSHAVSSLRRFIYRPPDRSAP NWNNTTGEVVVSMEPEEPVKKLETMVKLDATSDAVVKVDVAEDEKE RKKKKKGPSSEPEEEEPDESMLDWSKYFASIDTMKEQLRQHET SGTDLEEKEEMESAEGKGPMSKEKSRAAKEEKKKNQSPGPG QGSEAPEKKKAKIDELKVYPKELESEFDSFEDWLHTFNLLRGKTGD DEGGSTEERIVGRFKGSLCVYKVPDPEDVSREAGYDPTYGMFQGI PSNDPINVLRIVVVRATDLHPADINGKADPYIAIKLGKTDIRDKENYI SKQLNPVFGKSFIDEASFPMESMLTVAVYDWDLVGTDDLIGETKIDL ENRFYSKHRATCGIAQTYSIHGYNIRWDPMPKPSQILTRLCKEGKVD GPHFGPHGRVVRANRVFTGPSEIEDENGQRKPTDEHVALSALRH WEDI PRVGCRLVPEHVETRPLLNPDKPGIEQGRLELWDMFPMMP APGTPLDISPRKPKYELRVIVWNTDEVVLEDDDDFFTGEKSSDIFVR GWLKGQEDKQDQTDVHYHSLTGEGNFNWRYLFPFDYLAABEEKIV MSKKESMFSDTEYKIPARLTLQIWDADHFSADDFLGAI ELDLNR FPRGAKTAKQCTMEMATGEVDVPLVSIKQKRVKGVWPLLARNEN DEFELTGKVEAELHLLTAAEAENPVGLARNEPDPLEKPNRPDTAF VWFLNPLKSIKYLICTRYKWLIIKIVLALLGLLMLALFLYSLPGYMKV LLGA

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		CCAGGATTCCATGGGCAAAGCTGACTTCATGGGCCGGACCTTC GCCAAACCCCTGGTGAAGATGGCAGACGAGGCGTACTGCCAC CCCGCTTCCCACCTCAGCTCGAGTACTACCAGATCTACCGTGGC AACGCCACAGCTGGAGACCTGCTGGCGGCCCTTCGAGCTGCTGC AGATTGGACACGACAGGGAAGGCTGACCTGCCCCCATCAATGG CCCGCTGGACGTGGACCGAGGTCCCATCATGCCCGTGCCCATG GGCAATCCGGCCCGTGTCTCAGCAAGTACCGAGTGGAGGTGTGT TCTGGGGCTACGGGACCTAAAGCGGTGAACCTGGCCAGGT GGACCGGCCACGGTGGACATCGAGTGTGACGGGAAGGGGT GCAGTCGTCCCTGATCCACAATTATAAGAAGAACCCCAACTTCA ACACCCCTCGTCAAGTGGTTTGAAGTGGACCTCCAGAGAACGA GCTGTGCAACCCGCTTGAACATCCGTGTGGTGGACTGCCGG GCCTTCGGTCTGCTACACACTGGTGGGCTCCCATGCCGTGAGCT CCCTGCGACGCTTCATCTACCGGCCCCAGACCGCTCGGCCCC CAGCTGGAACACCAGGTGAGGCTTCTCCGGCGCTGCCGTGTG CTGTGCAATGGGGCTCCTCCTCTCACTCCACAGGGGAGGTTG TGGTGAATATGGAGCCAGAGGTACCCATCAAGAACTGGAGAC CATGGTGAAGCTGGACGCGACTTCTGAAGCTGTTGTCAAGGTG GATGTGGCTGAGGAGGAGAAGGAGAAGAAGAAGAAGAAGG GCACCTGCGGAGGAGCCAGAGGAGGAGGCCAGACGAGAGCA TGCTGGACTGGTGGTCCAAGTACTTTGCCTCCATTGACACCATG AAGGAGCAACTTCGACAACAAGAGCCCTCTGGAATTGACTTGA GGAGAAGGAGGAAGTGGACAATACCGAGGGCTGAAGGGGT AATGAAGGGCAAGGAGAAGGCAAGGGCTGCCAAGAGGAGAAG AAGAAGAAAACCTCAGAGCTCTGGCTCTGGCCAGGGGTCCGAGG CCCCAGAGAAGAAGAAACCAAGATTGATGAGCTTAAGGTATAC CCCAAGAGCTGGAGTCCGAGTTTGATAAATTTGAGGACTGGCT GCACACTTTCAACTTGCTTCGGGGCAAGACCGGGATGATGAG GATGGCTCCACCGAGGAGGAGCGCATTTGTGGGACGCTTCAAGG GCTCCTCTGCGTGTACAAAGTGCCACTCCAGAGGACGTGTC CCGGAAGCCGGCTACGACTCCACCTACGGCATGTTCCAGGGC ATCCCGAGCAATGACCCCATCAATGTGCTGGTCCGAGTCTATGT GGTCCGGGCCACGACCTGCACCTGCTGACATCAACGGCAAA GCTGACCCCTACATCGCCATCCGGCTAGGCAAGACTGACATCC GCGACAAGGAGAAGTACATCTCCAAGCAGCTCAACCTGTCTTT GGGAAGTCCTTTGACATCGAGGCTCCTTCCCATGGAATCCAT GCTGACGGTGGCTGTGTATGACTGGGACCTGGTGGGCACTGAT GACCTCATTTGGGAAACCAAGATCGACCTGGAGAACCGCTTCTA CAGCAAGCACCGCGCCACCTGCGGCATCGCCAGACCTACTCC ACACATGGCTACAATATCTGGCGGACCCCATGAAGCCAGCC AGATCCTGACCCGCTCTGCAAGAGCGCAAGTGGACGGGCC CCACTTTGGGCCCCCTGGGAGAGTGAAGGTGGCAACCGCGTC TTCCTGGGCCCCCTGAGATTGAGGACGAGAACGGTCAGAGGA AGCCACAGACGAGCATGTGGCGCTGTTGGCCCTGAGGCACTG GGAGGACATCCCCGCGCAGGCTGCCGCTGGTGCAGAGCA TGTGGAGACGAGGCCGCTGTCAACCCGACAAGCCGGGCATC GAGCAGGGCCGCTGGAGCTGTGGTGGACATGTTCCCATGG ACATGCCAGCCCCCTGGGACGCTCTGGACATCTCACCTCGGAA GCCCAGAAGTACGAGCTGCGGGTCTCATCTGGAACACAGAT GAGGTGGTCTTGGAGGACGACGACTTCTTACAGGGGAGAAGT CCAGTGACATCTTCTGAGGGGGTGGCTGAAGGGCCAGCAGGA GGACAAGCAGGACACAGACGTCCACTACCACTCCTCTCACTGGC GAGGGCAACTTCAACTGGCGCTACCTGTTCCTTTCGACTACCT GGCGGCGGAGGAGAAGATCGTCATCTCCAAGAAGGAGTCCATG TTCTCCTGGGACGAGACCGAGTACAAGATCCCCGCGCGGCTCA CCTGCAGATCTGGGATGCGGACCACTTCTCCGCTGACGACTT CTTGGGGCCATCGAGCTGGACCTGAACCGGTTCCTCGCGGG CGCAAAGACAGCCAAGCAGTGCCCATGGAGATGGCCACCGGG GAGGTGGACGTGCCCTCGTGTCTCATCTTCAAGCAAAAGCGCG TCAAAGGCTGGTGGCCCCCTCTGGCCCGCAATGAGAACGATGA GTTTGAGCTCACGGCAAGGTGGAGGCTGAGCTGCATTTACTG ACAGCAGAGGAGGACAGAGAAGAACCAAGTGGGCTGGCCCGC AATGAACCTGACCCCTAGAGAAACCAACCGGCCCGACACGA GCTTCATCTGGTTCCTGAACCTCTCAAGTGGGCTCGCTACTTC TTGTGGCACACGATCGCTGGCTGCTCCTCAAAGTGTGTGCTGCT CCTGTGCTGTCTCTCTCTCGCCCTGTTCTCTACTCTGTGC CTGGCTACCTGGTCAAGAAAATCCTCGGGGCTGAGCCAGTG GCCTCCTGGCCGGCCCGACAGGCTTCTGTGTTCTCTAAC CCTCTCAAGTCCATCAAGTACCTCATCTGCACCCGGTACAAGTG GCTCATCATCAAGATCGTGTGGCGCTGTTGGGGCTGCTCATGT TGGGGCTCTTCTCTACAGCTCCTGGCTACATGGTCAAAAAG CTCCTTGGGGCATGAAGGCCGCCAGCTCCCGCCAGCCGCTCCC CAGCCCTGCCGATTTCTTTTCAAGTGGCTTGGACTTTTCCCAT CTCCCTGGGGAGCCTGAGGAGCCAGCGTCCACTCTTCATGC CTTGGGCCGAGCCTGCCTCTGCTTGGGGGGCCGCTGTCTCT

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		CACTGCCCCAGGCTGCGGCTTGCCAGTCCCGCCCCCTCTGACC CCTGCCTGTGGCTGGGGAGCCTTGGATGGGGTGGGGACCTG GAATGGGTCTCTTTGCCCCACCTGGCTGAGGCGCCACCCTTC TTCAGGCCCCAGGCTCCAGAGGAAGACTCCTGAAACCTCCCCA GGTCTTCCAAGTACAGGATTGAAGCTTTAGTGAAATTAACCAAG GACCATGGGTCACTGCCAGGGCTTTAAAAGAAATGAACGAGC AAAAGGTATCCCCGCGTGACCCCTGCAGATAGCACCGGTCTTT GATCCGACAGAGGGCCAGACCCTGCCACAAGTCCCAGCGC GGCTGCTTCTGCCACTGTGGGCTCCACTTGGCTCCTCTCACTT CCCAGGGGGTGCCTGTCTGCTGTGGGTTTCCATGGCTTCC CAGAGCTCCCTCTGCCCCAGCCAGCGCTCCAGGCCAGCTGA GGAGCTGTGAGAAGCAGCAGAGGGGACTCCCATCCCGGGCA CACCTGTCTCTCCACCCCTGCCCTTGCCTTCCAGCCCTTT CAGCTGCAGCTGGGAGCTGGCCCTCAAGTGTGCCCTGCCT GTGTCTGGGTTTCTGTTGGCTGTTTTCTTTCTTGAGTGGTGAT TTTTCTCTAAATAAAAGAAGTCAAGCACTGAAAAAAAAAAAAA A
11	OTOF-202 transcript (NM_004802.3), human otoferlin transcript variant 2, 4954 bp, encodes the protein of SEQ ID NO: 2	CCGTGAGTTCTGCCAGGCCCTGTGAGCTCACAGAGCCACAG ACTCACAGCCAGAGGTGGCTTCTTCCTTCAGGAACGAAGAAC CCCCATGAACACCAACATCTCCAGGTTCTGAGAACAGAACCTGG GAAATTGATGACTTCCTCATGATGACCGATACTCAGGATGGCCC TAGCGAGAGCTCCAGATCATGAGGAAGAAGGCTGAACGACA TACAGGAGATGATCAAAACGGAGAAGTCTTACCTGAGCGTCG CCTGCGGGGCTCTTGGAGGAGCTGAGCTGTGGCTGTGCGG CTTCTCTCTCCCTCGCTGACAAGGACCAGGGCCACTCATCCGC ACCAGGCTTGACCGGGAGCGCCTCAAGTCTGCAATGAGGGAGC TGGAACACATGGGGCAGCAGGCCAGGATGTGCGGGCCAGG TGAAGCGGCACACGGTGCGGGACAAGCTGAGGCTGTGCCAGAA CTTCTCTGCAAGCTGCGCTTCTTGGCGGACGAGCCCGAGCAC AGCATTCCCGACATCTTATCTGGATGATGAGCAACAACAAGCG TGTCGCCTATGCCCGTGTGCCCTCCAAGGACCTGTCTTCTCCA TCGTGGAGGAGGAGACTGGCAAGGACTGCGCCAAGGTCAAGAC GCTCTTCTTAAGCTGCCAGGGAAGCGGGGCTTCGGCTCGGCA GGCTGGACAGTGCAGGCCAAGGTGGAGCTGTACCTGTGGCTGG GCCTCAGCAACAGCGCAAGGAGTTCTGTGCGGCTGCCCTG TGGCTTCCAGGAGGTCAAGGCAGCCAGGGCTGGGCTGCAT GCCTTCCACCCGTGAGCTGGTCTACCCAAGAAGCAGGCGT TCCAGCTCCGAGCGCACATGTACAGGCCCGCAGCCTCTTTGC CGCCGACAGCAGCGGACTCTCAGACCCCTTTGCCCGCGTCTTC TTCATCAATCAGAGTCAGTGACAGAGGTGCTGAATGAGACCT GTGTCCACCTGGGACCAGATGCTGGTGTTCGACAACCTGGAG CTCTATGGTGAAGCTCATGAGCTGAGGACGATCCGCCCATCAT TGTCATTGAAATCTATGACAGGATTCATGGGCAAGCTGACT TCATGGGCCGGACCTTCGCCAAACCCCTGGTGAAGATGGCAGA CGAGGCGTACTGCCCCACCCGCTTCCACCTCAGCTCGAGTAC TACCAGATCTACCGTGGCAACGCCACAGCTGGAGACTGCTGG CGGCCTTCGAGCTGTGAGATTGACAGCAGGGAAGGCTGA CCTGCCCCCATCAATGGCCCGGTGGAGCTGGACCGAGGTCCC ATCATGCCCGTGCCATGGGCATCCGGCCCGTGTCTAGCAAGT ACCGAGTGGAGGTGCTGTTCTGGGGCTACGGGACCTAAAGCG GGTGAACCTGGCCAGGTGGACCGGCCACGGGTGGACATCGA GTGTGCAGGAAGGGGTGCAGTCGTCCCTGATCCACAATTAT AAGAAGAACCCTCACTTCAACACCTCGTCAAGTGGTTGAAGT GGACCTCCAGAGAACGAGCTGCTGCACCCGCCCTTGAACATC CGTGTGGTGGACTGCCGGGCTTCGGTCGCTACACACTGGTGG GCTCCCATGCGCTCAGCTCCCTGCGACGCTTCATCTACCGGCC CCCAGACCGCTCGGCCCCAGCTGGAACACACCGGGGAGGT TGTGGTGACTATGGAGCCAGAGGTACCATCAAGAACTGGAG ACCATGGTGAAGCTGGACGCGACTTCTGAAGCTGTTGTCAAGGT GGATGTGGCTGAGGAGGAGAAGGAGAAGAAGAAGAAGAAG GGCACTGCGGAGGAGCCAGAGGAGGAGGAGCCAGACGAGAGC ATGCTGGACTGGTGGTCCAAGTACTTTGCCCTCCATTGACACCAT GAAGGAGCAACTTCGACAACAAGAGCCCTCTGGAATTGACTTGG AGGAGAAGGAGGAAGTGGACAATACCGAGGGCTGAAGGGGT CAATGAAGGGCAAGGAGAAGGCAAGGGCTGCCAAGAGGAGAA GAAGAAGAAACTCAGAGCTCTGGCTCTGGCCAGGGGTCCGAG GCCCCGAGAAGAAGAAACCAAGATTGATGAGCTTAAGGTATA CCCCAAGAGCTGGAGTCCGAGTTTGATAACTTTGAGGACTGGC TGCACACTTTCACCTTGCTTCGGGCAAGACCGGGGATGATGA GGATGGCTCCACCGAGGAGGAGCGATTGTGGGACGCTTCAAG GGCTCCCTCTGCGTGACAAAGTGCCACTCCAGAGGAGCTGT CCCGGGAAGCCGGTACGACTCCACCTACGGCATGTTCAGGG CATCCGAGCAATGACCCCATCAATGTGCTGGTCCGAGTCTATG TGGTCCGGGCCACGGACTGCACCTGTGACATCAACGGCAA

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		AGCTGACCCCTACATCGCCATCCGGCTAGGCAAGACTGACATC CGCGACAAGGAGAACTACATCTCCAAGCAGCTCAACCCGTCTT TGGGAAGTCCTTTGACATCGAGGCCCTCTTCCCCATGGAATCCA TGCTGACGGTGGCTGTGTATGACTGGGACCTGGTGGGCACTGA TGACCTCATTGGGGAACCAAGATCGACCTGGAGAACCCTTCT ACAGCAAGCACC CGCCACCTGCGGCATCGCCAGACCTACTC CACACATGGCTACAATATCTGGCGGGACCCCATGAAGCCCAGC CAGATCCTGACCCGCTCTGCAAAGACGGCAAGTGGACGGCC CCCACTTTGGGCCCTTGGGAGTGAAGGTGGCCCAACCGCT CTTCACTGGGCCCTCTGAGATTGAGGACGAGAAGGTGAGAGG AAGCCACAGACGAGCATGTGGCGCTGTGGCCCTGAGGCACT GGGAGGACATCCCCCGCGCAGGCTGCGCCTGGTGCCAGAGC ATGTGGAGACGAGGCCGCTGCTCAACCCGACAGCCGGGCAT CGAGCAGGGCCGCTGGAGCTGTGGGTGGACATGTTCCCATG GACATGCCAGCCCTGGGACGCTCTGACATCTCACCTCGGA AGCCCAAGAAGTACGAGCTGCGGGTCACTCTGGAACACAGA TGAGGTGGTCTTGGAGGACGACGACTTCTTACAGGGGAGAAG TCCAGTGACATCTTCTGTAGGGGGTGGCTGAAGGGCCAGCAGG AGGACAAGCAGGACACAGACGTCCTACTACCTCCCTCACTGG CGAGGGCAACTTCAACTGGCGCTACCTGTTCCCTTCGACTACC TGGCGGCGGAGGAGAAGATCGTCACTCTCAAGAAGGAGTCCAT GTTCTCTGGGACGAGACCGAGTACAAGATCCCGCGCGGCTC ACCCCTGCAGATCTGGGATGCGGACCACTTCTCCGCTGACGACT CCTGGGGGCCATCGAGCTGGACCTGAACCGGTTCCTCGGGGG CGCAAAGACAGCCAAAGCAGTGCACCATGGAGATGGCCACCGGG GAGGTGGACGTGCCCTCGTGTCTCATCTTCAAGCAAAAGCGCG TCAAAGGCTGGTGGCCCCCTCTGGCCCGCAATGAGAACGATGA GTTTGAGCTCACGGGCAAGGTGGAGGCTGAGCTGCATTTACTG ACAGCAGAGGAGGACAGAGAAGAACCCAGTGGGCCCTGGCCCGC AATGAACCTGACCCCTAGAGAAACCCAAACCGGCCCGACACGA GCTTCATCTGGTTCCTGAACCTCTCAAGTCGGCTCGCTACTTC TTGTGGCACACGTATCGCTGGCTGCTCCTCAAACCTGTTGCTGCT CCTGTCTGTCTCTCTCTCTCGCCCTGTTCTCTACTCTGTGC CTGGCTACCTGGTCAAGAAAATCTCGGGGCCCTGAGCCAGTG GCCTCTTGCCCGGCCCGACACGGCCTTCGTCTGGTTCCTCAAC CCTCTCAAGTCCATCAAGTACCTCATCTGCACCCGGTACAAGTG GCTCATCATCAAGATCGTGTGGCGCTGTTGGGGCTGCTCATGT TGGGGCTCTTCTCTACAGCCTCCCTGGCTACATGGTCAAAAAG CTCCTTGGGGCATGAAGGCCGCCAGCTCCCGCCAGCCGCTCCC CAGCCCTGCCCATTTCTCTTTCAGTGGCTTGGACTCTTTCCCAT CTCCCTGGGGAGCCTGAGGAGCCAGCGTCCACTCTTCATGC CTTGGGCCGAGCCTGCCTCTGCTTGGGGGGCCGCTGTCTCT CACTGCCCCAGGCTGCGGCTTGCCAGTCCCGCCCCCTTGACC CCTGCCTGTGGGTGGGGAGCCTTGGATGGGGTGGGGACCTG GAATGGGTCTCTTTGCCCCACCTGGCTGAGGCGCCACCCTTC TTCAGGCCCAGGCTCCAGAGGAAGACTCCTGAAACCTCCCCA GGTCTTCCAAGTACAGGATTGAAGCTTTAGTGAATTAACCAAG GACCATGGGTCAGTGCCAGGGCTTTAAAAGAATGAACGAGC AAAAGGTATCCCGCGGTGACCCCTGCAGATAGCACCGGTCTTT GATCCGCAGCAGGGGCCAGACCTGCCACAAAGTCCAGCGC GGCTGCTTCTGCCACTGCTGGGCTCCACTGGCTCCTCTCACTT CCCAGGGGGTGCCTGTCTGCTGTGGGTTCATGGCTTCC CAGAGCTCCCTCTGCCCCAGCCAGCGCCTCCAGGCCAGCTGA GGAGCTGTGAGAAGCAGCAGAGGGGACTCCCATCCCGGGCA CACCTGTCTCTCCACCCCTGCCCTTGGCCCTCCAGCCCTTT CAGCTGCAGCTGGGAGCTGGCCCGTCAAGTGTGCCCCGTGCT GTGTCTGGGTCTCTGTGTGGCTGTTTTCTTTCTTGTAGTGGTGAT TTTTCTCTAAATAAAGAAGTCAAGCACTGAAAAAAAAAAAAA A
12	OTOF-203 transcript (NM_194323.2), human otoferlin transcript variant 4, 4756 bp, encodes the protein of SEQ ID NO: 3	CCGTGAGTTCTGCCCAGGCCCTGTGAGCTCACCAGAGCCACAG ACTCACAGCCCAGAGGTGGCTTCTTCTTCAGGAACCTGAAGAAC CCCCATGAACACCAACATCTCCAGGTTCTGAGAACAGAACCTGG GAAATTGATGACTTCTCATGATGACCGATACTCAGGATGGCCC TAGCGAGAGCTCCAGATCATGAGGAAGAAGGCTGAACGACA TACAGGAGATGATCAAAACGGAGAAGTCTACCTTGAGCGTCG CCTGCGGGGCGTCTGGAGGAGCTGAGCTGTGGCTGTCTGCCG CTTCTCTCTCTCGCTGACAAGGACAGGGCCACTCATCCCGC ACCAGGCTTGACCGGGAGCGCCTCAAGTCTGCATGAGGGAGC TGGAACACATGGGGCAGCAGGCCAGGATGCTGCGGGCCAGG TGAAGCGGCACACGGTGCGGGACAAGCTGAGGCTGTGCCAGAA CTTCTCTGCAAGCTGCGCTTCTTGGCGACGAGCCCGAGCAC AGCATTCCCGACATCTTCATCTGGATGATGAGCAACAACAGCG TGTCGCTATGCCGTGTGCCCTCAAGGACCTGCTCTTCTCCA TCGTGGAGGAGGAGACTGGCAAGGACTGCGCCAAGGTCAAGAC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GCTCTTCCTTAAGCTGCCAGGGAAGCGGGGCTTCGGCTCGGCA GGCTGGACAGTGCAGGCCAAGGTGGAGCTGTACCTGTGGCTGG GCCTCAGCAAAACAGCGCAAGGAGTTCTGTGCGGCCTGCCCTG TGGCTTCCAGGAGGTCAAGGCAGCCCAGGGCCTGGGCCTGCAT GCCTTCCACCCGTCAGCCTGGTCTACACCAAGAAGCAGGCGT TCCAGCTCCGAGCGCACATGTACCAGGCCCGCAGCCTCTTTGC CGCCGACAGCAGCGGACTCTCAGACCCCTTTGCCCCGCTCTTC TTCATCAATCAGAGTCAGTGACAGAGGTGCTGAATGAGACCCT GTGTCCACCTGGGACCAGATGCTGGTGTTCGACAACCTGGAG CTCTATGGTGAAGCTCATGAGCTGAGGACGATCCGCCCATCAT TGTCATTGAAATCTATGACCAGGATTCATGGGCAAGCTGACT TCATGGGCCGGACTTTCGCCAAACCCCTGGTGAAGATGGCAGA CGAGGCGTACTGCCACCCCGCTTCCACCTCAGCTCGAGTAC TACCAGATCTACCGTGGCAACGCCACAGCTGGAGACCTGCTGG CGGCCTTCGAGCTGCTGCAGATTGGACAGCAGGGAAGGCTGA CCTGCCCCCATCAATGGCCCCGTGGAGCTGGACCGAGGTCCC ATCATGCCCGTGCCCATGGGCATCCGGCCCCGTGCTCAGCAAGT ACCGAGTGGAGGTGCTGTTCTGGGGCTACCGGACCTAAAGCG GGTGAACCTGGCCCAGGTGGACCGGCCACGGGTGGACATCGA GTGTGCAGGGAAGGGGTGCACTCGTCCCTGATCCACAATTAT AAGAAGAACCCCAACTTCAACACCCCTCGTCAAGTGGTTGAAGT GGACCTCCCAGAGAACGAGCTGCTGCACCCGCCCTTGAACATC CGTGTGGTGGACTGCCGGGCCCTCGGTGCTACACACTGGTGG GCTCCCATGCGCTCAGCTCCCTGCGACGCTTCATCTACCGGCC CCCAGACCGCTCGGCCCCAGCTGGAACACCACGGGGGAGGT TGTGGTGACTATGGAGCCAGAGGTACCCATCAAGAACTGGAG ACCATGGTGAAGCTGGACGCGACTTCTGAAGCTGTGTCAAGGT GGATGTGGCTGAGGAGGAGAAGGAGAAGAAGAAGAAGAAG GGCACTGCGGAGGAGCCAGAGGAGGAGGAGCCAGACGAGAGC ATGCTGGACTGGTGGTCCAAGTACTTTGCCTCCATTGACACCAT GAAGGAGCAACTTCGACAACAAGAGCCCTCTGGAATTGACTTGG AGGAGAAGGAGGAAGTGGACAATACCGAGGGCCTGAAGGGGT CAATGAAGGGCAAGGAGAAGGCAAGGGCTGCCAAGAGGAGAA GAAGAAGAAAACCTCAGAGCTCTGGCTCTGGCCAGGGGTCCGAG GCCCCGAGAAGAAGAAAACCAAGATTGATGAGCTTAAGGTATA CCCCAAAGAGCTGGAGTCCGAGTTTGATAACTTTGAGGACTGGC TGCACACTTTCAACTTGCTTCGGGGCAAGACCGGGGATGATGA GGATGGCTCCACCGAGGAGGAGCGCATTGTGGGACGCTTCAAG GGCTCCCTCTGCGTGTACAAAGTGCCACTCCAGAGGACGTGT CCCGGAAGCCGGCTACGACTCCACCTACGGCATGTTCCAGGG CATCCCGAGCAATGACCCCATCAATGTGCTGGTCCGAGTCTATG TGGTCCGGGCCACGGACCTGCACCTGCTGACATCAACGGCAA AGCTGACCCCTACATCGCCATCCGGCTAGGCAAGACTGACATC CGCGACAAGGAGAACTACATCTCCAAGCAGCTCAACCTGTCTT TGGGAAGTCTTTGACATCGAGGCTCCTTCCCCATGGAATCCA TGCTGACGGTGGCTGTGTATGACTGGGACCTGGTGGGCACTGA TGACCTCATTGGGGAAACCAAGATCGACCTGGAGAACCCTTCT ACAGCAAGCACCAGCCACCTGCGGCATCGCCAGACCTACTC CACACATGGCTACAATATCTGGCGGGACCCCATGAAGCCAGC CAGATCCTGACCCGCTCTGCAAGACGGCAAGTGGACGGCC CCCACTTTGGGCCCTTGGGAGAGTGAAGGTGGCCAACCGCGT CTTCACTGGGCCCTCTGAGATTGAGGACGAGAACGGTCAGAGG AAGCCACAGACGAGCATGTGGCGCTGTTGGCCCTGAGGCACT GGGAGGACATCCCCGCGCAGGCTGCGCCTGGTGCCAGAGC ATGTGGAGACGAGGCCGCTGCTCAACCCGACAAGCCGGGCAT CGAGCAGGGCCGCTGGAGCTGTGGGTGGACATGTTCCCATG GACATGCCAGCCCCCTGGGACGCTCTGGACATCTCACCTCGGA AGCCCAAGAAGTACGAGCTGCGGGTCACTCTGGAACACAGA TGAGGTGGTCTTGGAGGACGACGACTTCTTACAGGGGAGAAG TCCAGTGACATCTCTGTGAGGGGTGGCTGAAGGGCCAGCAGG AGGACAAGCAGGACACAGACGTCCACTACCCTCCCTCACTGG CGAGGGCAACTTCAACTGGCGCTACCTGTTCCCTTCGACTACC TGGCGGCGGAGGAGAAGATCGTATCTCCAAGAAGGAGTCCAT GTTCTCTGGGACGAGACCGAGTACAAGATCCCGCGCGGCTC ACCTTGCAGATCTGGGATGCGGACCACTTCTCCGCTGACGACTT CCTGGGGGCCATCGAGCTGGACCTGAACCGGTTCCTCGGGG CGCAAAGACAGCCAAGCAGTGCACCATGGAGATGGCCACCGGG GAGGTGGACGTGCCCTCGTGTCACTCTTCAAGCAAAAGCGCG TCAAAGGCTGGTGGCCCCCTCTGGCCCGCAATGAGAACGATGA GTTTGAGCTCACGGCAAGGTGGAGGCTGAGCTGCATTTACTG ACAGCAGAGGAGGACAGAGAAGAACCAAGTGGGCTTGCCCGC AATGAACCTGACCCCTAGAGAAACCAACCGGCCGACACGG CCTTCGTCTGGTTCCTCAACCTCTCAAGTCCATCAAGTACCTCA TCTGCAACCCGTTACAAGTGGCTCATCATCAAGATCGTGTGGCG CTGTTGGGGCTGCTCATGTTGGGGCTCTTCTCTACAGCCTCCC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		TGGCTACATGGTCAAAAAGCTCCTTGGGGCATGAAGGCCGCCA GCTCCCGCCAGCCGCTCCCGAGCCCTGCCGATTTCCTTTCAG TGGCTTGGAATCTTTCCATCTCCCTGGGAGCCTGAGGAGC CCAGCGTCCACTCTTCATGCTTGGGCGAGCCTGCCTCCTGC TTGCGGGGGCCGCTGTCTCACTGCCCGAGGCTGCGGCTTGC CCAGTCCCGCCCTCTGACCCCTGCCTGTGGGCTGGGAGCCT TGGATGGGGTGGGGACCTGGAATGGGTCTCTCTTGGCCACCT GGCTGAGGCGCCACCCTTCTTCAGGCCAGGCTCCAGAGGAAG ACTCCTGAAACCTCCCGAGTCTTCAAGTACAGGATTGAAGC TTTAGTGAAATTAACCAAGGACCATGGGTGAGTGCAGGAGCTT TAAAAAGAATGAACGAGCAAAAGGTATCCCGCCGTGACCCCTG CAGATAGCACCGGTCTTTGATCCGAGCAGGGGCCAGACCCCTG CCCACAAGTCCCGAGCGCGGTGCTTCTGCCACTGCTGGGCTCC ACTTGGCTCCTCTCACTTCCAGGGGGTGCCTGTCTGCTGTG GGGTTTCCATGGCTTCCAGAGCTCCCTCTGCCCGAGCCAGCG CCTCCAGGCCAGCTGAGGAGCTGTGAGAAGCAGCAGAGGGG ACTCCCATCCCGGGCACACCTGTCTCCACCCCTGCCCTG TTGCCCTTCCAGCCCTTTCAGCTGAGCTGGGAGCTGGCCCGT CAAGTGCTGCCCTGCCTGTGTCTGGGTTTCTGTTGGCTGTTT TCTTTTCTGAGTGGTGATTTTCTCTAAATAAAGAAGTCAAGC ACTGAAAAAAAAAAAAAAAA
13	OTOF-208 transcript (NM_194322.2), human otoferlin transcript variant 3, 3924 bp, encodes the protein of SEQ ID NO: 4	CCGTGAGTCTGCCAGGCCCTGTGAGCTCACCAGAGCCACAG ACTCACAGCCAGAGGTGGCTTCTTCTTCAGGAACTGAAGAAC CCCCATGAACACCAACATCTCCAGGTTCTGAGAACAGAACCTGG GAAATTGATGACTTCTCATGATGACCGATACTCAGGATGGCCC TAGCGAGAGCTCCAGATCATGAGGTCCCTCACTCCCTGATCA ACAGGGAGGAGGCATTTGGGGAGGCTGGGGAGGCGGGGCTGT GGCCAGCATCACCACTCTGATTACAGGAAGAAGGCCT GAACGACATACAGGAGATGATCAAAACGGAGAAGTCTACCTG AGCGTCGCTGCGGGGCTCCTGGAGGAGCTGAGCTGTGGCT GCTGCCGCTTCTCTCCCTCGCTGACAAGGACAGGGCCACTC ATCCCGCACCAGGCTTGACCGGGAGCGCTCAAGTCTGTCATG AGGGAGCTGGAACATGGGGCAGCAGGCCAGGATGCTGCGG GCCCAGGTGAAGCGGCACACGCTGCGGGACAAAGCTGAGGCTG TGCCAGAACTTCTGCAAGCTGCGCTTCTGGCGGACGAGC CCCAGCACAGCATTTCCGACATCTTCATCTGGATGATGAGCAAC AACAAGCGTGTGCTATGCCCGTGTGCCCTCAAGGACCTGC TCTTCTCCATCGTGGAGGAGGAGACTGGCAAGGACTGCGCCAA GGTCAGACGCTCTTCTTAAGCTGCCAGGGAAGCGGGGCTTC GGCTCGGCAGGCTGGACAGTGCCAGGCCAAGGTGGAGCTGTAC CTGTGGCTGGGCTCAGCAACAGCGCAAGGAGTTCCTGTGCG GCCTGCCCTGTGGCTTCCAGGAGGTCAAGGCAGCCAGGGCCT GGGCTGATGCTTCCACCCGTGAGCTGGTCTACACCAAG AAGCAGGCGTTCAGCTCCGAGCGCACATGTACCAGGCCGCA GCCTCTTTGCCGCGACAGCAGCGGACTCTCAGACCCCTTTGC CCGCTCTTCTTCATCAATCAGAGTCAAGTGACAGAGGTGCTGA ATGAGACCTGTGTCCACCTGGGACAGATGCTGGTGTTCGA CAACCTGGAGCTCTATGGTGAAGCTCATGAGCTGAGGAGCAT CCGCCATCATGTGCTTGAATCTATGACAGGATTCCATGGG CAAAGCTGACTTCATGGGCCGACCTTCGCCAAACCCCTGGTG AAGATGGCAGACGAGGCTACTGCCACCCCGCTTCCACCTC AGCTCGAGTACTACCAGATCTACCGTGGCAACGCCACAGCTGG AGACCTGCTGGCGGCTTCGAGCTGCTGCAGATTGGACAGCA GGGAAGGCTGACCTGCCCCCATCAATGGCCCGGTGGACGCTG GACCGAGGTCCATCATGCCGTGCCATGGGCATCCGGCCCG TGCTCAGCAAGTACCGAGTGGAGGTGCTGTTCTGGGGCTACG GGACCTAAAGCGGGTGAACCTGGCCAGGTGGACCGGCCACG GGTGGACATCGAGTGTGAGGGAAGGGGTGCAGTCTGCTCCT GATCCACAATTATAAGAAGAACCCCAACTTCAACACCTCGTCAA GTGGTTTGAAGTGGACCTCCAGAGAACGAGCTGCTGCACCCG CCTTGAACATCCGTGTGGTGGACTGCCGGGCTTCCGTGCT ACACACTGGTGGCTCCCATGCCGTGAGCTCCTGCGACGCTT CATCTACCGGCCCCAGACGCTCGGCCCCAGCTGGAACACC ACGGTCAGGCTTCTCCGCGCTGCCGTGTGCTGTCAATGGGG GCTCCTCCTCTCACTCCACAGGGGAGGTGTGGTGACTATGGA GCCAGAGGTACCCATCAAGAACTGGAGACCATGGTGAAGCTG GACGCGACTTCTGAAGCTGTTGTCAAGGTGGATGTGGCTGAGG AGGAGAAGGAGAAGAAGAAGAAGAAGGGCACTGCGGAGG AGCCAGAGGAGGAGGAGCCAGACGAGAGCATGCTGACTGGT GGTCCAAGTACTTTGCCCTCCATTGACACCATGAAGGAGCAACTT CGACAACAAGAGCCCTCTGGAATTGACTTGAGGAGAAGGAGG AAGTGGACAATACGAGGGCTGAAGGGGTCAATGAAGGGCAA GGAGAAGGCAAGGGCTGCCAAGAGGAGAAGAAGAAGAAACT CAGAGCTCTGGCTCTGGCCAGGGGTCAGAGGCCCGGAGAAG

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		CAGCAACAAGCTCATCGGGACCTTCCGCATGGTGTCTGCAGAAG GTGGTAGAGGAGAGCCATGTGGAGGTGACTGACACGCTGATTG ATGACAACAATGCTATCATCAAGACCAGCCTGTGCGTGGAGGT CGGTATCAGGCCACTGACGGCACAGTGGGCTCCTGGGACGATG GGGACTTCCTGGGAGATGAGTCTTCAAGAGGAAGAGAAGGA CAGCCAAGAGACGGATGGACTGCTCCAGGCTCCCGGCCAGC TCCCGGCCCCCAGGAGAGAAGAGCTTCCGGAGAGCCGGGAGG AGCGTGTTCTCCGCCATGAAGCTCGGCAAAAACCGGTCTCACA GGAGGAGCCCCAAGACCAGATGAACCGGCGGTGCTGGAGAT GGAAGACCTTGACCATCTGGCCATTCCGGCTAGGAGATGGACTG GATCCCGACTCGGTGTCTCTAGCCTCAGTCACAGCTCTCACCAC TAATGTCTCCAACAAGCGATCTAAGCCAGACATTAAGATGGAGC CAAGTGCTGGGCGGCCCATGGATTACCAAGTCCAGCATCACGGT GATCGAGGCCCGGCAGCTGGTGGGCTTGAAACATGGACCTGTG GTGTGCGTGGAGGTGGGTGACGACAGAAGTACACATCCATGA AGGAGTCCACTAACTGCCCTATTACAACGAGTACTTCGTCTTC GACTTCCATGTCTCTCCGGATGTGATGTTTGACAAAGATCATCAAG ATTTGCGGTGATTCACTCCAAGAACCTGCTGCGCAGTGGCACCT GGTGGGCTCCTTCAAAATGGACGTGGGAACCGTGTACTCGCAG CCAGAGCACCAAGTTCATCACAAGTGGGCCATCCTGTCTGACCC CGATGACATCTCCTCGGGCTGAAGGGCTACGTGAAGTGTGAC GTTGCCGTGGTGGGCAAGGGGACAACATCAAGACGCCCCACA AGGCCAATGAGACCGACGAAGATGACATTGAGGGGAAGTGTGCT GCTCCCCGAGGGGTGCCCCCGAACGCCAGTGGGCCCGGT CTATGTGAAAATTTACCGAGCAGAGGGGCTGCCCGTATGAACA CAAGCCTCATGGCCAATGTAAGAAGGGCTTTCATCGGTGAAAAC AAGGACCTCGTGGACCCCTACGTGCAAGTCTTCTTTGTGGCCA GAAGGGCAAGACTTCAGTGCAGAAGAGCAGCTATGAGCCCCTG TGGAATGAGCAGGTGCTTTACAGACCTTCCCCCACTCTG CAAACGCATGAAGTGCAGATCCGAGACTCGGACAAGGTCAAC GACGTGGCCATCGGCACCCACTTCATTGACCTGCGCAAGATTT TAATGACGGAGACAAAGGCTTCCCTGCCACACTGGGCCAGCC TGGGTGAACATGTACGGCTCCACACGTAACTACACGCTGCTGGA TGAGCATCAGGACCTGAACGAGGGCTTGGGGAGGGTGTGTCT CTTCCGGGCCCGGCTCCTGCTGGGCCCTGGCTGTGGAGATCGTA GACACCTCCAACCTGAGCTCACAGCTCCACAGAGGTGACAGG TGGAGCAGGCCACGCCCATCTCGGAGAGCTGTGAGGTAATAAT GGAAGAATTCTTTCTTTGGAGCCTTCTGGAGGCTCAATGA TCGACCGGAGAAACGGAGACAAGCCCATCACCTTTGAGGTCA CATAGGCAACTATGGGAACGAAGTTGATGGCCTGTCCCGGCC CAGCGGCTCGGCCCGGAAGGAGCGGGGATGAGGAAGAA GTAGACCTGATTAGAACGCAAGTGATGACGAGGCGGTGATG CCGGGGACCTGGCCTCAGTCTCCTCCACTCCACCAATGCGGCC CCAGGTCAACGACAGGAAGTACTTCCATCTGCCCTACCTGGAGC GAAAGCCTGCTATACATCAAGAGCTGGTGGCCGACAGCG CCGCCGCTCTACAATGCCAACATCATGGACCACATTTGCCGACA AGCTGGAAGAAGGCTGAACGACATACAGGAGATGATCAAAAC GGAGAAGTCTTACCTGAGCGTCTGCTGCGGGGCTCCTGGA GGAGCTGAGCTGTGGCTGCTGCCCTTCTCTCCTCGCTGAC AAGGACCAGGGCCACTCATCCGCAACAGGCTTGACCGGAGC GCCTCAAGTCTGCTGATGAGGAGCTGGAACCATGGGCGAGCA GGCCAGGATGCTGCGGGCCAGGTGAAGCGGCACACGGTGCG GGACAAGCTGAGGCTGTGCCAGAACTTCTGCAGAAGCTGCGC TTCCTGGCGGACGAGCCCCAGCACAGCATTCCCGACATCTTCAT CTGGATGATGAGCAACAAGCGTGTGCGCTATGCCCGTGTG CCTCCAAGGACCTGCTCTTCTCATCGTGGAGGAGGAGACTG GCAAGGACTGCGCCAAGGTCAAGACGCTTCTCTTAAGCTGCC AGGGAAGCGGGCTTCGGCTCGGCAGGCTGGACAGTGCAGGC CAAGGTGGAGCTGTACCTGTGGCTGGGCTCAGCAACAGCGC AAGGAGTTCTGTGCGGCTGCTTGTGGCTTCCAGGAGGTCA AGGCAGCCAGGGCTTGGGCTGCATGCTTCCACCCGCTCAG CCTGGTCTACACCAAGAAGCAGGCTTCCAGCTCCGAGCGCAC ATGTACAGGCCCGCAGCCTCTTTGCGCGGACAGCAGCGGAC TCTCAGACCCCTTTGCCCGGCTCTTCTCATCAATCAGAGTCAG TGCAAGAGGTGCTGAATGAGACCTGTGTCACCTGGGACC AGATGCTGGTGTTCGACAACCTGGAGCTCTATGGTGAAGCTCAT GAGCTGAGGGACGATCCGCCCATCATTGTCTATTGAAATCTATGA CCAGGATTCATGGGCAAGCTGACTTCATGGGCCGACCTTC GCCAAACCCCTGGTGAAGATGGCAGACGAGGCGTACTGCCAC CCCGCTTCCACCTCAGCTCAGTACTACCAGATCTACCGTGGC AACGCCACAGCTGGAGACCTGTGGCGGCTTCGAGCTGCTGC AGATTGGACAGCAGGGAAGGCTGACCTGCCCGCCATCAATGG CCCGGTGGAGCTGGACCGAGGTCCCATCATGCCGTGCCCATG GGCAATCCGGCCGTGCTCAGCAAGTACCGAGTGGAGGTGCTGT TCTGGGGCTACGGGACCTAAAGCGGGTGAACCTGGCCGAGGT

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GGACCGGCCACGGGTGGACATCGAGTGTGCAGGGAAGGGGGT GCAGTCGTCCCTGATCCACAATTATAAGAAGAACCCCAACTTCA ACACCCTCGTCAAGTGGTTTGAAGTGGACCTCCCAGAGAACGA GCTGCTGCACCCGCCCTTGAACATCCGTGTGGTGGACTGCCGG GCCTTCGGTCGCTACACACTGGTGGGCTCCCATGCCGTCAGCT CCCTGCGACGCTTCACTACCGGCCCCAGACCGCTCGGCCCC CAGCTGGAACACCACGGTCAGGCTTCTCCGGCGCTGCCGTGTG CTGTGCAATGGGGGCTCCTCCTCACTCCACAGGGGAGGTTG TGGTGACTATGGAGCCAGAGGTACCCATCAAGAACTGGAGAC CATGGTGAAGCTGGACGCGACTTCTGAAGCTGTTGTCAAGGTG GATGTGGCTGAGGAGGAGAAGGAGAAGAAGAAGAAGAAGG GCACTGCGGAGGAGCCAGAGGAGGAGGAGCCAGACGAGAGCA TGCTGGACTGGTGGTCCAAGTACTTTGGCTCCATTGACACCATG AAGGAGCAACTTCGACAACAAGAGCCCTCTGGAATTGACTTGA GGAGAAGGAGGAAGTGGACAATACCGAGGGCTGAAGGGGTC AATGAAGGGCAAGGAGAAGGCAAGGGCTGCCAAAGAGGAGAAG AAGAAGAAAACTCAGAGCTCTGGCTCTGGCCAGGGGTCCGAGG CCCCCGAGAAGAAGAAACCAAGATTGATGAGCTTAAGGTATAC CCCAAAGAGCTGGAGTCCGAGTTTGATAACTTTGAGGACTGGCT GCACACTTTCAACTTGCTTCGGGGCAAGACCGGGGATGATGAG GATGGCTCCACCGAGGAGGAGCGCATTGTGGACGCTTCAAGG GCTCCCTCTGCGTGTACAAAGTGCCACTCCAGAGGACGTGTC CCGGGAAGCCGGCTACGACTCCACCTACGGCATGTTCCAGGGC ATCCCGAGCAATGACCCCATCAATGTGCTGGTCCGAGTCTATGT GGTCCGGGCCACGGACCTGCACCCCTGCTGACATCAACGGCAAA GCTGACCCCTACATCGCCATCCGGCTAGGCAAGACTGACATCC GCGACAAGGAGAACTACATCTCCAAGCAGCTCAACCTGTCTTT GGGAAGTCCTTTGACATCGAGGCCCTCCTTCCCATGGAATCCAT GCTGACGGTGGCTGTGTATGACTGGGACCTGGTGGGCACTGAT GACCTCATTTGGGGAACCAAGATCGACCTGGAGAACCCTTCTA CAGCAAGCACCGGCCACCTGCGGCATCGCCAGACCTACTCC ACACATGGCTACAATATCTGGCGGACCCCATGAAGCCAGCC AGATCCTGACCCGCTCTGCAAAGACGGCAAGTGGACGGCCC CCACTTTGGGCCCCCTGGGAGAGTGAAGGTGGCCAACCGCGTC TTCACTGGGCCCTCTGAGATTGAGGACGAGAACGGTCAAGGA AGCCCAAGACGAGCATGTGGCGCTGTTGGCCCTGAGGCACGTG GGAGGACATCCCCGCGCAGGCTGCCGCTGGTGCCAGAGCA TGTGGAGACGAGCCGCTGCTCAACCCGACAAGCCGGGCATC GAGCAGGGCCGCTGGAGCTGTGGGTGGACATGTTCCCATGG ACATGCCAGCCCCCTGGGACGCTCTGGACATCTCACCTCGGAA GCCCAAGAAGTACGAGCTGCGGGTCACTATCTGGAACACAGAT GAGGTGGTCTTGGAGGACGACGACTTCTTCAAGGGGAGAAGT CCAGTGACATCTTCTGAGGGGGTGGCTGAAGGGCCAGCAGGA GGACAAGCAGGACACAGAGCTCCACTACCACTCCCTCACTGGC GAGGGCAACTTCAACTGGCGCTACCTGTTCCTTTCGACTACCT GGCGCGGAGGAGAAGATCGTCATCTCCAAGAAGGAGTCCATG TTCTCTGGGACGAGACCGAGTACAAGATCCCGCGCGGCTCA CCCTGCAGATCTGGGATGCGGACCACTTCTCCGCTGACGACTT CCTGGGGGCCATCGAGCTGGACCTGAACCGGTTCCCGCGGGG CGCAAAGACAGCCAAGCAGTGCAACCATGGAGATGGCCACGGG GAGGTGGACGTGCCCTCGTGTCTATCTTCAAGCAAAAGCGCG TCAAAGGCTGGTGGCCCCCTCTGGCCCGCAATGAGAACGATGA GTTTGAGCTACGGGCAAGGTGGAGGCTGAGCTGCATTACTG ACAGCAGAGGAGGACAGAGAAGAACCCAGTGGGCTTGGCCCGC AATGAACCTGACCCCTAGAGAAACCAACCGGCCGACACGG CCTTCGTCTGGTTCTCAACCTCTCAAGTCCATCAAGTACCTCA TCTGCACCCGGTACAAGTGGCTCATCATCAAGATCGTGTGGCG CTGTTGGGGCTGCTCATGTTGGGGCTCTTCTCTACAGCCTCCC TGGCTACATGGTCAAAAAGCTCCTTGGGGCATGAAGGCCGCCA GCTCCCGCCAGCCGCTCCCGAGCCTGCCGCAATTCTCTTTCAG TGGCTTGGACTCTTTCCCATCTCCCTGGGGAGCCTGAGGAGC CCAGCGTCCACTTTCATGCTTGGGCGGAGCCTGCCCTCCTGC TTGCGGGGGCCGCTGTCTCACTGCCCCAGGCTGCGGCTTGC CCAGTCCCGCCCCCTTGACCCCTGCTGTGGGCTGGGGAGCCT TGGATGGGGTGGGGACCTGGAATGGGTCTCTTTCGCCACCT GGCTGAGGCGCCACCCTTCTTCAAGGCCAGGCTCCAGAGGAAG ACTCCTGAAACCCCTCCCAGGTCTTCCAAGTACAGGATTGAAGC TTTAGTGAAATTAACCAAGGACCATGGGTGAGTCCAGGGCTT TAAAAGAAATGAACGAGCAAAAGGTATCCCGCCGTGACCCCTG CAGATAGCACCGGTCTTTGATCCGAGCAGGGGCCAGACCCCTG CCCAAGTCCAGCGCGGCTGCTTCTGCCACTGTGGGCTCC ACTTGGCTCCTCTCACTTCCAGGGGGTGGCTGTCTGCTGTG GGGTTCATGGCTTCCAGAGCTCCCTCTGCCCCAGCCAGCG CCTCCAGGCCAGCTGAGGAGCTGTGAGAAGCAGCAGAGGGG ACTCCCATCCCGGCCACACCTGTCTCTCCACCCCTGCCCCC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		TTGCCCTTCCAGCCCTTTTCAGCTGCAGCTGGGAGCTGGCCCGT CAAGTGCTGCCCTGCCTGTGTCTGGGTTTCTGTTGGCTGTTTT TCTTTTCTTGAGTGGTGATTTTCTCTAAATAAAGAAGTCAAGC ACTGAAAAAAAAAAAAAAAA
15	mOTOF-201_1 transcript (NM_031875.2), mouse otoferlin transcript variant 2, 7129 bp, encodes the protein of SEQ ID NO: 6	TTGGTTGCCTTGGTCTCTGTGGGCAGCAGCAGGAGGAGGCGGC AGCAGCCAGAGAAGAGGAGGCGTGTGAGCCACACTCCACCAG CGAGCTTCTTCCCGCTGCTCTGGAAGTCCCAGGCTCTCCCA CCAGCATGGCCCTGATTGTTCACTCAAGACTGTCTCAGAGCTC CGAGGCAAAGGTGACCGGATTGCCAAAGTCACTTTCCGAGGGC AGTCTTTCTACTCCCGGTCTGGAGAACTGCAGGGGTGTGGC TGACTTTGATGAGACGTTCCCGTGGCCAGTGGCCAGCAGCATC GACCGGAATGAAGTGTGGAGATTGAGATTTTCACTACAGCAA AGTCTTCAGCAACAAGCTGATAGGGACCTTCTGCATGGTGCTGC AGAAAGTGGTGGAGGAGAATCGGGTAGAGGTGACCGACACGCT GATGGATGACAGCAATGCTATCATCAAGACCAGCTGAGCATGG AGGTCCGGTATCAGGCCACAGATGGCACTGTGGGCCCTGGGA TGATGGAGACTTCTGGGAGATGAATCCCTCCAGGAGGAGAAG GACAGCCAGGAGACAGATGGGCTGCTACCTGGTTCGCGACCCA GCACCCGGATATCTGGCGAGAAGAGCTTTCGAGAGCGGGAAG GAGTGTGTCTCGGCCATGAAACTCGGCAAACTCGTCCACACA AAGAGGAGCCCCAAGACAAGATGAGCCAGCAGTGCTGGAGAT GGAGGACCTGGACCACCTAGCCATTAGCTGGGGATGGGCTG GATCCTGACTCCGTGTCTTAGCCCTCGGTCAACCGCTCTACCCAG CAATGTCTCCAAACAACGGTCTAAGCCAGATATTAAGATGGAGC CCAGTGCTGGAAGGCCCATGGATTACAGGTGAGCATCACAGT GATTGAGGCTCGGCAGCTGGTGGGCTTGAACATGGACCTGTG GTGTGTGTGGAGGTGGGTGATGACAAGAAATACAGTCAATGAA GGAGTCCACAACTGCCCTTACTACAACGAGTACTTTGTCTTCG ACTTCCATGTCTCTCTGATGTATGTTGACAAGATCATCAAGA TCTCGGTTATCCATTCTAAGAACCTGCTTCGGAGCGGCACCCCTG GTGGGTTCTTCAAAATGGATGTGGGACTGTGTATTCAGGCC TGAAACACAGTTCCATCACAATGGGCCATCCTGTGAGACCCCG ATGACATCTCTGCTGGGTTGAAGGGTTATGTAAAGTGTGATGTC GCTGTGTGGTGGCAAGGGAGACAACATCAAGACACCCACAAGG CCAACGAGACGGATGAGGACGACATTGAAGGGAACCTGCTGCT CCCCGAGGGCGTGCCCCCGAACCGCAGTGGGCACGGTTCTA TGTGAAAATTACCGAGCAGAGGGACTGCCCGGATGAACACAA GCCTCATGGCCAACGTGAAGAAGGCGTTATCGGTGAGAACAA GGACCTCGTGCACCCCTATGTGCAAGTCTTCTTTGCTGGACAAA AGGGCAAAACATCAGTGCAAGAGCAGCTATGAGCCGCTATG GAATGAGCAGGTCGTCTTACAGACTTGTTCCTCCCTCTGCA AACGCATGAAGTGCAGATCCGGGACTCTGACAAGGTCAATGAT GTGGCCATCGGCACCCACTTATCGACCTGCGCAAGATTTCAA CGATGGAGACAAAGGCTTCTGCTACCTCGGTCCAGCCTGG GTGAACATGTACGGCTCCACGCGCACTACACACTGCTGGACG AGCACAGGACTTGAATGAAGGCTGGGGAGGGTGTGTCTT CCGGGCCCGCTCATGTTGGGACTAGCTGTGGAGATCCTGGAC ACCTCCAACCCAGAGCTCACCAGCTCCACGGAGGTGCAAGTGG AGCAGGCCACGCTGTCTCGGAGAGCTGCACAGGAGAAATGGA AGAATTTTTTCTATTGGAGCCTTCTGGAAGCCTCAATGATTGA CCGGAATAATGGGACAAGCCAATTACCTTTGAGGTGACCATAG GAAACTACGGCAATGAAGTCGATGGTATGTCCCGGCCCTGAG GCCTCGGCCCGGAAAGAGCCTGGGGATGAAGAAGAGGTAGA CCTGATTCAGAACTCCAGTGACGATGAAGGTGACGAAGCCGGG GACCTGGCCTCGGTGTCTCCACCCACCTATGCGGCCCCAGA TCACGGACAGGAACATTTCCACCTGCCCTACCTGGAGCGCAAG CCCTGCATCTATATCAAGAGCTGGTGGCCTGACCAGAGGCGGC GCCTCTACAATGCCAACATCATGGATCACATTGCTGACAAGCTG GAAGAAGGCTGAATGATGTACAGGAGATGATCAAAACGGAGAA GTCTTACCCGGAGCGCCGCTGCGGGTGTGCTAGAGGAATC AGCTGTGGCTGCCACGCTTCTCTCCCTCCTCGGACAAGGACC AGGGCCGCTCGTCCCGCACAGGCTGGATCGAGAGCGTCTTAA GTCTGTATGAGGGAGTTGGAGAGCATGGGACAGCAGGCCAAG AGCCTGAGGGCTCAGGTGAAGCGGCACACTGTTCGGGACAAGC TGAGGTGATGCCAGAACTTCTGCAAGAAGCTACGCTTCTGGCG GATGAGCCCCAGCACAGCATTCCTGATGTGTTCAATTTGGATGAT GAGCAACAACAACGATATCGCCTATGCCCGGTGCTTCCAAAG ACCTGCTCTTCTCCATCGTGGAGGAGGAACCTGGGCAAGGACTG CGCCAAAGTCAAGACCCCTCTTCTGAAGCTGCCAGGGAAGAGG GGCTTCGGCTCGGAGGCTGGACAGTACAGGCCAAGCTGGAG CTCTACCTGTGGCTGGGCTCAGCAAGCAGCGAAAGGACTTCC TGTGTGCTGCTGCCCTGTGGCTTCGAGGAGTCAAGGCAGCCCA AGGCTGGGCTGCATTCTTTCGCCCATCAGCCTAGTCTACA CCAAGAAGCAAGCCTTCCAGTCCGAGCACACATGTATCAGGC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		CCGAAGCCTCTTTGCTGCTGACAGCAGTGGGCTCTCTGATCCCT TTGCCCGTGTCTTCTTCATCAACCAGAGCCAATGCACTGAGGTT CTAAACGAGACACTGTGTCCACCTGGGACCAGATGTGGTATT TGACAACTGGAGCTGTACGGTGAAGCTCACGAGTTACGAGAT GATCCCCCATCATTGTGATTGAAATCTACGACCAGGACAGCAT GGGCAAAGCCGACTTCATGGGCCGACCTTCGCCAAGCCCCTG GTGAAGATGGCAGATGAAGCATACTGCCACCTCGCTTCCCGC CGCAGCTTGAGTACTACCAGATCTACCGAGGCAGTGCCACTGC CGGAGACCTACTGGCTGCCTTCGAGCTGCTGCAGATTGGGCCA TCAGGGAAGGCTGACCTGCCACCCATCAATGGCCAGTGGACA TGGACAGAGGGCCATCATGCCTGTGCCCGTGGGAATCCGGCC AGTGCTCAGCAAGTACCGAGTGGAGGTGCTGTTCTGGGGCTG AGGGACCTAAAGAGGGTGAACCTGGCCAGGTGGACGACCAC GGGTGGACATCGAGTGTGCAGGAAAGGGGTACATCCTCCCT GATTACAATTATAAGAAGAACCCCACTTCAACACGCTGGTCAA GTGGTTTGAAGTGGACCTCCCGGAGAATGAGCTCCTGCACCCA CCCTTGAACATCCGAGTGGTAGATTGCCGGCCTTTGGACGATA CACCTGTGGTGGTTCACGCGAGTCAGCTCACTGAGGCGCTTC ATCTACCGACCTCCAGACCGCTCAGCCCCAACTGGAACACCA CAGTCAGGCTGCTCCGGGCTGCCACAGGCTGCGCAATGGGG GCCCCCTCTCTCGCCCCACAGGGGAGGTGTAGTAAGCATGGA GCCTGAGGAGCCAGTTAAGAAGCTGGAGACCATGGTGAACATG GATGCGACTTCTGATGCTGTGGTCAAGGTGGATGTGGCTGAAG ATGAGAAGGAAGGAAGAAGAGAAAAGAAAGGCCCGCTCAGA GGAGCCAGAGGAGGAAGAGCCCGATGAGAGCATGCTGGATTG GTGGTCCAAGTACTTCGCCTCCATCGACACAATGAAGGAGCAAC TTCGACAACATGAGACCTCTGGAACCTGACTTGAAGAGAGAG GAA GAGATGGAAGCGCTGAGGGCCTGAAGGGACCAATGAAGAGCA AGGAGAAGTCCAGAGCTGCAAGGAGGAGAGAAAAGAGAAAAA CCAGAGCCCTGGCCCTGGCCAGGGATCGGAGGCTCCTGAGAA GAAGAAAGCCAGATCGATGAGCTTAAGGTGTACCCCAAGGAG CTGGAATCGAGTTTGACAGCTTTGAGGACTGGCTGCACACCTT CAACCTGTTGAGGGGCAAGACGGGAGATGATGAGGATGGCTCC ACAGAGGAGGAGCGCATAGTAGGCCGATTCAAGGGCTCCCTCT GTGTGTACAAAGTGCCACTCCAGAAAGATGTATCTCGAGAAGCT GGCTATGATCCACCTATGGAATGTTCCAGGGCATCCCAAGCAA TGACCCCATCAATGTGCTGGTCCGAATCTATGTGGTCCGGGCCA CAGACCTGCACCCGGCCGACATCAATGGCAAAGCTGACCCTTA TATTGCCATCAAGTTAGGCAAGACCGACATCCGAGACAAGGAGA ACTACATCTCCAAGCAGCTCAACCTGTGTTTGGGAAGTCCTTT GACATTGAGGCCTCCTTCCCATGGAGTCCATGTTGACAGTGGC CGTGTACGACTGGGATCTGGTGGGCACTGATGACCTCATCGGA GAAACCAAGATTGACCTGGAACCCGCTTCTACAGCAAGCATCG CGCCACCTGCGGCATCGCACAGACCTATTCATACATGGCTACA ATATCTGAGGAGACCCATGAAGCCAGCCAGATCCTGACACG CCTCTGTAAAGAGGGCAAAGTGGACGGCCCCCACTTTGGTCCC CATGGGAGAGTGAGGGTTGCCAACCGTGTCTTACGGGGCCTT CAGAAATAGAGGATGAGAATGGTCAAGGAAGCCACAGATGA GCACGTGGCACTGTCTGCTCTGAGACACTGGGAGGACATCCCC CGGGTGGGCTGCCGCTTGTGCCGAACACGTGGAGACCAGG CCGCTGCTCAACCTGACAAGCCAGGCATTGAGCAGGGCCGCC TGGAGCTGTGGGTGGACATGTTCCCATGGACATGCCAGCCCC TGGGACACCTCTGGATATATCCCCAGGAAACCAAGAAGTACG AGCTGCGGGTCATCGTGTGAACACAGACGAGGTGGTCTTGA AGACGATGATTTCTTACGGGAGAGAAGTCCAGTGACATTTTGG TGAGGGGTGGCTGAAGGGCCAGCAGGAGGACAACAGGACA CAGATGTCCACTATCACTCCCTCACGGGGAGGGCACTTCAAC TGGAGATACCTCTTCCCCTTCGACTACCTAGCGGCCGAAGAGAA GATCGTTATGTCCAAAAGGAGTCTATGTTCTCCTGGGATGAGA CGGAGTACAAGATCCCTGCGCGGCTCACCTGCAGATCTGGGA CGCTGACCACCTTCTCGGCTGACGACTTCTGGGGGCTATCGAG CTGGACCTGAACCGGTTCCCGAGGGGCGCTAAGACAGCCAAGC AGTGACCATGGAGATGGCCACCGGGGAGGTGGACGTACCCCT GGTTTCCATCTTTAAACAGAAACGTGTCAAAGGCTGGTGGCCCC TCCTGGCCCGCAATGAGAATGATGAGTTTGAGCTCACAGGCAAA GTGGAGGCGGAGCTACACCTACTCACGGCAGAGGAGGAGAG AAGAACCCTGTGGGCCTGGCTCGCAATGAACCTGATCCCCTAG AAAAACCAATCGGCCGACACAAGCTTCATCTGGTTCTTGAAC CCTCTCAAGTCTGCCCGCTACTTCTGTGGCATACCTACCGCTG GCTACTCTCAAATTCCTGCTGCTCTTCTCCTGCTGCTGCTCTT CGCCCTGTTTCTCTACTCTGCTGCTGCTACCTGGCCAAGAAGA TCCTTGGGGCCTGAGCCCTGCAGTCGCTAGGCTGCGGCGCT GACACGGCATTCGTCTGGTTCCTGAACCCACTCAAATCTATCAA GTACCTCATCTGCACCCGGTACAAGTGGCTGATCATCAAGATCG TGCTGGCGCTGCTGGGGTGTCTATGCTGGCCCTCTTCTTTAC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		AGCCTCCCAGGCTACATGGTCAAGAAGCTCCTAGGGGCCTGAA GTGTGCCCCACCCAGCCCGCTCCAGCATCCCTCCAGGGGCTG CTGCGTATTTTGCTTCCCTCACCTGGACTCTCTCCCAACTCCCT GAGGAGCCCTCCACGCCTGCCAGCCTTGAGCAAGACACCTGC TTGCTGGACTTCATCCCCACCCACACCCAACTGTTGCTTGCC TGATCTTGTCAGGCTGCCCTGGGGTTTGGGGCACAGTTGGC CTCCAAAACAGATACCCTCTTGCTAAAGTACCAGGTTCTCTG CCCAACCCCAAGAGTGGTAGTGGCCCAACCTCCCTGTGCTTTC CAAATCTTGTCTAAGGCACCACTGAAATTAACCAAGAAACGCG GAGCAATGCCCAAGGCTCTGATGAGTAGGAACACGTGGAAAGC ACCAGGAATGCCAGCAGAGGCGAGGCGGCACACCTCTCTGCAG AGCATCCAGGCCGAGCGGCGGGCAGCGCCAGCTGCTTCTGC GCATGCTCTCCTCTTGGCTCTGCTTCTTCTCACAGTCACAGTCA CTTCACAGCTTAGCCTTGGGCTTCCCATCACTTCCAGGGGTGCC TCTGCCTTGGCCAGTGTGTGTGAGTAGTACACAGCTCCCAAGT GTGAATCAGGTGTAAGGCTGCTGAAGACTGACTGCCCTGTG CTTCTGCGCAGCAGCCACCCGAGTGTACACTTAAAGCGGTG CCCTTCTGCCTCTGTGGGCTGCTGGCTGCTGTTCTTCTTGA GTGTGATTTTTTTTTCTCTCCCTCAATAAAATAAATAAACTCTG AGAC
16	mOTOF-201 2 transcript (NM_001286421.1), mouse otoferlin transcript variant 3, 7129 bp, encodes the protein of SEQ ID NO: 7	TTGGTTGCCTTGGTCTCTGTGGGCAGCAGCAGGAGGAGGCGGC AGCAGCCAGAGAAGAGGGAGGCGTGTGAGCCACACTCCACCAG CGAGCTTCTTCCCGTGTCTTGGAACTGCCAGGCTCTCCCA CCAGCATGGCCCTGATTGTTACCTCAAGACTGTCTCAGAGCTC CGAGGCAAGGTGACCGGATTGCCAAAGTCACTTCCGAGGGC AGTCTTTCTACTCCCGGTCTTGGAGAACTGCGAGGGTGTGGC TGACTTTGATGAGACGTTCCGGTGGCCAGTGGCCAGCAGCATC GACCGGAATGAAGTGTGGAGATTGAGATTTTCAACTACAGCAA AGTCTTCAGCAACAAGCTGATAGGGACCTTCTGCATGGTGTGC AGAAAGTGGTGGAGGAGAATCGGGTAGAGGTGACCGACACGCT GATGGATGACAGCAATGCTATCATCAAGACCAGCCTGAGCATGG AGGTCCGGTATCAGGCCACAGATGGCACTGTGGGCCCTGGGA TGATGGAGACTTCTGGGAGATGAATCCCTCCAGGAGGAGAAG GACAGCCAGGAGACAGATGGGCTGCTACCTGGTTCCCGACCCA GCACCCGGATATCTGGCGAGAAGAGCTTTCGAGAGCGGGAAG GAGTGTGTTCTCGGCCATGAAACTCGGCAGAACTCGGTCCACAA AAGAGGAGCCCAAAGACAAGATGAGCCAGCAGTGTGGAGAT GGAGGACCTGGACCACCTAGCCATTGAGTGGGGATGGGCTG GATCCTGACTCCGTGTCTCTAGCCTCGGTCACCGCTCTCACCAG CAATGTCTCCAAACACGGTCTAAGCCAGATATTAAGATGGAGC CCAGTGTGGAAGGCCCATGGATTACAGGTGAGCATCACAGT GATTGAGGCTCGGCAGCTGGTGGGCTTGAACATGGACCCCTGTG GTGTGTGTGGAGGTGGGTGATGACAAGAAATACAGTCAATGAA GGAGTCCAAACTGCCCTTACTACAACGAGTACTTTGTCTTCG ACTTCCATGTCTCTCTGATGTATGTTTGACAAGATCATCAAGA TCTCGTTTATCCATTCTAAGAACCTGCTTCCGAGCGGCACCTG GTGGGTTCCTTCAAATGGATGTGGGGACTGTGTATCCAGCC TGAACACCAAGTTCATCAAAATGGGCCATCCTGTGACACCCG ATGACATCTCTGCTGGGTGAAGGTTATGTAAAGTGTGATGTC GCTGTGGTGGGCAAGGAGACAACATCAAGACACCCACAAGG CCAACGAGACGGATGAGGACGACATTGAAGGGAACCTTGCTGCT CCCCGAGGCGTGGCCCCGAACGGCAGTGGGCACGGTTCTA TGTGAAAATTTACCGAGCAGAGGGACTGCCCGGATGAACACAA GCCTCATGGCCAACGTGAAGAAGGCGTTTATCGGTGAGAACAA GGACCTCGTCGACCCCTATGTGCAAGTCTTCTTTGCTGGACAAA AGGGCAAAACATCAGTGCAAGAGCAGCTATGAGCCGCTATG GAATGAGCAGGTCGTCTTACAGACTTGTTCCTCCCACTCTGCA AACGATGAAGGTGAGATCCGGGACTCTGACAAGGTCAATGAT GTGGCCATCGGCACCCACTTCATCGACCTGCGCAAGATTTCCAA CGATGGAGACAAAGGCTTCTGCTTACCTCGGTCCAGCCTGG GTGAACATGTACGGCTCCACGCGCAACTACACACTGCTGGACG AGCACCAGGACTTGAATGAAGGCTTGGGGAGGGTGTGTCTT CCGGGCCCGCTCATGTTGGGACTAGCTGTGGAGATCCTGGAC ACCTCCAACCCAGAGCTCACCAGCTCCACGGAGGTGCAAGGTGG AGCAGGCCACGCTGTCTCGGAGAGCTGCACAGGAGAAATGGA AGAATTTTTTCTATTTGGAGCCTTCTTGAAGCCTCAATGATTGA CCGGAAAAATGGGACAAGCCAATTACCTTTGAGGTGACCATAG GAAACTACGGCAATGAAGTCGATGGTATGTCCCGGCCCTGAG GCCTCGGCCCGGAAAGAGCCTGGGGATGAAGAAGAGGTAGA CCTGATTCAAGACTCAGTGACGATGAAGGTGACGAAGCCGGG GACCTGGCCTCGGTGTCTTCCACCCACCTATGCGGCCCCAGA TCACGGACAGGAATAATTTCCACCTGCCCCTACCTGGAGCGCAAG CCCTGCATCTATATCAAGAGCTGGTGGCCTGACCAGAGGCGGC GCCTCTACAATGCCAACATCATGGATCACATTGCTGACAAGCTG

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GAAGAAGGCCTGAATGATGTACAGGAGATGATCAAAACGGAGAA GTCCCTACCCGGAGCGCCGCTGCGGGGTGTGCTAGAGGAACTC AGCTGTGGCTGCCACCGCTTCCTCTCCTCTCGGACAAGGACC AGGGCCGCTCGTCCCGCACCAGGCTGGATCGAGAGCGTCTTAA GTCCCTGTATGAGGGAGTTGGAGAGCATGGGACAGCAGGCCAAG AGCCTGAGGGCTCAGGTGAAGCGGCACACTGTTCGGGACAAGC TGAGGTCATGCCAGAACCTTCTGCAGAAGCTACGCTTCCTGGCG GATGAGCCCCAGCACAGCATTCTGTATGTTCATTGGATGAT GAGCAACAACAACGTATCGCCTATGCCCGCTGCCTTCCAAAG ACCTGCTCTTCTCCATCGTGGAGGAGGAACCTGGGCAAGGACTG CGCCAAAGTCAAGACCTCTTCTGAAGCTGCAGGGAAGAGG GGCTTCGGCTCGGCAGGCTGGACAGTACAGGCCAAGCTGGAG CTCTACCTGTGGCTGGGCCTCAGCAAGCAGCGAAAGGACTTCC TGTGTGGTCTGCCCTGTGGCTTCGAGGAGGTCAAGGCAGCCCA AGGCCTGGGCCTGCATTCTTCCGCCCATCAGCCTAGTCTACA CCAAGAAGCAAGCCTTCCAGCTCCGAGCACACATGTATCAGGC CCGAAGCCTCTTTGCTGCTGACAGCAGTGGGCTCTCTGATCCCT TTGCCCGTGTCTTCTTCAACAGAGCCATGCACTGAGGTT CTAAACGAGACACTGTGTCCACCTGGGACCAGATGCTGGTATT TGACAACCTGGAGCTGTACGGTGAAGCTCACGAGTTACGAGAT GATCCCCCATCATTGTCTTGAATCTACGACCAGGACAGCAT GGGCAAGCCGACTTCATGGGCCGACCTTCGCCAAGCCCCCTG GTGAAGATGGCAGATGAAGCATACTGCCACCTCGCTTCCCGC CGCAGCTTGAGTACTACAGATCTACCGAGGCAGTGCCACTGC CGGAGACCTACTGGCTGCCTTCGAGCTGCTGCAGATTGGGCCA TCAGGGAAGGCTGACCTGCCACCATCAATGGCCAGTGGACA TGGACAGAGGGCCCATCATGCCTGTGCCGTGGGAATCCGGCC AGTGCTCAGCAAGTACCAGTGGAGGTGCTGTTCTGGGGCCTG AGGGACCTAAGAGGGTGAACCTGGCCAGGTGGACCGACCAC GGGTGGACATCAGTGTGCAGGAAGGGGTACAATCCTCCCT GATTACAAATTATAAGAAGAACCCCACTTCAACACGCTGGTCAA GTGGTTTGAAGTGGACCTCCCGGAGAATGAGCTCCTGCACCCA CCCTTGAACATCCGAGTGGTAGATTGCCGGCCTTTGGACGATA CACCTGGTGGGTTCCACGCGAGTCAGCTCACTGAGGGCGCTTC ATCTACCGACCTCCAGACCGCTCAGCCCCCACTGGAACACCA CAGGGGAGGTGTAGTAAGCATGGAGCCTGAGGAGCCAGTTAA GAAGCTGGAGACCATGGTGAACCTGGATGCGACTTCTGATGCT GTGGTCAAGGTGGATGTGGCTGAAGATGAGAAGGAAGGAAGA AGAAGAAAAGAAAGGCCCGTCAGAGGAGCCAGAGGAGGAAGA GCCCCGATGAGAGCATGCTGGATTGGTGGTCCAAGTACTTCGCC TCCATCGACACAATGAAGGAGCAACTTCGACAACATGAGACCTC TGGAACCTGACTTGGAAGAGAAGGAAGAGATGGAAGCGCTGAG GGCCTGAAGGGACCAATGAAGAGCAAGGAGAAGTCCAGAGCTG CAAAGGAGGAGAAAAAGAAAAACAGAGCCCTGGCCCTGG CCAGGGATCGGAGGCTCCTGAGAAGAAAGCCAAAGATCGAT GAGCTTAAGGTGTACCCCAAGGAGCTGGAATCGGAGTTTGACA GCTTTGAGGACTGGCTGCACACCTTCAACCTGTTGAGGGGCAA GACGGGAGATGATGAGGATGGCTCCACAGAGGAGGAGCGCATA GTAGGCCGATTCAAGGGCTCCCTCTGTGTGTACAAAGTGCCACT CCAGAAGATGTATCTCGAGAAGCTGGCTATGATCCCACCTATG GAATGTTCCAGGGCATCCCAAGCAATGACCCCATCAATGTGCTG GTCCGAATCTATGTGGTCCGGGCCACAGACCTGCACCCGGCCG ACATCAATGGCAAAGCTGACCCCTATATTGCCATCAAGTTAGGC AAGACCGACATCCGAGACAAGGAGAACTACATCTCCAAGCAGCT CAACCTGTGTTTGGGAAGTCCTTTGACATTGAGGCCCTCCTTCC CCATGGAGTCCATGTGACAGTGGCCGTGACGACTGGGATCT GGTGGGCACTGATGACCTCATCGGAGAAACCAAGATTGACCTG GAAAACCGCTTCTACAGCAAGCATCGCGCCACCTGCGGCATCG CACAGACCTATTCCATACATGGCTACAATATCTGGAGGGACCCC ATGAAGCCAGCCAGATCCTGACACGCTCTGTAAAGAGGGCA AAGTGGACGGCCCCACTTTGGTCCCCATGGGAGAGTGAGGGT TGCCAACCGTGTCTTACGGGGCCTTCAGAAATAGAGGATGAGA ATGGTCAGAGGAAGCCACAGATGAGCAGTGGCACTGTCTGC TCTGAGACACTGGGAGGACATCCCCGGGTGGGCTGCCGCCTT GTGCCGAACACGTGGAGACCAGGCCGTGCTCAACCTGACA AGCCAGGCATTGAGCAGGGCCGCTGGAGCTGTGGGTGGACAT GTTCCCCATGGACATGCCAGCCCCCTGGGACACCTCTGGATATAT CCCCCAGGAAACCAAGAAGTACGAGCTGCGGGTCACTGTGTG GAACACAGACGAGGTGGTCTTGAAGACGATGATTCTTTCAGC GGAGAGAAGTCCAGTGACATTTTGTGAGGGGGTGGCTGAAGG GCCAGCAGGAGGACAAAAGGACACAGATGTCCACTATCACTC CTTACGGGGGAGGGCAACTTCACTGGAGATACCTCTTCCCC TTCGACTACCTAGCGGCCAAGAGAAGATCGTTATGTCCAAAAA GGAGTCTATGTTCTCTGGGATGAGACGGAGTACAAGATCCCTG CGCGCTCACCTGCAGATCTGGGACGCTGACCCTTCTCGGC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		TGACGACTTCCTGGGGGCTATCGAGCTGGACCTGAACCGGTTC CCGAGGGGCGCTAAGACAGCCAAGCAGTGCACCATGGAGATGG CCACCGGGGAGGTGGACGTACCCCTGGTTTCCATCTTTAAACAG AAACGTGTCAAAGGCTGGTGGCCCTCCTGGCCCGCAATGAGA ATGATGAGTTTGAGCTCACAGGCAAAGTGGAGGCGGAGCTACA CCTACTCACGGCAGAGGAGGAGAGAGAAGAACCTGTGGGCTG GCTCGCAATGAACCTGATCCCTAGAAAAACCAATCGGCCGGA CACAAGCTTCATCTGGTTCTTGAACCTCTCAAGTCTGCCCGCT ACTTCTGTGGCATACCTACCGCTGGCTACTCCTCAAATTCCTG CTGCTCTTCCCTCCTGCTGCTGCTCTTCCGCCCTGTTTCTCTACTCT CTGCCCTGGCTACCTGGCCAAGAAGATCCTTGGGGCTGAGCCC TGCAGTCGCCTAGGCCTGCCGGCTGACACGGCATTCGTCCTGG TTCTTGAACCCACTCAAATCTATCAAGTACCTCATCTGCACCCG GTACAAGTGGCTGATCATCAAGATCGTGTGGCGCTGCTGGGG CTGCTCATGCTGGCCCTCTTCTTTACAGCCTCCAGGCTACAT GGTCAAGAAGCTCCTAGGGGCTGAAGTGTGCCCCACCCAGC CCGCTCCAGCATCCCTCCAGGGGCTGCTGCGTATTTGCGCTTC CTACCTGGACTCTCTCCCACTCCCTGAGGAGCCCTCCACG CCTGCCAGCCTTGAGCAAGACACCTGCTTGCTGGACTTCATCCC CACCCACACCCAAACTGTTGCTTGCTGATCTTGTCCAGGCC TGCTTGGGGTTTGGGGCAGATTGGCTCCAAAACAGATACC CTCTTGCTAAAGTACCAGGTTCTCTGCCCAACCCCAAGAGTG GTAGTGGCCCAACCTCCCTGTGCTTTCCAAATCTTGCTTAAG GCACCAGTGAATTAACCAAGAAACGCGGAGCAATGCCAAGG CTCTGATGAGTAGGAACACGTGGAAGCACCAGGAATGCCAGC AGAGGCGAGGCGGCACACCTCTCTGCAGAGCATCCAGGCCGA GCGGCGGGCAGCGGCAGCTGCTTCTGCGCATGCTCTCTCTT GGCTCTGCTTCTTCTCACAGTCACAGTCACTTCACAGCTTAGC CTTGGGCTTCCCATCACTTCCAGGGGTGCTCTGCTTGGCCA GTGTGTGTGCTAGTACACAGCTCCAGTGTGAATCAGGTGT ACTGGCCGCTCTGAAGACTGACTGCCCTGTCTCTTCCGCGACA GCCACACCCAGTGTACACTTAAAGCGGTGCCCTTCTGCTCTG TGGGCTGCTGGCTGCTGCTTCTTCTTGTGAGTGTATTTTTTT TCTCTCCCTCAATAAAATAAATCAAACTCTGAGAC
17	mOTOF-202_1 transcript (NM_001100395.1), mouse otoferlin transcript variant 1, 6881 bp, encodes the protein of SEQ ID NO: 8	TTGGTTGCCTTGGTCTCTGTGGGCAGCAGCAGGAGGAGGCGGC AGCAGCCAGAGAAGAGGAGGCGGTGTGAGCCACACTCCACCAG CGAGCTTCTTCCGCTGCTCTGGAACCTCCAGGCTCTCCCA CCAGCATGGCCCTGATTGTTACCTCAAGACTGTCTCAGAGCTC CGAGGCAAAGGTGACCGGATTGCCAAAGTCACTTTCCGAGGGC AGTCTTTCTACTCCGGGTCTGGAGAACTGCGAGGGTGTGGC TGACTTTGATGAGACGTTCCGGTGGCCAGTGGCCAGCAGCATC GACCGGAATGAAGTGTGGAGATTGAGATTTTCAACTACAGCAA AGTCTTTCAGCAACAAGCTGATAGGGACCTTCTGCATGGTGTGC AGAAAGTGGTGGAGGAGAATCGGGTAGAGGTGACCGACACGCT GATGGATGACAGCAATGCTATCATCAAGACCAGCTGAGCATGG AGGTCCGGTATCAGGCCACAGATGGCACTGTGGGCCCTGGGA TGATGGAGACTTCTGGGAGATGAATCCCTCCAGGAGGAGAAG GACAGCCAGGAGACAGATGGGCTGCTACCTGGTTCCTGACCCA GCACCCGATATCTGGCGAGAAGAGCTTTCGACAGCAAAGGCAG AGAGAAGACCAGGGAGGAGAGATGGCGAGCACAAAGCGGG AAGGAGTGTGTTCTCGGCCATGAACTCGGCCAAACTCGGTCCC ACAAAGAGGAGCCCCAAAGACAAGATGAGCCAGCAGTGTGGA GATGGAGGACCTGGACCACCTAGCCATTGAGTGGGGATGGG CTGGATCCTGACTCCGTGTCTTAGCCTCGGTACCCGCTCTCAC CAGCAATGTCTCCAACAACCGGTCTAAGCCAGATATTAAGATGG AGCCAGTGTGGAAGGCCATGGATTACAGGTGAGCATCAC AGTGATTGAGGCTCGGCAGCTGGTGGGCTTGAACATGGACCT GTGGTGTGTGGAGGTGGGTGATGACAAGAAATACACGTCAAT GAAGGAGTCCAAAACCTGCCCTTACTACAACGAGTACTTTGTCT TCGACTTCCATGTCTCTCCTGATGTATGTTTGACAAGATCATCA AGATCTCGGTATCCATTCTAAGAACCTGCTTCCGAGCGGCACC CTGGTGGGTTCTTCAAATGGATGTGGGACTGTGTATCCCA GCCTGAACACCACTTCCATCAAAATGGGCATCTGTGACAGCC CCGATGACATCTCTGCTGGGTGAAGGGTTATGTAAGTGTGAT GTCGCTGTGGTGGGCAAGGAGACAAATCAAGACACCCACA AGGCCAACGAGACGATGAGGACGACATTGAAGGGAACCTGCT GCTCCCCGAGGGCGTGCCCCCGAACGGCAGTGGGCACGGTT CTATGTGAAAATTTACCGAGCAGAGGGACTGCCCGGATGAACA CAAGCCTCATGGCCAACGTGAAGAAGGCGTTTATCGGTGAGAA CAAGGACCTCGTCGACCCCTATGTGCAAGTCTTCTTTGTGGAC AAAAGGGCAAAACATCAGTGCAGAAGAGCAGCTATGAGCCGCT ATGGAATGAGCAGGTCGTCTTACAGACTTGTTCCTCCACTCT GCAAACGATGAAGGTGAGATCCGGGACTTGACAAGGTCAA TGATGTGCCCATCGGCACCCACTTCATCGACCTGCGCAAGATTT

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		CCAACGATGGAGACAAAGGCTTCTGCCTACCCCTCGGTCCAGC CTGGGTGAACATGTACGGCTCCACGCGCAACTACACACTGCTG GACGAGCACCAGGACTTGAATGAAGGCTTGGGGGAGGGTGTGT CCTTCCGGGCCCCCTCATGTTGGGACTAGCTGTGGAGATCCT GGACACCTCCAACCCAGAGCTCACCAGCTCCACGGAGGTGCAG GTGGAGCAGGCCACGCTGTCTCGGAGAGCTGCACAGGGAGA ATGGAAGAATTTTTCTATTGGAGCCTTCTTGAAGCCTCAATG ATTGACCGGAAAAATGGGGACAAGCCAATTACCTTGAGGTGAC CATAGGAACTACGGCAATGAAGTCGATGGTATGTCCCGGCC CTGAGGCTCGGCCCCGGAAAGAGCCTGGGGATGAAGAAGAG GTAGACCTGATTAGAACTCCAGTGACGATGAAGGTGACGAAGC CGGGGACCTGGCCTCGGTGTCTCCACCCACCTATGCGGCC CAGATCACGGACAGGAATATTTCCACCTGCCCTACCTGGAGCG CAAGCCCTGCATCTATATCAAGAGCTGGTGGCCTGACCAGAGG CGGCGCTCTACAATGCCAATCATGGATCACATTGCTGACAA GCTGGAAGAAGGCTGAATGATGTACAGGAGATGATCAAAACG GAGAAGTCTTACC CGAGCGCGCTCGGGGTGTGCTAGAG GAACTCAGCTGTGGTGCACCGCTTCTCTCCCTCTCGGACAA GGACCAGGGCCGCTCGTCCCGCACCAGGCTGGATCGAGAGCG TCTTAAGTCTGTATGAGGGAGTTGGAGAGCATGGGACAGCAG GCCAAGAGCCTGAGGGCTCAGGTGAAGCGGCACACTGTTCCGG ACAAGCTGAGGTCATGCCAGAACTTTCTGCAGAAGCTACGCTTC CTGGCGGATGAGCCCCAGCACAGCATTCTGATGTGTTCAATTG GATGATGAGCAACAACAACGTATCGCCTATGCCCGCGTGCCTT CCAAAGACCTGCTTCTTCCATCGTGGAGGAGGAAGTGGGCAA GGACTGCGCCAAAGTCAAGACCTTCTTCTGAAGCTGCCAGGG AAGAGGGCTTCGGCTCGGCAGGCTGGACAGTACAGGCCAAG CTGGAGCTCTACCTGTGGCTGGCCTCAGCAAGCAGCGAAAGG ACTTCTGTGTGGTCTGCCCTGTGGCTTCGAGGAGGTCAAGGC AGCCCAAGGCTTGGCCTGCATTCTTCCGCCCATCAGCCTA GTCTACACCAAGAAGCAAGCCTTCCAGCTCCGAGCACACATGTA TCAGGCCCGAAGCCTCTTTGCTGTGACAGCAGTGGGCTCTCT GATCCCTTTGCCCGTGTCTTCTCATCAACAGAGCCAATGCAC TGAGGTTCTAAACGAGACACTGTGTCCACCTGGGACCAGATGC TGGTATTTGACAACCTGGAGCTGTACGGTGAAGCTCACGAGTTA CGAGATGATCCCCCATCATTGTCTATTGAAATCTACGACCAGGA CAGCATGGGCAAAGCCGACTTCATGGGCGGACCTTCGCCAAG CCCCCTGGTGAAGATGGCAGATGAAGCATACTGCCACCTCGCTT CCGCGCGAGCTTGAGTACTACCAGATCTACCGAGGCAGTGCC ACTGCCGAGACCTACTGGCTGCCTTCGAGCTGCTGCAGATTG GGCCATCAGGGAAGGCTGACCTGCCACCCATCAATGGCCAGT GGACATGACAGAGGGCCATCATGCTGTGCCCGTGGGAATC CGGCCAGTGCTCAGCAAGTACCGAGTGGAGGTGCTGTTCTGGG GCCTGAGGGACCTAAAGAGGGTGAACCTGGCCCAGGTGGACC GACCACGGGTGGACATCGAGTGTGCAGGAAAGGGGTACAATC CTCCCTGATTCACAATTATAAGAAAGAACCCCACTTCAACACGCT GGTCAAGTGGTTGAAGTGGACCTCCCGGAGAATGAGCTCCTG CACCACCTTGAACATCCGAGTGGTAGATTGCCGGGCTTTG GACGATACACCTGGTGGGTTCCACGCGAGTCAGCTCACTGAG GCGCTTCATCTACCGACCTCCAGACCGCTCAGCCCCCACTGG AACACCACAGGGGAGGTTGTAGTAAGCATGGAGCCTGAGGAGC CAGTTAAGAAGCTGGAGACCATGGTGAAGCTGGATGCGACTTCT GATGCTGTGGTCAAGGTGGATGTGGCTGAAGATGAGAAGGAAA GGAAGAAGAAGAAAAGAAAGGCCGTGAGAGGAGCCAGAGGA GGAAGAGCCCGATGAGAGCATGCTGGATTGGTGGTCCAGTAC TTCGCCCTCCATCGACACAATGAAGGAGCAACTTCGACAACTGA GACCTCTGGAAGTGAAGTGAAGGAGGAAGAGATGGAAAGC GCTGAGGGCTGAAGGGACCAATGAAGAGCAAGGAGAAGTCCA GAGCTGCAAGGAGGAGAAAAAGAAAGAAAAACAGAGCCCTGG CCCTGGCCAGGGATCGGAGGCTCTGAGAGAAGAAAGCCAAAG ATCGATGAGCTTAAGGTGTACCCCAAGGAGCTGGAATCGGAGTT TGACAGCTTTGAGGACTGGCTGCACACCTTCAACCTGTTGAGGG GCAAGACGGGAGATGATGAGGATGGCTCCACAGAGGAGGAGC GCATAGTAGGCCGATTCAAGGGCTCCCTCTGTGTGTACAAAGTG CCACCTCCAGAAGATGTATCTCGAGAAGCTGGCTATGATCCAC CTATGGAATGTTCCAGGGCATCCCAAGCAATGACCCCATCAATG TGCTGGTCCGAATCTATGTGGTCCGGGCCACAGACCTGCACCC GGCCGACATCAATGGCAAAGCTGACCCCTATATTGCCATCAAGT TAGGCAAGACCACATCCGAGACAGGAGAACTACATCTCCAAG CAGCTCAACCTGTGTTTGGGAAGTCTTTGACATTGAGGCCTC CTTCCTCATGGAGTCCATGTTGACAGTGGCCGTGTACGACTGG GATCTGGTGGGCACTGATGACCTCATCGGAGAAACCAAGATTGA CCTGGAAACCGCTTCTACAGCAAGCATCGCGCCACCTGCAGGC ATCGCACAGACCTATTCATACATGGCTACAATATCTGGAGGGA CCCCATGAAGCCAGCCAGATCCTGACACGCTCTGTAAAGAG

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GGCAAAAGTGGACGGCCCCCACTTTGGTCCCCATGGGAGAGTGA GGGTTGCCAACCGTGTCTTACGGGGCCTTCAGAAATAGAGGA TGAGAATGGTCAGAGGAAGCCACAGATGAGCACGTGGCACTG TCTGCTCTGAGACACTGGGAGGACATCCCCGGGTGGGCTGCC GCCTTG TGCCGGAACAGTGGAGACCAGGCGCTGCTCAACCC TGACAAGCCAGGCATTGAGCAGGGCCGCTGGAGCTGTGGGTG GACATGTTCCCCATGGACATGCCAGCCCTGGGACACCTCTGG ATATATCCCCCAGGAAACCAAGAAGTACGAGCTGCGGGTCATC GTGTGGAACACAGACGAGGTGGTCTTGAAGACGATGATTTCTT CACGGGAGAGAAGTCCAGTGACATTTTGTGAGGGGTGGCTG AAGGGCCAGCAGGAGGACAAACAGGACACAGATGTCCACTATC ACTCCCTCACGGGGGAGGGCACTTCAACTGGAGATACCTCTTC CCCTTCGACTACCTAGCGGCCGAAGAGAAGATCGTTATGTCCAA AAAGGAGTCTATGTTCTCTGGGATGAGACGGAGTACAAGATCC CTGCGCGCTCACCTGCGATCTGGGACGCTGACCACTTCTC GGCTGACGACTTCTGGGGGCTATCGAGCTGGACCTGAACCGG TTCCCGAGGGGCGCTAAGACAGCCCAAGCAGTGACCATGGAGA TGGCCACCGGGGAGGTGGACGTACCCCTGGTTTCCATCTTTAA CAGAAACGTGTCAAAGGCTGGTGGCCCTCTGGCCCGCAATG AGAATGATGAGTTTGAGCTCACAGGCAAGTGGAGGCGGAGCT ACACCTACTCACGGCAGAGGAGGCAGAGAAGAACCTGTGGGC CTGGCTCGCAATGAACCTGATCCCTAGAAAAACCAACCGGCC TGACACGGCATTCTGCTGTTCTTGAACCCACTCAATCTATCA AGTACCTCATCTGCACCCGCTACAAGTGGCTGATCATCAAGATC GTGCTGGCGCTGCTGGGGCTGCTCATGCTGGCCCTCTTCTTT ACAGCCTCCAGGCTACATGGTCAAGAAGCTCTAGGGGCTG AAGTGTGCCCAACCCAGCCGCTCCAGCATCCCTCCAGGGGC TGCTGCGTATTTTGCTTCCCTCACCTGGACTCTCTCCAACTC CCTGAGGAGCCCTCCACGCTGCGAGCCTTGAGCAAGACACC TGCTTGCTGGACTTCATCCCCACCCACACCAAACTGTTGCTT GCCTGATCTTGTCCAGGCCTGCCCTGGGGTTTGGGGCACAGTT GGCCCTCAAAACAGATAACCTCTTGTCTAAGTACCAGGTTC TCTGCCCAACCCAGAGTGGTAGTGGCCCAACCTCCCTGTG CTTTCCAAATCTTGTCTTAAGGCACAGTGAAATTAACCAAGAAA CGCGGAGCAATGCCAAGGCTCTGATGAGTAGGAACACGTGGA AAGCACAGGAATGCCAGCAGAGGCGAGGCGGCACACCTCTCT GCAGAGCATCCAGGCCGAGCGGGCGGCGAGCGGCAGCTGCTT CTGCGCATGCTCTCTCTTGGCTCTGCTTCTTTCTCAGTCACA GTCACCTCACAGCTTAGCCTTGGGCTTCCATCACTTCCAGGGG TGCCCTCTGCCTTGGCCAGTGTGTGTCAGCTAGTACACAAGCT CCAAGTGTGAATCAGGTGTAAGTGGCCGCTGAAAGACTGACTGC CCTGTCTTCTGCGCAGCAGCCACCCGAGTGTACACTTAAAG CGGTGCCCTCTGCTCTGTGGGCTGCTGGCTGCTGTTCTTTT CTGAGTGTGATTTTTTTTTTCTCTCCCTCAATAAAATAAATCAAA CTCTGAGAC
18	mOTOF-202_2 transcript (NM_001313767.1), mouse otoferlin transcript variant 4, 6881 bp, encodes the protein of SEQ ID NO: 9	TTGGTTGCCTTGGTCTCTGTGGGCAGCAGCAGGAGGAGGCGGC AGCAGCCAGAGAAGAGGAGGCGTGTGAGCCACACTCCACCAG CGAGCTTCTTCCCGCTGCTCTGGAAGTGCCAGGCTCTCCCCA CCAGCATGGCCCTGATTGTTCACTCAAGACTGTCTCAGAGCTC CGAGGCAAGGTGACCGGATTGCCAAAGTCACTTTCCGAGGGC AGTCTTTCTACTCCCGGCTCTGGAGAAGTGCAGGGGTGTGGC TGACTTTGATGAGACGTTCCGGTGGCCAGTGGCCAGCAGCATC GACCGGAATGAAGTGTGGAGATTGAGATTTTCAACTACAGCAA AGTCTTCAGCAACAAGCTGATAGGGACCTTCTGCATGGTGTGTC AGAAAGTGGTGGAGGAGAATCGGGTAGAGGTGACCGACACGCT GATGGATGACAGCAATGCTATCATCAAGACCAGCCTGAGCATGG AGGTCCGGTATCAGGCCACAGATGGCACTGTGGGCCCCCTGGGA TGATGGAGACTTCTGGGAGATGAATCCCTCCAGGAGGAGAAG GACAGCCAGGAGACAGATGGGCTGTACTTGGTTCCCGACCCA GCACCCGGATATCTGGCGAGAAGAGCTTTTCGAGAGCGGGAAG GAGTGTGTTCTCGGCCATGAACCTCGGCAAACTCGGTCCACACA AAGAGGAGCCCCAAAGACAAGATGAGCCAGCAGTGTGGAGAT GGAGGACCTGGACCACCTAGCCATTACAGCTGGGGATGGGCTG GATCCTGACTCCGTGTCTCTAGCCTCGGTACCGCTCTCACCAG CAATGTCTCAACAACCGGTCTAAGCCAGATATTAGATGGAGC CCAGTGTGGAAGGCCCATGGATTACAGGTGAGCATCACAGT GATTGAGGCTCGGCAGCTGGTGGGCTTGAACATGGACCTGTG GTGTGTGTGGAGGTGGGTGATGACAAGAAATACACGTCAATGAA GGAGTCCCAAACTGCCCTTACTACAACGAGTACTTTGTCTTCG ACTTCCATGTCTCTCTGATGTGATGTTTGACAAGATCATCAAGA TCTCGGTTATCCATTCTAAGAACCTGCTTCGGAGCGGCACCTG GTGGGTTCTTCAAAATGGATGTGGGGACTGTGATTCCCAGCC TGAACACCAGTTCCATCACAATGGGCCATCTGTGACAGCCCG ATGACATCTCTGCTGGGTGAAGGTTATGTAAAGTGTGATGTC

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GCTGTGGTGGGCAAGGGAGACAACATCAAGACACCCACAAGG CCAACGAGACGGATGAGGACGACATTGAAGGGAACCTGCTGCT CCCCAGGGCGTGTCCCCCGAACGGCAGTGGGCACGGTCTA TGTGAAAATTTACCGAGCAGAGGGGACTGCCCCGGATGAACACAA GCCTCATGGCCAACGTGAAGAAGGCGTTATCGGTGAGAACAA GGACCTCGTCGACCCCTATGTGCAAGTCTTTTGTGGACAAA AGGGCAAAACATCAGTGCAGAAGAGCAGCTATGAGCCGCTATG GAATGAGCAGGTGCTTTCACAGACTTGTTCCTCCCTCAGCTGCA AACGCATGAAGGTGAGATCCGGGACTCTGACAGGTCAATGAT GTGGCCATCGGCACCCACTTCATCGACCTGCGCAAGATTTCCAA CGATGGAGACAAAGGCTTCCTGCCTACCCTCGGTCCAGCCTGG GTGAACATGTACGGCTCCACGCGCAACTACACACTGTCTGGACG AGCACCAGGACTTGAATGAAGGCTGGGGGAGGGTGTGTCCTT CCGGGCCCGCTCATGTTGGGACTAGCTGTGGAGATCCTGGAC ACCTCCAACCCAGAGCTCACCAGCTCCACGGAGGTGCAGGTGG AGCAGGCCACGCTGTCTCGGAGAGCTGCACAGGGAGAATGGA AGAATTTTTTCTATTGGAGCCTTCTTGAAGCCTCAATGATTGA CCGGAAAATAGGGGACAAGCCAATTACCTTTGAGGTGACCATAG GAAACTACGGCAATGAAGTCGATGGTATGTCCCGGCCCTGAG GCCTCGGCCCGGAAAGAGCCTGGGGATGAAGAAGAGGTAGA CCTGATTGAACTCCAGTGACGATGAAGGTGACGAAGCCGGG GACCTGGCCTCGGTGTCTCCACCCACCTATGCGGCCCCAGA TCACGGACAGGAACCTATTTCCACCTGCCCTACCTGGAGCGCAAG CCCTGCATCTATATCAAGAGCTGGTGGCTGACCAAGGCGGC GCCTCTACAATGCCAACATCATGGATCACATTGCTGACAAGCTG GAAGAAGGCTGAATGATGTACAGGAGATGATCAAAACGGAGAA GTCTTACCCGGAGCGCCGCTGCGGGGTGTGCTAGAGGAATC AGCTGTGGCTGCCACCGCTTCCTCTCCCTCTCGGACAAGGACC AGGGCCGCTCGTCCGACACAGGCTGGATCGAGAGCGTCTTAA GTCTGTATGAGGGAGTTGGAGAGCATGGGACAGCAGGCCAAG AGCCTGAGGGCTCAGGTGAAGCGGCACACTGTTCCGGACAAGC TGAGGTGATGCCAGAACTTTCTGCAGAAGCTACGCTTCTGGCG GATGAGCCCCAGCACAGCATTCTGATGTGTTCAATTTGGATGAT GAGCAACAACAAACGTATCGCCTATGCCCGCGTGCCCTTCCAAAG ACCTGCTCTTCTCATCGTGGAGGAGGAAGTGGGCAAGGACTG CGCCAAAGTCAAGACCCCTTCTCTGAAGCTGCCAGGGAAGAGG GGCTTCGGCTCGGCAGGCTGGACAGTACAGGCCAAGCTGGAG CTCTACCTGTGGCTGGGCCTCAGCAAGCAGCGAAGGACTTCC TGTGTGGTCTGCCCTGTGGCTTCGAGGAGGTCAAGGCAGCCCA AGGCCTGGGCTGCATTCTTCCGCCCATCAGCCTAGTCTACA CCAAGAAGCAAGCCTTCCAGCTCCGAGCACACATGTATCAGGC CCGAAGCCCTTTGTGTGTGACAGCAGTGGGCTCTCTGATCCCT TTGCCCGTGTCTTCTTCATCAACCAGAGCCAAATGCACTGAGGTT CTAACGAGACACTGTGTCCACCTGGGACCAAGTGTGGTATT TGACAACCTGGAGCTGTACGGTGAAGCTCACGAGTTACGAGAT GATCCCCCATCATTTGTCTTGAATCTACGACCAGGACAGCAT GGGCAAGCCGACTTCATGGGCCGACCTTCGCCAAGCCCTG GTGAAGATGGCAGATGAAGCATACTGCCACCTCGCTTCCCGC CGCAGCTTGAGTACTACAGATCTACCGAGGCAGTGCCACTGC CGGAGACCTACTGGCTGCCTTCGAGCTGCTGCAGATTGGGCCA TCAGGGAAGGCTGACCTGCCACCCATCAATGGCCAGTGGACA TGGACAGAGGGCCCATCATGCCTGTGCCGTGGGAATCCGGCC AGTGCTCAGCAAGTACCGAGTGGAGGTGCTGTTCTGGGGCTG AGGGACCTAAGAGGGTGAACCTGGCCAGGTGGACCGACCAC GGGTGGACATCGAGTGTGCAGGAAGGGGTACAATCCTCCCT GATTACAATTATAAGAAGAACCCTCACTTCAACACGCTGGTCAA GTGGTTGAAGTGGACCTCCCGGAGAATGAGCTCCTGCACCCA CCCTTGAACATCCGAGTGGTAGATTGCCGGGCTTTGGACGATA CACCCTGGTGGTTCCCACGAGTCAGCTCACTGAGGCGCTTC ATCTACCGACCTCCAGACCGCTCAGCCCCCACTGGAACACCA CAGGGGAGGTTGTAGTAAGCATGGAGCCTGAGGAGCCAGTTAA GAAGCTGGAGACCATGGTGAACCTGGATGCGACTTCTGATGCT GTGGTCAAGGTGGATGTGGCTGAAGATGAGAAGGAAGGAAGA AGAAGAAAAAGAAAGGCCCGTCAGAGGAGCCAGAGGAGGAAGA GCCGATGAGAGCATGCTGGATTGGTGGTCCAAGTACTTCGCC TCCATCGACACAATGAAGGAGCAACTTCGACAACATGAGACCTC TGGAACCTGACTTGGAAGAGAAGGAAGAGATGGAAGCGCTGAG GGCTTGAAGGGACCAATGAAGAGCAAGGAGAAGTCCAGAGCTG CAAAGGAGGAGAAAAAGAAAAACAGAGCCCTGGCCCTGG CCAGGGATCGGAGGCTCTGAGAAGAAAGGAAGCAAGATCGAT GAGCTTAAGGTGTACCCCAAGGAGCTGGAATCGGAGTTTGACA GCTTTGAGGACTGGCTGCACACCTTCAACCTGTTGAGGGGCAA GACGGGAGATGATGAGGATGGCTCCACAGAGGAGGAGCGCATA GTAGGCCGATTCAAGGGCTCCCTGTGTGTACAAGTGCCACT CCCAGAAGATGTATCTCGAGAAGCTGGCTATGATCCACCTATG

TABLE 2-continued

OTOF Sequences		
SEQ ID NO.	Sequence Name	Sequence
		GAATGTTCCAGGGCATCCCAAGCAATGACCCCATCAATGTGCTG GTCCGAATCTATGTGGTCCGGGCCACAGACCTGCACCCGGCCG ACATCAATGGCAAAGCTGACCCCTATATGCCATCAAGTTAGGC AAGACCGACATCCGAGACAAGGAGAACTACATCTCCAAGCAGCT CAACCCGTGTGTTTGGGAAGTCTTTGACATTGAGGCCTCCTTCC CCATGGAGTCCATGTGACAGTGGCCGTGTACGACTGGGATCT GGTGGGCACTGATGACCTCATCGGAGAAAACAGATTGACCTG GAAAACCGCTTCTACAGCAAGCATCGCGCCACCTGCGGCATCG CACAGACCTATTCCATACATGGCTACAATATCTGGAGGGACCCC ATGAAGCCAGCCAGATCCTGACACGCCTCTGTAAAGAGGGCA AAGTGGACGGCCCCACTTTGGTCCCATGGGAGAGTGAGGGT TGCCAACCGTGTCTTCACGGGGCCTTCAGAAATAGAGGATGAGA ATGGTCAGAGGAAGCCACAGATGAGCAGCTGGCCTGTCTGTC TCTGAGACACTGGGAGGACATCCCCGGGTGGGCTGCGCCTT GTGCCGGAACAGTGGAGACAGGCCGCTGCTCAACCTGACA AGCCAGGCATTGAGCAGGGCCGCTGGAGCTGTGGGTGGACAT GTTCCCATGGACATGCCAGCCCTGGGACACCTCTGGATATAT CCCCCAGGAAACCAAGAAGTACGAGCTGCGGGTCTCGTGTG GAACACAGACGAGGTGGTCTTGAAGACGATGATTTCTTCAG GGAGAGAAGTCCAGTGACATTTTGTGAGGGGTGGCTGAAGG GCCAGCAGGAGGACAAACAGGACACAGATGTCCACTATCACTC CCTCACGGGGGAGGGCAACTTCACTGGAGATACCTCTTCCCC TTCGACTACCTAGCGGCCGAAGAGAAGATCGTTATGTCCAAAA GGAGTCTATGTTCTCTGGGATGAGACGGAGTACAAGATCCCTG CGCGGCTCACCTGCAGATCTGGGACGCTGACCACTTCTCGGC TGACGACTTCTGGGGGCTATCGAGCTGGACCTGAACCGGTTT CCGAGGGGCGCTAAGACAGCCAAGCAGTGCACCATGGAGATGG CCACCGGGGAGGTGGACGTACCCCTGGTTTCCATCTTTAAACAG AAACGTGTCAAAGGCTGGTGGCCCTCTGGCCCGCAATGAGA ATGATGAGTTTGAAGTCAAGGCAAGTGGAGGCGGAGCTACA CCTACTCACGGCAGAGGAGGACAGAGAAGAACCTGTGGGCTG GCTCGCAATGAACCTGATCCCTAGAAAAACCAACCGGCTGA CACGGCATTCGTCTGGTTCCTGAACCACTCAAATCTATCAAGT ACCTCATCTGCACCCGGTACAAGTGGCTGATCATCAAGATCGTG CTGGCGCTGCTGGGGTGTCTATGCTGGCCCTCTTCCTTTACA GCCTCCAGGCTACATGGTCAAGAAGCTCTAGGGGCTGAAG TGTGCCCCACCCAGCCCGCTCCAGCATCCCTCCAGGGGCTGC TGCGTATTTTGCTTCCCTCACCTGGACTCTCTCCAACTCCCT GAGGAGCCCTCCACGCCTGCCAGCCTTGAGCAAGACACCTGC TTGCTGGACTTCATCCCCACCCACACCCAACTGTGCTTGCC TGATCTTGTCCAGGCTGCTGGGTTTGGGGCACAGTTGGC CTCCAAACAGATACCTCTTGTCTAAAGTACCAGGTTCTCTTG CCCAACCCCAAGAGTGGTAGTGGCCCAACCTCCTGTGCTTTT CAAACTTTGTCTTAAGGCACAGTGAAATTAACCAAGAAACGCG GAGCAATGCCAAGGCTCTGATGAGTAGGAACACGTGGAAAGC ACCAGGAATGCCAGCAGAGGCGAGCGGCACACCTCTCTGCAG AGCATCCAGGCGAGCGCGGGCAGCGGCCAGCTGCTTCTGC GCATGCTCTCTCTTGGCTCTGCTTCTTCTCACAGTCACAGTCA CTTCACAGCTTAGCCTTGGGCTTCCATCACTTCCAGGGGTGCC TCTGCTTGGCCAGTGTGTGACAGTAGTACACAGCTCCAAGT GTGAATCAGGTGACTGGCCGCTCTGAAGACTGACTGCCCTGTC CTTCTGCGCAGCAGCCACCCGAGTGTACACTTAAAGCGGTG CCCTTCTGCTCTGTGGGCTGCTGGCTGCTGTTCTTTCTTGA GTGTGATTTTTTTTCTCTCCCTCAATAAATAAATCAAACCTCTG AGAC

Expression of OTOF in Mammalian Cells

Mutations in OTOF have been linked to sensorineural hearing loss and auditory neuropathy. The compositions and methods described herein increase the expression of WT OTOF protein by administering a first nucleic acid vector that contains a polynucleotide encoding an N-terminal portion of an OTOF protein and a second nucleic acid vector that contains a polynucleotide encoding a C-terminal portion of an OTOF protein. In order to utilize nucleic acid vectors for therapeutic application in the treatment of sensorineural hearing loss and auditory neuropathy, they can be directed to the interior of the cell, and, in particular, to specific cell types. A wide array of methods has been established for the delivery of proteins to mammalian cells and for the stable expression of genes encoding proteins in mammalian cells.

Polynucleotides Encoding OTOF

One platform that can be used to achieve therapeutically effective intracellular concentrations of OTOF in mammalian cells is via the stable expression of the gene encoding OTOF (e.g., by integration into the nuclear or mitochondrial genome of a mammalian cell, or by episomal concatemer formation in the nucleus of a mammalian cell). The gene is a polynucleotide that encodes the primary amino acid sequence of the corresponding protein. In order to introduce exogenous genes into a mammalian cell, genes can be incorporated into a vector. Vectors can be introduced into a cell by a variety of methods, including transformation, transfection, transduction, direct uptake, projectile bombardment, and by encapsulation of the vector in a liposome. Examples of suitable methods of transfecting or transform-

ing cells include calcium phosphate precipitation, electroporation, microinjection, infection, lipofection and direct uptake. Such methods are described in more detail, for example, in Green, et al., *Molecular Cloning: A Laboratory Manual*, Fourth Edition (Cold Spring Harbor University Press, New York 2014); and Ausubel, et al., *Current Protocols in Molecular Biology* (John Wiley & Sons, New York 2015), the disclosures of each of which are incorporated herein by reference.

OTOF can also be introduced into a mammalian cell by targeting vectors containing portions of a gene encoding an OTOF protein to cell membrane phospholipids. For example, vectors can be targeted to the phospholipids on the extracellular surface of the cell membrane by linking the vector molecule to a VSV-G protein, a viral protein with affinity for all cell membrane phospholipids. Such a construct can be produced using methods well known to those of skill in the field.

Recognition and binding of the polynucleotide encoding an OTOF protein by mammalian RNA polymerase is important for gene expression. As such, one may include sequence elements within the polynucleotide that exhibit a high affinity for transcription factors that recruit RNA polymerase and promote the assembly of the transcription complex at the transcription initiation site. Such sequence elements include, e.g., a mammalian promoter, the sequence of which can be recognized and bound by specific transcription initiation factors and ultimately RNA polymerase.

Polynucleotides suitable for use in the compositions and methods described herein also include those that encode an OTOF protein downstream of a mammalian promoter (e.g., a polynucleotide that encodes an N-terminal portion of an OTOF protein downstream of a mammalian promoter). Promoters that are useful for the expression of an OTOF protein in mammalian cells include constitutive promoters, cochlear hair cell-specific promoters, and inner hair cell-specific promoters. Constitutive promoters include the CAG promoter, the cytomegalovirus (CMV) promoter, the smCBA promoter, the EF1 α promoter, and the PGK promoter. Cochlear hair cell-specific promoters include the Myosin 15 (Myo15) promoter, the Myosin 7A (Myo7A) promoter, the Myosin 6 (Myo6) promoter, the POU4F3 promoter, the Atonal BHLH Transcription Factor 1 (ATOH1) promoter, the LIM Homeobox 3 (LHX3) promoter, the $\alpha 9$ acetylcholine receptor ($\alpha 9$ AChR) promoter, and the $\alpha 10$ acetylcholine receptor ($\alpha 10$ AChR) promoter. Inner hair cell-specific promoters include the FGF8 promoter, the VGLUT3 promoter, and the OTOF promoter. Alternatively, promoters derived from viral genomes can also be used for the stable expression of these agents in mammalian cells. Examples of functional viral promoters that can be used to promote mammalian expression of these agents include adenovirus late promoter, vaccinia virus 7.5K promoter, SV40 promoter, tk promoter of HSV, mouse mammary tumor virus (MMTV) promoter, LTR promoter of HIV, promoter of moloney virus, Epstein barr virus (EBV) promoter, and the Rous sarcoma virus (RSV) promoter. Murine Myosin 15 Promoters

In some embodiments, the Myo15 promoter for use in the compositions and methods described herein includes nucleic acid sequences from regions of the murine Myo15 locus that are capable of expressing a transgene specifically in hair cells, or variants thereof, such as a nucleic acid sequences that have at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to regions of the murine Myo15 locus that are capable of expressing a transgene specifically in hair cells. These

regions include nucleic acid sequences immediately preceding the murine Myo15 translation start site and an upstream regulatory element that is located over 5 kb from the murine Myo15 translation start site. The Myo15 promoter for use in the compositions and methods described herein can optionally include a linker operably linking the regions of the murine Myo15 locus that are capable of expressing a transgene specifically in hair cells, or the regions of the murine Myo15 locus can be joined directly without an intervening linker.

In some embodiments, the Myo15 promoter for use in the compositions and methods described herein contains a first region (an upstream regulatory element) having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to a region containing the first non-coding exon of the murine Myo15 gene (nucleic acids from -6755 to -7209 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 24) or a functional portion or derivative thereof joined (e.g., operably linked) to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence immediately preceding the murine Myo15 translation start site (nucleic acids from -1 to -1157 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 25) or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 24 may have the sequence of nucleic acids from -7166 to -7091 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 26) and/or the sequence of nucleic acids from -7077 to -6983 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 27). The first region may contain the nucleic acid sequence of SEQ ID NO: 26 fused to the nucleic acid sequence of SEQ ID NO: 27 with no intervening nucleic acids, as set forth in SEQ ID NO: 28, or the first region may contain the nucleic acid sequence of SEQ ID NO: 27 fused to the nucleic acid sequence of SEQ ID NO: 26 with no intervening nucleic acids, as set forth in SEQ ID NO: 29. Alternatively, the first region may contain the sequences of SEQ ID NO: 26 and SEQ ID NO: 27 joined by the endogenous intervening nucleic acid sequence (e.g., the first region may have or include the sequence of nucleic acids from -7166 to -6983 with respect to the murine Myo15 translation start site, as set forth in SEQ ID NO: 30 and SEQ ID NO: 50) or a nucleic acid linker. In a murine Myo15 promoter in which the first region contains both SEQ ID NO: 26 and SEQ ID NO: 27, the two sequences can be included in any order (e.g., SEQ ID NO: 26 may be joined to (e.g., precede) SEQ ID NO: 27, or SEQ ID NO: 27 may be joined to (e.g., precede) SEQ ID NO: 26). The functional portion of SEQ ID NO: 25 may have the sequence of nucleic acids from -590 to -509 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 31) and/or the sequence of nucleic acids from -266 to -161 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 32). In some embodiments, the sequence containing SEQ ID NO: 31 has the sequence of SEQ ID NO: 51. In some embodiments, the sequence containing SEQ ID NO: 32 has the sequence of SEQ ID NO: 52. The second region may contain the nucleic acid sequence of SEQ ID NO: 31 fused to the nucleic acid sequence of SEQ ID NO: 32 with no intervening nucleic acids, as set forth in SEQ ID NO: 33, or the second region may contain the nucleic acid sequence of SEQ ID NO: 32 fused to the nucleic acid sequence of SEQ ID NO: 31 with no intervening nucleic acids, as set forth in SEQ ID NO: 34. The second region may contain the nucleic

acid sequence of SEQ ID NO: 51 fused to the nucleic acid sequence of SEQ ID NO: 52 with no intervening nucleic acids, as set forth in SEQ ID NO: 55, or the second region may contain the nucleic acid sequence of SEQ ID NO: 52 fused to the nucleic acid sequence of SEQ ID NO: 51 with no intervening nucleic acids. Alternatively, the second region may contain the sequences of SEQ ID NO: 31 and SEQ ID NO: 32 joined by the endogenous intervening nucleic acid sequence (e.g., the second region may have the sequence of nucleic acids from -590 to -161 with respect to the murine Myo15 translation start site, as set forth in SEQ ID NO: 35) or a nucleic acid linker. In a murine Myo15 promoter in which the second region contains both SEQ ID NO: 31 and SEQ ID NO: 32, the two sequences can be included in any order (e.g., SEQ ID NO: 31 may be joined to (e.g., precede) SEQ ID NO: 32, or SEQ ID NO: 32 may be joined to (e.g., precede) SEQ ID NO: 31).

The first region and the second region of the murine Myo15 promoter can be joined directly or can be joined by a nucleic acid linker. For example, the murine Myo15 promoter can contain the sequence of SEQ ID NO: 24 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 26-30 and 50, e.g., SEQ ID NOs 26 and 27) fused to the sequence of SEQ ID NO: 25 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 31-35, 51, 52, and 55, e.g., SEQ ID NOs 31 and 32) with no intervening nucleic acids. For example, the nucleic acid sequence of the murine Myo15 promoter that results from direct fusion of SEQ ID NO: 24 to SEQ ID NO: 25 is set forth in SEQ ID NO: 36. Alternatively, a linker can be used to join the sequence of SEQ ID NO: 24 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 26-30 and 50, e.g., SEQ ID NOs 26 and 27) to the sequence of SEQ ID NO: 25 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 31-35, 51, 52, and 55, e.g., SEQ ID NOs 31 and 32). Exemplary Myo15 promoters containing functional portions of both SEQ ID NO: 24 and SEQ ID NO: 25 are provided in SEQ ID NOs: 38, 39, 53, 54, 59, and 60.

The length of a nucleic acid linker for use in a murine Myo15 promoter described herein can be about 5 kb or less (e.g., about 5 kb, 4.5 kb, 4 kb, 3.5 kb, 3 kb, 2.5 kb, 2 kb, 1.5 kb, 1 kb, 900 bp, 800 bp, 700 bp, 600 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp, 250 bp, 200 bp, 150 bp, 100 bp, 90 bp, 80 bp, 70 bp, 60 bp, 50 bp, 40 bp, 30 bp, 25 bp, 20 bp, 15 bp, 10 bp, 5 bp, 4 bp, 3 bp, 2 bp, or less). Nucleic acid linkers that can be used in the murine Myo15 promoter described herein do not disrupt the ability of the murine Myo15 promoter of the invention to induce transgene expression in hair cells.

In some embodiments, the sequence of SEQ ID NO: 24 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 26-30 and 50, e.g., SEQ ID NOs 26 and 27) is joined (e.g., operably linked) to the sequence of SEQ ID NO: 25 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 31-35, 51, 52, and 55, e.g., SEQ ID NOs 31 and 32), and, in some embodiments, the order of the regions is reversed (e.g., the sequence of SEQ ID NO: 25 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 31-35, 51, 52, and 55, e.g., SEQ ID NOs 31 and 32) is joined (e.g., operably linked) to the sequence of SEQ ID NO: 24 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 26-30 and 50, e.g., SEQ ID NOs 26 and 27)). For example, the nucleic acid sequence of a murine Myo15 promoter that results from direct fusion of SEQ ID NO: 25 to SEQ ID NO: 24 is set forth in SEQ ID NO: 37.

An example of a murine Myo15 promoter in which a functional portion or derivative of SEQ ID NO: 25 precedes a functional portion or derivative of SEQ ID NO: 24 is provided in SEQ ID NO: 58. Regardless of order, the sequence of SEQ ID NO: 24 or a functional portion or derivative thereof and the sequence of SEQ ID NO: 25 or a functional portion or derivative thereof can be joined by direct fusion or a nucleic acid linker, as described above.

In some embodiments, the murine Myo15 promoter for use in the compositions and methods described herein contains a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to a region containing the first non-coding exon of the murine Myo15 gene (nucleic acids from -6755 to -7209 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 24) or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 24 may have the sequence of nucleic acids from -7166 to -7091 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 26) and/or the sequence of nucleic acids from -7077 to -6983 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 27). The murine Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 26 fused to the nucleic acid sequence of SEQ ID NO: 27 with no intervening nucleic acids, as set forth in SEQ ID NO: 28, or the murine Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 27 fused to the nucleic acid sequence of SEQ ID NO: 26 with no intervening nucleic acids, as set forth in SEQ ID NO: 29. Alternatively, the murine Myo15 promoter may contain the sequences of SEQ ID NO: 26 and SEQ ID NO: 27 joined by the endogenous intervening nucleic acid sequence (e.g., the first region may have or include the sequence of nucleic acids from -7166 to -6983 with respect to the murine Myo15 translation start site, as set forth in SEQ ID NO: 30 and SEQ ID NO: 50) or a nucleic acid linker. In a murine Myo15 promoter that contains both SEQ ID NO: 26 and SEQ ID NO: 27, the two sequences can be included in any order (e.g., SEQ ID NO: 26 may be joined to (e.g., precede) SEQ ID NO: 27, or SEQ ID NO: 27 may be joined to (e.g., precede) SEQ ID NO: 26).

In some embodiments, the murine Myo15 promoter for use in the compositions and methods described herein contains a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence immediately upstream of the murine Myo15 translation start site (nucleic acids from -1 to -1157 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 25) or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 25 may have the sequence of nucleic acids from -590 to -509 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 31) and/or the sequence of nucleic acids from -266 to -161 with respect to the murine Myo15 translation start site (set forth in SEQ ID NO: 32). In some embodiments, the sequence containing SEQ ID NO: 31 has the sequence of SEQ ID NO: 51. In some embodiments, the sequence containing SEQ ID NO: 32 has the sequence of SEQ ID NO: 52. The murine Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 31 fused to the nucleic acid sequence of SEQ ID NO: 32 with no intervening nucleic acids, as set forth in SEQ ID NO: 33, or the murine Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 32 fused to the nucleic acid sequence of SEQ ID NO: 31 with no intervening nucleic acids, as set forth in SEQ ID NO: 34. The murine Myo15 promoter may contain the

nucleic acid sequence of SEQ ID NO: 51 fused to the nucleic acid sequence of SEQ ID NO: 52 with no intervening nucleic acids, as set forth in SEQ ID NO: 55, or the murine Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 52 fused to the nucleic acid sequence of SEQ ID NO: 51 with no intervening nucleic acids. Alternatively, the murine Myo15 promoter may contain the sequences of SEQ ID NO: 31 and SEQ ID NO: 32 joined by the endogenous intervening nucleic acid sequence (e.g., the second region may have the sequence of nucleic acids from -590 to -161 with respect to the murine Myo15 translation start site, as set forth in SEQ ID NO: 35) or a nucleic acid linker. In a murine Myo15 promoter that contains both SEQ ID NO: 31 and SEQ ID NO: 32, the two sequences can be included in any order (e.g., SEQ ID NO: 31 may be joined to (e.g., precede) SEQ ID NO: 32, or SEQ ID NO: 32 may be joined to (e.g., precede) SEQ ID NO: 31).

In some embodiments, the murine Myo15 promoter for use in the compositions and methods described herein contains a functional portion or derivative of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to a region containing the first non-coding exon of the Myo15 gene (nucleic acids from -6755 to -7209 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 24) flanked on both sides by a functional portion or derivative of a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence immediately upstream of the murine Myo15 translation start site (nucleic acids from -1 to -1157 with respect to the murine Myo15 translation start site, the sequence of which is set forth in SEQ ID NO: 25). For example, a functional portion or derivative of SEQ ID NO: 25, such as SEQ ID NO: 31 or 51 may be directly fused or joined by a nucleic acid linker to a portion of SEQ ID NO: 24, such as any one of SEQ ID NOs: 26-30 and 50, which is directly fused or joined by a nucleic acid linker to a different functional portion of SEQ ID NO: 25, such as SEQ ID NO: 32 or 52. In other embodiments, a functional portion or derivative of SEQ ID NO: 25, such as SEQ ID NO: 32 or 52 may be directly fused or joined by a nucleic acid linker to a portion of SEQ ID NO: 24, such as any one of SEQ ID NOs: 26-30 and 50, which is directly fused or joined by a nucleic acid linker to a different functional portion of SEQ ID NO: 25, such as SEQ ID NO: 31 or 51. For example, polynucleotides having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence of SEQ ID NOs: 51, 50, and 52 can be fused to produce a polynucleotide having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence of SEQ ID NO: 56. In some embodiments, polynucleotides having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence of SEQ ID NOs: 52, 50, and 51 can be fused to produce a polynucleotide having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequence of SEQ ID NO: 57.

Human Myosin 15 Promoters

In some embodiments, the Myo15 promoter for use in the compositions and methods described herein includes nucleic acid sequences from regions of the human Myo15 locus that are capable of expressing a transgene specifically in hair cells, or variants thereof, such as a nucleic acid sequences that have at least 85% sequence identity (e.g., 85%, 90%,

95%, 96%, 97%, 98%, 99%, or more, sequence identity) to regions of the human Myo15 locus that are capable of expressing a transgene specifically in hair cells. The Myo15 promoter for use in the compositions and methods described herein can optionally include a linker operably linking the regions of the human Myo15 locus that are capable of expressing a transgene specifically in hair cells, or the regions of the human Myo15 locus can be joined directly without an intervening linker.

In some embodiments, the Myo15 promoter for use in the compositions and methods described herein contains a first region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the sequence set forth in SEQ ID NO: 40 or a functional portion or derivative thereof joined (e.g., operably linked) to a second region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the sequence set forth in SEQ ID NO: 41 or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 40 may have the sequence set forth in SEQ ID NO: 42. The functional portion of SEQ ID NO: 41 may have the sequence set forth in SEQ ID NO: 43 and/or the sequence set forth in SEQ ID NO: 44. The second region may contain the nucleic acid sequence of SEQ ID NO: 43 fused to the nucleic acid sequence of SEQ ID NO: 44 with no intervening nucleic acids, as set forth in SEQ ID NO: 45, or the second region may contain the nucleic acid sequence of SEQ ID NO: 44 fused to the nucleic acid sequence of SEQ ID NO: 43 with no intervening nucleic acids, as set forth in SEQ ID NO: 46. Alternatively, the second region may contain the sequences of SEQ ID NO: 43 and SEQ ID NO: 44 joined by the endogenous intervening nucleic acid sequence (as set forth in SEQ ID NO: 47) or a nucleic acid linker. In a human Myo15 promoter in which the second region contains both SEQ ID NO: 43 and SEQ ID NO: 44, the two sequences can be included in any order (e.g., SEQ ID NO: 43 may be joined to (e.g., precede) SEQ ID NO: 44, or SEQ ID NO: 44 may be joined to (e.g., precede) SEQ ID NO: 43).

The first region and the second region of the human Myo15 promoter can be joined directly or can be joined by a nucleic acid linker. For example, the human Myo15 promoter can contain the sequence of SEQ ID NO: 40 or a functional portion or derivative thereof (e.g., SEQ ID NO: 42) fused to the sequence of SEQ ID NO: 41 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 43-47, e.g., SEQ ID NOs: 43 and/or 44) with no intervening nucleic acids. Alternatively, a linker can be used to join the sequence of SEQ ID NO: 40 or a functional portion or derivative thereof (e.g., SEQ ID NO: 42) to the sequence of SEQ ID NO: 41 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 43-47, e.g., SEQ ID NOs: 43 and/or 44). Exemplary human Myo15 promoters containing functional portions of both SEQ ID NO: 40 and SEQ ID NO: 41 are provided in SEQ ID NOs: 48 and 49.

In some embodiments, the sequence of SEQ ID NO: 40 or a functional portion or derivative thereof (e.g., SEQ ID NO: 42) is joined (e.g., operably linked) to the sequence of SEQ ID NO: 41 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 43-47, e.g., SEQ ID NOs: 43 and 44), and, in some embodiments, the order of the regions is reversed (e.g., the sequence of SEQ ID NO: 41 or a functional portion or derivative thereof (e.g., any one or more of SEQ ID NOs: 43-47, e.g., SEQ ID NOs: 43 and/or 44) is joined (e.g., operably linked) to the sequence of SEQ ID NO: 40 or a functional portion or derivative thereof (e.g.,

SEQ ID NO: 42). Regardless of order, the sequence of SEQ ID NO: 40 or a functional portion or derivative thereof and the sequence of SEQ ID NO: 41 or a functional portion or derivative thereof can be joined by direct fusion or a nucleic acid linker, as described above.

In some embodiments, the human Myo15 promoter for use in the compositions and methods described herein contains a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to a region containing the sequence set forth in SEQ ID NO: 40 or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 40 may have the sequence of nucleic acids set forth in SEQ ID NO: 42.

In some embodiments, the human Myo15 promoter for use in the compositions and methods described herein contains a region having at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the sequence set forth in SEQ ID NO: 41 or a functional portion or derivative thereof. The functional portion of SEQ ID NO: 41 may have the sequence set forth in SEQ ID NO: 43 and/or the sequence set forth in SEQ ID NO: 44. The human Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 43 fused to the nucleic acid sequence of SEQ ID NO: 44 with no intervening nucleic acids, as set forth in SEQ ID NO: 45, or the human

Myo15 promoter may contain the nucleic acid sequence of SEQ ID NO: 44 fused to the nucleic acid sequence of SEQ ID NO: 43 with no intervening nucleic acids, as set forth in SEQ ID NO: 46. Alternatively, the human Myo15 promoter may contain the sequences of SEQ ID NO: 43 and SEQ ID NO: 44 joined by the endogenous intervening nucleic acid sequence (e.g., as set forth in SEQ ID NO: 47) or a nucleic acid linker. In a human Myo15 promoter that contains both SEQ ID NO: 43 and SEQ ID NO: 44, the two sequences can be included in any order (e.g., SEQ ID NO: 43 may be joined to (e.g., precede) SEQ ID NO: 44, or SEQ ID NO: 44 may be joined to (e.g., precede) SEQ ID NO: 43).

The length of a nucleic acid linker for use in a human Myo15 promoter described herein can be about 5 kb or less (e.g., about 5 kb, 4.5 kb, 4 kb, 3.5 kb, 3 kb, 2.5 kb, 2 kb, 1.5 kb, 1 kb, 900 bp, 800 bp, 700 bp, 600 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp, 250 bp, 200 bp, 150 bp, 100 bp, 90 bp, 80 bp, 70 bp, 60 bp, 50 bp, 40 bp, 30 bp, 25 bp, 20 bp, 15 bp, 10 bp, 5 bp, 4 bp, 3 bp, 2 bp, or less). Nucleic acid linkers that can be used in the human Myo15 promoters described herein do not disrupt the ability of the human Myo15 promoter of the invention to induce transgene expression in hair cells.

The foregoing Myo15 promoter sequences are summarized in Table 3, below.

TABLE 3

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
24	Region containing non-coding exon 1 of Myo15 (-6755 to -7209)	CTGCAGCTCAGCCTACTTGTGCTTCCAGGCTGTTCCTAGT TCCCATGTGCTGCTGTGCTTCCAGAGACAAAACAGGAA TAATAGATGTCATTAAATATACATTGGGCCCCAGGCGGTCAAT GTGGCAGCCTGAGCCTCCTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCCAACAAACAGCATGTCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCCCTTAAGCTTCTGC CACTGGCTCCGGCATTGCAGAGAGAAGAGAAGGGCGGCA GAGCTGAACCTTAGCCTTGCCCTTCTGGGTACCCCTTCTGAGC CTCACTGTCTTCTGTGAGATGGGCAAGTGCGGGTGTGACTC CTTGGCAACGGTGTACACCAGGGCAGGTAAAGTTGTAGTTA TTTGTGGGTACACCAGGACTGTTAAAGTGTAACAT
25	Region immediately preceding the translation start site of Myo15 (-1 to -1157)	GGTCTCACCCAGCATTTTCACTTCTAATAAGTTCAATGTGAT ACGGCACCTTTCTAAAAATTAGTTTTCAGGGAATAGGGTTCA AAACTGGTAGTGGTAGGGTCCATTCTCACGACCCCGAGGCT GCTAACCCCTGACCAAGCTACCTATTACTTACCTCCTCTTCT CCTCCTCCTCTTCTCCTTCTCCTGTCTCCCTCTTCTCCTTCT CCTCCTTCTCCTCCTCCTCCTCCTCCTTGGCTGTGATCAG ATCCAGAGCCTGAATGAGCCTCCTGACCCACACCCCACTA GCATGGGCTGCAAGTGCCAGAGTCCCTCCTGCTCCTA AACTGCCCAGCCGATCCATTAGTCTTCTTCTTCCAGTGA AAGAAGCAGGCACAGCCTGTCCCTCCCGTTCTACAGAAAGG AAGCTACAGCACAGGGAGGGCCAAAGGCCTTCTGGGACTA GACAGTTGATCAACAGCAGGACTGGAGAGCTGGGCTCCATTT TTGTTCTTGGTGCCCTGCCCTCCCATGACCTGCAGAGAC ATTGAGCCTGCCCAGGCTTTATGAGGTGGGAGCTGGGCTCTC CCTGATGTATTATTGAGCTCCCTGGAGTTGGCCAGCTCCTGT TACACTGGCCACAGCCCTGGGATCCGCTTCTCACTTCTAGT TTCCCTTCAAGGTAATGTGGTGGGTCATGATCATTCTATCCT GGCTTCAGGGACCTGACTCCACTTTGGGGCCATTGAGGGG TCTAGGGTAGATGATGTCCCTGTGGGGATTAAATGTCTCTGC TCTGTAAACTGAGCTAGCTGAGATCCAGGAGGCTTGCCCA GAGACAGCAAGTTGTTGCCATGGTGAAGTTAAAGCCAGGTTG CTGCCCCAGCACAGGCTCCAGTCTACCTCACTAGAAAAC AACACCCAGGCCTTTCCACCCTCTCAAAGGTGAACCCA AGGCTGGTCTAGAGAATGAATTATGGATCCTCGCTGTCCGTG CCACCCAGCTAGTCCAGCGGCTCAGACACTGAGGAGAGAC TGATGTTTCACTACAAGCAAAAGACCTAGCTGGTCTCCAA GCAGTGTCTCCAAGTCCCTGAACCTGTGACACCTGCCCCAG GCATCATCAGGCACAGAGGGCCACC
26	Portion of SEQ ID NO: 24 (-7166 to -7091)	CCCATGTGCTGCTTGTGCTTCCAGAGACAAAACAGGAAT AATAGATGTCATTAAATATACATTGGGCCCCAGG

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
27	Portion of SEQ ID NO: 24 (-7077 to -6983)	AGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGACATAGGA CCCCAACAAACAGCATGCAGGTTGGGAGCCAGCCACAGGA CCCAGGTAAGGG
28	Portion of SEQ ID NO: 24 (SEQ ID NO: 26 fused to SEQ ID NO: 27)	CCCATGTCAGCTGCTTGTGCTTTCCAGAGACAAAACAGGAAT AATAGATGTCATTAAATATACATTGGGCCCCAGGAGCCTGAG CCTCCTTTCCATCTCTGTGGAGGCAGACATAGGACCCCAAC AAACAGCATGCAGGTTGGGAGCCAGCCACAGGACCCAGGTA AGGG
29	Portion of SEQ ID NO: 24 (SEQ ID NO: 27 fused to SEQ ID NO: 26)	AGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGACATAGGA CCCCAACAAACAGCATGCAGGTTGGGAGCCAGCCACAGGA CCCAGGTAAGGGCCCATGTCAGCTGCTTGTGCTTTCCAGAGA CAAAACAGGAATAATAGATGTCATTAAATATACATTGGCCCC AGG
30	Portion of SEQ ID NO: 24 (-7166 to -6983)	CCCATGTCAGCTGCTTGTGCTTTCCAGAGACAAAACAGGAAT AATAGATGTCATTAAATATACATTGGGCCCCAGGCGGTCAAT GTGGCAGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGG
31	Portion of SEQ ID NO: 25 (-590 to -509)	TGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTAGCTCC CTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCCCTG
32	Portion of SEQ ID NO: 25 (-266 to -161)	CACAGGCCTCCAGTCTACCCCTCACTAGAAAAACACCCAG GCACTTTCCACCACCTCTCAAAGGTGAAACCCAAGGCTGGTC TAGAGAATGAATTATGGATCCT
33	Portion of SEQ ID NO: 25 (SEQ ID NO: 31 fused to SEQ ID NO: 32)	TGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTAGCTCC CTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCCCTGC ACAGGCCTCCAGTCTACCCCTCACTAGAAAAACACCCAGG CACTTTCCACCACCTCTCAAAGGTGAAACCCAAGGCTGGTCT AGAGAATGAATTATGGATCCT
34	Portion of SEQ ID NO: 25 (SEQ ID NO: 32 fused to SEQ ID NO: 31)	CACAGGCCTCCAGTCTACCCCTCACTAGAAAAACACCCAG GCACTTTCCACCACCTCTCAAAGGTGAAACCCAAGGCTGGTC TAGAGAATGAATTATGGATCCTTGAGGTGGGAGCTGGGCTCT CCCTGATGTATTATTAGCTCCCTGGAGTTGGCCAGCTCCTG TTACACTGGCCACAGCCCTG
35	Portion of SEQ ID NO: 25 (-590 to -161)	TGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTAGCTCC CTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCCCTGG GCATCCGCTTCTCACTTCTAGTTTCCCTCCAAGGTAATGTG GTGGGTCATGATCATTCTATCCTGGCTTCAGGGACCTGACTC CACTTTGGGGCCATTGAGGGGTCTAGGGTAGATGATGTCC CCCTGTGGGGATTAATGTCTCTGTAAAACCTGAGCTAGC TGAGATCCAGGAGGGCTTGGCCAGAGACAGCAAGTTGTGTC CATGGTGACTTTAAAGCCAGGTGCTGCCCCAGCACAGGCCT CCCAGTCTACCCCTCACTAGAAAAACACCCAGGCACTTTCC ACCACCTCTCAAAGGTGAAACCCAAGGCTGGTCTAGAGAATG AATTATGGATCCT
36	SEQ ID NO: 24 fused to SEQ ID NO: 25	CTGCAGCTCAGCCTACTACTTGCTTTCCAGGCTGTTCCTAGT TCCCATGTCAGCTGCTTGTGCTTCCAGAGACAAAACAGGAA TAATAGATGTCATTAAATATACATTGGGCCCCAGGCGTCAAT GTGGCAGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTAAGCTTCTGTC CACTGGCTCCGGCATTGCAGAGAGAAGAGAAGGGGCGGCA GAGCTGAACCTTAGCCTTGCTTCCCTGGGTACCTTCTGAGC CTCACTGTCTTCTGTGAGATGGGCAAGTGCGGGTGTGACTC CTTGGCAACGGTGTTACACCAGGGCAGGTAAGTTGTAGTTA TTTGTGGGGTACACAGGACTGTTAAAGGTGTAACATATGGTC TCACCCAGCATTTCACTTCTAATAAGTTCAAATGTGATACGG CACCTTTCTAAAAATAGTTTTCAGGAAATAGGGTTCAAAAC TGGTAGTGGTAGGGTCCATTCTACGACCCCAAGGCTGCT AACCTTGACCAAGCTACCTATTACTTACCCTCCTCTTTCTCCT CCTCCTCTTTCTCCTTCTCCTGCTTCCCCTCTTCTCTCCT CCCTTCTCTCCTCCTCCTCCTCCTTGGCTGTGATCAGATC CAGAGCCTGAATGAGCCTCCTGACCCACACCCCACTAGC ATGGGCCTGCAAGTGCCAGAGTCCCTCCTGCCTCCTAAAC TGCCAGCCGATCCATTAGCTCTCCTTCTTCCAGTGAAAG AAGCAGGCACAGCCTGTCCCTCCCGTTCTACAGAAAGGAAG

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
		CTACAGCACAGGGAGGGCCAAAGGCCTTCCTGGGACTAGAC AGTTGATCAACAGCAGGACTGGAGAGCTGGGCTCCATTTTGG TTCCTTGGTGCCCTGCCCTCCCATGACCTGCAGAGACATT CAGCCTGCCAGGCTTTATGAGGTGGGAGCTGGGCTCTCCCT GATGTATTATTAGCTCCCTGGAGTTGGCCAGCTCCTGTAC ACTGGCCACAGCCCTGGGCATCCGCTTCTCACTTCTAGTTTC CCCTCCAAGGTAATGTGGTGGGTCATGATCATTCTATCCTGG CTTCAGGGACCTGACTCCACTTTGGGGCATTGAGGGGTC TAGGGTAGATGATGTCCCTGTGGGGATTAATGTCTGCTC TGTAAGCTGAGCTAGCTGAGATCCAGGAGGGCTTGGCCAG AGACAGCAAGTTGTTGCCATGGTGACTTAAAGCCAGGTTGC TGCCCCAGCACAGGCTCCAGTCTACCTCACTAGAAAAACA ACACCCAGGCACCTTCCACCACTCTCAAAGGTGAAACCAA GGCTGGTCTAGAGAATGAATTATGGATCCTCGCTGTCGTGC CAGCCAGCTAGTCCAGCGGCTCAGACACTGAGGAGAGACT GTAGGTTAGCTACAAGCAAAAGACCTAGCTGGTCTCCAAG CAGTGTCTCAAGTCCCTGAACCTGTGACACCTGCCCCAGG CATCATCAGGCACAGAGGGCCACC
37	SEQ ID NO: 25 fused to SEQ ID NO: 24	GGTCTCACCCAGCATTTTCACTTCTAATAAGTTCAAATGTGAT ACGGCACCTTTCTAAAAATTAGTTTTCAGGGAATAGGGTTCA AACTGGTAGTGGTAGGGTCCATTCTCAGACCCCCAGGCCT GCTAACCTTGACCAAGCTACCTATTACTTACCTCCTCTTCT CCTCCTCCTTCTTCTCCTTCTCCTGCTTCCCTCTTCTTCTC CCTCCCTTCTCCTCCTCCTCCTCCCTCCTTGGCTGTGATCAG ATCCAGAGCCTGAATGAGCCTCCTGACCCACACCCCACTA GCATGGGCCTGCAAGTGCCCAAGTCCCTCCTGCTCCTA AACTGCCCAGCCGATCCATTAGCTCTTCTTCTTCTCCAGTGA AAGAAGCAGGCACAGCCTGTCCCTCCGTTCTACAGAAAGG AAGCTACAGCACAGGGAGGGCCAAAGGCCTTCTGGGACTA GACAGTTGATCAACAGCAGGACTGGAGAGCTGGGCTCCATT TTGTTCTTGGTGCCCTGCCCTCCCATGACCTGCAGAGAC ATTAGCCTGCCCAGGCTTTATGAGGTGGGAGCTGGGCTCTC CCTGATGTATTATTAGCTCCCTGGAGTTGGCCAGCTCCTGT TACACTGGCCACAGCCCTGGGCATCCGCTTCTCACTTCTAGT TTCCCTTCCAAGGTAATGTGGTGGGTCATGATCATTCTATCCT GGCTTCAGGGACCTGACTCCACTTTGGGGCATTGAGGGG TCTAGGGTAGATGATGTCCCTGTGGGGATTAATGTCTGCTG TCTGTAAACTGAGCTAGCTGAGATCCAGGAGGGCTTGCCA GAGACAGCAAGTTGTTGCCATGGTGACTTAAAGCCAGGTTG CTGCCCCAGCACAGGCTCCAGTCTACCTCACTAGAAAAAC AACACCCAGGCACCTTCCACCACTCTCAAAGGTGAAACCA AGGCTGGTCTAGAGAATGAATTATGGATCCTCGCTGTCCGTG CCACCCAGCTAGTCCAGCGGCTCAGACACTGAGGAGAGAC TGAGGTTAGCTACAAGCAAAAGACCTAGCTGGTCTCAA GCAGTGTCTCAAGTCCCTGAACCTGTGACACCTGCCCCAG GCATCATCAGGCACAGAGGGCCACCTGCAGCTCAGCCTAC TACTTGCTTCCAGGCTGTTCCTAGTTCCCATGTAGCTGCTT GTGCTTTCCAGAGACAAAACAGGAATAATAGATGTCTTAAAT ATACATTGGGCCCCAGGCGTCAATGTGGCAGCCTGAGCCT CCTTTCATCTCTGTGGAGGCAGACATAGGACCCCCAACAAA CAGCATGCAGGTTGGGAGCCAGCCACAGGACCCAGGTAAGG GGCCCTGGGTCTTAAAGCTTCTGCCACTGGCTCCGGCATTG CAGAGAGAAGAGAAGGGCGGCAGAGCTGAACCTTAGCCTT GCCTTCTGGGTACCTTCTGAGCCTCACTGTCTTCTGTGAG ATGGGCAAAGTGCAGGTGTGACTCCTTGGCAACGGTGTTAC ACCAGGGCAGGTAAGTTGTAGTTATTTGTGGGTACACCAG GACTGTTAAAGGTGTAACAT
38	Portion of SEQ ID NO: 24 that contains SEQ ID NO: 26 and SEQ ID NO: 27 fused to portion of SEQ ID NO: 25 that contains SEQ ID NO: 31 and SEQ ID NO: 32	CTGCAGCTCAGCCTACTACTGCTTTCCAGGCTGTTCCTAGT TCCCATGTGAGCTGCTTGTGCTTCCAGAGACAAAACAGGAA TAATAGATGTCTTAAATATACATTGGGCCCCAGGCGGTCAAT GTGGCAGCCTGAGCCTCCTTTCATCTCTGTGGAGGCAGAC ATAGGACCCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTAAAGCTTCTGC CACTGGCTCCGGCATTGCAGAGAGAAGAGAAGGGGCGGCA GACTGGAGAGCTGGGCTCCATTTTGTTCCTTGGTGCCCTGC CCCTCCCATGACCTGCAGAGACATTAGCCTGCCAGGCTTT ATGAGGTGGGAGCTGGGCTCTCCTGATGTATTATTAGCTC CCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCCCTG GGCATCCGCTTCTCACTTCTAGTTTCCCTCCAAGGTAATGT GGTGGGTGATGATCATCTATCTGGCTTCAGGACCTGACT CCACTTTGGGGCCATTGAGGGGCTAGGGTAGATGATGTC CCCCTGTGGGATTAATGCTCTGCTCTGTAAGCTGAGCTAG CTGAGATCCAGGAGGGCTTGGCCAGAGACAGCAAGTTGTTG

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
		CCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACAGGC CTCCCAGTCTACCCCTCACTAGAAAAACAACCCAGGCACCTT CCACCACCTCTCAAAGGTGAAACCCAGGCTGGTCTAGAGAA TGAATTATGGATCCTCGCTGTCCGTGCCACCCAGCTAGTCCC AGCGGCTCAGACACTGAGGAGAGACTGTAGGTTAGCTACA AGCAAAAGACCTAGCTGGTCTCCAAGCAGTGTCTCAAGTC CCTGAACCTGTGACACCTGCCCCAGGCATCATCAGGCACAG AGGGCCACC
39	Portion of SEQ ID NO: 24 that contains SEQ ID NO: 26 and SEQ ID NO: 27 fused to portion of SEQ ID NO: 25 that contains SEQ ID NO: 31 and SEQ ID NO: 32	CTGCAGCTCAGCCTACTACTTGTCTTCCAGGCTGTTCTAGT TCCCATGTGAGCTGCTTGTGCTTTCAGAGACAAAACAGGAA TAATAGATGTCAATAAATATACATTGGGCCCCAGGCGGTCAAT GTGGCAGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAACAGCATGCGAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGCCCCCTGGTCTTTTATGAGGT GGGAGCTGGGCTCTCCCTGATGTATTATTCAGCTCCCTGGAG TTGGCCAGCTCCTGTTACACTGGCCACAGCCCTGGGCATCC GCTGCCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACA GGCCTCCCAGTCTACCCCTCACTAGAAAAACAACCCAGGCAC TTTCCACCCTCTCAAAGGTGAAACCCAGGCTGGTCTAGA GAATGAATTATGGATCCTCGCTGTCCGTGCCACCCAGCTAGT CCCAGCGGCTCAGACACTG
40	Region 1 of the human Myo15 promoter	GTATGCCTTTTGAGATGGATGCAGCAGGTTCTGTGAGGCTGC CAGGAGGGGTAGAGTTCCCGGGGGCTCGGGCCCCGCTGG AGTGTGGAGCAGGCCCATGCTCAGCTCTCCAGGCTGTTCTG GGCTCCCCGTGTCAGCTGCTCACTCCTTCCAGAGACAAAACA GGAATAATAGACATCATTAATATACATAGGGCCCCAGGCGG TCGGCGTGGTGGGCTGGGCTCCCTTCC
41	Region 2 of human Myo15 promoter	TGCCCTGCCTTCTGAGCCGGCAGCCTGGCTCCCCACCCCAT GTATTATTCAGCTCCTGAGAGCCAGCCAGCTCCTGTACACT GACCGCAGCCAGCAGCTGCTCTGCCATTTCCCTCCTCCC TTGCCTAGGACCTAGAGGGTTCAAAGTTCTCCTCCAAGATGA CTTGGTGGGCTTTGGCCATCCACCCTAGGCCCACTTCTG GCCCAGTGCAAGGTGTGCTGGTGATTTAGGGCAGGTGGCATT CCATCTCTGTGGCTCAATGTCTTCTCTGTGAAGCCGAAGTG ACCCAAGGGCTCCCTTCATGGGTTGAGCCAGCTGTGGCCC AGGGAGGGCCTAACAGGATGAGCACTGATGTTGCCATGAC GACTCCGAGGCCAGAATGTCTCCCCCAGCACAGGCCTCATA GGCAGGCTTCCCATCCTGGTAAACAACCCACACACTTTC TACTACTGCTCTAGGGTGAACCCAGGCGCTCTAGAGGAGA TGAATTATGGATCCGCCCCCTCCGGAATCCTGGCTCGGCCCTC CCCACGCCACCCAGGGCCAGTCGGGTCTGCTCACAGCCGA GGAGGCCGCGTGTCCAGCCGCGGGCAAGAGACAGAGCAGG TCCCTGTGTCTCCAAGTCCCTGAGCCCCGTGACACCGGCCCC AGGCCCTGTAGAGAGCAGGCAGCCACC
42	Portion of SEQ ID NO: 40	CCCCTGTGCTGCTGCTCACTCCTTTCAGAGACAAAACAGGAA TAATAGACATCATTAATATACATAGGGCCCCAGG
43	Portion of SEQ ID NO: 41	TGAGCCGGCAGCCTGGCTCCCCACCCCATGTATTATTCAGCT CCTGAGAGCCAGCCAGCTCCTGTACACTGACCGCAGCCC
44	Portion of SEQ ID NO: 41	CACAGGCCTCATAGGCAGGCTTCCCCATCCTGGTAAACAACA CCACACACTTTCTACTACTGCTCTAGGGTGAAACCCAAGGC GCTCTAGAGGAGATGAATTATGGATCC
45	Portion of SEQ ID NO: 41 (SEQ ID NO: 43 fused to SEQ ID NO: 44)	TGAGCCGGCAGCCTGGCTCCCCACCCCATGTATTATTCAGCT CCTGAGAGCCAGCCAGCTCCTGTACACTGACCGCAGCCCC ACAGGCCTCATAGGCAGGCTTCCCCATCCTGGTAAACAACAC CCACACACTTTCTACTACTGCTCTAGGGTGAAACCCAAGGCG CTCTAGAGGAGATGAATTATGGATCC
46	Portion of SEQ ID NO: 41 (SEQ ID NO: 44 fused to SEQ ID NO: 43)	CACAGGCCTCATAGGCAGGCTTCCCCATCCTGGTAAACAACA CCACACACTTTCTACTACTGCTCTAGGGTGAAACCCAAGGC GCTCTAGAGGAGATGAATTATGGATCCTGAGCCGCGCAGCT GGTCCCCACCCCATGTATTATTCAGCTCCTGAGAGCCAGCC AGCTCCTGTACACTGACCGCAGCCC
47	Portion of SEQ ID NO: 41 (contiguous sequence including SEQ ID NO: 43 and SEQ ID NO: 44)	TGAGCCGGCAGCCTGGCTCCCCACCCCATGTATTATTCAGCT CCTGAGAGCCAGCCAGCTCCTGTACACTGACCGCAGCCCC GCACCTGCTCTGCCATTCCCCCTCCTCCCTTGCTTAGGACCT AGAGGGTTCAAAGTTCTCCTCCAAGATGACTTGGTGGGCTTT GGCCATCCACCCTAGGCCCACTTCTGCCCCAGTGCAAGT

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
		GTGCTGGTGATTAGGGCAGGTGGCATTCCATCTCTGTGGCT CAATGTCTTCTCTGTGAAGCCGAAGTGACCCAAGGGCTCCC TTCATGGGGTTGAGCCAGCTGTGGCCCAGGGAGGGCTAAC CAGGATGAGCACTGATGTTGCCATGACGACTCCGAGGCCAG AATGTCTCCCCAGCACAGGCCTCATAGGCAGGCTTCCCCAT CCTGGTAAACAACACCCACACACTTTCTACTACTGCTCTAGG GTGAAACCAAGCGCTCTAGAGGAGATGAATTATGGATCC
48	Polynucleotide containing SEQ ID NO: 40 and SEQ ID NO: 41	GTATGCCTTTTGAGATGGATGCAGCAGGTTCTGTGAGGCTGC CAGGAGGGGTAGAGTTCCCGGGGGCTCGGGCCCCGCTGG AGTGTGGAGCAGGCCCATGCTCAGCTCTCCAGGCTGTTTCT GGCTCCCCGTGTCAGCTGCTCACTCTTTCAGAGACAAAACA GGAATAATAGACATCATTAAATATACATAGGGCCCCAGGCGG TCGGCGTGGTGGGCTGGGCTCCCTTCCCCATAACACTGAG CTGCTCTGTGGGCCAACCGTGCTCTGGGCCAGCCAGAGG ACCCCATGAGGCGGCATGCAGGCGGGGAGCAGGCCACAG AACGCAGGTAAGGAGACCTTAGCCTAGAGTCTTGGGGTCT GTCACTGGCCACCCTCGCATCCAGGCTGCAGGAACTGAG GCCCAGAGAGGACAAGGACTTTCCTGGACCCACACAGCCAG TCAGTGACAGAGCCTAGGGTCTGAGCCAGGCCTGACCCAAC CTCCATTTCTGCCTCTCTACCCCTGCCCGCCCCAACACAC ACACACACACAAGTGAGATTCCACTGAAACGCCCTTCTTGC CCTGCCTTCTGAGCCGGCAGCCTGGCTCCCCACCCATGTA TTATTGAGCTCCTGAGAGCCAGCCAGCTCCTGTTACACTGAC CGCAGCCCAGCACCTGCTCTGCCATTCCCCCTCTCCCTTGC CTAGGACCTAGAGGGTTCAAAGTTCTCCTCCAAGATGACTTG GTGGGCTTTGGCCATCCACCCTAGGCCCACTTCTGGCCC AGTGCAGGTGTGCTGGTGATTAGGGCAGGTGGCATTCAT CTCTGTGGCTCAATGTCTTCTCTGTGAAGCCGAAGTGACCC AAGGGCTCCCTTCATGGGGTTGAGCCAGCTGTGGCCAGGG AGGGCCTAACAGGATGAGCACTGATGTTGCCATGACGACT CCGAGGCCAGAAATGTCTCCCCAGCACAGGCCTCATAGGCA GGCTTCCCCATCTGGTAAACAACACCCACACACTTTCTACT ACTGCTCTAGGGTGAACCAAGGCGCTCTAGAGGAGATGA ATTATGGATCCGCCCTCCCGGAATCCTGGCTCGGCCCTCCC CACGCCACCCAGGGCCAGTCGGGTCTGCTCACAGCCCGAG GAGGCCGCGTGTCCAGCCGCGGGCAGAGACAGAGCAGGT CCCTGTGTCTCAAGTCCCTGAGCCCGTGACACCGGCCCA GGCCCTGTAGAGAGCAGGCAGCCACC
49	Polynucleotide containing SEQ ID NO: 42, SEQ ID NO: 43, and SEQ ID NO: 44	GCAGGCCCATGCTCAGCTCTCCAGGCTGTTCTGTGGCTCCCC TGTCAGCTGCTCACTCTTCCAGAGACAAAACAGGAATAAT AGACATCATTAAATATACATAGGGCCCCAGGCGGTCTGGCGTG GTGGGCTGGGCTCCCTTCCCCATAACACTGAGCTGCTCTG CTGGGCCAACCGTGCTCTGGGCCAGCCAGAGGACCCCCAT GAGGCGGCATGCAGGCGGGGAGCAGGCCACAGAACGCAGG TAAGGAGACCTTGCTTCTGAGCCGGCAGCCTGGCTCCCCA CCCCATGTATTATTAGCTCCTGAGAGCCAGCCAGCTCCTGT TACACTGACCCGAGCCAGCCTGCTCTGCCATTCCCCCTC CTCCCTTGCTAGGACCTAGAGGGTTCAAAGTTCTCTCCAA GATGACTTGGTGGGCTTTGGCCATCGGGCCTAACAGGATG AGCACTGATGTTGCCATGACGACTCCGAGGCCAGAATGTCTC CCCCAGCACAGGCCTCATAGGCAGGCTTCCCCATCCTGGTA AACAACACCCACACACTTTCTACTACTGCTCTAGGGTGAAC CCAAGGCGCTCTAGAGGAGATGAATTATGGATCCGCCCTCC CGGAATCCTGGCTCGGCCCTCCCCAGC
50	Portion of SEQ ID NO: 24 that contains SEQ ID NO: 26 and SEQ ID NO: 27	CTGCAGCTCAGCCTACTACTTGCTTTCCAGGCTGTTCTAGT TCCCATGTGCTGCTTGTGCTTTCAGAGACAAAACAGGAA TAATAGATGTCATTAAATATACATTGGGCCCCAGGCGGTCAAT GTGGCAGCTGAGCCTCCTTTCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGCCCTGGGCTCT
51	Portion of SEQ ID NO: 25 that contains SEQ ID NO: 31	TTTATGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTCAG CTCCCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCC CTGGGCATCCGC
52	Portion of SEQ ID NO: 25 that contains SEQ ID NO: 32	TGCCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACAG GCCTCCCACTTACCCTCACTAGAAAACAACCCAGGCACT TTCACCACTCTCAAAGGTGAACCCAAGGCTGGTCTAGAG AATGAATTATGGATCCTCGCTGCTGCCACCCAGCTAGTC CCAGCGGCTCAGACACTG

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
53	SEQ ID NO: 50 fused to SEQ ID NO: 51	CTGCAGCTCAGCCTACTACTTGCTTTCCAGGCTGTTCCCTAGT TCCCATGTGCTGCTGTGCTTTCCAGAGACAAAACAGGAA TAATAGATGTCATTAAATATACATTGGGCCCCAGGCGTCAAT GTGGCAGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTTTATGAGGT GGGAGCTGGGCTCTCCCTGATGTATTATTCAGCTCCCTGGAG TTGGCCAGCTCCTGTTACACTGGCCACAGCCCTGGGCATCC GC
54	SEQ ID NO: 50 fused to SEQ ID NO: 52	CTGCAGCTCAGCCTACTACTTGCTTTCCAGGCTGTTCCCTAGT TCCCATGTGCTGCTGTGCTTTCCAGAGACAAAACAGGAA TAATAGATGTCATTAAATATACATTGGGCCCCAGGCGTCAAT GTGGCAGCCTGAGCCTCCTTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTTGCCATGGTG ACTTTAAAGCCAGGTTGCTGCCCCAGCACAGGCTCCAGTC TACCCTCACTAGAAAACAACCCAGGCACCTTCCACCACCT CTCAAAGGTGAACCCAGGCTGGTCTAGAGAATGAATTATG GATCCTCGCTGTCCGTGCCACCCAGCTAGTCCAGCGGCTC AGACACTG
55	SEQ ID NO: 51 fused to SEQ ID NO: 52	TTTATGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTCAG CTCCCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCC CTGGGCATCCGCTGCCATGGTGACTTTAAAGCCAGGTTGCTG CCCCAGCACAGGCTCCAGTCCTACCCTCACTAGAAAACAAC ACCCAGGCACCTTCCACCACCTCTCAAAGGTGAAACCCAGG CTGGTCTAGAGAATGAATTATGGATCCTCGCTGTCCGTGCCA CCCAGCTAGTCCAGCGGCTCAGACACTG
56	SEQ ID NO: 51 fused to SEQ ID NO: 50, which is fused to SEQ ID NO: 52	TTTATGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTCAG CTCCCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCC CTGGGCATCCGCTGCCATGGTGACTTTAAAGCCAGGTTGCTG GCTGTTCCCTAGTTCCCATGTCAGCTGCTTGTGCTTTCCAGAG ACAAAACAGGAATAATAGATGTCATTAAATATACATTGGGCCC CAGGCGGTCAATGTGGCAGCTGAGCCTCCTTCCATCTCTG TGGAGGCAGACATAGGACCCCCAACAAACAGCATGCAGGTT GGGAGCCAGCCACAGGACCCAGGTAAGGGGCCCTGGGTCC TTTGCCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACA GGCCTCCAGTCCTACCCTCACTAGAAAACAACCCAGGCAC TTTCCACCCTCTCAAAGGTGAAACCCAGGCTGGTCTAGA GAATGAATTATGGATCCTCGCTGTCCGTGCCACCCAGCTAGT CCCAGCGGCTCAGACACTG
57	SEQ ID NO: 52 fused to SEQ ID NO: 50, which is fused to SEQ ID NO: 51	TGCCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACAG GCCTCCAGTCTACCCTCACTAGAAAACAACCCAGGCACCT TTCCACCCTCTCAAAGGTGAAACCCAGGCTGGTCTAGAG AATGAATTATGGATCCTCGCTGTCCGTGCCACCCAGCTAGTC CCAGCGGCTCAGACACTGCTGCAGCTCAGCCTACTACTTGCT TTCCAGGCTGTTCTTAGTTCCCATGTCAGCTGCTTGTGCTTTC CAGAGACAAAACAGGAATAATAGATGTCATTAAATATACATTG GGCCCCAGGCGGTCAATGTGGCAGCCTGAGCCTCCTTTCCA TCTCTGTGGAGGCAGACATAGGACCCCCAACAAACAGCATG CAGGTTGGGAGCCAGCCACAGGACCCAGGTAAGGGGCCCT GGGTCTTTTATGAGGTGGGAGCTGGGCTCTCCCTGATGTA TTATTAGCTCCTGGAGTTGGCCAGCTCCTGTTACACTGGC CAGACCCCTGGGCATCCGC
58	SEQ ID NO: 51 fused to SEQ ID NO: 52, which is fused to SEQ ID NO: 50	TTTATGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTCAG CTCCCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCC CTGGGCATCCGCTGCCATGGTGACTTTAAAGCCAGGTTGCTG CCCCAGCACAGGCTCCAGTCCTACCCTCACTAGAAAACAAC ACCCAGGCACCTTCCACCACCTCTCAAAGGTGAAACCCAGG CTGGTCTAGAGAATGAATTATGGATCCTCGCTGTCCGTGCCA CCCAGCTAGTCCAGCGGCTCAGACACTGCTGCAGTCAAGC CTACTACTTGCTTTCCAGGCTGTTCTAGTTCCTATGTCAGCT GCTTGTGCTTTCCAGAGACAAAACAGGAATAATAGATGTCATT AAATATACATTGGGCCCCAGGCGTCAATGTGGCAGCCTGA GCCTCCTTTCCATCTCTGTGGAGGCAGACATAGGACCCCCAA CAAACAGCATGCAGGTTGGGAGCCAGCCACAGGACCCAGGT AAGGGCCCTGGGTCTT
59	Portion of SEQ ID NO: 24 that contains SEQ ID NO: 26 and SEQ ID	TGCAGCTCAGCCTACTACTTGCTTTCCAGGCTGTTCCCTAGTT CCCATGTGCTGCTGTGCTTTCCAGAGACAAAACAGGAAT AATAGATGTCATTAAATATACATTGGGCCCCAGGCGTCAAT

TABLE 3-continued

Exemplary nucleotide sequences for use in the Myo15 promoter described herein

SEQ ID NO.	Description of nucleic acid sequence	Nucleic Acid Sequence
	NO: 27 fused to portion of SEQ ID NO: 25 that contains SEQ ID NO: 31 and SEQ ID NO: 32	GTGGCAGCCTGAGCCTCCTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTAAGCTTCTGC CACTGGCTCCGGCATTGCAGAGAGAAGAGAGGGGCGGCA GACTGGAGAGCTGGGCTCCATTTTGTTCCTTGGTCCCTGC CCCTCCCCATGACCTGCAGAGACATTACGCTGCCAGGCTTT ATGAGGTGGGAGCTGGGCTCTCCCTGATGTATTATTACGCTC CCTGGAGTTGGCCAGCTCCTGTTACACTGGCCACAGCCCTG GGCATCCGCTTCTCACTTCTAGTTTCCCTCCAAGGTAATGT GGTGGGTCAATGATCATTCTATCCTGGCTTCAGGACCTGACT CCACTTTGGGGCCATTCGAGGGGTCTAGGGTAGATGATGC CCCCTGTGGGGATTAATGTCTCTGTAAAACCTGAGCTAG CTGAGATCCAGGAGGGCTTGGCCAGAGACAGCAAGTTGTTG CCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACAGGC CTCCAGTCTACCTCACTAGAAAACAACCCAGGCACCTTT CCACCACCTCTCAAAGGTGAAACCAAGGCTGGTCTAGAGAA TGAATTATGGATCCTCGCTGTCGTCGCCACCCAGCTAGTCCC AGCGGCTCAGACACTGAGGAGAGACTGTAGGTTAGCTACA AGCAAAAAGACCTAGCTGGTCTCCAAGCAGTGTCTCCAAGTC CCTGAACCTGTGACACCTGCCCCAGGCATCATCAGGCACAG AGGGCCACC
60	Portion of SEQ ID NO: 24 that contains SEQ ID NO: 26 and SEQ ID NO: 27 fused to portion of SEQ ID NO: 25 that contains SEQ ID NO: 31 and SEQ ID NO: 32	TGCAGCTCAGCCTACTACTTGCTTTCAGGCTGTTCTTAGTT CCCATGTGAGCTGCTTGTGCTTCCAGAGACAAAACAGGAAT AATAGATGTCATTAAATATACATTTGGGCCCCAGGCGGTCAAT GTGGCAGCCTGAGCCTCCTTCCATCTCTGTGGAGGCAGAC ATAGGACCCCAACAAACAGCATGCAGGTTGGGAGCCAGCC ACAGGACCCAGGTAAGGGGCCCTGGGTCTTTTATGAGGT GGGAGCTGGGCTCTCCCTGATGTATTATTACGCTCCCTGGAG TTGGCCAGCTCCTGTTACACTGGCCACAGCCCTGGGCATCC GCTGCCATGGTGACTTTAAAGCCAGGTTGCTGCCCCAGCACA GGCCTCCAGTCTACCTCACTAGAAAACAACCCAGGCAC TTTCCACCCTCTCAAAGGTGAAACCAAGGCTGGTCTAGA GAATGAATTATGGATCCTCGCTGTCGTCGCCACCCAGCTAGT CCCAGGGCTCAGACACTG

Additional Myo15 promoters useful in conjunction with the compositions and methods described herein include nucleic acid molecules that have at least 85% sequence identity (e.g., 85%, 90%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity) to the nucleic acid sequences set forth in Table 3, as well as functional portions or derivatives of the nucleic acid sequences set forth in Table 3. The Myo15 promoters listed in Table 3 are characterized in U.S. Provisional Application Nos. 62/663,679, 62/802,874, 62/928,311, and 62/965,773 and International Application No. PCT/US2019/029336, which are incorporated herein by reference.

Once a polynucleotide encoding OTOF has been incorporated into the nuclear DNA of a mammalian cell or stabilized in an episomal monomer or concatemer, the transcription of this polynucleotide can be induced by methods known in the art. For example, expression can be induced by exposing the mammalian cell to an external chemical reagent, such as an agent that modulates the binding of a transcription factor and/or RNA polymerase to the mammalian promoter and thus regulates gene expression. The chemical reagent can serve to facilitate the binding of RNA polymerase and/or transcription factors to the mammalian promoter, e.g., by removing a repressor protein that has bound the promoter. Alternatively, the chemical reagent can serve to enhance the affinity of the mammalian promoter for RNA polymerase and/or transcription factors such that the rate of transcription of the gene located downstream of the promoter is increased in the presence of the chemical reagent. Examples of chemical reagents that

potentiate polynucleotide transcription by the above mechanisms include tetracycline and doxycycline. These reagents are commercially available (Life Technologies, Carlsbad, Calif.) and can be administered to a mammalian cell in order to promote gene expression according to established protocols.

Other DNA sequence elements that may be included in the nucleic acid vectors for use in the compositions and methods described herein include enhancer sequences. Enhancers represent another class of regulatory elements that induce a conformational change in the polynucleotide containing the gene of interest such that the DNA adopts a three-dimensional orientation that is favorable for binding of transcription factors and RNA polymerase at the transcription initiation site. Thus, polynucleotides for use in the compositions and methods described herein include those that encode an OTOF protein and additionally include a mammalian enhancer sequence. Many enhancer sequences are now known from mammalian genes, and examples include enhancers from the genes that encode mammalian globin, elastase, albumin, α -fetoprotein, and insulin. Enhancers for use in the compositions and methods described herein also include those that are derived from the genetic material of a virus capable of infecting a eukaryotic cell. Examples include the SV40 enhancer on the late side of the replication origin (bp 100-270), the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, and adenovirus enhancers. Additional enhancer sequences that induce activation of eukaryotic gene transcription are disclosed in Yaniv, et al.,

101

Nature 297:17 (1982). An enhancer may be spliced into a vector containing a polynucleotide encoding an OTOF protein, for example, at a position 5' or 3' to this gene. In a preferred orientation, the enhancer is positioned at the 5' side of the promoter, which in turn is located 5' relative to the polynucleotide encoding an OTOF protein.

The nucleic acid vectors described herein may include a Woodchuck Posttranscriptional Regulatory Element (WPRE). The WPRE acts at the mRNA level, by promoting nuclear export of transcripts and/or by increasing the efficiency of polyadenylation of the nascent transcript, thus increasing the total amount of mRNA in the cell. The addition of the WPRE to a vector can result in a substantial improvement in the level of transgene expression from several different promoters, both in vitro and in vivo. The WPRE can be located in the second nucleic acid vector between the polynucleotide encoding a C-terminal portion of an OTOF protein and the poly(A) sequence. In the compositions and methods described herein, the WPRE can have the sequence:

(SEQ ID NO: 23)
 GATCCAATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTC
 TTAACATATGTTGCTCCTTTTACGCTATGTGGATACGCTGCTTTAATGCCTT
 TGTATCATGCTATTGCTTCCCGTATGGCTTTTCTCTCTCTGTATA
 AATCCTGGTTGCTGTCTCTTTATGAGGAGTTGTGCCCGTTGTTCAGGCAAC
 GTGGCGTGGTGTGCACTGTGTTTGTGACGCAACCCCACTGGTTGGGGCA
 TTGCCACCACCTGTGAGCTCCTTCCGGGACTTTCGCTTTCCTCTCCCTA
 TTGCCACGGCGGAATCATCGCCGCTGCTTGTCCCGCTGTGACAGGGG
 CTCGGCTGTTGGGCACTGACAATTCCTGGTGTGTCGGGGAATCATCGT
 CCTTCTCTGGCTGTGCTGCTGTGTTGCCACCTGGATTCTGCGCGGACGT
 CCTTCTGCTACGTCCCTTCGGCCCTCAATCCAGCGGACCTTCTTCCCGCG
 GCCTGCTCGCGCTCTGCGGCCTCTCCGCTCTTCGA.

The WPRE can also have the sequence:

(SEQ ID NO: 61)
 AATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTCTTAAC
 TATGTTGCTCCTTTTACGCTATGTGGATACGCTGCTTTAATGCCTTTGTAT
 CATGCTATTGCTTCCCGTATGGCTTTTCTCTCTCTGTATAAATCC
 TGGTTAGTTCTTGCCACGGCGGAATCATCGCCGCTGCTTGCCTGCTGC
 TGACAGGGGCTCGGCTGTTGGGCACTGACAATTCCTGGTGTATTATTGT
 GAAATTTGTGATGCTATTGCTTTATTGTAAACCATCTAGCTTTATTGTGA
 AATTTGTGATGCTATTGCTTTATTGTAAACCATCTAGCTTTATTGTGA
 AGTTAACACAACAATTGCATTCATTTTATGTTTTCAGGTTTCAGGGGAGAT
 GTGGGAGGTTTTTTAA.

In some embodiments, the nucleic acid vectors for use in the compositions and methods described herein include a reporter sequence, which can be useful in verifying OTOF gene expression, for example, in specific cells and tissues (e.g., in cochlear hair cells). Reporter sequences that may be provided in a transgene include DNA sequences encoding β -lactamase, β -galactosidase (LacZ), alkaline phosphatase, thymidine kinase, green fluorescent protein (GFP), chloram-

102

phenicol acetyltransferase (CAT), luciferase, and others well known in the art. When associated with regulatory elements which drive their expression, the reporter sequences provide signals detectable by conventional means, including enzymatic, radiographic, colorimetric, fluorescence or other spectrographic assays, fluorescent activating cell sorting assays and immunological assays, including enzyme linked immunosorbent assay (ELISA), radioimmunoassay (RIA), and immunohistochemistry. For example, where the marker sequence is the LacZ gene, the presence of the vector carrying the signal is detected by assays for β -galactosidase activity. Where the transgene is green fluorescent protein or luciferase, the vector carrying the signal may be measured visually by color or light production in a luminometer.

15 Overlapping Dual Vectors

One approach for expressing large proteins in mammalian cells involves the use of overlapping dual vectors. This approach is based on the use of two nucleic acid vectors, each of which contains a portion of a polynucleotide that encodes a protein of interest and has a defined region of sequence overlap with the other polynucleotide. Homologous recombination can occur at the region of overlap and lead to the formation of a single nucleic acid molecule that encodes the full-length protein of interest.

25 Overlapping dual vectors for use in the methods and compositions described herein contain at least one kilobase (kb) of overlapping sequence (e.g., 1 kb, 1.5 kb, 2 kb, 2.5 kb, 3 kb or more of overlapping sequence). The nucleic acid vectors are designed such that the overlapping region is centered at an OTOF exon boundary, with an equal amount of overlap on either side of the boundary. The boundaries are chosen based on the size of the promoter and the locations of the portions of the polynucleotide that encode OTOF C2 domains. Overlapping regions are centered on exon boundaries that occur outside of the portion of the polynucleotide that encodes the C2C domain (e.g., after the portion of the polynucleotide that encodes the C2C domain). Exon boundaries within the portion of the polynucleotide that encodes the C2D domain can be selected as the center of the overlapping region, or exon boundaries located after the portion of the polynucleotide that encodes the C2D domain and before the portion of the polynucleotide that encodes the C2E domain can serve as the center of an overlapping region. The nucleic acid vectors for use in the methods and compositions described herein are also designed such that approximately half of the OTOF gene is contained within each vector (e.g., each vector contains a polynucleotide that encodes approximately half of the OTOF protein).

One exemplary overlapping dual vector system includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-28 and the 500 kb immediately 3' of the exon 28/29 boundary of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5, or mouse OTOF, e.g., SEQ ID NO: 6); and a second nucleic acid vector containing the 500 kb immediately 5' of the exon 28/29 boundary and exons 29-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5, or mouse OTOF, e.g., SEQ ID NO: 6) and a poly(A) sequence (e.g., a bovine growth hormone (bGH) poly(A) signal sequence). In this overlapping dual vector system, the overlapping sequence is centered at the exon 28/29 boundary, which is after the portion of the polynucleotide that encodes the C2D domain. Another exemplary overlapping dual vector system includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-24 and the 500 kb immediately 3' of the exon 24/25 boundary of a polynucleotide

encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5, or mouse OTOF, e.g., SEQ ID NO: 6); and a second nucleic acid vector containing the 500 kb immediately 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5, or mouse OTOF, e.g., SEQ ID NO: 6) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). In this overlapping dual vector system, the overlapping sequence is centered at the exon 24/25 boundary, which is within the portion of the polynucleotide that encodes the C2D domain. The two exon boundaries described above can be used with any promoter that is a similar size to the CAG promoter (e.g., the CMV promoter or smCBA promoter), such as promoters that are 1 kb or shorter (e.g., approximately 1 kb, 950 bp, 900 bp, 850 bp, 800 bp, 750 bp, 700 bp, 650 bp, 600 bp, 550 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp or shorter). For example, in either of the foregoing dual vector systems, the CMV promoter or the smCBA promoter, can be used in the place of the CAG promoter. A Myo15 promoter having a sequence that is 1 kb or shorter (e.g., a Myo15 promoter described hereinabove, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in place of the CAG promoter. Alternatively, a different exon boundary can be chosen that is within or after the portion of the polynucleotide that encodes the C2D domain and before the portion of the polynucleotide that encodes the C2E domain. The nucleic acid vectors containing promoters of this size can optionally contain OTOF UTRs. For example, in the foregoing overlapping dual vector system in which the overlapping region is centered at the exon 28/29 boundary of OTOF, the second nucleic acid vector can contain the full length OTOF 3' UTR (e.g., the 1035 bp human OTOF 3' UTR in dual vector systems encoding human OTOF, or the 1001 bp mouse OTOF 3' UTR in dual vector systems encoding mouse OTOF). In the foregoing overlapping dual vector system in which the overlapping region is centered at the exon 24/25 boundary of OTOF, neither the first nor the second nucleic acid vector contains an OTOF UTR.

In some embodiments, the first nucleic acid vector in the overlapping dual vector system contains a long promoter (e.g., a promoter that is longer than 1 kb, e.g., 1.1 kb, 1.25 kb, 1.5 kb, 1.75 kb, 2 kb, 2.5 kb, 3 kb or longer). In such overlapping dual vector systems, the overlapping region is centered at an exon boundary that is located after the portion of the polynucleotide that encodes the C2C domain and before the portion of the polynucleotide that encodes the C2D domain. For example, an overlapping dual vector system for use in the methods and compositions described herein includes a first nucleic acid vector containing a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked exons 1-21 and the 500 kb immediately 3' of the exon 21/22 boundary of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5); and a second nucleic acid vector containing the 500 kb immediately 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). In this overlapping dual vector system, neither the first nor the second nucleic acid vector includes an OTOF UTR. A short promoter (e.g., a CMV promoter, CAG promoter, smCBA promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in this dual vector system (e.g., a dual vector system in which the overlapping region

is centered at the exon 21/22 boundary). If a short promoter is used, additional elements, such as a 5' OTOF UTR, can be included in the first vector (e.g., the vector containing exons 1-21 and the 500 kb immediately 3' of the exon 21/22 boundary of a polynucleotide encoding an OTOF protein). Trans-Splicing Dual Vectors

A second approach for expressing large proteins in mammalian cells involves the use of trans-splicing dual vectors. In this approach, two nucleic acid vectors are used that contain distinct nucleic acid sequences, and the polynucleotide encoding the N-terminal portion of the protein of interest and the polynucleotide encoding the C-terminal portion of the protein of interest do not overlap. Instead, the first nucleic acid vector includes a splice donor sequence 3' of the polynucleotide encoding the N-terminal portion of the protein of interest, and the second nucleic acid vector includes a splice acceptor sequence 5' of the polynucleotide encoding the C-terminal portion of the protein of interest. When the first and second nucleic acids are present in the same cell, their ITRs can concatemerize, forming a single nucleic acid structure in which the concatemerized ITRs are positioned between the splice donor and splice acceptor. Trans-splicing then occurs during transcription, producing a nucleic acid molecule in which the polynucleotides encoding the N-terminal and C-terminal portions of the protein of interest are contiguous, thereby forming the full-length coding sequence.

Trans-splicing dual vectors for use in the methods and compositions described herein are designed such that approximately half of the OTOF gene is contained within each vector (e.g., each vector contains a polynucleotide that encodes approximately half of the OTOF protein). The determination of how to split the polynucleotide sequence between the two nucleic acid vectors is made based on the size of the promoter and the locations of the portions of the polynucleotide that encode the OTOF C2 domains. When a short promoter is used in the trans-splicing dual vector system (e.g., a promoter that is 1 kb or shorter, e.g., approximately 1 kb, 950 bp, 900 bp, 850 bp, 800 bp, 750 bp, 700 bp, 650 bp, 600 bp, 550 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp or shorter), such as a CAG promoter, a CMV promoter, a smCBA promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter (e.g., a Myo15 promoter described hereinabove, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) the OTOF polynucleotide sequence is divided between the two nucleic acid vectors at an exon boundary that occurs after the portion of the polynucleotide that encodes the C2D domain and before the portion of the polynucleotide that encodes the C2E domain, for example, the exon 26/27 boundary. The nucleic acid vectors containing promoters of this size can optionally contain OTOF UTRs (e.g., both the 5' and 3' OTOF UTRs, e.g., full-length UTRs). When a long promoter is used in the trans-splicing dual vector system (e.g., a promoter that is longer than 1 kb, e.g., 1.1 kb, 1.25 kb, 1.5 kb, 1.75 kb, 2 kb, 2.5 kb, 3 kb or longer), such as a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36), the OTOF polynucleotide sequence will be divided between the two nucleic acid vectors at an exon boundary that occurs after the portion of the polynucleotide that encodes the C2C domain, and either before the portion of the polynucleotide that encodes the C2D domain, such as the exon 19/20 boundary, or within the portion of the polynucleotide that encodes the C2D domain, such as the exon 25/26 boundary. A short promoter (e.g., a CMV promoter, smCBA promoter, CAG promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the

sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the dual vector systems designed for large promoters, in which case additional elements (e.g., OTOF UTR sequences) may be included in the first vector (e.g., the vector containing the portion of the polynucleotide that encodes the C2C domain).

One exemplary trans-splicing dual vector system that uses a short promoter includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-26 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a splice donor sequence 3' of the polynucleotide sequence; and a second nucleic acid vector containing a splice acceptor sequence 5' of exons 27-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). An alternative trans-splicing dual vector system includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-28 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a splice donor sequence 3' of the polynucleotide sequence; and a second nucleic acid vector containing a splice acceptor sequence 5' of exons 29-48 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). The CMV promoter, smCBA promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter (e.g., a Myo15 promoter described hereinabove, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can be used in place of the CAG promoter either of the foregoing dual vector systems. These nucleic acid vectors can also contain full length 5' and 3' OTOF UTRs in the first and second nucleic acid vectors, respectively (e.g., the first nucleic acid vector can contain the 5' human OTOF UTR (127 bp) in dual vector systems encoding human OTOF, or the 5' mouse UTR (134 bp) in dual vector systems encoding mouse OTOF; and the second nucleic acid vector can contain the 3' human OTOF UTR (1035 bp) in dual vector systems encoding human OTOF, or the 3' mouse OTOF UTR (1001 bp) in dual vector systems encoding mouse OTOF).

An exemplary trans-splicing dual vector system that uses a long promoter includes a first nucleic acid vector containing a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a splice donor sequence 3' of the polynucleotide sequence; and a second nucleic acid vector containing a splice acceptor sequence 5' of exons 20-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). Alternatively, the trans-splicing dual vector system can include a first nucleic acid vector containing a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-20 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a splice donor sequence 3' of the polynucleotide sequence; and a second nucleic acid vector containing a splice acceptor sequence 5' of exons 21-48 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). Neither the first nor the second nucleic acid vector in either of the foregoing Myo15 promoter trans-splicing dual vector systems contains an OTOF UTR. A short promoter (e.g., a CMV promoter, smCBA promoter, CAG promoter, or a Myo15 promoter having a sequence that

is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the foregoing dual vector systems designed for large promoters. If these dual vector systems contain a short promoter, they may also include a 5' OTOF UTR in the first vector.

To accommodate an OTOF UTR, the OTOF coding sequence can be divided in a different position. For example, in a trans-splicing dual vector system in which the first nucleic acid vector contains a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-25 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a splice donor sequence 3' of the polynucleotide sequence; and the second nucleic acid vector contains a splice acceptor sequence 5' of exons 26-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence), the second nucleic acid can also contain a full length OTOF 3' UTR (e.g., the 1035 bp human OTOF 3' UTR). For mouse OTOF, the trans-splicing dual vector system can contain a 3' UTR if the first nucleic acid vector contains a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-24 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a splice donor sequence 3' of the polynucleotide sequence; and the second nucleic acid vector contains a splice acceptor sequence 5' of exons 25-48 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6) and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). In this dual vector system, the second nucleic acid can also contain a full length OTOF 3' UTR (e.g., the 1001 bp mouse OTOF 3' UTR). A short promoter (e.g., a CMV promoter, smCBA promoter, CAG promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the foregoing dual vector systems designed for large promoters. If these dual vector systems contain a short promoter, they may also include a 5' OTOF UTR in the first vector.

Dual Hybrid Vectors

A third approach for expressing large proteins in mammalian cells involves the use of dual hybrid vectors. This approach combines elements of the overlapping dual vector strategy and the trans-splicing strategy in that it features both an overlapping region at which homologous recombination can occur and splice donor and splice acceptor sequences. In dual hybrid vector systems, the overlapping region is a recombinogenic region that is contained in both the first and second nucleic acid vectors, rather than a portion of the polynucleotide sequence encoding the protein of interest—the polynucleotide encoding the N-terminal portion of the protein of interest and the polynucleotide encoding the C-terminal portion of the protein of interest do not overlap in this approach. The recombinogenic region is 3' of the splice donor sequence in the first nucleic acid vector and 5' of the splice acceptor sequence in the second nucleic acid sequence. The first and second nucleic acid sequences can then join to form a single sequence based on one of two mechanisms: 1) recombination at the overlapping region, or 2) concatemerization of the ITRs. The remaining recombinogenic region(s) and/or the concatemerized ITRs can be removed by splicing, leading to the formation of a contiguous polynucleotide sequence that encodes the full-length protein of interest.

107

Recombinogenic regions that can be used in the compositions and methods described herein include the F1 phage AK gene having a sequence of: GGGATTTTGCCGAT-TTCGGCCTATTGGTTAA AAAATGAGCTGATT-TAACAAAAAATTTAACGCGAATTTTAACAAAAT (SEQ ID NO: 19) and alkaline phosphatase (AP) gene fragments as described in U.S. Pat. No. 8,236,557, which are incorporated herein by reference. In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 62)
 CCCCCGGTGCGCGCGTCGGTGGTGCCGCGGGGGCGCCAGGTCGCAGGC
 GGTGTAGGGCTCCAGGCAGGCGGCGAAGGCCATGACGTGCGCTATGAAGGT
 CTGCTCCTGCACGCGTGAACAGGTGCGCTGCGGGCCGCGCGCAACAC
 CGCCACGTCCTCGCCTGCGTGGGTCTCTTCGTCCAGGGGCACTGCTGACTG
 CTGCCGATACTCGGGGCTCCCGCTCTCGCTCTCGGTAACATCCGGCCGGGC
 GCCGTCTTGAGCACATAGCCTGGACCGTTTCCGTATAGGAGGACCGTGTA
 GGCCCTTCCTGTCCCGGGCCTTGCCAGCGGCCAGCCGATGAAGGAGCTCCC
 TCGCAGGGGGTAGCCTCCGAAGGAGAAGACGTGGGAGTGGTCGGCAGTGAC
 GAGGCTCAGCGTGTCTCTCGCTGGTGGTGGCCCGCCCTCTCAATGGC
 GTCGTCGAACATGATCGTCTCAGTCAGTGCCCGTAAGCCCTGCTTTTCATG
 ATGACCATGGTCGATGCGACCACTCCACGAAGAGGAAGAAGCCGCGGGG
 GTGTCTGCTCAGCAGGCGCAGGGCAGCCTCTGTCATCTCCATCAGGGAGGG
 GTCCAGTGTGGAGTCTCGTGGATCTCGTATTTTCATGTCTCCAGGCTCAAA
 GAGACCATGAGATGGGTACAGACGGGTCCAGGAAGCCTGCATGAGCTC
 AGTGCGGTTCACACGTACCGGGCACCCCTGGCGTTCGCGGAGCCATTCTG
 CACCAGATTCTTCCCGTCCAGCCTGGTCCCACCTTGGCTGTAGTCATCTGG
 GTACTCAGGGTCTGGGGTCCCATGCGAATCATGTACTTTTCGGCTCCA.
 In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 63)
 CCCCCGGTGCGCGCGTCGGTGGTGCCGCGGGGGCGCCAGGTCGCAGGC
 GGTGTAGGGCTCCAGGCAGGCGGCGAAGGCCATGACGTGCGCTATGAAGGT
 CTGCTCCTGCACGCGTGAACAGGTGCGCTGCGGGCCGCGCGCAACAC
 CGCCACGTCCTCGCCTGCGTGGGTCTCTTCGTCCAGGGGCACTGCTGACTG
 CTGCCGATACTCGGGGCTCCCGCTCTCGCTCTCGGTAACATCCGGCCGGGC
 GCCGTCTTGAGCACATAGCCTGGACCGTTTCCGTATAGGAGGACCGTGTA
 GGCCCTTCCTGTCCCGGGCCTTGCCAGCGGCCAGCCGATGAAGGAGCTCCC
 TCGCAGGGGGTAGCCTCCGAAGGAGAAGACGTGGGAGTGGTCGGCAGTGAC
 GAGGCTCAGCGTGTCTCTCG CTGGTGA.
 In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 64)
 GCTGGCCCCCTCTCAATGGCGTCGTCGAACATGATCGTCTCAGTCAGTG
 CCCGTAAGCCCTGCTTTTCATGATGACCATGGTCGATGCGACCACTCCA

108

-continued

CGAAGAGGAAGAAGCCGCGGGGTGTCTGCTCAGCAGGCGCAGGGCAGCCT
 CTGTCTATCTCCATCAGGAGGGGTCCAGTGTGGAGTCTCGGTGGATCTCGT
 5 ATTTTCATGTCTCCAGGCTCAAAGAGACCCATGAGATGGGTACAGACGGGT
 CCAGGGAAGCCTGCATGAGCTCAGTGCAGTTCCACACGTACCGGGCACCCCT
 GCGGTCGCGGAGCCATTCTGACACAGATTCTTCCCGTCCAGCCTGGTCC
 10 CACCTTGGCTGTAGTCATCTGGGTACTCAGGGTCTGGGGTTCATGCGAA
 ACATGTACTTTTCGGCTCCA.
 In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 65)
 CCCCCGGTGCGCGCGTCGGTGGTGCCGCGGGGGCGCCAGGTCGCAGGC
 GGTGTAGGGCTCCAGGCAGGCGGCGAAGGCCATGACGTGCGCTATGAAGGT
 20 CTGCTCCTGCACGCGTGAACAGGTGCGCTGCGGGCCGCGCGCAACAC
 CGCCACGTCCTCGCCTGCGTGGGTCTCTTCGTCCAGGGGCACTGCTGACTG
 CTGCCGATACTCGGGGCTCCCGCTCTCGCTCTCGGTAACATCCGGCCGGGC
 25 GCCGTCTTGAGCACATAGCCTGGACCGTTTC
 In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 66)
 CGTATAGGAGGACCGTGTAGGCCTTCTGTCCCGGGCCTTGCCAGCGGCCA
 GCCCGATGAAGGAGCTCCCTCGCAGGGGGTAGCCTCCGAAGGAGAAGACGT
 35 GGGAGTGGTCGGCAGTGACGAGGCTCAGCGTGTCTCTCTCGTGGTGGCT
 GGCCCGCCCTCTCAATGGCGTCGTCGAACATGATCGTCTCAGTCAGTGCCC
 GGTAAGCCCTGCTTTTCATGATGACCATGGTCGATGCGACCACTCCACGA
 AGAGGAAGAAGCCGCGGGGTGTCTGCTCAGCAGG.
 In some embodiments, the AP gene fragment has the sequence of:

(SEQ ID NO: 67)
 CGCAGGGCAGCCTCTGTCTATCTCCATCAGGAGGGGTCCAGTGTGGAGTCT
 CGGTGGATCTCGTATTTTCATGTCTCCAGGCTCAAAGAGACCCATGAGATGG
 50 GTCACAGACGGGTCCAGGAAGCCTGCATGAGCTCAGTGCAGTTCCACACG
 TACCGGGCACCTGGCGTTCGCCGAGCCATTCTGACACAGATTCTTCCCG
 TCCAGCCTGGTCCCACCTTGGCTGTAGTCATCTGGGTACTCAGGGTCTGGG
 GTTCCCATGCGAAACATGTACTTTTCGGCTCCA.
 In some embodiments, the AP gene fragment has the sequence of:

An exemplary splice donor sequence for use in the methods and compositions described herein (e.g., in trans-splicing and dual hybrid approaches) has the sequence: GTAAGTATCAAGGTTACAAGAC AGGTTTAAAGGA-GACCAATAGAACTGGGCTTGTGCGAGACAGAGAA-GACTCTTGCGTTTCT (SEQ ID NO: 20). An exemplary splice acceptor sequence for use in the methods and compositions described herein (e.g., in trans-splicing and dual hybrid approaches) has the sequence: GATAGGCACCTAT-TGG TCTTACTGACATCCACTTTGCCTTTCTCTC-CACAG (SEQ ID NO: 21). The splice donor sequence GTAAGTATCAAGGTTACAAGACAGGTTTAAAGGA-

GACCAATAGAACTGGGCTTGTCGAGACAGAGAA GACTCTTGCGTTTCTGA (SEQ ID NO: 68) and the splice acceptor sequence TAGGCACCTATTGGTCTTACTGACATCCACTTTGCCTTCTCTCCACAG (SEQ ID NO: 69) can also be used in the methods and compositions described herein. Additional examples of splice donor and splice acceptor sequences are known in the art.

Dual hybrid vectors for use in the methods and compositions described herein are designed such that approximately half of the OTOF gene is contained within each vector (e.g., each vector contains a polynucleotide that encodes approximately half of the OTOF protein). The determination of how to split the polynucleotide sequence between the two nucleic acid vectors is made based on the size of the promoter and the locations of the portions of the polynucleotide that encode the OTOF C2 domains. When a short promoter is used in the dual hybrid vector system (e.g., a promoter that is 1 kb or shorter, e.g., approximately 1 kb, 950 bp, 900 bp, 850 bp, 800 bp, 750 bp, 700 bp, 650 bp, 600 bp, 550 bp, 500 bp, 450 bp, 400 bp, 350 bp, 300 bp or shorter), such as CAG, CMV, smCBA, or a Myo15 promoter having a sequence that is 1 kb or shorter (e.g., a Myo15 promoter described hereinabove, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60), the OTOF polynucleotide sequence is divided between the two nucleic acid vectors at an exon boundary that occurs after the portion of the polynucleotide that encodes the C2D domain and before the portion of the polynucleotide that encodes C2E domain, for example, the exon 26/27 boundary. The nucleic acid vectors containing promoters of this size can optionally contain OTOF UTRs (e.g., full-length 5' and 3' UTRs). When a long promoter is used in the dual hybrid vector system (e.g., a promoter that is longer than 1 kb, e.g., 1.1 kb, 1.25 kb, 1.5 kb, 1.75 kb, 2 kb, 2.5 kb, 3 kb or longer), such as a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36), the OTOF polynucleotide sequence will be divided between the two nucleic acid vectors at an exon boundary that occurs after the portion of the polynucleotide that encodes the C2C domain, and either before the portion of the polynucleotide that encodes the C2D domain, such as the exon 19/20 boundary, or within the portion of the polynucleotide that encodes the C2D domain, such as the exon 25/26 boundary. A short promoter (e.g., a CMV promoter, CAG promoter, smCBA promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the dual vector systems designed for large promoters, in which case additional elements (e.g., OTOF UTR sequences) may be included in the first vector (e.g., the vector containing the portion of the polynucleotide that encodes the C2C domain).

One exemplary dual hybrid vector system that uses a short promoter includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-26 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and a second nucleic acid vector containing a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 27-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). The first and second nucleic acid vectors can also contain the full length 5' and 3' OTOF UTRs, respectively

(e.g., the 127 bp human OTOF 5' UTR can be included in the first nucleic acid vector, and the 1035 bp human OTOF 3' UTR can be included in the second nucleic acid vector). Another exemplary dual hybrid vector system that uses a short promoter includes a first nucleic acid vector containing a CAG promoter operably linked to exons 1-28 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and a second nucleic acid vector containing a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 29-48 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). The first and second nucleic acid vectors can also contain the full length 5' and 3' OTOF UTRs, respectively (e.g., the 134 bp mouse OTOF 5' UTR can be included in the first nucleic acid vector, and the 1001 bp mouse OTOF 3' UTR can be included in the second nucleic acid vector). The CMV promoter, smCBA promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter (e.g., a Myo15 promoter described hereinabove, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can be used in place of the CAG promoter either of the foregoing dual vector systems.

An exemplary dual hybrid vector system that uses a long promoter includes a first nucleic acid vector containing a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and a second nucleic acid vector containing a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 20-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). Another exemplary dual hybrid vector system that uses a long promoter includes a first nucleic acid vector containing a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-20 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and a second nucleic acid vector containing a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 21-48 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). Neither the first nor the second nucleic acid vector in either of the foregoing Myo15 promoter dual hybrid vector systems contains an OTOF UTR. A short promoter (e.g., a CMV promoter, smCBA promoter, CAG promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the foregoing dual vector systems designed for large promoters. If these dual vector systems contain a short promoter, they may also include a 5' OTOF UTR in the first vector.

To accommodate an OTOF UTR, the OTOF coding sequence can be divided in a different position. For example, in a dual hybrid vector system in which the first nucleic acid

vector contains a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-25 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and the second nucleic acid vector contains a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 26-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence), the second nucleic acid can also contain a full-length OTOF 3' UTR (e.g., the 1035 bp human OTOF UTR). For mouse OTOF, the dual hybrid vector system can contain a 3' UTR if the first nucleic acid vector contains a Myo15 promoter that is longer than 1 kb (e.g., SEQ ID NO: 36) operably linked to exons 1-24 of a polynucleotide encoding an OTOF protein (e.g., mouse OTOF, e.g., SEQ ID NO: 6), a splice donor sequence 3' of the polynucleotide sequence, and a recombinogenic region 3' of the splice donor sequence; and the second nucleic acid vector contains a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 25-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), and a poly(A) sequence (e.g., a bGH poly(A) signal sequence). In this dual hybrid vector system, the second nucleic acid can also

contain a full-length OTOF 3' UTR (e.g., the 1001 bp mouse OTOF UTR). A short promoter (e.g., a CMV promoter, smCBA promoter, CAG promoter, or a Myo15 promoter having a sequence that is 1 kb or shorter, e.g., a Myo15 promoter having the sequence of any one of SEQ ID NOs: 38, 39, or 49-60) can also be used in the foregoing dual vector systems designed for large promoters. If these dual vector systems contain a short promoter, they may also include a 5' OTOF UTR in the first vector.

The dual hybrid vectors used in the methods and compositions described herein can optionally include a degradation signal sequence in both the first and second nucleic acid vectors. The degradation signal sequence can be included to prevent or reduce the expression of portions of the OTOF protein from polynucleotides that failed to recombine and/or undergo splicing. The degradation signal sequence is positioned 3' of the recombinogenic region in the first nucleic acid vector, and is positioned between the recombinogenic region and the splice acceptor in the second nucleic acid vector. A degradation signal sequence that can be used in the compositions and methods described herein has the sequence of:

(SEQ ID NO: 22)
GCCTGCAAGAACTGGTTCAGCAGCCTGAGCCATTCTGATCCACCTG.

Exemplary pairs of overlapping, trans-splicing, and dual hybrid vectors are described in Table 4 below.

TABLE 4

Exemplary pairs of overlapping, trans-splicing, and hybrid dual vectors for use in the methods and compositions described herein		
Vector Pair Number	Vector Type	Vector Pair
1	Overlapping	First nucleic acid vector contains: CAG promoter operably linked to exons 1-24 and the 500 kb 3' of the exon 24/25 boundary of a polynucleotide encoding a human OTOF protein Second nucleic acid vector contains: the 500 kb 5' of the exon 24/25 boundary and exons 25-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence
2	Overlapping	First nucleic acid vector contains: CAG promoter operably linked to exons 1-28 and the 500 kb 3' of the exon 28/29 boundary of a polynucleotide encoding a human OTOF protein Second nucleic acid vector contains: the 500 kb 5' of the exon 28/29 boundary and exons 29-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence
3	Overlapping	First nucleic acid vector contains: Myo15 promoter operably linked to exons 1-21 and the 500 kb 3' of the exon 21/22 boundary of a polynucleotide encoding a human OTOF protein Second nucleic acid vector contains: the 500 kb 5' of the exon 21/22 boundary and exons 22-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence
4	Trans-splicing	First nucleic acid vector contains: CAG promoter operably linked to exons 1-26 of a polynucleotide encoding a human OTOF protein and a splice donor sequence 3' of the polynucleotide Second nucleic acid vector contains: a splice acceptor sequence 5' of exons 27-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence
5	Trans-splicing	First nucleic acid vector contains: Myo15 promoter operably linked to exons 1-19 of a polynucleotide encoding a human OTOF protein and a splice donor sequence 3' of the polynucleotide Second nucleic acid vector contains: a splice acceptor sequence 5' of exons 20-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence
6	Trans-splicing	First nucleic acid vector contains: Myo15 promoter operably linked to exons 1-25 of a polynucleotide encoding a human OTOF protein and a splice donor sequence 3' of the polynucleotide Second nucleic acid vector contains: a splice acceptor sequence 5' of exons 26-48 of a polynucleotide encoding a human OTOF protein and a bGH poly(A) sequence

TABLE 4-continued

Exemplary pairs of overlapping, trans-splicing, and hybrid dual vectors for use in the methods and compositions described herein		
Vector Pair Number	Vector Type	Vector Pair
7	Hybrid	First nucleic acid vector contains: CAG promoter operably linked to exons 1-26 of a polynucleotide encoding a human OTOF protein, a splice donor sequence 3' of the polynucleotide, and a recombinogenic region 3' of the splice donor sequence Second nucleic acid vector contains: a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, exons 27-48 of a polynucleotide encoding a human OTOF protein 3' of the splice acceptor sequence, and a bGH poly(A) sequence
8	Hybrid	First nucleic acid vector contains: Myo15 promoter operably linked to exons 1-19 of a polynucleotide encoding a human OTOF protein, a splice donor sequence 3' of the polynucleotide, and a recombinogenic region 3' of the splice donor sequence Second nucleic acid vector contains: a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, exons 20-48 of a polynucleotide encoding a human OTOF protein 3' of the splice acceptor sequence, and a bGH poly(A) sequence
9	Hybrid	First nucleic acid vector contains: Myo15 promoter operably linked to exons 1-25 of a polynucleotide encoding a human OTOF protein, a splice donor sequence 3' of the polynucleotide, and a recombinogenic region 3' of the splice donor sequence Second nucleic acid vector contains: a recombinogenic region, a splice acceptor sequence 3' of the recombinogenic region, exons 26-48 of a polynucleotide encoding a human OTOF protein 3' of the splice acceptor sequence, and a bGH poly(A) sequence

Vectors for the Expression of OTOF

In addition to achieving high rates of transcription and translation, stable expression of an exogenous gene in a mammalian cell can be achieved by integration of the polynucleotide containing the gene into the nuclear genome of the mammalian cell. A variety of vectors for the delivery and integration of polynucleotides encoding exogenous proteins into the nuclear DNA of a mammalian cell have been developed. Examples of expression vectors are disclosed in, e.g., WO 1994/011026 and are incorporated herein by reference. Expression vectors for use in the compositions and methods described herein contain a polynucleotide sequence that encodes a portion of OTOF, as well as, e.g., additional sequence elements used for the expression of these agents and/or the integration of these polynucleotide sequences into the genome of a mammalian cell. Certain vectors that can be used for the expression of OTOF include plasmids that contain regulatory sequences, such as promoter and enhancer regions, which direct gene transcription. Other useful vectors for expression of OTOF contain polynucleotide sequences that enhance the rate of translation of these genes or improve the stability or nuclear export of the mRNA that results from gene transcription. These sequence elements include, e.g., 5' and 3' untranslated regions and a polyadenylation signal site in order to direct efficient transcription of the gene carried on the expression vector. The expression vectors suitable for use with the compositions and methods described herein may also contain a polynucleotide encoding a marker for selection of cells that contain such a vector. Examples of a suitable marker include genes that encode resistance to antibiotics, such as ampicillin, chloramphenicol, kanamycin, or nourseothricin.

AAV Vectors for Nucleic Acid Delivery

In some embodiments, nucleic acids of the compositions and methods described herein are incorporated into recombinant AAV (rAAV) vectors and/or virions in order to facilitate their introduction into a cell. rAAV vectors useful in the compositions and methods described herein are

recombinant nucleic acid constructs that include (1) a heterologous sequence to be expressed (e.g., a polynucleotide encoding an N-terminal or C-terminal portion of an OTOF protein) and (2) viral sequences that facilitate stability and expression of the heterologous genes. The viral sequences may include those sequences of AAV that are required in cis for replication and packaging (e.g., functional ITRs) of the DNA into a virion. Such rAAV vectors may also contain marker or reporter genes. Useful rAAV vectors have one or more of the AAV WT genes deleted in whole or in part but retain functional flanking ITR sequences. The AAV ITRs may be of any serotype suitable for a particular application. For use in the methods and compositions described herein, the ITRs can be AAV2 ITRs. Methods for using rAAV vectors are described, for example, in Tal et al., J. Biomed. Sci. 7:279 (2000), and Monahan and Samulski, Gene Delivery 7:24 (2000), the disclosures of each of which are incorporated herein by reference as they pertain to AAV vectors for gene delivery.

The nucleic acids and vectors described herein can be incorporated into a rAAV virion in order to facilitate introduction of the nucleic acid or vector into a cell. The capsid proteins of AAV compose the exterior, non-nucleic acid portion of the virion and are encoded by the AAV cap gene. The cap gene encodes three viral coat proteins, VP1, VP2 and VP3, which are required for virion assembly. The construction of rAAV virions has been described, for instance, in U.S. Pat. Nos. 5,173,414; 5,139,941; 5,863,541; 5,869,305; 6,057,152; and 6,376,237; as well as in Rabinowitz et al., J. Virol. 76:791 (2002) and Bowles et al., J. Virol. 77:423 (2003), the disclosures of each of which are incorporated herein by reference as they pertain to AAV vectors for gene delivery.

rAAV virions useful in conjunction with the compositions and methods described herein include those derived from a variety of AAV serotypes including AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74,

Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, and PHP.S. For targeting cochlear hair cells, AAV1, AAV2, AAV6, AAV9, Anc80, Anc80L65, DJ/9, 7m8, and PHP.B may be particularly useful. Serotypes evolved for transduction of the retina may also be used in the methods and compositions described herein. The first and second nucleic acid vectors in the compositions and methods described herein may have the same serotype or different serotypes. Construction and use of AAV vectors and AAV proteins of different serotypes are described, for instance, in Chao et al., *Mol. Ther.* 2:619 (2000); Davidson et al., *Proc. Natl. Acad. Sci. USA* 97:3428 (2000); Xiao et al., *J. Virol.* 72:2224 (1998); Halbert et al., *J. Virol.* 74:1524 (2000); Halbert et al., *J. Virol.* 75:6615 (2001); and Auricchio et al., *Hum. Molec. Genet.* 10:3075 (2001), the disclosures of each of which are incorporated herein by reference as they pertain to AAV vectors for gene delivery.

Also useful in conjunction with the compositions and methods described herein are pseudotyped rAAV vectors. Pseudotyped vectors include AAV vectors of a given serotype (e.g., AAV9) pseudotyped with a capsid gene derived from a serotype other than the given serotype (e.g., AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, etc.). Techniques involving the construction and use of pseudotyped rAAV virions are known in the art and are described, for instance, in Duan et al., *J. Virol.* 75:7662 (2001); Halbert et al., *J. Virol.* 74:1524 (2000); Zolotukhin et al., *Methods*, 28:158 (2002); and Auricchio et al., *Hum. Molec. Genet.* 10:3075 (2001).

AAV virions that have mutations within the virion capsid may be used to infect particular cell types more effectively than non-mutated capsid virions. For example, suitable AAV mutants may have ligand insertion mutations for the facilitation of targeting AAV to specific cell types. The construction and characterization of AAV capsid mutants including insertion mutants, alanine screening mutants, and epitope tag mutants is described in Wu et al., *J. Virol.* 74:8635 (2000). Other rAAV virions that can be used in methods described herein include those capsid hybrids that are generated by molecular breeding of viruses as well as by exon shuffling. See, e.g., Soong et al., *Nat. Genet.*, 25:436 (2000) and Kolman and Stemmer, *Nat. Biotechnol.* 19:423 (2001). **Pharmaceutical Compositions**

The nucleic acid vectors described herein may be incorporated into a vehicle for administration into a patient, such as a human patient suffering from sensorineural hearing loss or auditory neuropathy, as described herein. Pharmaceutical compositions containing vectors, such as viral vectors, that contain a polynucleotide encoding a portion of an OTOF protein can be prepared using methods known in the art. For example, such compositions can be prepared using, e.g., physiologically acceptable carriers, excipients or stabilizers (Remington's *Pharmaceutical Sciences* 16th edition, Osol, A. Ed. (1980); incorporated herein by reference), and in a desired form, e.g., in the form of lyophilized formulations or aqueous solutions.

Mixtures of the nucleic acid vectors (e.g., viral vectors) described herein may be prepared in water suitably mixed with one or more excipients, carriers, or diluents. Dispersions may also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations may contain a preservative to prevent the growth of microorganisms. The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions (described in U.S. Pat. No. 5,466,

468, the disclosure of which is incorporated herein by reference). In any case the formulation may be sterile and may be fluid to the extent that easy syringability exists. Formulations may be stable under the conditions of manufacture and storage and may be preserved against the contaminating action of microorganisms, such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.

For example, a solution containing a pharmaceutical composition described herein may be suitably buffered, if necessary, and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous, and intraperitoneal administration. In this connection, sterile aqueous media that can be employed will be known to those of skill in the art in light of the present disclosure. For example, one dosage may be dissolved in 1 ml of isotonic NaCl solution and either added to 1000 ml of hypodermoclysis fluid or injected at the proposed site of infusion. Some variation in dosage will necessarily occur depending on the condition of the subject being treated. For local administration to the inner ear, the composition may be formulated to contain a synthetic perilymph solution. An exemplary synthetic perilymph solution includes 20-200 mM NaCl, 1-5 mM KCl, 0.1-10 mM CaCl_2 , 1-10 mM glucose, and 2-50 mM HEPES, with a pH between about 6 and 9 and an osmolality of about 300 mOsm/kg. The person responsible for administration will, in any event, determine the appropriate dose for the individual subject. Moreover, for human administration, preparations may meet sterility, pyrogenicity, general safety, and purity standards as required by FDA Office of Biologics standards.

Methods of Treatment

The compositions described herein may be administered to a subject with sensorineural hearing loss or auditory neuropathy by a variety of routes, such as local administration to the inner ear (e.g., administration into the perilymph or endolymph, e.g., through the oval window, round window, or horizontal canal, e.g., administration to a cochlear hair cell), intravenous, parenteral, intradermal, transdermal, intramuscular, intranasal, subcutaneous, percutaneous, intratracheal, intraperitoneal, intraarterial, intravascular, inhalation, perfusion, lavage, and oral administration. The most suitable route for administration in any given case will depend on the particular composition administered, the patient, pharmaceutical formulation methods, administration methods (e.g., administration time and administration route), the patient's age, body weight, sex, severity of the disease being treated, the patient's diet, and the patient's excretion rate. Compositions may be administered once, or more than once (e.g., once annually, twice annually, three times annually, bi-monthly, monthly, or bi-weekly). In some embodiments, the first and second nucleic acid vectors are

117

administered simultaneously (e.g., in one composition). In some embodiments, the first and second nucleic acid vectors are administered sequentially (e.g., the second nucleic acid vector is administered immediately after the first nucleic acid vector, or 5 minutes, 10 minutes, 15 minutes, 20 minutes, 25 minutes, 30 minutes, 45 minutes, 1 hour, 2 hours, 8 hours, 12 hours, 1 day, 2 days, 7 days, two weeks, 1 month or more after the first nucleic acid vector). The first and second nucleic acid vector can have the same serotype or different serotypes (e.g., AAV serotypes).

Subjects that may be treated as described herein are subjects having or at risk of developing sensorineural hearing loss or auditory neuropathy. The compositions and methods described herein can be used to treat subjects having a mutation in OTOF (e.g., a mutation that reduces OTOF function or expression, or an OTOF mutation associated with sensorineural hearing loss), subjects having a family history of autosomal recessive sensorineural hearing loss or deafness (e.g., a family history of OTOF-related hearing loss), or subjects whose OTOF mutational status and/or OTOF activity level is unknown. The methods described herein may include a step of screening a subject for a mutation in OTOF prior to treatment with or administration of the compositions described herein. A subject can be screened for an OTOF mutation using standard methods known to those of skill in the art (e.g., genetic testing). The methods described herein may also include a step of assessing hearing in a subject prior to treatment with or administration of the compositions described herein. Hearing can be assessed using standard tests, such as audiometry, ABR, electrocochleography (ECOG), and otoacoustic emissions. The compositions and methods described herein may also be administered as a preventative treatment to patients at risk of developing hearing loss or auditory neuropathy, e.g., patients who have a family history of inherited hearing loss or patients carrying an OTOF mutation who do not yet exhibit hearing loss or impairment.

Treatment may include administration of a composition containing the nucleic acid vectors (e.g., AAV viral vectors) described herein in various unit doses. Each unit dose will ordinarily contain a predetermined-quantity of the therapeutic composition. The quantity to be administered, and the particular route of administration and formulation, are within the skill of those in the clinical arts. A unit dose need not be administered as a single injection but may include continuous infusion over a set period of time. Dosing may be performed using a syringe pump to control infusion rate in order to minimize damage to the cochlea. In cases in which the nucleic acid vectors are AAV vectors (e.g., AAV1, AAV2, AAV2quad(Y-F), AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, rh10, rh39, rh43, rh74, Anc80, Anc80L65, DJ/8, DJ/9, 7m8, PHP.B, PHP.eb, or PHP.S vectors), the viral vectors may be administered to the patient at a dose of, for example, from about 1×10^{10} vector genomes (VG) to 1×10^{15} VG (e.g., 1×10^{10} VG, 2×10^{10} VG, 3×10^{10} VG, 4×10^{10} VG, 5×10^{10} VG, 6×10^{10} VG, 7×10^{10} VG, 8×10^{10} VG, 9×10^{10} VG, 1×10^{11} VG, 2×10^{11} VG, 3×10^{11} VG, 4×10^{11} VG, 5×10^{11} VG, 6×10^{11} VG, 7×10^{11} VG, 8×10^{11} VG, 9×10^{11} VG, 1×10^{12} VG, 2×10^{12} VG, 3×10^{12} VG, 4×10^{12} VG, 5×10^{12} VG, 6×10^{12} VG, 7×10^{12} VG, 8×10^{12} VG, 9×10^{12} VG, 1×10^{13} VG, 2×10^{13} VG, 3×10^{13} VG, 4×10^{13} VG, 5×10^{13} VG, 6×10^{13} VG, 7×10^{13} VG, 8×10^{13} VG, 9×10^{13} VG, 1×10^{14} VG, 2×10^{14} VG, 3×10^{14} VG, 4×10^{14} VG, 5×10^{14} VG, 6×10^{14} VG, 7×10^{14} VG, 8×10^{14} VG, 9×10^{14} VG, 1×10^{15} VG) in a volume of 1 μ L to 200 μ L (e.g., 1, 2, 3, 5, 6, 7, 8, 9, 10, 15, 20, 25,

118

30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 μ L).

The compositions described herein are administered in an amount sufficient to improve hearing, increase WT OTOF expression (e.g., expression in a cochlear hair cell, e.g., an inner hair cell), or increase OTOF function. Hearing may be evaluated using standard hearing tests (e.g., audiometry, ABR, electrocochleography (ECOG), and otoacoustic emissions) and may be improved by 5% or more (e.g., 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100% or more) compared to hearing measurements obtained prior to treatment. In some embodiments, the compositions are administered in an amount sufficient to improve the subject's ability to understand speech. The compositions described herein may also be administered in an amount sufficient to slow or prevent the development or progression of sensorineural hearing loss or auditory neuropathy (e.g., in subjects who carry a mutation in OTOF or have a family history of autosomal recessive hearing loss but do not exhibit hearing impairment, or in subjects exhibiting mild to moderate hearing loss). OTOF expression may be evaluated using immunohistochemistry, Western blot analysis, quantitative real-time PCR, or other methods known in the art for detection protein or mRNA, and may be increased by 5% or more (e.g., 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100% or more) compared to OTOF expression prior to administration of the compositions described herein. OTOF function may be evaluated directly (e.g., using electrophysiological methods or imaging methods to assess exocytosis) or indirectly based on hearing tests, and may be increased by 5% or more (e.g., 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100% or more) compared to OTOF function prior to administration of the compositions described herein. These effects may occur, for example, within 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, 6 weeks, 7 weeks, 8 weeks, 9 weeks, 10 weeks, 15 weeks, 20 weeks, 25 weeks, or more, following administration of the compositions described herein. The patient may be evaluated 1 month, 2 months, 3 months, 4 months, 5 months, 6 months or more following administration of the composition depending on the dose and route of administration used for treatment. Depending on the outcome of the evaluation, the patient may receive additional treatments. Kits

The compositions described herein can be provided in a kit for use in treating sensorineural hearing loss or auditory neuropathy (e.g., hearing loss associated with a mutation in OTOF). Compositions may include nucleic acid vectors described herein (e.g., a first nucleic acid vector containing a polynucleotide that encodes and N-terminal portion of an OTOF protein and a second nucleic acid vector containing a polynucleotide that encodes a C-terminal portion of an OTOF protein), optionally packaged in an AAV virus capsid (e.g., AAV1, AAV9, Anc80L65, DJ/9, or 7m8). The kit can further include a package insert that instructs a user of the kit, such as a physician, to perform the methods described herein. The kit may optionally include a syringe or other device for administering the composition.

EXAMPLES

The following examples are put forth so as to provide those of ordinary skill in the art with a description of how the compositions and methods described herein may be used, made, and evaluated, and are intended to be purely exem-

plary of the invention and are not intended to limit the scope of what the inventors regard as their invention.

Example 1: Generation of Nucleic Acid Vectors that Recombine to Produce Full-Length OTOF

Gene fragments were synthesized and sub-cloned into an AAV2/cis-plasmid using restriction enzyme sites. Plasmids were maxi prepped to generate 1 mg of transfection grade plasmid. Inner ear-derived HEI-OC1 cells were seeded into a 12-well tissue culture dish 24 hours before plasmid transfection at a density of 200,000 cells/well. One microgram of each plasmid was transfected using Lipofectamine 3000 according to standard manufacturer's protocol. For wells that received both 5' and 3' plasmids, 1 µg of each was transfected for a total of 2 µg of DNA. As a positive control, full-length Otoferlin cDNA was also transfected. Cells were incubated with plasmid for 48 hours.

For PCR to check for recombination at the DNA level, genomic DNA was extracted from each well using Qiagen's DNeasy Blood and Tissue kit, according to standard manufacturer's protocol. PCR primers were designed to anneal to the plasmid outside of the region of overlap or splicing to generate an amplicon of ~1200 bp. PCR was performed using My Taq 2× mastermix according to manufacturer's recommendations: annealing temperature of 58° C., elongation step of 30 seconds, and cycle number of 35×. Ten microliters of PCR product was run on a pre-cast 1.2% agarose E-gel and imaged on a BioRad gel doc imaging station. Both dual hybrid vectors (FIG. 20A) and overlapping vectors (FIG. 20B) showed evidence of recombination when the 5' and 3' plasmids were transfected together.

For immunofluorescence to check for recombination and generation of protein, cells were fixed with cold 4% PFA for 20 minutes at room temperature. Cells were washed three times with PBS and then permeabilized in a blocking solution of PBS with 10% normal donkey serum and 0.01% TritonX100. Cells were incubated in primary antibody overnight (mouse-anti-Otoferlin, Abcam ab53233) at a concentration of 1:1000 at 4° C. Cells were washed three times with PBS and incubated in secondary antibody for three hours at room temperature (donkey-anti-mouse Alexa Fluor 647, ThermoFisher A-31571). Cells were washed three times in PBS and stained with DAPI for 15 minutes at room temperature. Cells were imaged using a Zeiss inverted Apotome microscope. Increased staining was observed in cells transfected with both 5' and 3' plasmids compared to transfection of the 5' or 3' plasmid alone, indicating that the dual hybrid vector (FIG. 21A), trans-splicing vector (FIG. 21B), and overlapping vector (FIG. 21C) systems recombined in HEI-OC1 cells and generated OTOF protein.

Example 2: Administration of a Composition Containing Dual Hybrid Vectors that Express OTOF to Mice Restores Electrophysiological Signatures of Hearing Function

Homozygous (HOM) OTOF-Q828X mice (a mouse model of human OTOF mutation p.Gln828Ter) were either left untreated or treated (by injection through the round window membrane) with 4E10 (4×10^{10}) vector genomes (vg)/ear of an AAV1-Myo15 (SEQ ID NO: 38)-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector system in which exons 1-20 of the polynucleotide encoding the N-terminal portion of the OTOF protein and exons 21-46 and 48 of the polynucleotide encoding the C-terminal portion of the OTOF protein were delivered in separate vectors (FIG.

22A). An AP recombinogenic region (SEQ ID NO: 65) was included in both vectors of the dual hybrid vector system. Auditory brainstem response (ABR) thresholds were used to assess hearing function. Untreated animals (untreated Otof HOM) showed no detectable recovery in hearing function, whereas treated animals exhibited a robust recovery, which was consistent from four weeks post-treatment (Otof HOM at 4 weeks after treatment) to eight weeks post-treatment (Otof HOM at 8-11 weeks after treatment). ABR thresholds in heterozygous animals (Otof HET) were also tested.

In a separate set of experiments, homozygous OTOF-Q828X mice were either left untreated or treated (by injection through the round window membrane) with 4E10 (4×10^{10}) vg/ear of an AAV1-truncated chimeric CMV-chicken β -actin (smCBA)-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector system as described above (FIG. 22B). The first vector contained exons 1-20 of the polynucleotide encoding the N-terminal portion of the OTOF protein and exons 21-46 and 48 of the polynucleotide encoding the C-terminal portion of the OTOF protein and both vectors contained an AP recombinogenic region (SEQ ID NO: 65). Untreated animals had no detectable recovery in hearing function, whereas treated animals exhibited a robust recovery at 4 weeks post-treatment (Otof HOM at 4 weeks after treatment). When these same animals were evaluated at 8 weeks post-treatment (Otof HOM at 8 weeks after treatment), ABR thresholds increased, suggesting less durable recovery with the smCBA promoter. ABR thresholds in heterozygous animals were also tested.

In yet another set of experiments, homozygous OTOF-Q828X mice were either left untreated or treated (by injection through the round window membrane) with an AAV1-smCBA-hOTOF (isoform 5, SEQ ID NO: 5) dual hybrid vector, as described above, at either 8E9 (8×10^9) vg/ear (low dose), 1.6E10 (1.6×10^{10}) vg/ear (mid dose), or 6.4E10 (6.4×10^{10}) vg/ear (high dose). The first vector contained exons 1-20 of the polynucleotide encoding the N-terminal portion of the OTOF protein and exons 21-46 and 48 of the polynucleotide encoding the C-terminal portion of the OTOF protein and both vectors contained an AP recombinogenic region (SEQ ID NO: 65). ABR thresholds were used to assess hearing function at four weeks and eight weeks post-treatment (FIG. 22C). A dose-dependent recovery in ABR was observed at both timepoints. When comparing the eight weeks versus the four weeks timepoint, recovery of hearing function was steady for the low and mid doses but decreased for the high dose animals. ABR thresholds in heterozygous animals were also tested.

Example 3: Administration of a Composition Containing Overlapping Dual Vectors that Express OTOF to a Subject with Sensorineural Hearing Loss

According to the methods disclosed herein, a physician of skill in the art can treat a patient, such as a human patient, with sensorineural hearing loss (e.g., sensorineural hearing loss associated with a mutation in OTOF) so as to improve or restore hearing. To this end, a physician of skill in the art can administer to the human patient a composition containing a first AAV vector (e.g., AAV1 or AAV9) containing a Myo15 promoter (e.g., SEQ ID NO: 36) operably linked to exons 1-21 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and the 500 kb immediately 3' of the exon 21/22 boundary, and a second AAV vector (e.g., AAV1 or AAV9) containing the 500 kb immediately 5' of the exon 21/22 boundary and

121

exons 22-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a bGH poly(A) sequence. The composition containing the overlapping dual AAV vectors may be administered to the patient, for example, by local administration to the inner ear (e.g., injection into the perilymph), to treat sensorineural hearing loss.

Following administration of the composition to a patient, a practitioner of skill in the art can monitor the expression of OTOF, and the patient's improvement in response to the therapy, by a variety of methods. For example, a physician can monitor the patient's hearing by performing standard tests, such as audiometry, ABR, electrocochleography (ECOG), and otoacoustic emissions following administration of the composition. A finding that the patient exhibits improved hearing in one or more of the tests following administration of the composition compared to hearing test results prior to administration of the composition indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

Example 4: Administration of a Composition
Containing Trans-Splicing Dual Vectors that
Express OTOF to a Subject with Sensorineural
Hearing Loss

According to the methods disclosed herein, a physician of skill in the art can treat a patient, such as a human patient, with sensorineural hearing loss (e.g., sensorineural hearing loss associated with a mutation in OTOF) so as to improve or restore hearing. To this end, a physician of skill in the art can administer to the human patient a composition containing a first AAV vector (e.g., AAV1 or AAV9) containing a Myo15 promoter (e.g., SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a splice donor sequence (e.g., SEQ ID NO: 20 or SEQ ID NO: 68) 3' of the polynucleotide sequence, and a second AAV vector (e.g., AAV1 or AAV9) containing a splice acceptor sequence (e.g., SEQ ID NO: 21 or SEQ ID NO: 69) 5' of exons 20-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5) and a bGH poly(A) sequence. The composition containing the trans-splicing dual AAV vectors may be administered to the patient, for example, by local administration to the inner ear (e.g., injection into the perilymph), to treat sensorineural hearing loss.

Following administration of the composition to a patient, a practitioner of skill in the art can monitor the expression of OTOF, and the patient's improvement in response to the therapy, by a variety of methods. For example, a physician can monitor the patient's hearing by performing standard tests, such as audiometry, ABR, electrocochleography (ECOG), and otoacoustic emissions following administration of the composition. A finding that the patient exhibits improved hearing in one or more of the tests following administration of the composition compared to hearing test results prior to administration of the composition indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

122

Example 5: Administration of a Composition
Containing Dual Hybrid Vectors that Express
OTOF to a Subject with Sensorineural Hearing
Loss

According to the methods disclosed herein, a physician of skill in the art can treat a patient, such as a human patient, with sensorineural hearing loss (e.g., sensorineural hearing loss associated with a mutation in OTOF) so as to improve or restore hearing. To this end, a physician of skill in the art can administer to the human patient a composition containing a first AAV vector (e.g., AAV1 or AAV9) containing a Myo15 promoter (e.g., SEQ ID NO: 36) operably linked to exons 1-19 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), a splice donor sequence (e.g., SEQ ID NO: 20 or SEQ ID NO: 68) 3' of the polynucleotide sequence, and an F1 phage recombinogenic region (e.g., an F1 phage AK gene, SEQ ID NO: 19) 3' of the splice donor sequence, and a second nucleic acid vector containing an F1 phage recombinogenic region (e.g., an F1 phage AK gene, SEQ ID NO: 19), a splice acceptor sequence (e.g., SEQ ID NO: 21 or SEQ ID NO: 69) 3' of the recombinogenic region, a polynucleotide 3' of the splice acceptor sequence that contains exons 20-48 of a polynucleotide encoding an OTOF protein (e.g., human OTOF, e.g., SEQ ID NO: 1 or SEQ ID NO: 5), and a bGH poly(A) sequence. The first and second dual hybrid AAV vectors can optionally include a degradation signal sequence (e.g., SEQ ID NO: 22) positioned 3' of the recombinogenic region in the first nucleic acid vector, and positioned between the recombinogenic region and the splice acceptor sequence in the second nucleic acid vector. The composition containing the dual hybrid AAV vectors may be administered to the patient, for example, by local administration to the inner ear (e.g., injection into the perilymph), to treat sensorineural hearing loss.

Following administration of the composition to a patient, a practitioner of skill in the art can monitor the expression of OTOF, and the patient's improvement in response to the therapy, by a variety of methods. For example, a physician can monitor the patient's hearing by performing standard tests, such as audiometry, ABR, electrocochleography (ECOG), and otoacoustic emissions following administration of the composition. A finding that the patient exhibits improved hearing in one or more of the tests following administration of the composition compared to hearing test results prior to administration of the composition indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

OTHER EMBODIMENTS

Various modifications and variations of the described invention will be apparent to those skilled in the art without departing from the scope and spirit of the invention. Although the invention has been described in connection with specific embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention that are obvious to those skilled in the art are intended to be within the scope of the invention. Other embodiments are in the claims.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 69

<210> SEQ ID NO 1

<211> LENGTH: 1997

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 1

```

Met Ala Leu Leu Ile His Leu Lys Thr Val Ser Glu Leu Arg Gly Arg
1          5          10          15

Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser
20        25        30

Arg Val Leu Glu Asn Cys Glu Asp Val Ala Asp Phe Asp Glu Thr Phe
35        40        45

Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Met Leu Glu Ile
50        55        60

Gln Val Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr
65        70        75        80

Phe Arg Met Val Leu Gln Lys Val Val Glu Glu Ser His Val Glu Val
85        90        95

Thr Asp Thr Leu Ile Asp Asp Asn Asn Ala Ile Ile Lys Thr Ser Leu
100       105       110

Cys Val Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Ser Trp
115       120       125

Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Glu Lys
130       135       140

Asp Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Ser
145       150       155       160

Arg Pro Pro Gly Glu Lys Ser Phe Arg Arg Ala Gly Arg Ser Val Phe
165       170       175

Ser Ala Met Lys Leu Gly Lys Asn Arg Ser His Lys Glu Glu Pro Gln
180       185       190

Arg Pro Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu
195       200       205

Ala Ile Arg Leu Gly Asp Gly Leu Asp Pro Asp Ser Val Ser Leu Ala
210       215       220

Ser Val Thr Ala Leu Thr Thr Asn Val Ser Asn Lys Arg Ser Lys Pro
225       230       235       240

Asp Ile Lys Met Glu Pro Ser Ala Gly Arg Pro Met Asp Tyr Gln Val
245       250       255

Ser Ile Thr Val Ile Glu Ala Arg Gln Leu Val Gly Leu Asn Met Asp
260       265       270

Pro Val Val Cys Val Glu Val Gly Asp Asp Lys Lys Tyr Thr Ser Met
275       280       285

Lys Glu Ser Thr Asn Cys Pro Tyr Tyr Asn Glu Tyr Phe Val Phe Asp
290       295       300

Phe His Val Ser Pro Asp Val Met Phe Asp Lys Ile Ile Lys Ile Ser
305       310       315       320

Val Ile His Ser Lys Asn Leu Leu Arg Ser Gly Thr Leu Val Gly Ser
325       330       335

Phe Lys Met Asp Val Gly Thr Val Tyr Ser Gln Pro Glu His Gln Phe
340       345       350

His His Lys Trp Ala Ile Leu Ser Asp Pro Asp Asp Ile Ser Ser Gly
355       360       365

```

Leu 370	Lys	Gly	Tyr	Val	Lys	Cys 375	Asp	Val	Ala	Val	Val 380	Gly	Lys	Gly	Asp
Asn 385	Ile	Lys	Thr	Pro	His 390	Lys	Ala	Asn	Glu	Thr	Asp 395	Glu	Asp	Asp	Ile 400
Glu	Gly	Asn	Leu	Leu 405	Leu	Pro	Glu	Gly	Val 410	Pro	Pro	Glu	Arg	Gln 415	Trp
Ala	Arg	Phe	Tyr	Val	Lys	Ile	Tyr	Arg	Ala	Glu	Gly	Leu	Pro	Arg	Met
Asn	Thr	Ser	Leu	Met	Ala	Asn	Val 440	Lys	Lys	Ala	Phe	Ile 445	Gly	Glu	Asn
Lys	Asp 450	Leu	Val	Asp	Pro	Tyr 455	Val	Gln	Val	Phe	Phe 460	Ala	Gly	Gln	Lys
Gly 465	Lys	Thr	Ser	Val	Gln 470	Lys	Ser	Ser	Tyr	Glu 475	Pro	Leu	Trp	Asn	Glu 480
Gln	Val	Val	Phe	Thr 485	Asp	Leu	Phe	Pro	Pro 490	Leu	Cys	Lys	Arg	Met 495	Lys
Val	Gln	Ile	Arg 500	Asp	Ser	Asp	Lys	Val	Asn 505	Asp	Val	Ala	Ile 510	Gly	Thr
His	Phe 515	Ile	Asp	Leu	Arg	Lys	Ile 520	Ser	Asn	Asp	Gly	Asp 525	Lys	Gly	Phe
Leu	Pro 530	Thr	Leu	Gly	Pro	Ala 535	Trp	Val	Asn	Met	Tyr 540	Gly	Ser	Thr	Arg
Asn 545	Tyr	Thr	Leu	Leu	Asp 550	Glu	His	Gln	Asp	Leu 555	Asn	Glu	Gly	Leu	Gly 560
Glu	Gly	Val	Ser	Phe 565	Arg	Ala	Arg	Leu	Leu 570	Leu	Gly	Leu	Ala	Val 575	Glu
Ile	Val	Asp	Thr 580	Ser	Asn	Pro	Glu	Leu 585	Thr	Ser	Ser	Thr	Glu 590	Val	Gln
Val	Glu 595	Gln	Ala	Thr	Pro	Ile	Ser 600	Glu	Ser	Cys	Ala	Gly 605	Lys	Met	Glu
Glu	Phe 610	Phe	Leu	Phe	Gly	Ala 615	Phe	Leu	Glu	Ala	Ser 620	Met	Ile	Asp	Arg
Arg 625	Asn	Gly	Asp	Lys	Pro 630	Ile	Thr	Phe	Glu	Val 635	Thr	Ile	Gly	Asn	Tyr 640
Gly	Asn	Glu	Val	Asp 645	Gly	Leu	Ser	Arg	Pro 650	Gln	Arg	Pro	Arg	Pro 655	Arg
Lys	Glu	Pro	Gly 660	Asp	Glu	Glu	Glu	Val 665	Asp	Leu	Ile	Gln	Asn	Ala	Ser
Asp	Asp 675	Glu	Ala	Gly	Asp	Ala	Gly 680	Asp	Leu	Ala	Ser	Val 685	Ser	Ser	Thr
Pro	Pro 690	Met	Arg	Pro	Gln	Val 695	Thr	Asp	Arg	Asn	Tyr 700	Phe	His	Leu	Pro
Tyr 705	Leu	Glu	Arg	Lys	Pro 710	Cys	Ile	Tyr	Ile 715	Lys	Ser	Trp	Trp	Pro	Asp 720
Gln	Arg	Arg	Arg	Leu 725	Tyr	Asn	Ala	Asn	Ile 730	Met	Asp	His	Ile	Ala	Asp 735
Lys	Leu	Glu	Glu 740	Gly	Leu	Asn	Asp	Ile 745	Gln	Glu	Met	Ile	Lys	Thr	Glu
Lys	Ser 755	Tyr	Pro	Glu	Arg	Arg	Leu 760	Arg	Gly	Val	Leu 765	Glu	Glu	Leu	Ser
Cys 770	Gly	Cys	Cys	Arg	Phe	Leu 775	Ser	Leu	Ala	Asp	Lys 780	Asp	Gln	Gly	His
Ser	Ser	Arg	Thr	Arg	Leu	Asp	Arg	Glu	Arg	Leu	Lys	Ser	Cys	Met	Arg

-continued

785	790	795	800
Glu Leu Glu Asn Met Gly Gln Gln Ala Arg Met Leu Arg Ala Gln Val			
805		810	815
Lys Arg His Thr Val Arg Asp Lys Leu Arg Leu Cys Gln Asn Phe Leu			
820		825	830
Gln Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln His Ser Ile Pro Asp			
835		840	845
Ile Phe Ile Trp Met Met Ser Asn Asn Lys Arg Val Ala Tyr Ala Arg			
850		855	860
Val Pro Ser Lys Asp Leu Leu Phe Ser Ile Val Glu Glu Glu Thr Gly			
865		870	875
Lys Asp Cys Ala Lys Val Lys Thr Leu Phe Leu Lys Leu Pro Gly Lys			
885		890	895
Arg Gly Phe Gly Ser Ala Gly Trp Thr Val Gln Ala Lys Val Glu Leu			
900		905	910
Tyr Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys Glu Phe Leu Cys Gly			
915		920	925
Leu Pro Cys Gly Phe Gln Glu Val Lys Ala Ala Gln Gly Leu Gly Leu			
930		935	940
His Ala Phe Pro Pro Val Ser Leu Val Tyr Thr Lys Lys Gln Ala Phe			
945		950	955
Gln Leu Arg Ala His Met Tyr Gln Ala Arg Ser Leu Phe Ala Ala Asp			
965		970	975
Ser Ser Gly Leu Ser Asp Pro Phe Ala Arg Val Phe Phe Ile Asn Gln			
980		985	990
Ser Gln Cys Thr Glu Val Leu Asn Glu Thr Leu Cys Pro Thr Trp Asp			
995		1000	1005
Gln Met Leu Val Phe Asp Asn Leu Glu Leu Tyr Gly Glu Ala His			
1010		1015	1020
Glu Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu Ile Tyr Asp			
1025		1030	1035
Gln Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr Phe Ala			
1040		1045	1050
Lys Pro Leu Val Lys Met Ala Asp Glu Ala Tyr Cys Pro Pro Arg			
1055		1060	1065
Phe Pro Pro Gln Leu Glu Tyr Tyr Gln Ile Tyr Arg Gly Asn Ala			
1070		1075	1080
Thr Ala Gly Asp Leu Leu Ala Ala Phe Glu Leu Leu Gln Ile Gly			
1085		1090	1095
Pro Ala Gly Lys Ala Asp Leu Pro Pro Ile Asn Gly Pro Val Asp			
1100		1105	1110
Val Asp Arg Gly Pro Ile Met Pro Val Pro Met Gly Ile Arg Pro			
1115		1120	1125
Val Leu Ser Lys Tyr Arg Val Glu Val Leu Phe Trp Gly Leu Arg			
1130		1135	1140
Asp Leu Lys Arg Val Asn Leu Ala Gln Val Asp Arg Pro Arg Val			
1145		1150	1155
Asp Ile Glu Cys Ala Gly Lys Gly Val Gln Ser Ser Leu Ile His			
1160		1165	1170
Asn Tyr Lys Lys Asn Pro Asn Phe Asn Thr Leu Val Lys Trp Phe			
1175		1180	1185
Glu Val Asp Leu Pro Glu Asn Glu Leu Leu His Pro Pro Leu Asn			
1190		1195	1200

-continued

Ile Arg Val Val Asp Cys Arg Ala Phe Gly Arg Tyr Thr Leu Val 1205 1210 1215	
Gly Ser His Ala Val Ser Ser Leu Arg Arg Phe Ile Tyr Arg Pro 1220 1225 1230	
Pro Asp Arg Ser Ala Pro Ser Trp Asn Thr Thr Val Arg Leu Leu 1235 1240 1245	
Arg Arg Cys Arg Val Leu Cys Asn Gly Gly Ser Ser Ser His Ser 1250 1255 1260	
Thr Gly Glu Val Val Val Thr Met Glu Pro Glu Val Pro Ile Lys 1265 1270 1275	
Lys Leu Glu Thr Met Val Lys Leu Asp Ala Thr Ser Glu Ala Val 1280 1285 1290	
Val Lys Val Asp Val Ala Glu Glu Glu Lys Glu Lys Lys Lys Lys 1295 1300 1305	
Lys Lys Gly Thr Ala Glu Glu Pro Glu Glu Glu Glu Pro Asp Glu 1310 1315 1320	
Ser Met Leu Asp Trp Trp Ser Lys Tyr Phe Ala Ser Ile Asp Thr 1325 1330 1335	
Met Lys Glu Gln Leu Arg Gln Gln Glu Pro Ser Gly Ile Asp Leu 1340 1345 1350	
Glu Glu Lys Glu Glu Val Asp Asn Thr Glu Gly Leu Lys Gly Ser 1355 1360 1365	
Met Lys Gly Lys Glu Lys Ala Arg Ala Ala Lys Glu Glu Lys Lys 1370 1375 1380	
Lys Lys Thr Gln Ser Ser Gly Ser Gly Gln Gly Ser Glu Ala Pro 1385 1390 1395	
Glu Lys Lys Lys Pro Lys Ile Asp Glu Leu Lys Val Tyr Pro Lys 1400 1405 1410	
Glu Leu Glu Ser Glu Phe Asp Asn Phe Glu Asp Trp Leu His Thr 1415 1420 1425	
Phe Asn Leu Leu Arg Gly Lys Thr Gly Asp Asp Glu Asp Gly Ser 1430 1435 1440	
Thr Glu Glu Glu Arg Ile Val Gly Arg Phe Lys Gly Ser Leu Cys 1445 1450 1455	
Val Tyr Lys Val Pro Leu Pro Glu Asp Val Ser Arg Glu Ala Gly 1460 1465 1470	
Tyr Asp Ser Thr Tyr Gly Met Phe Gln Gly Ile Pro Ser Asn Asp 1475 1480 1485	
Pro Ile Asn Val Leu Val Arg Val Tyr Val Val Arg Ala Thr Asp 1490 1495 1500	
Leu His Pro Ala Asp Ile Asn Gly Lys Ala Asp Pro Tyr Ile Ala 1505 1510 1515	
Ile Arg Leu Gly Lys Thr Asp Ile Arg Asp Lys Glu Asn Tyr Ile 1520 1525 1530	
Ser Lys Gln Leu Asn Pro Val Phe Gly Lys Ser Phe Asp Ile Glu 1535 1540 1545	
Ala Ser Phe Pro Met Glu Ser Met Leu Thr Val Ala Val Tyr Asp 1550 1555 1560	
Trp Asp Leu Val Gly Thr Asp Asp Leu Ile Gly Glu Thr Lys Ile 1565 1570 1575	
Asp Leu Glu Asn Arg Phe Tyr Ser Lys His Arg Ala Thr Cys Gly 1580 1585 1590	

-continued

Ile	Ala	Gln	Thr	Tyr	Ser	Thr	His	Gly	Tyr	Asn	Ile	Trp	Arg	Asp
1595						1600					1605			
Pro	Met	Lys	Pro	Ser	Gln	Ile	Leu	Thr	Arg	Leu	Cys	Lys	Asp	Gly
1610						1615					1620			
Lys	Val	Asp	Gly	Pro	His	Phe	Gly	Pro	Pro	Gly	Arg	Val	Lys	Val
1625						1630					1635			
Ala	Asn	Arg	Val	Phe	Thr	Gly	Pro	Ser	Glu	Ile	Glu	Asp	Glu	Asn
1640						1645					1650			
Gly	Gln	Arg	Lys	Pro	Thr	Asp	Glu	His	Val	Ala	Leu	Leu	Ala	Leu
1655						1660					1665			
Arg	His	Trp	Glu	Asp	Ile	Pro	Arg	Ala	Gly	Cys	Arg	Leu	Val	Pro
1670						1675					1680			
Glu	His	Val	Glu	Thr	Arg	Pro	Leu	Leu	Asn	Pro	Asp	Lys	Pro	Gly
1685						1690					1695			
Ile	Glu	Gln	Gly	Arg	Leu	Glu	Leu	Trp	Val	Asp	Met	Phe	Pro	Met
1700						1705					1710			
Asp	Met	Pro	Ala	Pro	Gly	Thr	Pro	Leu	Asp	Ile	Ser	Pro	Arg	Lys
1715						1720					1725			
Pro	Lys	Lys	Tyr	Glu	Leu	Arg	Val	Ile	Ile	Trp	Asn	Thr	Asp	Glu
1730						1735					1740			
Val	Val	Leu	Glu	Asp	Asp	Asp	Phe	Phe	Thr	Gly	Glu	Lys	Ser	Ser
1745						1750					1755			
Asp	Ile	Phe	Val	Arg	Gly	Trp	Leu	Lys	Gly	Gln	Gln	Glu	Asp	Lys
1760						1765					1770			
Gln	Asp	Thr	Asp	Val	His	Tyr	His	Ser	Leu	Thr	Gly	Glu	Gly	Asn
1775						1780					1785			
Phe	Asn	Trp	Arg	Tyr	Leu	Phe	Pro	Phe	Asp	Tyr	Leu	Ala	Ala	Glu
1790						1795					1800			
Glu	Lys	Ile	Val	Ile	Ser	Lys	Lys	Glu	Ser	Met	Phe	Ser	Trp	Asp
1805						1810					1815			
Glu	Thr	Glu	Tyr	Lys	Ile	Pro	Ala	Arg	Leu	Thr	Leu	Gln	Ile	Trp
1820						1825					1830			
Asp	Ala	Asp	His	Phe	Ser	Ala	Asp	Asp	Phe	Leu	Gly	Ala	Ile	Glu
1835						1840					1845			
Leu	Asp	Leu	Asn	Arg	Phe	Pro	Arg	Gly	Ala	Lys	Thr	Ala	Lys	Gln
1850						1855					1860			
Cys	Thr	Met	Glu	Met	Ala	Thr	Gly	Glu	Val	Asp	Val	Pro	Leu	Val
1865						1870					1875			
Ser	Ile	Phe	Lys	Gln	Lys	Arg	Val	Lys	Gly	Trp	Trp	Pro	Leu	Leu
1880						1885					1890			
Ala	Arg	Asn	Glu	Asn	Asp	Glu	Phe	Glu	Leu	Thr	Gly	Lys	Val	Glu
1895						1900					1905			
Ala	Glu	Leu	His	Leu	Leu	Thr	Ala	Glu	Glu	Ala	Glu	Lys	Asn	Pro
1910						1915					1920			
Val	Gly	Leu	Ala	Arg	Asn	Glu	Pro	Asp	Pro	Leu	Glu	Lys	Pro	Asn
1925						1930					1935			
Arg	Pro	Asp	Thr	Ser	Phe	Ile	Trp	Phe	Leu	Asn	Pro	Leu	Lys	Ser
1940						1945					1950			
Ala	Arg	Tyr	Phe	Leu	Trp	His	Thr	Tyr	Arg	Trp	Leu	Leu	Leu	Lys
1955						1960					1965			
Leu	Leu	Leu	Leu	Leu	Leu	Leu	Leu	Leu	Leu	Leu	Ala	Leu	Phe	Leu
1970						1975					1980			
Tyr	Ser	Val	Pro	Gly	Tyr	Leu	Val	Lys	Lys	Ile	Leu	Gly	Ala	

-continued

1985	1990	1995
<210> SEQ ID NO 2		
<211> LENGTH: 1230		
<212> TYPE: PRT		
<213> ORGANISM: Homo sapiens		
<400> SEQUENCE: 2		
Met Ile Lys Thr Glu Lys Ser Tyr Pro Glu Arg Arg Leu Arg Gly Val		
1	5	10 15
Leu Glu Glu Leu Ser Cys Gly Cys Cys Arg Phe Leu Ser Leu Ala Asp		
	20	25 30
Lys Asp Gln Gly His Ser Ser Arg Thr Arg Leu Asp Arg Glu Arg Leu		
	35	40 45
Lys Ser Cys Met Arg Glu Leu Glu Asn Met Gly Gln Gln Ala Arg Met		
	50	55 60
Leu Arg Ala Gln Val Lys Arg His Thr Val Arg Asp Lys Leu Arg Leu		
	65	70 75 80
Cys Gln Asn Phe Leu Gln Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln		
	85	90 95
His Ser Ile Pro Asp Ile Phe Ile Trp Met Met Ser Asn Asn Lys Arg		
	100	105 110
Val Ala Tyr Ala Arg Val Pro Ser Lys Asp Leu Leu Phe Ser Ile Val		
	115	120 125
Glu Glu Glu Thr Gly Lys Asp Cys Ala Lys Val Lys Thr Leu Phe Leu		
	130	135 140
Lys Leu Pro Gly Lys Arg Gly Phe Gly Ser Ala Gly Trp Thr Val Gln		
	145	150 155 160
Ala Lys Val Glu Leu Tyr Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys		
	165	170 175
Glu Phe Leu Cys Gly Leu Pro Cys Gly Phe Gln Glu Val Lys Ala Ala		
	180	185 190
Gln Gly Leu Gly Leu His Ala Phe Pro Pro Val Ser Leu Val Tyr Thr		
	195	200 205
Lys Lys Gln Ala Phe Gln Leu Arg Ala His Met Tyr Gln Ala Arg Ser		
	210	215 220
Leu Phe Ala Ala Asp Ser Ser Gly Leu Ser Asp Pro Phe Ala Arg Val		
	225	230 235 240
Phe Phe Ile Asn Gln Ser Gln Cys Thr Glu Val Leu Asn Glu Thr Leu		
	245	250 255
Cys Pro Thr Trp Asp Gln Met Leu Val Phe Asp Asn Leu Glu Leu Tyr		
	260	265 270
Gly Glu Ala His Glu Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu		
	275	280 285
Ile Tyr Asp Gln Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr		
	290	295 300
Phe Ala Lys Pro Leu Val Lys Met Ala Asp Glu Ala Tyr Cys Pro Pro		
	305	310 315 320
Arg Phe Pro Pro Gln Leu Glu Tyr Tyr Gln Ile Tyr Arg Gly Asn Ala		
	325	330 335
Thr Ala Gly Asp Leu Leu Ala Ala Phe Glu Leu Leu Gln Ile Gly Pro		
	340	345 350
Ala Gly Lys Ala Asp Leu Pro Pro Ile Asn Gly Pro Val Asp Val Asp		
	355	360 365

-continued

Arg Gly Pro Ile Met Pro Val Pro Met Gly Ile Arg Pro Val Leu Ser	
370 375 380	
Lys Tyr Arg Val Glu Val Leu Phe Trp Gly Leu Arg Asp Leu Lys Arg	
385 390 395 400	
Val Asn Leu Ala Gln Val Asp Arg Pro Arg Val Asp Ile Glu Cys Ala	
405 410 415	
Gly Lys Gly Val Gln Ser Ser Leu Ile His Asn Tyr Lys Lys Asn Pro	
420 425 430	
Asn Phe Asn Thr Leu Val Lys Trp Phe Glu Val Asp Leu Pro Glu Asn	
435 440 445	
Glu Leu Leu His Pro Pro Leu Asn Ile Arg Val Val Asp Cys Arg Ala	
450 455 460	
Phe Gly Arg Tyr Thr Leu Val Gly Ser His Ala Val Ser Ser Leu Arg	
465 470 475 480	
Arg Phe Ile Tyr Arg Pro Pro Asp Arg Ser Ala Pro Ser Trp Asn Thr	
485 490 495	
Thr Gly Glu Val Val Val Thr Met Glu Pro Glu Val Pro Ile Lys Lys	
500 505 510	
Leu Glu Thr Met Val Lys Leu Asp Ala Thr Ser Glu Ala Val Val Lys	
515 520 525	
Val Asp Val Ala Glu Glu Glu Lys Glu Lys Lys Lys Lys Lys Gly	
530 535 540	
Thr Ala Glu Glu Pro Glu Glu Glu Pro Asp Glu Ser Met Leu Asp	
545 550 555 560	
Trp Trp Ser Lys Tyr Phe Ala Ser Ile Asp Thr Met Lys Glu Gln Leu	
565 570 575	
Arg Gln Gln Glu Pro Ser Gly Ile Asp Leu Glu Glu Lys Glu Glu Val	
580 585 590	
Asp Asn Thr Glu Gly Leu Lys Gly Ser Met Lys Gly Lys Glu Lys Ala	
595 600 605	
Arg Ala Ala Lys Glu Glu Lys Lys Lys Lys Thr Gln Ser Ser Gly Ser	
610 615 620	
Gly Gln Gly Ser Glu Ala Pro Glu Lys Lys Lys Pro Lys Ile Asp Glu	
625 630 635 640	
Leu Lys Val Tyr Pro Lys Glu Leu Glu Ser Glu Phe Asp Asn Phe Glu	
645 650 655	
Asp Trp Leu His Thr Phe Asn Leu Leu Arg Gly Lys Thr Gly Asp Asp	
660 665 670	
Glu Asp Gly Ser Thr Glu Glu Glu Arg Ile Val Gly Arg Phe Lys Gly	
675 680 685	
Ser Leu Cys Val Tyr Lys Val Pro Leu Pro Glu Asp Val Ser Arg Glu	
690 695 700	
Ala Gly Tyr Asp Ser Thr Tyr Gly Met Phe Gln Gly Ile Pro Ser Asn	
705 710 715 720	
Asp Pro Ile Asn Val Leu Val Arg Val Tyr Val Val Arg Ala Thr Asp	
725 730 735	
Leu His Pro Ala Asp Ile Asn Gly Lys Ala Asp Pro Tyr Ile Ala Ile	
740 745 750	
Arg Leu Gly Lys Thr Asp Ile Arg Asp Lys Glu Asn Tyr Ile Ser Lys	
755 760 765	
Gln Leu Asn Pro Val Phe Gly Lys Ser Phe Asp Ile Glu Ala Ser Phe	
770 775 780	
Pro Met Glu Ser Met Leu Thr Val Ala Val Tyr Asp Trp Asp Leu Val	

-continued

785	790	795	800
Gly Thr Asp Asp Leu Ile Gly Glu Thr Lys Ile Asp Leu Glu Asn Arg	805	810	815
Phe Tyr Ser Lys His Arg Ala Thr Cys Gly Ile Ala Gln Thr Tyr Ser	820	825	830
Thr His Gly Tyr Asn Ile Trp Arg Asp Pro Met Lys Pro Ser Gln Ile	835	840	845
Leu Thr Arg Leu Cys Lys Asp Gly Lys Val Asp Gly Pro His Phe Gly	850	855	860
Pro Pro Gly Arg Val Lys Val Ala Asn Arg Val Phe Thr Gly Pro Ser	865	870	875
Glu Ile Glu Asp Glu Asn Gly Gln Arg Lys Pro Thr Asp Glu His Val	885	890	895
Ala Leu Leu Ala Leu Arg His Trp Glu Asp Ile Pro Arg Ala Gly Cys	900	905	910
Arg Leu Val Pro Glu His Val Glu Thr Arg Pro Leu Leu Asn Pro Asp	915	920	925
Lys Pro Gly Ile Glu Gln Gly Arg Leu Glu Leu Trp Val Asp Met Phe	930	935	940
Pro Met Asp Met Pro Ala Pro Gly Thr Pro Leu Asp Ile Ser Pro Arg	945	950	955
Lys Pro Lys Lys Tyr Glu Leu Arg Val Ile Ile Trp Asn Thr Asp Glu	965	970	975
Val Val Leu Glu Asp Asp Asp Phe Phe Thr Gly Glu Lys Ser Ser Asp	980	985	990
Ile Phe Val Arg Gly Trp Leu Lys Gly Gln Gln Glu Asp Lys Gln Asp	995	1000	1005
Thr Asp Val His Tyr His Ser Leu Thr Gly Glu Gly Asn Phe Asn	1010	1015	1020
Trp Arg Tyr Leu Phe Pro Phe Asp Tyr Leu Ala Ala Glu Glu Lys	1025	1030	1035
Ile Val Ile Ser Lys Lys Glu Ser Met Phe Ser Trp Asp Glu Thr	1040	1045	1050
Glu Tyr Lys Ile Pro Ala Arg Leu Thr Leu Gln Ile Trp Asp Ala	1055	1060	1065
Asp His Phe Ser Ala Asp Asp Phe Leu Gly Ala Ile Glu Leu Asp	1070	1075	1080
Leu Asn Arg Phe Pro Arg Gly Ala Lys Thr Ala Lys Gln Cys Thr	1085	1090	1095
Met Glu Met Ala Thr Gly Glu Val Asp Val Pro Leu Val Ser Ile	1100	1105	1110
Phe Lys Gln Lys Arg Val Lys Gly Trp Trp Pro Leu Leu Ala Arg	1115	1120	1125
Asn Glu Asn Asp Glu Phe Glu Leu Thr Gly Lys Val Glu Ala Glu	1130	1135	1140
Leu His Leu Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro Val Gly	1145	1150	1155
Leu Ala Arg Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn Arg Pro	1160	1165	1170
Asp Thr Ser Phe Ile Trp Phe Leu Asn Pro Leu Lys Ser Ala Arg	1175	1180	1185
Tyr Phe Leu Trp His Thr Tyr Arg Trp Leu Leu Leu Lys Leu Leu	1190	1195	1200

-continued

Leu Leu Leu Leu Leu Leu Leu Leu Ala Leu Phe Leu Tyr Ser
1205 1210 1215

Val Pro Gly Tyr Leu Val Lys Lys Ile Leu Gly Ala
1220 1225 1230

<210> SEQ ID NO 3

<211> LENGTH: 1230

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 3

Met Ile Lys Thr Glu Lys Ser Tyr Pro Glu Arg Arg Leu Arg Gly Val
1 5 10 15

Leu Glu Glu Leu Ser Cys Gly Cys Cys Arg Phe Leu Ser Leu Ala Asp
20 25 30

Lys Asp Gln Gly His Ser Ser Arg Thr Arg Leu Asp Arg Glu Arg Leu
35 40 45

Lys Ser Cys Met Arg Glu Leu Glu Asn Met Gly Gln Gln Ala Arg Met
50 55 60

Leu Arg Ala Gln Val Lys Arg His Thr Val Arg Asp Lys Leu Arg Leu
65 70 75 80

Cys Gln Asn Phe Leu Gln Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln
85 90 95

His Ser Ile Pro Asp Ile Phe Ile Trp Met Met Ser Asn Asn Lys Arg
100 105 110

Val Ala Tyr Ala Arg Val Pro Ser Lys Asp Leu Leu Phe Ser Ile Val
115 120 125

Glu Glu Glu Thr Gly Lys Asp Cys Ala Lys Val Lys Thr Leu Phe Leu
130 135 140

Lys Leu Pro Gly Lys Arg Gly Phe Gly Ser Ala Gly Trp Thr Val Gln
145 150 155 160

Ala Lys Val Glu Leu Tyr Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys
165 170 175

Glu Phe Leu Cys Gly Leu Pro Cys Gly Phe Gln Glu Val Lys Ala Ala
180 185 190

Gln Gly Leu Gly Leu His Ala Phe Pro Pro Val Ser Leu Val Tyr Thr
195 200 205

Lys Lys Gln Ala Phe Gln Leu Arg Ala His Met Tyr Gln Ala Arg Ser
210 215 220

Leu Phe Ala Ala Asp Ser Ser Gly Leu Ser Asp Pro Phe Ala Arg Val
225 230 235 240

Phe Phe Ile Asn Gln Ser Gln Cys Thr Glu Val Leu Asn Glu Thr Leu
245 250 255

Cys Pro Thr Trp Asp Gln Met Leu Val Phe Asp Asn Leu Glu Leu Tyr
260 265 270

Gly Glu Ala His Glu Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu
275 280 285

Ile Tyr Asp Gln Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr
290 295 300

Phe Ala Lys Pro Leu Val Lys Met Ala Asp Glu Ala Tyr Cys Pro Pro
305 310 315 320

Arg Phe Pro Pro Gln Leu Glu Tyr Tyr Gln Ile Tyr Arg Gly Asn Ala
325 330 335

Thr Ala Gly Asp Leu Leu Ala Ala Phe Glu Leu Leu Gln Ile Gly Pro

-continued

340					345					350					
Ala	Gly	Lys	Ala	Asp	Leu	Pro	Pro	Ile	Asn	Gly	Pro	Val	Asp	Val	Asp
		355					360					365			
Arg	Gly	Pro	Ile	Met	Pro	Val	Pro	Met	Gly	Ile	Arg	Pro	Val	Leu	Ser
		370				375					380				
Lys	Tyr	Arg	Val	Glu	Val	Leu	Phe	Trp	Gly	Leu	Arg	Asp	Leu	Lys	Arg
		385			390					395					400
Val	Asn	Leu	Ala	Gln	Val	Asp	Arg	Pro	Arg	Val	Asp	Ile	Glu	Cys	Ala
				405					410					415	
Gly	Lys	Gly	Val	Gln	Ser	Ser	Leu	Ile	His	Asn	Tyr	Lys	Lys	Asn	Pro
			420					425					430		
Asn	Phe	Asn	Thr	Leu	Val	Lys	Trp	Phe	Glu	Val	Asp	Leu	Pro	Glu	Asn
		435					440					445			
Glu	Leu	Leu	His	Pro	Pro	Leu	Asn	Ile	Arg	Val	Val	Asp	Cys	Arg	Ala
		450				455					460				
Phe	Gly	Arg	Tyr	Thr	Leu	Val	Gly	Ser	His	Ala	Val	Ser	Ser	Leu	Arg
		465			470					475					480
Arg	Phe	Ile	Tyr	Arg	Pro	Pro	Asp	Arg	Ser	Ala	Pro	Ser	Trp	Asn	Thr
				485					490					495	
Thr	Gly	Glu	Val	Val	Val	Thr	Met	Glu	Pro	Glu	Val	Pro	Ile	Lys	Lys
			500					505					510		
Leu	Glu	Thr	Met	Val	Lys	Leu	Asp	Ala	Thr	Ser	Glu	Ala	Val	Val	Lys
		515					520					525			
Val	Asp	Val	Ala	Glu	Glu	Glu	Lys	Glu	Lys	Lys	Lys	Lys	Lys	Lys	Gly
		530				535					540				
Thr	Ala	Glu	Glu	Pro	Glu	Glu	Glu	Glu	Pro	Asp	Glu	Ser	Met	Leu	Asp
					550					555					560
Trp	Trp	Ser	Lys	Tyr	Phe	Ala	Ser	Ile	Asp	Thr	Met	Lys	Glu	Gln	Leu
				565					570					575	
Arg	Gln	Gln	Glu	Pro	Ser	Gly	Ile	Asp	Leu	Glu	Glu	Lys	Glu	Glu	Val
			580					585					590		
Asp	Asn	Thr	Glu	Gly	Leu	Lys	Gly	Ser	Met	Lys	Gly	Lys	Glu	Lys	Ala
		595					600					605			
Arg	Ala	Ala	Lys	Glu	Glu	Lys	Lys	Lys	Lys	Thr	Gln	Ser	Ser	Gly	Ser
		610				615					620				
Gly	Gln	Gly	Ser	Glu	Ala	Pro	Glu	Lys	Lys	Lys	Pro	Lys	Ile	Asp	Glu
					630					635					640
Leu	Lys	Val	Tyr	Pro	Lys	Glu	Leu	Glu	Ser	Glu	Phe	Asp	Asn	Phe	Glu
				645					650					655	
Asp	Trp	Leu	His	Thr	Phe	Asn	Leu	Leu	Arg	Gly	Lys	Thr	Gly	Asp	Asp
			660					665					670		
Glu	Asp	Gly	Ser	Thr	Glu	Glu	Glu	Arg	Ile	Val	Gly	Arg	Phe	Lys	Gly
		675					680					685			
Ser	Leu	Cys	Val	Tyr	Lys	Val	Pro	Leu	Pro	Glu	Asp	Val	Ser	Arg	Glu
						695					700				
Ala	Gly	Tyr	Asp	Ser	Thr	Tyr	Gly	Met	Phe	Gln	Gly	Ile	Pro	Ser	Asn
					710					715					720
Asp	Pro	Ile	Asn	Val	Leu	Val	Arg	Val	Tyr	Val	Val	Arg	Ala	Thr	Asp
				725					730					735	
Leu	His	Pro	Ala	Asp	Ile	Asn	Gly	Lys	Ala	Asp	Pro	Tyr	Ile	Ala	Ile
			740					745					750		
Arg	Leu	Gly	Lys	Thr	Asp	Ile	Arg	Asp	Lys	Glu	Asn	Tyr	Ile	Ser	Lys
		755					760					765			

-continued

Gln Leu Asn Pro Val Phe Gly Lys Ser Phe Asp Ile Glu Ala Ser Phe
 770 775 780
 Pro Met Glu Ser Met Leu Thr Val Ala Val Tyr Asp Trp Asp Leu Val
 785 790 795 800
 Gly Thr Asp Asp Leu Ile Gly Glu Thr Lys Ile Asp Leu Glu Asn Arg
 805 810 815
 Phe Tyr Ser Lys His Arg Ala Thr Cys Gly Ile Ala Gln Thr Tyr Ser
 820 825 830
 Thr His Gly Tyr Asn Ile Trp Arg Asp Pro Met Lys Pro Ser Gln Ile
 835 840 845
 Leu Thr Arg Leu Cys Lys Asp Gly Lys Val Asp Gly Pro His Phe Gly
 850 855 860
 Pro Pro Gly Arg Val Lys Val Ala Asn Arg Val Phe Thr Gly Pro Ser
 865 870 875 880
 Glu Ile Glu Asp Glu Asn Gly Gln Arg Lys Pro Thr Asp Glu His Val
 885 890 895
 Ala Leu Leu Ala Leu Arg His Trp Glu Asp Ile Pro Arg Ala Gly Cys
 900 905 910
 Arg Leu Val Pro Glu His Val Glu Thr Arg Pro Leu Leu Asn Pro Asp
 915 920 925
 Lys Pro Gly Ile Glu Gln Gly Arg Leu Glu Leu Trp Val Asp Met Phe
 930 935 940
 Pro Met Asp Met Pro Ala Pro Gly Thr Pro Leu Asp Ile Ser Pro Arg
 945 950 955 960
 Lys Pro Lys Lys Tyr Glu Leu Arg Val Ile Ile Trp Asn Thr Asp Glu
 965 970 975
 Val Val Leu Glu Asp Asp Asp Phe Phe Thr Gly Glu Lys Ser Ser Asp
 980 985 990
 Ile Phe Val Arg Gly Trp Leu Lys Gly Gln Gln Glu Asp Lys Gln Asp
 995 1000 1005
 Thr Asp Val His Tyr His Ser Leu Thr Gly Glu Gly Asn Phe Asn
 1010 1015 1020
 Trp Arg Tyr Leu Phe Pro Phe Asp Tyr Leu Ala Ala Glu Glu Lys
 1025 1030 1035
 Ile Val Ile Ser Lys Lys Glu Ser Met Phe Ser Trp Asp Glu Thr
 1040 1045 1050
 Glu Tyr Lys Ile Pro Ala Arg Leu Thr Leu Gln Ile Trp Asp Ala
 1055 1060 1065
 Asp His Phe Ser Ala Asp Asp Phe Leu Gly Ala Ile Glu Leu Asp
 1070 1075 1080
 Leu Asn Arg Phe Pro Arg Gly Ala Lys Thr Ala Lys Gln Cys Thr
 1085 1090 1095
 Met Glu Met Ala Thr Gly Glu Val Asp Val Pro Leu Val Ser Ile
 1100 1105 1110
 Phe Lys Gln Lys Arg Val Lys Gly Trp Trp Pro Leu Leu Ala Arg
 1115 1120 1125
 Asn Glu Asn Asp Glu Phe Glu Leu Thr Gly Lys Val Glu Ala Glu
 1130 1135 1140
 Leu His Leu Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro Val Gly
 1145 1150 1155
 Leu Ala Arg Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn Arg Pro
 1160 1165 1170

-continued

```

Asp Thr  Ala Phe Val Trp Phe  Leu Asn Pro Leu Lys  Ser Ile Lys
1175                      1180                      1185

Tyr Leu   Ile Cys Thr Arg Tyr  Lys Trp Leu Ile Ile  Lys Ile Val
1190                      1195                      1200

Leu Ala   Leu Leu Gly Leu Leu  Met Leu Gly Leu Phe  Leu Tyr Ser
1205                      1210                      1215

Leu Pro   Gly Tyr Met Val Lys  Lys Leu Leu Gly Ala
1220                      1225                      1230

```

```

<210> SEQ ID NO 4
<211> LENGTH: 1307
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 4

```

```

Met Met  Thr Asp Thr Gln Asp Gly Pro Ser Glu Ser Ser Gln Ile Met
1          5              10              15

Arg Ser  Leu Thr Pro Leu Ile Asn Arg Glu Glu Ala Phe Gly Glu Ala
20        25              30

Gly Glu  Ala Gly Leu Trp Pro Ser Ile Thr His Thr Pro Asp Ser Gln
35        40              45

Glu Glu  Gly Leu Asn Asp Ile Gln Glu Met Ile Lys Thr Glu Lys Ser
50        55              60

Tyr Pro  Glu Arg Arg Leu Arg Gly Val Leu Glu Leu Ser Cys Gly
65        70              75              80

Cys Cys  Arg Phe Leu Ser Leu Ala Asp Lys Asp Gln Gly His Ser Ser
85        90              95

Arg Thr  Arg Leu Asp Arg Glu Arg Leu Lys Ser Cys Met Arg Glu Leu
100       105             110

Glu Asn  Met Gly Gln Gln Ala Arg Met Leu Arg Ala Gln Val Lys Arg
115       120             125

His Thr  Val Arg Asp Lys Leu Arg Leu Cys Gln Asn Phe Leu Gln Lys
130       135             140

Leu Arg  Phe Leu Ala Asp Glu Pro Gln His Ser Ile Pro Asp Ile Phe
145       150             155             160

Ile Trp  Met Met Ser Asn Asn Lys Arg Val Ala Tyr Ala Arg Val Pro
165       170             175

Ser Lys  Asp Leu Leu Phe Ser Ile Val Glu Glu Glu Thr Gly Lys Asp
180       185             190

Cys Ala  Lys Val Lys Thr Leu Phe Leu Lys Leu Pro Gly Lys Arg Gly
195       200             205

Phe Gly  Ser Ala Gly Trp Thr Val Gln Ala Lys Val Glu Leu Tyr Leu
210       215             220

Trp Leu  Gly Leu Ser Lys Gln Arg Lys Glu Phe Leu Cys Gly Leu Pro
225       230             235             240

Cys Gly  Phe Gln Glu Val Lys Ala Ala Gln Gly Leu Gly Leu His Ala
245       250             255

Phe Pro  Pro Val Ser Leu Val Tyr Thr Lys Lys Gln Ala Phe Gln Leu
260       265             270

Arg Ala  His Met Tyr Gln Ala Arg Ser Leu Phe Ala Ala Asp Ser Ser
275       280             285

Gly Leu  Ser Asp Pro Phe Ala Arg Val Phe Phe Ile Asn Gln Ser Gln
290       295             300

Cys Thr  Glu Val Leu Asn Glu Thr Leu Cys Pro Thr Trp Asp Gln Met
305       310             315             320

```

-continued

Leu Val Phe Asp Asn Leu Glu Leu Tyr Gly Glu Ala His Glu Leu Arg
 325 330 335
 Asp Asp Pro Pro Ile Ile Val Ile Glu Ile Tyr Asp Gln Asp Ser Met
 340 345 350
 Gly Lys Ala Asp Phe Met Gly Arg Thr Phe Ala Lys Pro Leu Val Lys
 355 360 365
 Met Ala Asp Glu Ala Tyr Cys Pro Pro Arg Phe Pro Pro Gln Leu Glu
 370 375 380
 Tyr Tyr Gln Ile Tyr Arg Gly Asn Ala Thr Ala Gly Asp Leu Leu Ala
 385 390 395 400
 Ala Phe Glu Leu Leu Gln Ile Gly Pro Ala Gly Lys Ala Asp Leu Pro
 405 410 415
 Pro Ile Asn Gly Pro Val Asp Val Asp Arg Gly Pro Ile Met Pro Val
 420 425 430
 Pro Met Gly Ile Arg Pro Val Leu Ser Lys Tyr Arg Val Glu Val Leu
 435 440 445
 Phe Trp Gly Leu Arg Asp Leu Lys Arg Val Asn Leu Ala Gln Val Asp
 450 455 460
 Arg Pro Arg Val Asp Ile Glu Cys Ala Gly Lys Gly Val Gln Ser Ser
 465 470 475 480
 Leu Ile His Asn Tyr Lys Lys Asn Pro Asn Phe Asn Thr Leu Val Lys
 485 490 495
 Trp Phe Glu Val Asp Leu Pro Glu Asn Glu Leu Leu His Pro Pro Leu
 500 505 510
 Asn Ile Arg Val Val Asp Cys Arg Ala Phe Gly Arg Tyr Thr Leu Val
 515 520 525
 Gly Ser His Ala Val Ser Ser Leu Arg Arg Phe Ile Tyr Arg Pro Pro
 530 535 540
 Asp Arg Ser Ala Pro Ser Trp Asn Thr Thr Val Arg Leu Leu Arg Arg
 545 550 555 560
 Cys Arg Val Leu Cys Asn Gly Gly Ser Ser Ser His Ser Thr Gly Glu
 565 570 575
 Val Val Val Thr Met Glu Pro Glu Val Pro Ile Lys Lys Leu Glu Thr
 580 585 590
 Met Val Lys Leu Asp Ala Thr Ser Glu Ala Val Val Lys Val Asp Val
 595 600 605
 Ala Glu Glu Glu Lys Glu Lys Lys Lys Lys Lys Lys Gly Thr Ala Glu
 610 615 620
 Glu Pro Glu Glu Glu Glu Pro Asp Glu Ser Met Leu Asp Trp Trp Ser
 625 630 635 640
 Lys Tyr Phe Ala Ser Ile Asp Thr Met Lys Glu Gln Leu Arg Gln Gln
 645 650 655
 Glu Pro Ser Gly Ile Asp Leu Glu Glu Lys Glu Glu Val Asp Asn Thr
 660 665 670
 Glu Gly Leu Lys Gly Ser Met Lys Gly Lys Glu Lys Ala Arg Ala Ala
 675 680 685
 Lys Glu Glu Lys Lys Lys Lys Thr Gln Ser Ser Gly Ser Gly Gln Gly
 690 695 700
 Ser Glu Ala Pro Glu Lys Lys Lys Pro Lys Ile Asp Glu Leu Lys Val
 705 710 715 720
 Tyr Pro Lys Glu Leu Glu Ser Glu Phe Asp Asn Phe Glu Asp Trp Leu
 725 730 735

His	Thr	Phe	Asn	Leu	Leu	Arg	Gly	Lys	Thr	Gly	Asp	Asp	Glu	Asp	Gly
			740					745					750		
Ser	Thr	Glu	Glu	Glu	Arg	Ile	Val	Gly	Arg	Phe	Lys	Gly	Ser	Leu	Cys
		755				760					765				
Val	Tyr	Lys	Val	Pro	Leu	Pro	Glu	Asp	Val	Ser	Arg	Glu	Ala	Gly	Tyr
		770				775					780				
Asp	Ser	Thr	Tyr	Gly	Met	Phe	Gln	Gly	Ile	Pro	Ser	Asn	Asp	Pro	Ile
					790					795					800
Asn	Val	Leu	Val	Arg	Val	Tyr	Val	Val	Arg	Ala	Thr	Asp	Leu	His	Pro
				805					810					815	
Ala	Asp	Ile	Asn	Gly	Lys	Ala	Asp	Pro	Tyr	Ile	Ala	Ile	Arg	Leu	Gly
			820					825					830		
Lys	Thr	Asp	Ile	Arg	Asp	Lys	Glu	Asn	Tyr	Ile	Ser	Lys	Gln	Leu	Asn
		835					840					845			
Pro	Val	Phe	Gly	Lys	Ser	Phe	Asp	Ile	Glu	Ala	Ser	Phe	Pro	Met	Glu
		850				855					860				
Ser	Met	Leu	Thr	Val	Ala	Val	Tyr	Asp	Trp	Asp	Leu	Val	Gly	Thr	Asp
					870					875					880
Asp	Leu	Ile	Gly	Glu	Thr	Lys	Ile	Asp	Leu	Glu	Asn	Arg	Phe	Tyr	Ser
				885					890					895	
Lys	His	Arg	Ala	Thr	Cys	Gly	Ile	Ala	Gln	Thr	Tyr	Ser	Thr	His	Gly
			900					905					910		
Tyr	Asn	Ile	Trp	Arg	Asp	Pro	Met	Lys	Pro	Ser	Gln	Ile	Leu	Thr	Arg
		915				920						925			
Leu	Cys	Lys	Asp	Gly	Lys	Val	Asp	Gly	Pro	His	Phe	Gly	Pro	Pro	Gly
		930				935					940				
Arg	Val	Lys	Val	Ala	Asn	Arg	Val	Phe	Thr	Gly	Pro	Ser	Glu	Ile	Glu
					950					955					960
Asp	Glu	Asn	Gly	Gln	Arg	Lys	Pro	Thr	Asp	Glu	His	Val	Ala	Leu	Leu
				965					970					975	
Ala	Leu	Arg	His	Trp	Glu	Asp	Ile	Pro	Arg	Ala	Gly	Cys	Arg	Leu	Val
			980					985					990		
Pro	Glu	His	Val	Glu	Thr	Arg	Pro	Leu	Leu	Asn	Pro	Asp	Lys	Pro	Gly
		995				1000						1005			
Ile	Glu	Gln	Gly	Arg	Leu	Glu	Leu	Trp	Val	Asp	Met	Phe	Pro	Met	
	1010					1015					1020				
Asp	Met	Pro	Ala	Pro	Gly	Thr	Pro	Leu	Asp	Ile	Ser	Pro	Arg	Lys	
	1025					1030					1035				
Pro	Lys	Lys	Tyr	Glu	Leu	Arg	Val	Ile	Ile	Trp	Asn	Thr	Asp	Glu	
	1040					1045					1050				
Val	Val	Leu	Glu	Asp	Asp	Asp	Phe	Phe	Thr	Gly	Glu	Lys	Ser	Ser	
	1055					1060					1065				
Asp	Ile	Phe	Val	Arg	Gly	Trp	Leu	Lys	Gly	Gln	Gln	Glu	Asp	Lys	
	1070					1075					1080				
Gln	Asp	Thr	Asp	Val											

-continued

1145	1150	1155
Leu Asp Leu Asn Arg Phe Pro Arg Gly Ala Lys Thr Ala Lys Gln		
1160	1165	1170
Cys Thr Met Glu Met Ala Thr Gly Glu Val Asp Val Pro Leu Val		
1175	1180	1185
Ser Ile Phe Lys Gln Lys Arg Val Lys Gly Trp Trp Pro Leu Leu		
1190	1195	1200
Ala Arg Asn Glu Asn Asp Glu Phe Glu Leu Thr Gly Lys Val Glu		
1205	1210	1215
Ala Glu Leu His Leu Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro		
1220	1225	1230
Val Gly Leu Ala Arg Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn		
1235	1240	1245
Arg Pro Asp Thr Ser Phe Ile Trp Phe Leu Asn Pro Leu Lys Ser		
1250	1255	1260
Ala Arg Tyr Phe Leu Trp His Thr Tyr Arg Trp Leu Leu Leu Lys		
1265	1270	1275
Leu Leu Leu Leu Leu Leu Leu Leu Leu Leu Ala Leu Phe Leu		
1280	1285	1290
Tyr Ser Val Pro Gly Tyr Leu Val Lys Lys Ile Leu Gly Ala		
1295	1300	1305

<210> SEQ ID NO 5

<211> LENGTH: 1997

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Met Ala Leu Leu Ile His Leu Lys Thr Val Ser Glu Leu Arg Gly Arg		
1	5	10 15
Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser		
20	25	30
Arg Val Leu Glu Asn Cys Glu Asp Val Ala Asp Phe Asp Glu Thr Phe		
35	40	45
Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Met Leu Glu Ile		
50	55	60
Gln Val Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr		
65	70	75 80
Phe Arg Met Val Leu Gln Lys Val Val Glu Glu Ser His Val Glu Val		
85	90	95
Thr Asp Thr Leu Ile Asp Asp Asn Asn Ala Ile Ile Lys Thr Ser Leu		
100	105	110
Cys Val Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Ser Trp		
115	120	125
Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Glu Lys		
130	135	140
Asp Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Ser		
145	150	155 160
Arg Pro Pro Gly Glu Lys Ser Phe Arg Arg Ala Gly Arg Ser Val Phe		
165	170	175
Ser Ala Met Lys Leu Gly Lys Asn Arg Ser His Lys Glu Glu Pro Gln		
180	185	190
Arg Pro Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu		
195	200	205

Ala 210	Ile	Arg	Leu	Gly	Asp	Gly 215	Leu	Asp	Pro	Asp	Ser 220	Val	Ser	Leu	Ala
Ser 225	Val	Thr	Ala	Leu	Thr	Thr	Asn	Val	Ser	Asn	Lys	Arg	Ser	Lys	Pro 240
Asp	Ile	Lys	Met	Glu 245	Pro	Ser	Ala	Gly	Arg 250	Pro	Met	Asp	Tyr	Gln 255	Val
Ser	Ile	Thr	Val 260	Ile	Glu	Ala	Arg	Gln 265	Leu	Val	Gly	Leu	Asn	Met	Asp
Pro	Val	Val	Cys 275	Val	Glu	Val	Gly 280	Asp	Asp	Lys	Lys	Tyr 285	Thr	Ser	Met
Lys 290	Glu	Ser	Thr	Asn	Cys	Pro	Tyr	Tyr	Asn	Glu	Tyr	Phe	Val	Phe	Asp
Phe 305	His	Val	Ser	Pro	Asp 310	Val	Met	Phe	Asp	Lys 315	Ile	Ile	Lys	Ile	Ser 320
Val	Ile	His	Ser	Lys 325	Asn	Leu	Leu	Arg	Ser 330	Gly	Thr	Leu	Val	Gly 335	Ser
Phe	Lys	Met	Asp 340	Val	Gly	Thr	Val	Tyr	Ser 345	Gln	Pro	Glu	His	Gln	Phe
His	His	Lys 355	Trp	Ala	Ile	Leu	Ser 360	Asp	Pro	Asp	Asp	Ile 365	Ser	Ser	Gly
Leu 370	Lys	Gly	Tyr	Val	Lys	Cys 375	Asp	Val	Ala	Val	Val	Gly 380	Lys	Gly	Asp
Asn 385	Ile	Lys	Thr	Pro	His 390	Lys	Ala	Asn	Glu	Thr 395	Asp	Glu	Asp	Asp	Ile 400
Glu	Gly	Asn	Leu	Leu 405	Leu	Pro	Glu	Gly	Val 410	Pro	Pro	Glu	Arg	Gln 415	Trp
Ala	Arg	Phe	Tyr 420	Val	Lys	Ile	Tyr	Arg 425	Ala	Glu	Gly	Leu	Pro	Arg	Met
Asn	Thr	Ser 435	Leu	Met	Ala	Asn 440	Val	Lys	Lys	Ala	Phe	Ile 445	Gly	Glu	Asn
Lys 450	Asp	Leu	Val	Asp	Pro	Tyr 455	Val	Gln	Val	Phe	Phe	Ala	Gly	Gln	Lys
Gly 465	Lys	Thr	Ser	Val	Gln 470	Lys	Ser	Ser	Tyr	Glu 475	Pro	Leu	Trp	Asn	Glu 480
Gln	Val	Val	Phe	Thr 485	Asp	Leu	Phe	Pro	Pro 490	Leu	Cys	Lys	Arg	Met 495	Lys
Val	Gln	Ile	Arg 500	Asp	Ser	Asp	Lys	Val	Asn 505	Asp	Val	Ala	Ile 510	Gly	Thr
His	Phe	Ile 515	Asp	Leu	Arg	Lys 520	Ile	Ser	Asn	Asp	Gly	Asp 525	Lys	Gly	Phe
Leu	Pro 530	Thr	Leu	Gly	Pro	Ala 535	Trp	Val	Asn	Met	Tyr 540	Gly	Ser	Thr	Arg
Asn 545	Tyr	Thr	Leu	Leu	Asp 550	Glu	His	Gln	Asp	Leu 555	Asn	Glu	Gly	Leu	Gly 560
Glu	Gly	Val	Ser	Phe 565	Arg	Ala	Arg	Leu	Leu 570	Leu	Gly	Leu	Ala	Val 575	Glu
Ile	Val	Asp 580	Thr	Ser	Asn	Pro	Glu	Leu 585	Thr	Ser	Ser	Thr	Glu	Val 590	Gln
Val	Glu	Gln 595	Ala	Thr	Pro	Ile 600	Ser	Glu	Ser	Cys	Ala	Gly 605	Lys	Met	Glu
Glu 610	Phe	Phe	Leu	Phe	Gly	Ala 615	Phe	Leu	Glu	Ala	Ser 620	Met	Ile	Asp	Arg
Arg	Asn	Gly	Asp	Lys	Pro	Ile	Thr	Phe	Glu	Val	Thr	Ile	Gly	Asn	Tyr

-continued

625	630	635	640
Gly Asn Glu Val Asp 645	Gly Leu Ser Arg 650	Pro Gln Arg Pro Arg Pro Arg 655	
Lys Glu Pro Gly Asp Glu Glu Glu Val Asp 660	Leu Ile Gln Asn Ala Ser 665		
Asp Asp Glu Ala Gly Asp Ala Gly Asp Leu Ala Ser Val Ser Ser Thr 675		680	685
Pro Pro Met Arg Pro Gln Val Thr Asp Arg Asn Tyr Phe His Leu Pro 690		695	700
Tyr Leu Glu Arg Lys Pro Cys Ile Tyr Ile Lys Ser Trp Trp Pro Asp 705		710	715
Gln Arg Arg Arg Leu Tyr Asn Ala Asn Ile Met Asp His Ile Ala Asp 725		730	735
Lys Leu Glu Glu Gly Leu Asn Asp Ile Gln Glu Met Ile Lys Thr Glu 740		745	750
Lys Ser Tyr Pro Glu Arg Arg Leu Arg Gly Val Leu Glu Glu Leu Ser 755		760	765
Cys Gly Cys Cys Arg Phe Leu Ser Leu Ala Asp Lys Asp Gln Gly His 770		775	780
Ser Ser Arg Thr Arg Leu Asp Arg Glu Arg Leu Lys Ser Cys Met Arg 785		790	795
Glu Leu Glu Asn Met Gly Gln Gln Ala Arg Met Leu Arg Ala Gln Val 805		810	815
Lys Arg His Thr Val Arg Asp Lys Leu Arg Leu Cys Gln Asn Phe Leu 820		825	830
Gln Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln His Ser Ile Pro Asp 835		840	845
Ile Phe Ile Trp Met Met Ser Asn Asn Lys Arg Val Ala Tyr Ala Arg 850		855	860
Val Pro Ser Lys Asp Leu Leu Phe Ser Ile Val Glu Glu Glu Thr Gly 865		870	875
Lys Asp Cys Ala Lys Val Lys Thr Leu Phe Leu Lys Leu Pro Gly Lys 885		890	895
Arg Gly Phe Gly Ser Ala Gly Trp Thr Val Gln Ala Lys Val Glu Leu 900		905	910
Tyr Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys Glu Phe Leu Cys Gly 915		920	925
Leu Pro Cys Gly Phe Gln Glu Val Lys Ala Ala Gln Gly Leu Gly Leu 930		935	940
His Ala Phe Pro Pro Val Ser Leu Val Tyr Thr Lys Lys Gln Ala Phe 945		950	955
Gln Leu Arg Ala His Met Tyr Gln Ala Arg Ser Leu Phe Ala Ala Asp 965		970	975
Ser Ser Gly Leu Ser Asp Pro Phe Ala Arg Val Phe Phe Ile Asn Gln 980		985	990
Ser Gln Cys Thr Glu Val Leu Asn Glu Thr Leu Cys Pro Thr Trp Asp 995		1000	1005
Gln Met Leu Val Phe Asp Asn Leu Glu Leu Tyr Gly Glu Ala His 1010		1015	1020
Glu Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu Ile Tyr Asp 1025		1030	1035
Gln Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr Phe Ala 1040		1045	1050

-continued

Lys Pro	Leu Val	Lys Met	Ala	Asp Glu	Ala Tyr	Cys	Pro Pro	Arg
1055			1060			1065		
Phe Pro	Pro Gln	Leu Glu	Tyr	Tyr Gln	Ile Tyr	Arg	Gly Asn	Ala
1070			1075			1080		
Thr Ala	Gly Asp	Leu Leu	Ala	Ala Phe	Glu Leu	Leu	Gln Ile	Gly
1085			1090			1095		
Pro Ala	Gly Lys	Ala Asp	Leu	Pro Pro	Ile Asn	Gly	Pro Val	Asp
1100			1105			1110		
Val Asp	Arg Gly	Pro Ile	Met	Pro Val	Pro Met	Gly	Ile Arg	Pro
1115			1120			1125		
Val Leu	Ser Lys	Tyr Arg	Val	Glu Val	Leu Phe	Trp	Gly Leu	Arg
1130			1135			1140		
Asp Leu	Lys Arg	Val Asn	Leu	Ala Gln	Val Asp	Arg	Pro Arg	Val
1145			1150			1155		
Asp Ile	Glu Cys	Ala Gly	Lys	Gly Val	Gln Ser	Ser	Leu Ile	His
1160			1165			1170		
Asn Tyr	Lys Lys	Asn Pro	Asn	Phe Asn	Thr Leu	Val	Lys Trp	Phe
1175			1180			1185		
Glu Val	Asp Leu	Pro Glu	Asn	Glu Leu	Leu His	Pro	Pro Leu	Asn
1190			1195			1200		
Ile Arg	Val Val	Asp Cys	Arg	Ala Phe	Gly Arg	Tyr	Thr Leu	Val
1205			1210			1215		
Gly Ser	His Ala	Val Ser	Ser	Leu Arg	Arg Phe	Ile	Tyr Arg	Pro
1220			1225			1230		
Pro Asp	Arg Ser	Ala Pro	Ser	Trp Asn	Thr Thr	Val	Arg Leu	Leu
1235			1240			1245		
Arg Arg	Cys Arg	Val Leu	Cys	Asn Gly	Gly Ser	Ser	Ser His	Ser
1250			1255			1260		
Thr Gly	Glu Val	Val Val	Thr	Met Glu	Pro Glu	Val	Pro Ile	Lys
1265			1270			1275		
Lys Leu	Glu Thr	Met Val	Lys	Leu Asp	Ala Thr	Ser	Glu Ala	Val
1280			1285			1290		
Val Lys	Val Asp	Val Ala	Glu	Glu Glu	Lys Glu	Lys	Lys Lys	Lys
1295			1300			1305		
Lys Lys	Gly Thr	Ala Glu	Glu	Pro Glu	Glu Glu	Glu	Pro Asp	Glu
1310			1315			1320		
Ser Met	Leu Asp	Trp Trp	Ser	Lys Tyr	Phe Ala	Ser	Ile Asp	Thr
1325			1330			1335		
Met Lys	Glu Gln	Leu Arg	Gln	Gln Glu	Pro Ser	Gly	Ile Asp	Leu
1340			1345			1350		
Glu Glu	Lys Glu	Glu Val	Asp	Asn Thr	Glu Gly	Leu	Lys Gly	Ser
1355			1360			1365		
Met Lys	Gly Lys	Glu Lys	Ala	Arg Ala	Ala Lys	Glu	Glu Lys	Lys
1370			1375			1380		
Lys Lys	Thr Gln	Ser Ser	Gly	Ser Gly	Gln Gly	Ser	Glu Ala	Pro
1385			1390			1395		
Glu Lys	Lys Lys	Pro Lys	Ile	Asp Glu	Leu Lys	Val	Tyr Pro	Lys
1400			1405			1410		
Glu Leu	Glu Ser	Glu Phe	Asp	Asn Phe	Glu Asp	Trp	Leu His	Thr
1415			1420			1425		
Phe Asn	Leu Leu	Arg Gly	Lys	Thr Gly	Asp Asp	Glu	Asp Gly	Ser
1430			1435			1440		

-continued

Thr	Glu	Glu	Glu	Arg	Ile	Val	Gly	Arg	Phe	Lys	Gly	Ser	Leu	Cys
1445						1450					1455			
Val	Tyr	Lys	Val	Pro	Leu	Pro	Glu	Asp	Val	Ser	Arg	Glu	Ala	Gly
1460						1465					1470			
Tyr	Asp	Ser	Thr	Tyr	Gly	Met	Phe	Gln	Gly	Ile	Pro	Ser	Asn	Asp
1475						1480					1485			
Pro	Ile	Asn	Val	Leu	Val	Arg	Val	Tyr	Val	Val	Arg	Ala	Thr	Asp
1490						1495					1500			
Leu	His	Pro	Ala	Asp	Ile	Asn	Gly	Lys	Ala	Asp	Pro	Tyr	Ile	Ala
1505						1510					1515			
Ile	Arg	Leu	Gly	Lys	Thr	Asp	Ile	Arg	Asp	Lys	Glu	Asn	Tyr	Ile
1520						1525					1530			
Ser	Lys	Gln	Leu	Asn	Pro	Val	Phe	Gly	Lys	Ser	Phe	Asp	Ile	Glu
1535						1540					1545			
Ala	Ser	Phe	Pro	Met	Glu	Ser	Met	Leu	Thr	Val	Ala	Val	Tyr	Asp
1550						1555					1560			
Trp	Asp	Leu	Val	Gly	Thr	Asp	Asp	Leu	Ile	Gly	Glu	Thr	Lys	Ile
1565						1570					1575			
Asp	Leu	Glu	Asn	Arg	Phe	Tyr	Ser	Lys	His	Arg	Ala	Thr	Cys	Gly
1580						1585					1590			
Ile	Ala	Gln	Thr	Tyr	Ser	Thr	His	Gly	Tyr	Asn	Ile	Trp	Arg	Asp
1595						1600					1605			
Pro	Met	Lys	Pro	Ser	Gln	Ile	Leu	Thr	Arg	Leu	Cys	Lys	Asp	Gly
1610						1615					1620			
Lys	Val	Asp	Gly	Pro	His	Phe	Gly	Pro	Pro	Gly	Arg	Val	Lys	Val
1625						1630					1635			
Ala	Asn	Arg	Val	Phe	Thr	Gly	Pro	Ser	Glu	Ile	Glu	Asp	Glu	Asn
1640						1645					1650			
Gly	Gln	Arg	Lys	Pro	Thr	Asp	Glu	His	Val	Ala	Leu	Leu	Ala	Leu
1655						1660					1665			
Arg	His	Trp	Glu	Asp	Ile	Pro	Arg	Ala	Gly	Cys	Arg	Leu	Val	Pro
1670						1675					1680			
Glu	His	Val	Glu	Thr	Arg	Pro	Leu	Leu	Asn	Pro	Asp	Lys	Pro	Gly
1685						1690					1695			
Ile	Glu	Gln	Gly	Arg	Leu	Glu	Leu	Trp	Val	Asp	Met	Phe	Pro	Met
1700						1705					1710			
Asp	Met	Pro	Ala	Pro	Gly	Thr	Pro	Leu	Asp	Ile	Ser	Pro	Arg	Lys
1715						1720					1725			
Pro	Lys	Lys	Tyr	Glu	Leu	Arg	Val	Ile	Ile	Trp	Asn	Thr	Asp	Glu
1730						1735					1740			
Val	Val	Leu	Glu	Asp	Asp	Asp	Phe	Phe	Thr	Gly	Glu	Lys	Ser	Ser
1745						1750					1755			
Asp	Ile	Phe	Val	Arg	Gly	Trp	Leu	Lys	Gly	Gln	Gln	Glu	Asp	Lys
1760						1765					1770			
Gln	Asp	Thr	Asp	Val	His	Tyr	His	Ser	Leu	Thr	Gly	Glu	Gly	Asn
1775						1780					1785			
Phe	Asn	Trp	Arg	Tyr	Leu	Phe	Pro	Phe	Asp	Tyr	Leu	Ala	Ala	Glu
1790						1795					1800			
Glu	Lys	Ile	Val	Ile	Ser	Lys	Lys	Glu	Ser	Met	Phe	Ser	Trp	Asp
1805						1810					1815			
Glu	Thr	Glu	Tyr	Lys	Ile	Pro	Ala	Arg	Leu	Thr	Leu	Gln	Ile	Trp
1820						1825					1830			
Asp	Ala	Asp	His	Phe	Ser	Ala	Asp	Asp	Phe	Leu	Gly	Ala	Ile	Glu

-continued

1835	1840	1845
Leu Asp Leu Asn Arg Phe Pro Arg Gly Ala Lys Thr Ala Lys Gln		
1850	1855	1860
Cys Thr Met Glu Met Ala Thr Gly Glu Val Asp Val Pro Leu Val		
1865	1870	1875
Ser Ile Phe Lys Gln Lys Arg Val Lys Gly Trp Trp Pro Leu Leu		
1880	1885	1890
Ala Arg Asn Glu Asn Asp Glu Phe Glu Leu Thr Gly Lys Val Glu		
1895	1900	1905
Ala Glu Leu His Leu Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro		
1910	1915	1920
Val Gly Leu Ala Arg Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn		
1925	1930	1935
Arg Pro Asp Thr Ala Phe Val Trp Phe Leu Asn Pro Leu Lys Ser		
1940	1945	1950
Ile Lys Tyr Leu Ile Cys Thr Arg Tyr Lys Trp Leu Ile Ile Lys		
1955	1960	1965
Ile Val Leu Ala Leu Leu Gly Leu Leu Met Leu Gly Leu Phe Leu		
1970	1975	1980
Tyr Ser Leu Pro Gly Tyr Met Val Lys Lys Leu Leu Gly Ala		
1985	1990	1995

<210> SEQ ID NO 6

<211> LENGTH: 1997

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 6

Met Ala Leu Ile Val His Leu Lys Thr Val Ser Glu Leu Arg Gly Lys		
1	5	10 15
Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser		
20	25	30
Arg Val Leu Glu Asn Cys Glu Gly Val Ala Asp Phe Asp Glu Thr Phe		
35	40	45
Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Val Leu Glu Ile		
50	55	60
Gln Ile Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr		
65	70	75 80
Phe Cys Met Val Leu Gln Lys Val Val Glu Glu Asn Arg Val Glu Val		
85	90	95
Thr Asp Thr Leu Met Asp Asp Ser Asn Ala Ile Ile Lys Thr Ser Leu		
100	105	110
Ser Met Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Pro Trp		
115	120	125
Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Lys Asp		
130	135	140
Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Thr Arg		
145	150	155 160
Ile Ser Gly Glu Lys Ser Phe Arg Arg Ala Gly Arg Ser Val Phe Ser		
165	170	175
Ala Met Lys Leu Gly Lys Thr Arg Ser His Lys Glu Glu Pro Gln Arg		
180	185	190
Gln Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu Ala		
195	200	205

Ile 210	Gln	Leu	Gly	Asp	Gly	Leu	Asp	Pro	Asp	Ser	Val	Ser	Leu	Ala	Ser
Val 225	Thr	Ala	Leu	Thr	Ser	Asn	Val	Ser	Asn	Lys	Arg	Ser	Lys	Pro	Asp 240
Ile	Lys	Met	Glu	Pro	Ser	Ala	Gly	Arg	Pro	Met	Asp	Tyr	Gln	Val	Ser 255
Ile	Thr	Val	Ile	Glu	Ala	Arg	Gln	Leu	Val	Gly	Leu	Asn	Met	Asp	Pro 270
Val	Val	Cys	Val	Glu	Val	Gly	Asp	Asp	Lys	Lys	Tyr	Thr	Ser	Met	Lys 285
Glu	Ser	Thr	Asn	Cys	Pro	Tyr	Tyr	Asn	Glu	Tyr	Phe	Val	Phe	Asp	Phe 300
His 305	Val	Ser	Pro	Asp	Val	Met	Phe	Asp	Lys	Ile	Ile	Lys	Ile	Ser	Val 320
Ile	His	Ser	Lys	Asn	Leu	Leu	Arg	Ser	Gly	Thr	Leu	Val	Gly	Ser	Phe 335
Lys	Met	Asp	Val	Gly	Thr	Val	Tyr	Ser	Gln	Pro	Glu	His	Gln	Phe	His 350
His	Lys	Trp	Ala	Ile	Leu	Ser	Asp	Pro	Asp	Asp	Ile	Ser	Ala	Gly	Leu 365
Lys	Gly	Tyr	Val	Lys	Cys	Asp	Val	Ala	Val	Val	Gly	Lys	Gly	Asp	Asn 380
Ile 385	Lys	Thr	Pro	His	Lys	Ala	Asn	Glu	Thr	Asp	Glu	Asp	Asp	Ile	Glu 400
Gly	Asn	Leu	Leu	Leu	Pro	Glu	Gly	Val	Pro	Pro	Glu	Arg	Gln	Trp	Ala 415
Arg	Phe	Tyr	Val	Lys	Ile	Tyr	Arg	Ala	Glu	Gly	Leu	Pro	Arg	Met	Asn 430
Thr	Ser	Leu	Met	Ala	Asn	Val	Lys	Lys	Ala	Phe	Ile	Gly	Glu	Asn	Lys 445
Asp	Leu	Val	Asp	Pro	Tyr	Val	Gln	Val	Phe	Phe	Ala	Gly	Gln	Lys	Gly 460
Lys 465	Thr	Ser	Val	Gln	Lys	Ser	Ser	Tyr	Glu	Pro	Leu	Trp	Asn	Glu	Gln 480
Val	Val	Phe	Thr	Asp	Leu	Phe	Pro	Pro	Leu	Cys	Lys	Arg	Met	Lys	Val 495
Gln	Ile	Arg	Asp	Ser	Asp	Lys	Val	Asn	Asp	Val	Ala	Ile	Gly	Thr	His 510
Phe	Ile	Asp	Leu	Arg	Lys	Ile	Ser	Asn	Asp	Gly	Asp	Lys	Gly	Phe	Leu 525
Pro	Thr	Leu	Gly	Pro	Ala	Trp	Val	Asn	Met	Tyr	Gly	Ser	Thr	Arg	Asn 540
Tyr 545	Thr	Leu	Leu	Asp	Glu	His	Gln	Asp	Leu	Asn	Glu	Gly	Leu	Gly	Glu 560
Gly	Val	Ser	Phe	Arg	Ala	Arg	Leu	Met	Leu	Gly	Leu	Ala	Val	Glu	Ile 575
Leu	Asp	Thr	Ser	Asn	Pro	Glu	Leu	Thr	Ser	Ser	Thr	Glu	Val	Gln	Val 590
Glu	Gln	Ala	Thr	Pro	Val	Ser	Glu	Ser	Cys	Thr	Gly	Arg	Met	Glu	Glu 605
Phe 610	Phe	Leu	Phe	Gly	Ala	Phe	Leu	Glu	Ala	Ser	Met	Ile	Asp	Arg	Lys 620
Asn	Gly	Asp	Lys	Pro	Ile	Thr	Phe	Glu	Val	Thr	Ile	Gly	Asn	Tyr	Gln

-continued

625	630	635	640
Asn Glu Val Asp Gly Met Ser Arg Pro Leu Arg Pro Arg Pro Arg Lys			
645		650	655
Glu Pro Gly Asp Glu Glu Glu Val Asp Leu Ile Gln Asn Ser Ser Asp			
660		665	670
Asp Glu Gly Asp Glu Ala Gly Asp Leu Ala Ser Val Ser Ser Thr Pro			
675		680	685
Pro Met Arg Pro Gln Ile Thr Asp Arg Asn Tyr Phe His Leu Pro Tyr			
690		695	700
Leu Glu Arg Lys Pro Cys Ile Tyr Ile Lys Ser Trp Trp Pro Asp Gln			
705		710	715
Arg Arg Arg Leu Tyr Asn Ala Asn Ile Met Asp His Ile Ala Asp Lys			
725		730	735
Leu Glu Glu Gly Leu Asn Asp Val Gln Glu Met Ile Lys Thr Glu Lys			
740		745	750
Ser Tyr Pro Glu Arg Arg Leu Arg Gly Val Leu Glu Glu Leu Ser Cys			
755		760	765
Gly Cys His Arg Phe Leu Ser Leu Ser Asp Lys Asp Gln Gly Arg Ser			
770		775	780
Ser Arg Thr Arg Leu Asp Arg Glu Arg Leu Lys Ser Cys Met Arg Glu			
785		790	795
Leu Glu Ser Met Gly Gln Gln Ala Lys Ser Leu Arg Ala Gln Val Lys			
805		810	815
Arg His Thr Val Arg Asp Lys Leu Arg Ser Cys Gln Asn Phe Leu Gln			
820		825	830
Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln His Ser Ile Pro Asp Val			
835		840	845
Phe Ile Trp Met Met Ser Asn Asn Lys Arg Ile Ala Tyr Ala Arg Val			
850		855	860
Pro Ser Lys Asp Leu Leu Phe Ser Ile Val Glu Glu Glu Leu Gly Lys			
865		870	875
Asp Cys Ala Lys Val Lys Thr Leu Phe Leu Lys Leu Pro Gly Lys Arg			
885		890	895
Gly Phe Gly Ser Ala Gly Trp Thr Val Gln Ala Lys Leu Glu Leu Tyr			
900		905	910
Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys Asp Phe Leu Cys Gly Leu			
915		920	925
Pro Cys Gly Phe Glu Glu Val Lys Ala Ala Gln Gly Leu Gly Leu His			
930		935	940
Ser Phe Pro Pro Ile Ser Leu Val Tyr Thr Lys Lys Gln Ala Phe Gln			
945		950	955
Leu Arg Ala His Met Tyr Gln Ala Arg Ser Leu Phe Ala Ala Asp Ser			
965		970	975
Ser Gly Leu Ser Asp Pro Phe Ala Arg Val Phe Phe Ile Asn Gln Ser			
980		985	990
Gln Cys Thr Glu Val Leu Asn Glu Thr Leu Cys Pro Thr Trp Asp Gln			
995		1000	1005
Met Leu Val Phe Asp Asn Leu Glu Leu Tyr Gly Glu Ala His Glu			
1010		1015	1020
Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu Ile Tyr Asp Gln			
1025		1030	1035
Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr Phe Ala Lys			
1040		1045	1050

-continued

Pro Leu Val Lys Met Ala Asp	Glu Ala Tyr Cys Pro	Pro Arg Phe
1055	1060	1065
Pro Pro Gln Leu Glu Tyr Tyr	Gln Ile Tyr Arg Gly	Ser Ala Thr
1070	1075	1080
Ala Gly Asp Leu Leu Ala Ala	Phe Glu Leu Leu Gln	Ile Gly Pro
1085	1090	1095
Ser Gly Lys Ala Asp Leu Pro	Pro Ile Asn Gly Pro	Val Asp Met
1100	1105	1110
Asp Arg Gly Pro Ile Met Pro	Val Pro Val Gly Ile	Arg Pro Val
1115	1120	1125
Leu Ser Lys Tyr Arg Val Glu	Val Leu Phe Trp Gly	Leu Arg Asp
1130	1135	1140
Leu Lys Arg Val Asn Leu Ala	Gln Val Asp Arg Pro	Arg Val Asp
1145	1150	1155
Ile Glu Cys Ala Gly Lys Gly	Val Gln Ser Ser Leu	Ile His Asn
1160	1165	1170
Tyr Lys Lys Asn Pro Asn Phe	Asn Thr Leu Val Lys	Trp Phe Glu
1175	1180	1185
Val Asp Leu Pro Glu Asn Glu	Leu Leu His Pro Pro	Leu Asn Ile
1190	1195	1200
Arg Val Val Asp Cys Arg Ala	Phe Gly Arg Tyr Thr	Leu Val Gly
1205	1210	1215
Ser His Ala Val Ser Ser Leu	Arg Arg Phe Ile Tyr	Arg Pro Pro
1220	1225	1230
Asp Arg Ser Ala Pro Asn Trp	Asn Thr Thr Val Arg	Leu Leu Arg
1235	1240	1245
Gly Cys His Arg Leu Arg Asn	Gly Gly Pro Ser Ser	Arg Pro Thr
1250	1255	1260
Gly Glu Val Val Val Ser Met	Glu Pro Glu Glu Pro	Val Lys Lys
1265	1270	1275
Leu Glu Thr Met Val Lys Leu	Asp Ala Thr Ser Asp	Ala Val Val
1280	1285	1290
Lys Val Asp Val Ala Glu Asp	Glu Lys Glu Arg Lys	Lys Lys Lys
1295	1300	1305
Lys Lys Gly Pro Ser Glu Glu	Pro Glu Glu Glu Glu	Pro Asp Glu
1310	1315	1320
Ser Met Leu Asp Trp Trp Ser	Lys Tyr Phe Ala Ser	Ile Asp Thr
1325	1330	1335
Met Lys Glu Gln Leu Arg Gln	His Glu Thr Ser Gly	Thr Asp Leu
1340	1345	1350
Glu Glu Lys Glu Glu Met Glu	Ser Ala Glu Gly Leu	Lys Gly Pro
1355	1360	1365
Met Lys Ser Lys Glu Lys Ser	Arg Ala Ala Lys Glu	Glu Lys Lys
1370	1375	1380
Lys Lys Asn Gln Ser Pro Gly	Pro Gly Gln Gly Ser	Glu Ala Pro
1385	1390	1395
Glu Lys Lys Lys Ala Lys Ile	Asp Glu Leu Lys Val	Tyr Pro Lys
1400	1405	1410
Glu Leu Glu Ser Glu Phe Asp	Ser Phe Glu Asp Trp	Leu His Thr
1415	1420	1425
Phe Asn Leu Leu Arg Gly Lys	Thr Gly Asp Asp Glu	Asp Gly Ser
1430	1435	1440

-continued

Thr	Glu	Glu	Glu	Arg	Ile	Val	Gly	Arg	Phe	Lys	Gly	Ser	Leu	Cys
1445						1450					1455			
Val	Tyr	Lys	Val	Pro	Leu	Pro	Glu	Asp	Val	Ser	Arg	Glu	Ala	Gly
1460						1465					1470			
Tyr	Asp	Pro	Thr	Tyr	Gly	Met	Phe	Gln	Gly	Ile	Pro	Ser	Asn	Asp
1475						1480					1485			
Pro	Ile	Asn	Val	Leu	Val	Arg	Ile	Tyr	Val	Val	Arg	Ala	Thr	Asp
1490						1495					1500			
Leu	His	Pro	Ala	Asp	Ile	Asn	Gly	Lys	Ala	Asp	Pro	Tyr	Ile	Ala
1505						1510					1515			
Ile	Lys	Leu	Gly	Lys	Thr	Asp	Ile	Arg	Asp	Lys	Glu	Asn	Tyr	Ile
1520						1525					1530			
Ser	Lys	Gln	Leu	Asn	Pro	Val	Phe	Gly	Lys	Ser	Phe	Asp	Ile	Glu
1535						1540					1545			
Ala	Ser	Phe	Pro	Met	Glu	Ser	Met	Leu	Thr	Val	Ala	Val	Tyr	Asp
1550						1555					1560			
Trp	Asp	Leu	Val	Gly	Thr	Asp	Asp	Leu	Ile	Gly	Glu	Thr	Lys	Ile
1565						1570					1575			
Asp	Leu	Glu	Asn	Arg	Phe	Tyr	Ser	Lys	His	Arg	Ala	Thr	Cys	Gly
1580						1585					1590			
Ile	Ala	Gln	Thr	Tyr	Ser	Ile	His	Gly	Tyr	Asn	Ile	Trp	Arg	Asp
1595						1600					1605			
Pro	Met	Lys	Pro	Ser	Gln	Ile	Leu	Thr	Arg	Leu	Cys	Lys	Glu	Gly
1610						1615					1620			
Lys	Val	Asp	Gly	Pro	His	Phe	Gly	Pro	His	Gly	Arg	Val	Arg	Val
1625						1630					1635			
Ala	Asn	Arg	Val	Phe	Thr	Gly	Pro	Ser	Glu	Ile	Glu	Asp	Glu	Asn
1640						1645					1650			
Gly	Gln	Arg	Lys	Pro	Thr	Asp	Glu	His	Val	Ala	Leu	Ser	Ala	Leu
1655						1660					1665			
Arg	His	Trp	Glu	Asp	Ile	Pro	Arg	Val	Gly	Cys	Arg	Leu	Val	Pro
1670						1675					1680			
Glu	His	Val	Glu	Thr	Arg	Pro	Leu	Leu	Asn	Pro	Asp	Lys	Pro	Gly
1685						1690					1695			
Ile	Glu	Gln	Gly	Arg	Leu	Glu	Leu	Trp	Val	Asp	Met	Phe	Pro	Met
1700						1705					1710			
Asp	Met	Pro	Ala	Pro	Gly	Thr	Pro	Leu	Asp	Ile	Ser	Pro	Arg	Lys
1715						1720					1725			
Pro	Lys	Lys	Tyr	Glu	Leu	Arg	Val	Ile	Val	Trp	Asn	Thr	Asp	Glu
1730						1735					1740			
Val	Val	Leu	Glu	Asp	Asp	Asp	Phe	Phe	Thr	Gly	Glu	Lys	Ser	Ser
1745						1750					1755			
Asp	Ile	Phe	Val	Arg	Gly	Trp	Leu	Lys	Gly	Gln	Gln	Glu	Asp	Lys
1760						1765					1770			
Gln	Asp	Thr	Asp	Val	His	Tyr	His	Ser	Leu	Thr	Gly	Glu	Gly	Asn
1775						1780					1785			
Phe	Asn	Trp	Arg	Tyr	Leu	Phe	Pro	Phe	Asp	Tyr	Leu	Ala	Ala	Glu
1790						1795					1800			
Glu	Lys	Ile	Val	Met	Ser	Lys	Lys	Glu	Ser	Met	Phe	Ser	Trp	Asp
1805						1810					1815			
Glu	Thr	Glu	Tyr	Lys	Ile	Pro	Ala	Arg	Leu	Thr	Leu	Gln	Ile	Trp
1820						1825					1830			
Asp	Ala	Asp	His	Phe	Ser	Ala	Asp	Asp	Phe	Leu	Gly	Ala	Ile	Glu

-continued

1835	1840	1845
Leu Asp Leu Asn Arg Phe Pro Arg Gly Ala Lys Thr Ala Lys Gln		
1850	1855	1860
Cys Thr Met Glu Met Ala Thr Gly Glu Val Asp Val Pro Leu Val		
1865	1870	1875
Ser Ile Phe Lys Gln Lys Arg Val Lys Gly Trp Trp Pro Leu Leu		
1880	1885	1890
Ala Arg Asn Glu Asn Asp Glu Phe Glu Leu Thr Gly Lys Val Glu		
1895	1900	1905
Ala Glu Leu His Leu Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro		
1910	1915	1920
Val Gly Leu Ala Arg Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn		
1925	1930	1935
Arg Pro Asp Thr Ser Phe Ile Trp Phe Leu Asn Pro Leu Lys Ser		
1940	1945	1950
Ala Arg Tyr Phe Leu Trp His Thr Tyr Arg Trp Leu Leu Leu Lys		
1955	1960	1965
Phe Leu Leu Leu Phe Leu Leu Leu Leu Leu Phe Ala Leu Phe Leu		
1970	1975	1980
Tyr Ser Leu Pro Gly Tyr Leu Ala Lys Lys Ile Leu Gly Ala		
1985	1990	1995

<210> SEQ ID NO 7

<211> LENGTH: 1977

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 7

Met Ala Leu Ile Val His Leu Lys Thr Val Ser Glu Leu Arg Gly Lys		
1	5	10 15
Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser		
20	25	30
Arg Val Leu Glu Asn Cys Glu Gly Val Ala Asp Phe Asp Glu Thr Phe		
35	40	45
Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Val Leu Glu Ile		
50	55	60
Gln Ile Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr		
65	70	75 80
Phe Cys Met Val Leu Gln Lys Val Val Glu Glu Asn Arg Val Glu Val		
85	90	95
Thr Asp Thr Leu Met Asp Asp Ser Asn Ala Ile Ile Lys Thr Ser Leu		
100	105	110
Ser Met Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Pro Trp		
115	120	125
Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Lys Asp		
130	135	140
Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Thr Arg		
145	150	155 160
Ile Ser Gly Glu Lys Ser Phe Arg Arg Ala Gly Arg Ser Val Phe Ser		
165	170	175
Ala Met Lys Leu Gly Lys Thr Arg Ser His Lys Glu Glu Pro Gln Arg		
180	185	190
Gln Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu Ala		
195	200	205

Ile 210	Gln 215	Leu	Gly	Asp	Gly	Leu 215	Asp	Pro	Asp	Ser	Val 220	Ser	Leu	Ala	Ser
Val 225	Thr	Ala	Leu	Thr	Ser	Asn 230	Val	Ser	Asn	Lys 235	Arg	Ser	Lys	Pro	Asp 240
Ile	Lys	Met	Glu	Pro	Ser	Ala	Gly	Arg	Pro	Met	Asp	Tyr	Gln	Val	Ser 255
Ile	Thr	Val	Ile	Glu	Ala	Arg	Gln	Leu	Val	Gly	Leu	Asn	Met	Asp	Pro 270
Val	Val	Cys	Val	Glu	Val	Gly	Asp	Asp	Lys	Lys	Tyr	Thr	Ser	Met	Lys 285
Glu	Ser	Thr	Asn	Cys	Pro	Tyr	Tyr	Asn	Glu	Tyr	Phe	Val	Phe	Asp	Phe 300
His 305	Val	Ser	Pro	Asp	Val	Met	Phe	Asp	Lys	Ile	Ile	Lys	Ile	Ser	Val 320
Ile	His	Ser	Lys	Asn	Leu	Leu	Arg	Ser	Gly	Thr	Leu	Val	Gly	Ser	Phe 335
Lys	Met	Asp	Val	Gly	Thr	Val	Tyr	Ser	Gln	Pro	Glu	His	Gln	Phe	His 350
His	Lys	Trp	Ala	Ile	Leu	Ser	Asp	Pro	Asp	Asp	Ile	Ser	Ala	Gly	Leu 365
Lys	Gly	Tyr	Val	Lys	Cys	Asp	Val	Ala	Val	Val	Gly	Lys	Gly	Asp	Asn 380
Ile 385	Lys	Thr	Pro	His	Lys	Ala	Asn	Glu	Thr	Asp	Glu	Asp	Asp	Ile	Glu 400
Gly	Asn	Leu	Leu	Leu	Pro	Glu	Gly	Val	Pro	Pro	Glu	Arg	Gln	Trp	Ala 415
Arg	Phe	Tyr	Val	Lys	Ile	Tyr	Arg	Ala	Glu	Gly	Leu	Pro	Arg	Met	Asn 430
Thr	Ser	Leu	Met	Ala	Asn	Val	Lys	Lys	Ala	Phe	Ile	Gly	Glu	Asn	Lys 445
Asp	Leu	Val	Asp	Pro	Tyr	Val	Gln	Val	Phe	Phe	Ala	Gly	Gln	Lys	Gly 460
Lys 465	Thr	Ser	Val	Gln	Lys	Ser	Ser	Tyr	Glu	Pro	Leu	Trp	Asn	Glu	Gln 480
Val	Val	Phe	Thr	Asp	Leu	Phe	Pro	Pro	Leu	Cys	Lys	Arg	Met	Lys	Val 495
Gln	Ile	Arg	Asp	Ser	Asp	Lys	Val	Asn	Asp	Val	Ala	Ile	Gly	Thr	His 510
Phe	Ile	Asp	Leu	Arg	Lys	Ile	Ser	Asn	Asp	Gly	Asp	Lys	Gly	Phe	Leu 525
Pro	Thr	Leu	Gly	Pro	Ala	Trp	Val	Asn	Met	Tyr	Gly	Ser	Thr	Arg	Asn 540
Tyr 545	Thr	Leu	Leu	Asp	Glu	His	Gln	Asp	Leu	Asn	Glu	Gly	Leu	Gly	Glu 560
Gly	Val	Ser	Phe	Arg	Ala	Arg	Leu	Met	Leu	Gly	Leu	Ala	Val	Glu	Ile 575
Leu	Asp	Thr	Ser	Asn	Pro	Glu	Leu	Thr	Ser	Ser	Thr	Glu	Val	Gln	Val 590
Glu	Gln	Ala	Thr	Pro	Val	Ser	Glu	Ser	Cys	Thr	Gly	Arg	Met	Glu	Glu 605
Phe 610	Phe	Leu	Phe	Gly	Ala	Phe	Leu	Glu	Ala	Ser	Met	Ile	Asp	Arg	Lys 620
Asn	Gly	Asp	Lys	Pro	Ile	Thr	Phe	Glu	Val	Thr	Ile	Gly	Asn	Tyr	Gln

-continued

625	630	635	640
Asn Glu Val Asp Gly Met Ser Arg Pro Leu Arg Pro Arg Pro Arg Lys			
645		650	655
Glu Pro Gly Asp Glu Glu Glu Val Asp Leu Ile Gln Asn Ser Ser Asp			
660		665	670
Asp Glu Gly Asp Glu Ala Gly Asp Leu Ala Ser Val Ser Ser Thr Pro			
675		680	685
Pro Met Arg Pro Gln Ile Thr Asp Arg Asn Tyr Phe His Leu Pro Tyr			
690		695	700
Leu Glu Arg Lys Pro Cys Ile Tyr Ile Lys Ser Trp Trp Pro Asp Gln			
705	710	715	720
Arg Arg Arg Leu Tyr Asn Ala Asn Ile Met Asp His Ile Ala Asp Lys			
725		730	735
Leu Glu Glu Gly Leu Asn Asp Val Gln Glu Met Ile Lys Thr Glu Lys			
740		745	750
Ser Tyr Pro Glu Arg Arg Leu Arg Gly Val Leu Glu Glu Leu Ser Cys			
755		760	765
Gly Cys His Arg Phe Leu Ser Leu Ser Asp Lys Asp Gln Gly Arg Ser			
770	775	780	
Ser Arg Thr Arg Leu Asp Arg Glu Arg Leu Lys Ser Cys Met Arg Glu			
785	790	795	800
Leu Glu Ser Met Gly Gln Gln Ala Lys Ser Leu Arg Ala Gln Val Lys			
805		810	815
Arg His Thr Val Arg Asp Lys Leu Arg Ser Cys Gln Asn Phe Leu Gln			
820		825	830
Lys Leu Arg Phe Leu Ala Asp Glu Pro Gln His Ser Ile Pro Asp Val			
835		840	845
Phe Ile Trp Met Met Ser Asn Asn Lys Arg Ile Ala Tyr Ala Arg Val			
850	855	860	
Pro Ser Lys Asp Leu Leu Phe Ser Ile Val Glu Glu Glu Leu Gly Lys			
865	870	875	880
Asp Cys Ala Lys Val Lys Thr Leu Phe Leu Lys Leu Pro Gly Lys Arg			
885		890	895
Gly Phe Gly Ser Ala Gly Trp Thr Val Gln Ala Lys Leu Glu Leu Tyr			
900		905	910
Leu Trp Leu Gly Leu Ser Lys Gln Arg Lys Asp Phe Leu Cys Gly Leu			
915		920	925
Pro Cys Gly Phe Glu Glu Val Lys Ala Ala Gln Gly Leu Gly Leu His			
930	935	940	
Ser Phe Pro Pro Ile Ser Leu Val Tyr Thr Lys Lys Gln Ala Phe Gln			
945	950	955	960
Leu Arg Ala His Met Tyr Gln Ala Arg Ser Leu Phe Ala Ala Asp Ser			
965		970	975
Ser Gly Leu Ser Asp Pro Phe Ala Arg Val Phe Phe Ile Asn Gln Ser			
980		985	990
Gln Cys Thr Glu Val Leu Asn Glu Thr Leu Cys Pro Thr Trp Asp Gln			
995		1000	1005
Met Leu Val Phe Asp Asn Leu Glu Leu Tyr Gly Glu Ala His Glu			
1010		1015	1020
Leu Arg Asp Asp Pro Pro Ile Ile Val Ile Glu Ile Tyr Asp Gln			
1025		1030	1035
Asp Ser Met Gly Lys Ala Asp Phe Met Gly Arg Thr Phe Ala Lys			
1040		1045	1050

-continued

Pro Leu Val Lys Met Ala Asp	Glu Ala Tyr Cys Pro	Pro Arg Phe
1055	1060	1065
Pro Pro Gln Leu Glu Tyr Tyr	Gln Ile Tyr Arg Gly	Ser Ala Thr
1070	1075	1080
Ala Gly Asp Leu Leu Ala Ala	Phe Glu Leu Leu Gln	Ile Gly Pro
1085	1090	1095
Ser Gly Lys Ala Asp Leu Pro	Pro Ile Asn Gly Pro	Val Asp Met
1100	1105	1110
Asp Arg Gly Pro Ile Met Pro	Val Pro Val Gly Ile	Arg Pro Val
1115	1120	1125
Leu Ser Lys Tyr Arg Val Glu	Val Leu Phe Trp Gly	Leu Arg Asp
1130	1135	1140
Leu Lys Arg Val Asn Leu Ala	Gln Val Asp Arg Pro	Arg Val Asp
1145	1150	1155
Ile Glu Cys Ala Gly Lys Gly	Val Gln Ser Ser Leu	Ile His Asn
1160	1165	1170
Tyr Lys Lys Asn Pro Asn Phe	Asn Thr Leu Val Lys	Trp Phe Glu
1175	1180	1185
Val Asp Leu Pro Glu Asn Glu	Leu Leu His Pro Pro	Leu Asn Ile
1190	1195	1200
Arg Val Val Asp Cys Arg Ala	Phe Gly Arg Tyr Thr	Leu Val Gly
1205	1210	1215
Ser His Ala Val Ser Ser Leu	Arg Arg Phe Ile Tyr	Arg Pro Pro
1220	1225	1230
Asp Arg Ser Ala Pro Asn Trp	Asn Thr Thr Gly Glu	Val Val Val
1235	1240	1245
Ser Met Glu Pro Glu Glu Pro	Val Lys Lys Leu Glu	Thr Met Val
1250	1255	1260
Lys Leu Asp Ala Thr Ser Asp	Ala Val Val Lys Val	Asp Val Ala
1265	1270	1275
Glu Asp Glu Lys Glu Arg Lys	Lys Lys Lys Lys Lys	Gly Pro Ser
1280	1285	1290
Glu Glu Pro Glu Glu Glu Glu	Pro Asp Glu Ser Met	Leu Asp Trp
1295	1300	1305
Trp Ser Lys Tyr Phe Ala Ser	Ile Asp Thr Met Lys	Glu Gln Leu
1310	1315	1320
Arg Gln His Glu Thr Ser Gly	Thr Asp Leu Glu Glu	Lys Glu Glu
1325	1330	1335
Met Glu Ser Ala Glu Gly Leu	Lys Gly Pro Met Lys	Ser Lys Glu
1340	1345	1350
Lys Ser Arg Ala Ala Lys Glu	Glu Lys Lys Lys Lys	Asn Gln Ser
1355	1360	1365
Pro Gly Pro Gly Gln Gly Ser	Glu Ala Pro Glu Lys	Lys Lys Ala
1370	1375	1380
Lys Ile Asp Glu Leu Lys Val	Tyr Pro Lys Glu Leu	Glu Ser Glu
1385	1390	1395
Phe Asp Ser Phe Glu Asp Trp	Leu His Thr Phe Asn	Leu Leu Arg
1400	1405	1410
Gly Lys Thr Gly Asp Asp Glu	Asp Gly Ser Thr Glu	Glu Glu Arg
1415	1420	1425
Ile Val Gly Arg Phe Lys Gly	Ser Leu Cys Val Tyr	Lys Val Pro
1430	1435	1440

-continued

Leu Pro 1445	Glu Asp Val Ser Arg 1450	Glu Ala Gly Tyr Asp 1455	Pro Thr Tyr
Gly Met 1460	Phe Gln Gly Ile Pro 1465	Ser Asn Asp Pro Ile 1470	Asn Val Leu
Val Arg 1475	Ile Tyr Val Val Arg 1480	Ala Thr Asp Leu His 1485	Pro Ala Asp
Ile Asn 1490	Gly Lys Ala Asp Pro 1495	Tyr Ile Ala Ile Lys 1500	Leu Gly Lys
Thr Asp 1505	Ile Arg Asp Lys Glu 1510	Asn Tyr Ile Ser Lys 1515	Gln Leu Asn
Pro Val 1520	Phe Gly Lys Ser Phe 1525	Asp Ile Glu Ala Ser 1530	Phe Pro Met
Glu Ser 1535	Met Leu Thr Val Ala 1540	Val Tyr Asp Trp Asp 1545	Leu Val Gly
Thr Asp 1550	Asp Leu Ile Gly Glu 1555	Thr Lys Ile Asp Leu 1560	Glu Asn Arg
Phe Tyr 1565	Ser Lys His Arg Ala 1570	Thr Cys Gly Ile Ala 1575	Gln Thr Tyr
Ser Ile 1580	His Gly Tyr Asn Ile 1585	Trp Arg Asp Pro Met 1590	Lys Pro Ser
Gln Ile 1595	Leu Thr Arg Leu Cys 1600	Lys Glu Gly Lys Val 1605	Asp Gly Pro
His Phe 1610	Gly Pro His Gly Arg 1615	Val Arg Val Ala Asn 1620	Arg Val Phe
Thr Gly 1625	Pro Ser Glu Ile Glu 1630	Asp Glu Asn Gly Gln 1635	Arg Lys Pro
Thr Asp 1640	Glu His Val Ala Leu 1645	Ser Ala Leu Arg His 1650	Trp Glu Asp
Ile Pro 1655	Arg Val Gly Cys Arg 1660	Leu Val Pro Glu His 1665	Val Glu Thr
Arg Pro 1670	Leu Leu Asn Pro Asp 1675	Lys Pro Gly Ile Glu 1680	Gln Gly Arg
Leu Glu 1685	Leu Trp Val Asp Met 1690	Phe Pro Met Asp Met 1695	Pro Ala Pro
Gly Thr 1700	Pro Leu Asp Ile Ser 1705	Pro Arg Lys Pro Lys 1710	Lys Tyr Glu
Leu Arg 1715	Val Ile Val Trp Asn 1720	Thr Asp Glu Val Val 1725	Leu Glu Asp
Asp Asp 1730	Phe Phe Thr Gly Glu 1735	Lys Ser Ser Asp Ile 1740	Phe Val Arg
Gly Trp 1745	Leu Lys Gly Gln Gln 1750	Glu Asp Lys Gln Asp 1755	Thr Asp Val
His Tyr 1760	His Ser Leu Thr Gly 1765	Glu Gly Asn Phe Asn 1770	Trp Arg Tyr
Leu Phe 1775	Pro Phe Asp Tyr Leu 1780	Ala Ala Glu Glu Lys 1785	Ile Val Met
Ser Lys 1790	Lys Glu Ser Met Phe 1795	Ser Trp Asp Glu Thr 1800	Glu Tyr Lys
Ile Pro 1805	Ala Arg Leu Thr Leu 1810	Gln Ile Trp Asp Ala 1815	Asp His Phe
Ser Ala 1820	Asp Asp Phe Leu Gly 1825	Ala Ile Glu Leu Asp 1830	Leu Asn Arg
Phe Pro	Arg Gly Ala Lys Thr	Ala Lys Gln Cys Thr	Met Glu Met

-continued

1835	1840	1845
Ala Thr Gly Glu Val Asp Val Pro Leu Val Ser Ile Phe Lys Gln		
1850	1855	1860
Lys Arg Val Lys Gly Trp Trp Pro Leu Leu Ala Arg Asn Glu Asn		
1865	1870	1875
Asp Glu Phe Glu Leu Thr Gly Lys Val Glu Ala Glu Leu His Leu		
1880	1885	1890
Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro Val Gly Leu Ala Arg		
1895	1900	1905
Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn Arg Pro Asp Thr Ser		
1910	1915	1920
Phe Ile Trp Phe Leu Asn Pro Leu Lys Ser Ala Arg Tyr Phe Leu		
1925	1930	1935
Trp His Thr Tyr Arg Trp Leu Leu Leu Lys Phe Leu Leu Leu Phe		
1940	1945	1950
Leu Leu Leu Leu Leu Phe Ala Leu Phe Leu Tyr Ser Leu Pro Gly		
1955	1960	1965
Tyr Leu Ala Lys Lys Ile Leu Gly Ala		
1970	1975	

<210> SEQ ID NO 8

<211> LENGTH: 1992

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 8

Met Ala Leu Ile Val His Leu Lys Thr Val Ser Glu Leu Arg Gly Lys		
1	5	10 15
Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser		
20	25	30
Arg Val Leu Glu Asn Cys Glu Gly Val Ala Asp Phe Asp Glu Thr Phe		
35	40	45
Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Val Leu Glu Ile		
50	55	60
Gln Ile Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr		
65	70	75 80
Phe Cys Met Val Leu Gln Lys Val Val Glu Glu Asn Arg Val Glu Val		
85	90	95
Thr Asp Thr Leu Met Asp Asp Ser Asn Ala Ile Ile Lys Thr Ser Leu		
100	105	110
Ser Met Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Pro Trp		
115	120	125
Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Lys Asp		
130	135	140
Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Thr Arg		
145	150	155 160
Ile Ser Gly Glu Lys Ser Phe Arg Ser Lys Gly Arg Glu Lys Thr Lys		
165	170	175
Gly Gly Arg Asp Gly Glu His Lys Ala Gly Arg Ser Val Phe Ser Ala		
180	185	190
Met Lys Leu Gly Lys Thr Arg Ser His Lys Glu Glu Pro Gln Arg Gln		
195	200	205
Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu Ala Ile		
210	215	220

-continued

Gln	Leu	Gly	Asp	Gly	Leu	Asp	Pro	Asp	Ser	Val	Ser	Leu	Ala	Ser	Val	225	230	235	240
Thr	Ala	Leu	Thr	Ser	Asn	Val	Ser	Asn	Lys	Arg	Ser	Lys	Pro	Asp	Ile	245	250	255	
Lys	Met	Glu	Pro	Ser	Ala	Gly	Arg	Pro	Met	Asp	Tyr	Gln	Val	Ser	Ile	260	265	270	
Thr	Val	Ile	Glu	Ala	Arg	Gln	Leu	Val	Gly	Leu	Asn	Met	Asp	Pro	Val	275	280	285	
Val	Cys	Val	Glu	Val	Gly	Asp	Asp	Lys	Lys	Tyr	Thr	Ser	Met	Lys	Glu	290	295	300	
Ser	Thr	Asn	Cys	Pro	Tyr	Tyr	Asn	Glu	Tyr	Phe	Val	Phe	Asp	Phe	His	305	310	315	320
Val	Ser	Pro	Asp	Val	Met	Phe	Asp	Lys	Ile	Ile	Lys	Ile	Ser	Val	Ile	325	330	335	
His	Ser	Lys	Asn	Leu	Leu	Arg	Ser	Gly	Thr	Leu	Val	Gly	Ser	Phe	Lys	340	345	350	
Met	Asp	Val	Gly	Thr	Val	Tyr	Ser	Gln	Pro	Glu	His	Gln	Phe	His	His	355	360	365	
Lys	Trp	Ala	Ile	Leu	Ser	Asp	Pro	Asp	Asp	Ile	Ser	Ala	Gly	Leu	Lys	370	375	380	
Gly	Tyr	Val	Lys	Cys	Asp	Val	Ala	Val	Val	Gly	Lys	Gly	Asp	Asn	Ile	385	390	395	400
Lys	Thr	Pro	His	Lys	Ala	Asn	Glu	Thr	Asp	Glu	Asp	Asp	Ile	Glu	Gly	405	410	415	
Asn	Leu	Leu	Leu	Pro	Glu	Gly	Val	Pro	Pro	Glu	Arg	Gln	Trp	Ala	Arg	420	425	430	
Phe	Tyr	Val	Lys	Ile	Tyr	Arg	Ala	Glu	Gly	Leu	Pro	Arg	Met	Asn	Thr	435	440	445	
Ser	Leu	Met	Ala	Asn	Val	Lys	Lys	Ala	Phe	Ile	Gly	Glu	Asn	Lys	Asp	450	455	460	
Leu	Val	Asp	Pro	Tyr	Val	Gln	Val	Phe	Phe	Ala	Gly	Gln	Lys	Gly	Lys	465	470	475	480
Thr	Ser	Val	Gln	Lys	Ser	Ser	Tyr	Glu	Pro	Leu	Trp	Asn	Glu	Gln	Val	485	490	495	
Val	Phe	Thr	Asp	Leu	Phe	Pro	Pro	Leu	Cys	Lys	Arg	Met	Lys	Val	Gln	500	505	510	
Ile	Arg	Asp	Ser	Asp	Lys	Val	Asn	Asp	Val	Ala	Ile	Gly	Thr	His	Phe	515	520	525	
Ile	Asp	Leu	Arg	Lys	Ile	Ser	Asn	Asp	Gly	Asp	Lys	Gly	Phe	Leu	Pro	530	535	540	
Thr	Leu	Gly	Pro	Ala	Trp	Val	Asn	Met	Tyr	Gly	Ser	Thr	Arg	Asn	Tyr	545	550	555	560
Thr	Leu	Leu	Asp	Glu	His	Gln	Asp	Leu	Asn	Glu	Gly	Leu	Gly	Glu	Gly	565	570	575	
Val	Ser	Phe	Arg	Ala	Arg	Leu	Met	Leu	Gly	Leu	Ala	Val	Glu	Ile	Leu	580	585	590	
Asp	Thr	Ser	Asn	Pro	Glu	Leu	Thr	Ser	Ser	Thr	Glu	Val	Gln	Val	Glu	595	600	605	
Gln	Ala	Thr	Pro	Val	Ser	Glu	Ser	Cys	Thr	Gly	Arg	Met	Glu	Glu	Phe	610	615	620	
Phe	Leu	Phe	Gly	Ala	Phe	Leu	Glu	Ala	Ser	Met	Ile	Asp	Arg	Lys	Asn	625	630	635	640
Gly	Asp	Lys	Pro	Ile	Thr	Phe	Glu	Val	Thr	Ile	Gly	Asn	Tyr	Gly	Asn				

-continued

645										650					655				
Glu	Val	Asp	Gly	Met	Ser	Arg	Pro	Leu	Arg	Pro	Arg	Pro	Arg	Lys	Glu				
			660				665			670									
Pro	Gly	Asp	Glu	Glu	Glu	Val	Asp	Leu	Ile	Gln	Asn	Ser	Ser	Asp	Asp				
			675				680			685									
Glu	Gly	Asp	Glu	Ala	Gly	Asp	Leu	Ala	Ser	Val	Ser	Ser	Thr	Pro	Pro				
			690	695						700									
Met	Arg	Pro	Gln	Ile	Thr	Asp	Arg	Asn	Tyr	Phe	His	Leu	Pro	Tyr	Leu				
705			710						715			720							
Glu	Arg	Lys	Pro	Cys	Ile	Tyr	Ile	Lys	Ser	Trp	Trp	Pro	Asp	Gln	Arg				
			725						730			735							
Arg	Arg	Leu	Tyr	Asn	Ala	Asn	Ile	Met	Asp	His	Ile	Ala	Asp	Lys	Leu				
			740			745						750							
Glu	Glu	Gly	Leu	Asn	Asp	Val	Gln	Glu	Met	Ile	Lys	Thr	Glu	Lys	Ser				
			755			760						765							
Tyr	Pro	Glu	Arg	Arg	Leu	Arg	Gly	Val	Leu	Glu	Glu	Leu	Ser	Cys	Gly				
770						775			780										
Cys	His	Arg	Phe	Leu	Ser	Leu	Ser	Asp	Lys	Asp	Gln	Gly	Arg	Ser	Ser				
785			790						795			800							
Arg	Thr	Arg	Leu	Asp	Arg	Glu	Arg	Leu	Lys	Ser	Cys	Met	Arg	Glu	Leu				
			805						810			815							
Glu	Ser	Met	Gly	Gln	Gln	Ala	Lys	Ser	Leu	Arg	Ala	Gln	Val	Lys	Arg				
			820						825			830							
His	Thr	Val	Arg	Asp	Lys	Leu	Arg	Ser	Cys	Gln	Asn	Phe	Leu	Gln	Lys				
			835			840						845							
Leu	Arg	Phe	Leu	Ala	Asp	Glu	Pro	Gln	His	Ser	Ile	Pro	Asp	Val	Phe				
850						855			860										
Ile	Trp	Met	Met	Ser	Asn	Asn	Lys	Arg	Ile	Ala	Tyr	Ala	Arg	Val	Pro				
865			870						875			880							
Ser	Lys	Asp	Leu	Leu	Phe	Ser	Ile	Val	Glu	Glu	Glu	Leu	Gly	Lys	Asp				
			885						890			895							
Cys	Ala	Lys	Val	Lys	Thr	Leu	Phe	Leu	Lys	Leu	Pro	Gly	Lys	Arg	Gly				
			900			905						910							
Phe	Gly	Ser	Ala	Gly	Trp	Thr	Val	Gln	Ala	Lys	Leu	Glu	Leu	Tyr	Leu				
			915			920						925							
Trp	Leu	Gly	Leu	Ser	Lys	Gln	Arg	Lys	Asp	Phe	Leu	Cys	Gly	Leu	Pro				
930						935			940										
Cys	Gly	Phe	Glu	Glu	Val	Lys	Ala	Ala	Gln	Gly	Leu	Gly	Leu	His	Ser				
945			950						955			960							
Phe	Pro	Pro	Ile	Ser	Leu	Val	Tyr	Thr	Lys	Lys	Gln	Ala	Phe	Gln	Leu				
			965						970			975							
Arg	Ala	His	Met	Tyr	Gln	Ala	Arg	Ser	Leu	Phe	Ala	Ala	Asp	Ser	Ser				
			980			985						990							
Gly	Leu	Ser	Asp	Pro	Phe	Ala	Arg	Val	Phe	Phe	Ile	Asn	Gln	Ser	Gln				
			995			1000						1005							
Cys	Thr	Glu	Val	Leu	Asn	Glu	Thr	Leu	Cys	Pro	Thr	Trp	Asp	Gln					
1010						1015						1020							
Met	Leu	Val	Phe	Asp	Asn	Leu	Glu	Leu	Tyr	Gly	Glu	Ala	His	Glu					
1025						1030						1035							
Leu	Arg	Asp	Asp	Pro	Pro	Ile	Ile	Val	Ile	Glu	Ile	Tyr	Asp	Gln					
1040						1045						1050							
Asp	Ser	Met	Gly	Lys	Ala	Asp	Phe	Met	Gly	Arg	Thr	Phe	Ala	Lys					
1055						1060						1065							

-continued

Pro Leu Val Lys Met Ala Asp	Glu Ala Tyr Cys Pro	Pro Arg Phe
1070	1075	1080
Pro Pro Gln Leu Glu Tyr Tyr	Gln Ile Tyr Arg Gly	Ser Ala Thr
1085	1090	1095
Ala Gly Asp Leu Leu Ala Ala	Phe Glu Leu Leu Gln	Ile Gly Pro
1100	1105	1110
Ser Gly Lys Ala Asp Leu Pro	Pro Ile Asn Gly Pro	Val Asp Met
1115	1120	1125
Asp Arg Gly Pro Ile Met Pro	Val Pro Val Gly Ile	Arg Pro Val
1130	1135	1140
Leu Ser Lys Tyr Arg Val Glu	Val Leu Phe Trp Gly	Leu Arg Asp
1145	1150	1155
Leu Lys Arg Val Asn Leu Ala	Gln Val Asp Arg Pro	Arg Val Asp
1160	1165	1170
Ile Glu Cys Ala Gly Lys Gly	Val Gln Ser Ser Leu	Ile His Asn
1175	1180	1185
Tyr Lys Lys Asn Pro Asn Phe	Asn Thr Leu Val Lys	Trp Phe Glu
1190	1195	1200
Val Asp Leu Pro Glu Asn Glu	Leu Leu His Pro Pro	Leu Asn Ile
1205	1210	1215
Arg Val Val Asp Cys Arg Ala	Phe Gly Arg Tyr Thr	Leu Val Gly
1220	1225	1230
Ser His Ala Val Ser Ser Leu	Arg Arg Phe Ile Tyr	Arg Pro Pro
1235	1240	1245
Asp Arg Ser Ala Pro Asn Trp	Asn Thr Thr Gly Glu	Val Val Val
1250	1255	1260
Ser Met Glu Pro Glu Glu Pro	Val Lys Lys Leu Glu	Thr Met Val
1265	1270	1275
Lys Leu Asp Ala Thr Ser Asp	Ala Val Val Lys Val	Asp Val Ala
1280	1285	1290
Glu Asp Glu Lys Glu Arg Lys	Lys Lys Lys Lys Lys	Gly Pro Ser
1295	1300	1305
Glu Glu Pro Glu Glu Glu Glu	Pro Asp Glu Ser Met	Leu Asp Trp
1310	1315	1320
Trp Ser Lys Tyr Phe Ala Ser	Ile Asp Thr Met Lys	Glu Gln Leu
1325	1330	1335
Arg Gln His Glu Thr Ser Gly	Thr Asp Leu Glu Glu	Lys Glu Glu
1340	1345	1350
Met Glu Ser Ala Glu Gly Leu	Lys Gly Pro Met Lys	Ser Lys Glu
1355	1360	1365
Lys Ser Arg Ala Ala Lys Glu	Glu Lys Lys Lys Lys	Asn Gln Ser
1370	1375	1380
Pro Gly Pro Gly Gln Gly Ser	Glu Ala Pro Glu Lys	Lys Lys Ala
1385	1390	1395
Lys Ile Asp Glu Leu Lys Val	Tyr Pro Lys Glu Leu	Glu Ser Glu
1400	1405	1410
Phe Asp Ser Phe Glu Asp Trp	Leu His Thr Phe Asn	Leu Leu Arg
1415	1420	1425
Gly Lys Thr Gly Asp Asp Glu	Asp Gly Ser Thr Glu	Glu Glu Arg
1430	1435	1440
Ile Val Gly Arg Phe Lys Gly	Ser Leu Cys Val Tyr	Lys Val Pro
1445	1450	1455

-continued

Leu Pro 1460	Glu Asp Val Ser Arg 1465	Glu Ala Gly Tyr Asp 1470	Pro Thr Tyr
Gly Met 1475	Phe Gln Gly Ile Pro 1480	Ser Asn Asp Pro Ile 1485	Asn Val Leu
Val Arg 1490	Ile Tyr Val Val Arg 1495	Ala Thr Asp Leu His 1500	Pro Ala Asp
Ile Asn 1505	Gly Lys Ala Asp Pro 1510	Tyr Ile Ala Ile Lys 1515	Leu Gly Lys
Thr Asp 1520	Ile Arg Asp Lys Glu 1525	Asn Tyr Ile Ser Lys 1530	Gln Leu Asn
Pro Val 1535	Phe Gly Lys Ser Phe 1540	Asp Ile Glu Ala Ser 1545	Phe Pro Met
Glu Ser 1550	Met Leu Thr Val Ala 1555	Val Tyr Asp Trp Asp 1560	Leu Val Gly
Thr Asp 1565	Asp Leu Ile Gly Glu 1570	Thr Lys Ile Asp Leu 1575	Glu Asn Arg
Phe Tyr 1580	Ser Lys His Arg Ala 1585	Thr Cys Gly Ile Ala 1590	Gln Thr Tyr
Ser Ile 1595	His Gly Tyr Asn Ile 1600	Trp Arg Asp Pro Met 1605	Lys Pro Ser
Gln Ile 1610	Leu Thr Arg Leu Cys 1615	Lys Glu Gly Lys Val 1620	Asp Gly Pro
His Phe 1625	Gly Pro His Gly Arg 1630	Val Arg Val Ala Asn 1635	Arg Val Phe
Thr Gly 1640	Pro Ser Glu Ile Glu 1645	Asp Glu Asn Gly Gln 1650	Arg Lys Pro
Thr Asp 1655	Glu His Val Ala Leu 1660	Ser Ala Leu Arg His 1665	Trp Glu Asp
Ile Pro 1670	Arg Val Gly Cys Arg 1675	Leu Val Pro Glu His 1680	Val Glu Thr
Arg Pro 1685	Leu Leu Asn Pro Asp 1690	Lys Pro Gly Ile Glu 1695	Gln Gly Arg
Leu Glu 1700	Leu Trp Val Asp Met 1705	Phe Pro Met Asp Met 1710	Pro Ala Pro
Gly Thr 1715	Pro Leu Asp Ile Ser 1720	Pro Arg Lys Pro Lys 1725	Lys Tyr Glu
Leu Arg 1730	Val Ile Val Trp Asn 1735	Thr Asp Glu Val Val 1740	Leu Glu Asp
Asp Asp 1745	Phe Phe Thr Gly Glu 1750	Lys Ser Ser Asp Ile 1755	Phe Val Arg
Gly Trp 1760	Leu Lys Gly Gln Gln 1765	Glu Asp Lys Gln Asp 1770	Thr Asp Val
His Tyr 1775	His Ser Leu Thr Gly 1780	Glu Gly Asn Phe Asn 1785	Trp Arg Tyr
Leu Phe 1790	Pro Phe Asp Tyr Leu 1795	Ala Ala Glu Glu Lys 1800	Ile Val Met
Ser Lys 1805	Lys Glu Ser Met Phe 1810	Ser Trp Asp Glu Thr 1815	Glu Tyr Lys
Ile Pro 1820	Ala Arg Leu Thr Leu 1825	Gln Ile Trp Asp Ala 1830	Asp His Phe
Ser Ala 1835	Asp Asp Phe Leu Gly 1840	Ala Ile Glu Leu Asp 1845	Leu Asn Arg
Phe Pro	Arg Gly Ala Lys Thr	Ala Lys Gln Cys Thr	Met Glu Met

-continued

1850	1855	1860
Ala Thr Gly Glu Val Asp Val Pro Leu Val Ser Ile Phe Lys Gln		
1865	1870	1875
Lys Arg Val Lys Gly Trp Trp Pro Leu Leu Ala Arg Asn Glu Asn		
1880	1885	1890
Asp Glu Phe Glu Leu Thr Gly Lys Val Glu Ala Glu Leu His Leu		
1895	1900	1905
Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro Val Gly Leu Ala Arg		
1910	1915	1920
Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn Arg Pro Asp Thr Ala		
1925	1930	1935
Phe Val Trp Phe Leu Asn Pro Leu Lys Ser Ile Lys Tyr Leu Ile		
1940	1945	1950
Cys Thr Arg Tyr Lys Trp Leu Ile Ile Lys Ile Val Leu Ala Leu		
1955	1960	1965
Leu Gly Leu Leu Met Leu Ala Leu Phe Leu Tyr Ser Leu Pro Gly		
1970	1975	1980
Tyr Met Val Lys Lys Leu Leu Gly Ala		
1985	1990	

<210> SEQ ID NO 9

<211> LENGTH: 1977

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 9

Met Ala Leu Ile Val His Leu Lys Thr Val Ser Glu Leu Arg Gly Lys		
1	5	10 15
Gly Asp Arg Ile Ala Lys Val Thr Phe Arg Gly Gln Ser Phe Tyr Ser		
20	25	30
Arg Val Leu Glu Asn Cys Glu Gly Val Ala Asp Phe Asp Glu Thr Phe		
35	40	45
Arg Trp Pro Val Ala Ser Ser Ile Asp Arg Asn Glu Val Leu Glu Ile		
50	55	60
Gln Ile Phe Asn Tyr Ser Lys Val Phe Ser Asn Lys Leu Ile Gly Thr		
65	70	75 80
Phe Cys Met Val Leu Gln Lys Val Val Glu Glu Asn Arg Val Glu Val		
85	90	95
Thr Asp Thr Leu Met Asp Asp Ser Asn Ala Ile Ile Lys Thr Ser Leu		
100	105	110
Ser Met Glu Val Arg Tyr Gln Ala Thr Asp Gly Thr Val Gly Pro Trp		
115	120	125
Asp Asp Gly Asp Phe Leu Gly Asp Glu Ser Leu Gln Glu Glu Lys Asp		
130	135	140
Ser Gln Glu Thr Asp Gly Leu Leu Pro Gly Ser Arg Pro Ser Thr Arg		
145	150	155 160
Ile Ser Gly Glu Lys Ser Phe Arg Arg Ala Gly Arg Ser Val Phe Ser		
165	170	175
Ala Met Lys Leu Gly Lys Thr Arg Ser His Lys Glu Glu Pro Gln Arg		
180	185	190
Gln Asp Glu Pro Ala Val Leu Glu Met Glu Asp Leu Asp His Leu Ala		
195	200	205
Ile Gln Leu Gly Asp Gly Leu Asp Pro Asp Ser Val Ser Leu Ala Ser		
210	215	220

-continued

Val Thr Ala Leu Thr Ser Asn Val Ser Asn Lys Arg Ser Lys Pro Asp	225	230	235	240
Ile Lys Met Glu Pro Ser Ala Gly Arg Pro Met Asp Tyr Gln Val Ser	245	250	255	
Ile Thr Val Ile Glu Ala Arg Gln Leu Val Gly Leu Asn Met Asp Pro	260	265	270	
Val Val Cys Val Glu Val Gly Asp Asp Lys Lys Tyr Thr Ser Met Lys	275	280	285	
Glu Ser Thr Asn Cys Pro Tyr Tyr Asn Glu Tyr Phe Val Phe Asp Phe	290	295	300	
His Val Ser Pro Asp Val Met Phe Asp Lys Ile Ile Lys Ile Ser Val	305	310	315	320
Ile His Ser Lys Asn Leu Leu Arg Ser Gly Thr Leu Val Gly Ser Phe	325	330	335	
Lys Met Asp Val Gly Thr Val Tyr Ser Gln Pro Glu His Gln Phe His	340	345	350	
His Lys Trp Ala Ile Leu Ser Asp Pro Asp Asp Ile Ser Ala Gly Leu	355	360	365	
Lys Gly Tyr Val Lys Cys Asp Val Ala Val Val Gly Lys Gly Asp Asn	370	375	380	
Ile Lys Thr Pro His Lys Ala Asn Glu Thr Asp Glu Asp Asp Ile Glu	385	390	395	400
Gly Asn Leu Leu Leu Pro Glu Gly Val Pro Pro Glu Arg Gln Trp Ala	405	410	415	
Arg Phe Tyr Val Lys Ile Tyr Arg Ala Glu Gly Leu Pro Arg Met Asn	420	425	430	
Thr Ser Leu Met Ala Asn Val Lys Lys Ala Phe Ile Gly Glu Asn Lys	435	440	445	
Asp Leu Val Asp Pro Tyr Val Gln Val Phe Phe Ala Gly Gln Lys Gly	450	455	460	
Lys Thr Ser Val Gln Lys Ser Ser Tyr Glu Pro Leu Trp Asn Glu Gln	465	470	475	480
Val Val Phe Thr Asp Leu Phe Pro Pro Leu Cys Lys Arg Met Lys Val	485	490	495	
Gln Ile Arg Asp Ser Asp Lys Val Asn Asp Val Ala Ile Gly Thr His	500	505	510	
Phe Ile Asp Leu Arg Lys Ile Ser Asn Asp Gly Asp Lys Gly Phe Leu	515	520	525	
Pro Thr Leu Gly Pro Ala Trp Val Asn Met Tyr Gly Ser Thr Arg Asn	530	535	540	
Tyr Thr Leu Leu Asp Glu His Gln Asp Leu Asn Glu Gly Leu Gly Glu	545	550	555	560
Gly Val Ser Phe Arg Ala Arg Leu Met Leu Gly Leu Ala Val Glu Ile	565	570	575	
Leu Asp Thr Ser Asn Pro Glu Leu Thr Ser Ser Thr Glu Val Gln Val	580	585	590	
Glu Gln Ala Thr Pro Val Ser Glu Ser Cys Thr Gly Arg Met Glu Glu	595	600	605	
Phe Phe Leu Phe Gly Ala Phe Leu Glu Ala Ser Met Ile Asp Arg Lys	610	615	620	
Asn Gly Asp Lys Pro Ile Thr Phe Glu Val Thr Ile Gly Asn Tyr Gly	625	630	635	640
Asn Glu Val Asp Gly Met Ser Arg Pro Leu Arg Pro Arg Pro Arg Lys				

-continued

645								650				655			
Glu	Pro	Gly	Asp	Glu	Glu	Glu	Val	Asp	Leu	Ile	Gln	Asn	Ser	Ser	Asp
			660					665					670		
Asp	Glu	Gly	Asp	Glu	Ala	Gly	Asp	Leu	Ala	Ser	Val	Ser	Ser	Thr	Pro
			675				680					685			
Pro	Met	Arg	Pro	Gln	Ile	Thr	Asp	Arg	Asn	Tyr	Phe	His	Leu	Pro	Tyr
			690			695					700				
Leu	Glu	Arg	Lys	Pro	Cys	Ile	Tyr	Ile	Lys	Ser	Trp	Trp	Pro	Asp	Gln
705					710					715					720
Arg	Arg	Arg	Leu	Tyr	Asn	Ala	Asn	Ile	Met	Asp	His	Ile	Ala	Asp	Lys
			725						730					735	
Leu	Glu	Glu	Gly	Leu	Asn	Asp	Val	Gln	Glu	Met	Ile	Lys	Thr	Glu	Lys
			740					745					750		
Ser	Tyr	Pro	Glu	Arg	Arg	Leu	Arg	Gly	Val	Leu	Glu	Glu	Leu	Ser	Cys
			755				760					765			
Gly	Cys	His	Arg	Phe	Leu	Ser	Leu	Ser	Asp	Lys	Asp	Gln	Gly	Arg	Ser
	770					775					780				
Ser	Arg	Thr	Arg	Leu	Asp	Arg	Glu	Arg	Leu	Lys	Ser	Cys	Met	Arg	Glu
785					790					795					800
Leu	Glu	Ser	Met	Gly	Gln	Gln	Ala	Lys	Ser	Leu	Arg	Ala	Gln	Val	Lys
			805						810						815
Arg	His	Thr	Val	Arg	Asp	Lys	Leu	Arg	Ser	Cys	Gln	Asn	Phe	Leu	Gln
			820					825						830	
Lys	Leu	Arg	Phe	Leu	Ala	Asp	Glu	Pro	Gln	His	Ser	Ile	Pro	Asp	Val
		835					840						845		
Phe	Ile	Trp	Met	Met	Ser	Asn	Asn	Lys	Arg	Ile	Ala	Tyr	Ala	Arg	Val
	850					855					860				
Pro	Ser	Lys	Asp	Leu	Leu	Phe	Ser	Ile	Val	Glu	Glu	Glu	Leu	Gly	Lys
865					870					875					880
Asp	Cys	Ala	Lys	Val	Lys	Thr	Leu	Phe	Leu	Lys	Leu	Pro	Gly	Lys	Arg
			885						890					895	
Gly	Phe	Gly	Ser	Ala	Gly	Trp	Thr	Val	Gln	Ala	Lys	Leu	Glu	Leu	Tyr
			900					905						910	
Leu	Trp	Leu	Gly	Leu	Ser	Lys	Gln	Arg	Lys	Asp	Phe	Leu	Cys	Gly	Leu
		915					920						925		
Pro	Cys	Gly	Phe	Glu	Glu	Val	Lys	Ala	Ala	Gln	Gly	Leu	Gly	Leu	His
	930						935					940			
Ser	Phe	Pro	Pro	Ile	Ser	Leu	Val	Tyr	Thr	Lys	Lys	Gln	Ala	Phe	Gln
945					950					955					960
Leu	Arg	Ala	His	Met	Tyr	Gln	Ala	Arg	Ser	Leu	Phe	Ala	Ala	Asp	Ser
			965						970					975	
Ser	Gly	Leu	Ser	Asp	Pro	Phe	Ala	Arg	Val	Phe	Phe	Ile	Asn	Gln	Ser
		980						985					990		
Gln	Cys	Thr	Glu	Val	Leu	Asn	Glu	Thr	Leu	Cys	Pro	Thr	Trp	Asp	Gln
		995					1000						1005		
Met	Leu	Val	Phe	Asp	Asn	Leu	Glu	Leu	Tyr	Gly	Glu	Ala	His	Glu	
	1010						1015					1020			
Leu	Arg	Asp	Asp	Pro	Pro	Ile	Ile	Val	Ile	Glu	Ile	Tyr	Asp	Gln	
	1025						1030					1035			
Asp	Ser	Met	Gly	Lys	Ala	Asp	Phe	Met	Gly	Arg	Thr	Phe	Ala	Lys	
	1040						1045					1050			
Pro	Leu	Val	Lys	Met	Ala	Asp	Glu	Ala	Tyr	Cys	Pro	Pro	Arg	Phe	
	1055						1060					1065			

-continued

Pro	Pro	Gln	Leu	Glu	Tyr	Tyr	Gln	Ile	Tyr	Arg	Gly	Ser	Ala	Thr
1070						1075					1080			
Ala	Gly	Asp	Leu	Leu	Ala	Ala	Phe	Glu	Leu	Leu	Gln	Ile	Gly	Pro
1085						1090					1095			
Ser	Gly	Lys	Ala	Asp	Leu	Pro	Pro	Ile	Asn	Gly	Pro	Val	Asp	Met
1100						1105					1110			
Asp	Arg	Gly	Pro	Ile	Met	Pro	Val	Pro	Val	Gly	Ile	Arg	Pro	Val
1115						1120					1125			
Leu	Ser	Lys	Tyr	Arg	Val	Glu	Val	Leu	Phe	Trp	Gly	Leu	Arg	Asp
1130						1135					1140			
Leu	Lys	Arg	Val	Asn	Leu	Ala	Gln	Val	Asp	Arg	Pro	Arg	Val	Asp
1145						1150					1155			
Ile	Glu	Cys	Ala	Gly	Lys	Gly	Val	Gln	Ser	Ser	Leu	Ile	His	Asn
1160						1165					1170			
Tyr	Lys	Lys	Asn	Pro	Asn	Phe	Asn	Thr	Leu	Val	Lys	Trp	Phe	Glu
1175						1180					1185			
Val	Asp	Leu	Pro	Glu	Asn	Glu	Leu	Leu	His	Pro	Pro	Leu	Asn	Ile
1190						1195					1200			
Arg	Val	Val	Asp	Cys	Arg	Ala	Phe	Gly	Arg	Tyr	Thr	Leu	Val	Gly
1205						1210					1215			
Ser	His	Ala	Val	Ser	Ser	Leu	Arg	Arg	Phe	Ile	Tyr	Arg	Pro	Pro
1220						1225					1230			
Asp	Arg	Ser	Ala	Pro	Asn	Trp	Asn	Thr	Thr	Gly	Glu	Val	Val	Val
1235						1240					1245			
Ser	Met	Glu	Pro	Glu	Glu	Pro	Val	Lys	Lys	Leu	Glu	Thr	Met	Val
1250						1255					1260			
Lys	Leu	Asp	Ala	Thr	Ser	Asp	Ala	Val	Val	Lys	Val	Asp	Val	Ala
1265						1270					1275			
Glu	Asp	Glu	Lys	Glu	Arg	Lys	Lys	Lys	Lys	Lys	Lys	Gly	Pro	Ser
1280						1285					1290			
Glu	Glu	Pro	Glu	Glu	Glu	Glu	Pro	Asp	Glu	Ser	Met	Leu	Asp	Trp
1295						1300					1305			
Trp	Ser	Lys	Tyr	Phe	Ala	Ser	Ile	Asp	Thr	Met	Lys	Glu	Gln	Leu
1310						1315					1320			
Arg	Gln	His	Glu	Thr	Ser	Gly	Thr	Asp	Leu	Glu	Glu	Lys	Glu	Glu
1325						1330					1335			
Met	Glu	Ser	Ala	Glu	Gly	Leu	Lys	Gly	Pro	Met	Lys	Ser	Lys	Glu
1340						1345					1350			
Lys	Ser	Arg	Ala	Ala	Lys	Glu	Glu	Lys	Lys	Lys	Lys	Asn	Gln	Ser
1355						1360					1365			
Pro	Gly	Pro	Gly	Gln	Gly	Ser	Glu	Ala	Pro	Glu	Lys	Lys	Lys	Ala
1370						1375					1380			
Lys	Ile	Asp	Glu	Leu	Lys	Val	Tyr	Pro	Lys	Glu	Leu	Glu	Ser	Glu
1385						1390					1395			
Phe	Asp	Ser	Phe	Glu	Asp	Trp	Leu	His	Thr	Phe	Asn	Leu	Leu	Arg
1400						1405					1410			
Gly	Lys	Thr	Gly	Asp	Asp	Glu	Asp	Gly	Ser	Thr	Glu	Glu	Glu	Arg
1415						1420					1425			
Ile	Val	Gly	Arg	Phe	Lys	Gly	Ser	Leu	Cys	Val	Tyr	Lys	Val	Pro
1430						1435					1440			
Leu	Pro	Glu	Asp	Val	Ser	Arg	Glu	Ala	Gly	Tyr	Asp	Pro	Thr	Tyr
1445						1450					1455			

-continued

Gly Met 1460	Phe Gln Gly Ile 1465	Ser Asn Asp 1470	Asn Val Leu
Val Arg 1475	Ile Tyr Val Val 1480	Ala Thr Asp 1485	Pro Ala Asp
Ile Asn 1490	Gly Lys Ala Asp 1495	Tyr Ile Ala Ile 1500	Leu Gly Lys
Thr Asp 1505	Ile Arg Asp Lys 1510	Asn Tyr Ile Ser 1515	Gln Leu Asn
Pro Val 1520	Phe Gly Lys Ser 1525	Asp Ile Glu Ala 1530	Phe Pro Met
Glu Ser 1535	Met Leu Thr Val 1540	Val Tyr Asp Trp 1545	Leu Val Gly
Thr Asp 1550	Asp Leu Ile Gly 1555	Thr Lys Ile Asp 1560	Glu Asn Arg
Phe Tyr 1565	Ser Lys His Arg 1570	Thr Cys Gly Ile 1575	Gln Thr Tyr
Ser Ile 1580	His Gly Tyr Asn 1585	Trp Arg Asp Pro 1590	Lys Pro Ser
Gln Ile 1595	Leu Thr Arg Leu 1600	Lys Glu Gly Lys 1605	Asp Gly Pro
His Phe 1610	Gly Pro His Gly 1615	Val Arg Val Ala 1620	Arg Val Phe
Thr Gly 1625	Pro Ser Glu Ile 1630	Asp Glu Asn Gly 1635	Arg Lys Pro
Thr Asp 1640	Glu His Val Ala 1645	Ser Ala Leu Arg 1650	Trp Glu Asp
Ile Pro 1655	Arg Val Gly Cys 1660	Leu Val Pro Glu 1665	Val Glu Thr
Arg Pro 1670	Leu Leu Asn Pro 1675	Lys Pro Gly Ile 1680	Gln Gly Arg
Leu Glu 1685	Leu Trp Val Asp 1690	Phe Pro Met Asp 1695	Pro Ala Pro
Gly Thr 1700	Pro Leu Asp Ile 1705	Pro Arg Lys Pro 1710	Lys Tyr Glu
Leu Arg 1715	Val Ile Val Trp 1720	Thr Asp Glu Val 1725	Leu Glu Asp
Asp Asp 1730	Phe Phe Thr Gly 1735	Lys Ser Ser Asp 1740	Phe Val Arg
Gly Trp 1745	Leu Lys Gly Gln 1750	Glu Asp Lys Gln 1755	Thr Asp Val
His Tyr 1760	His Ser Leu Thr 1765	Glu Gly Asn Phe 1770	Trp Arg Tyr
Leu Phe 1775	Pro Phe Asp Tyr 1780	Ala Ala Glu Glu 1785	Ile Val Met
Ser Lys 1790	Lys Glu Ser Met 1795	Ser Trp Asp Glu 1800	Glu Tyr Lys
Ile Pro 1805	Ala Arg Leu Thr 1810	Gln Ile Trp Asp 1815	Asp His Phe
Ser Ala 1820	Asp Asp Phe Leu 1825	Ala Ile Glu Leu 1830	Leu Asn Arg
Phe Pro 1835	Arg Gly Ala Lys 1840	Ala Lys Gln Cys 1845	Met Glu Met
Ala Thr	Gly Glu Val Asp Val	Pro Leu Val Ser Ile	Phe Lys Glu

-continued

1850	1855	1860
Lys Arg Val Lys Gly Trp Trp Pro Leu Leu Ala Arg Asn Glu Asn		
1865	1870	1875
Asp Glu Phe Glu Leu Thr Gly Lys Val Glu Ala Glu Leu His Leu		
1880	1885	1890
Leu Thr Ala Glu Glu Ala Glu Lys Asn Pro Val Gly Leu Ala Arg		
1895	1900	1905
Asn Glu Pro Asp Pro Leu Glu Lys Pro Asn Arg Pro Asp Thr Ala		
1910	1915	1920
Phe Val Trp Phe Leu Asn Pro Leu Lys Ser Ile Lys Tyr Leu Ile		
1925	1930	1935
Cys Thr Arg Tyr Lys Trp Leu Ile Ile Lys Ile Val Leu Ala Leu		
1940	1945	1950
Leu Gly Leu Leu Met Leu Ala Leu Phe Leu Tyr Ser Leu Pro Gly		
1955	1960	1965
Tyr Met Val Lys Lys Leu Leu Gly Ala		
1970	1975	

<210> SEQ ID NO 10

<211> LENGTH: 7171

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 10

```

atcggaggagg ggtcgggagg agggaggagga ggcagcggca gagaagagag aggcgtgtga      60
gccgtgctcc accggctagc tccttcccgc tgctcctgcc tggcagtgcc aggcagccca      120
caccagcatg gccttgctca tccacctcaa gacagtctcg gagctgcggg gcaggggcca      180
ccggatcgcc aaagtgaact tccgagggca atcctttctac tctcgggtcc tggagaactg      240
tgaggatgtg gctgactttg atgagacatt tcggtggccg gtggccagca gcatcgacag      300
aaatgagatg ctggagattc aggttttcaa ctacagcaaa gtcttcagca acaagctcat      360
cgggaccttc cgcattgtgc tgcagaaggt ggtagaggag agccatgtgg aggtgactga      420
cacgtgatt gatgacaaca atgctatcat caagaccagc ctgtgcgtgg aggtccggta      480
tcaggccact gacggcacag tgggtcctcg ggacgatggg gacttcctgg gagatgagtc      540
tcttcaagag gaagagaagg acagccaaga gacggatgga ctgtcccag gctcccggcc      600
cagctcccg ccccccaggag agaagagctt ccggagagcc gggaggagcg tgttctccgc      660
catgaagctc ggcaaaaacc ggtctcacia ggaggagccc caaagaccag atgaaccggc      720
ggtgtcggag atggaagacc ttgaccatct ggccattcgg ctaggagatg gactggatcc      780
cgactcggtg tctctagcct cagtcacagc tctcaccact aatgtctcca acaagcgatc      840
taagccagac attaagatgg agccaagtgc tgggcggccc atggattacc aggtcagcat      900
cacgggtgatc gagggccggc agctggtggg cttgaacatg gaccctgtgg tgtgcgtgga      960
ggtgggtgac gacaagaagt acacatccat gaaggagtcc actaactgcc cctattacaa     1020
cgagtacttc gtcttcgact tccatgtctc tccgatgtc atgtttgaca agatcatcaa     1080
gatttcggtg attcactcca agaacctgct gcgcagtggc accctggtgg gctccttcaa     1140
aatggacgtg ggaaccgtgt actcgcagcc agagcaccag ttccatcaca agtgggcca      1200
cctgtctgac cccgatgaca tctcctcggg gctgaagggc tacgtgaagt gtgacgttgc     1260
cgtggtgggc aaaggggaca acatcaagac gcccacaag gccaatgaga ccgacgaaga     1320
tgacattgag gggaaacttg tgctccccga gggggtgccc ccgaacgcc agtgggccc      1380

```

-continued

gttctatgtg aaaatttacc gagcagaggg gctgccccgt atgaacacaa gectcatggc	1440
caatgtaaag aaggctttca tcggtgaaaa caaggacctc gtggacccct acgtgcaagt	1500
cttctttgct ggccagaagg gcaagacttc agtgacagaag agcagctatg agccccctgtg	1560
gaatgagcag gtcgtcttta cagacctctt cccccactc tgcaaacgca tgaaggtgca	1620
gatccgagac tcggacaagg tcaacgacgt ggccatcggc acccacttca ttgacctgcg	1680
caagatttct aatgacggag acaaaaggctt cctgcccaca ctgggcccag cctgggtgaa	1740
catgtacggc tccacacgta actacacgct gctggatgag catcaggacc tgaacgaggg	1800
cctgggggag ggtgtgtcct tccgggcccg gctcctgctg ggcttgctg tggagatcgt	1860
agacacctcc aacctgagc tcaccagctc cacagagggtg caggtggagc aggccacgcc	1920
catctcggag agctgtgcag gtaaaatgga agaattcttt ctctttggag ccttctgga	1980
ggcctcaatg atcgaccgga gaaacggaga caagccctc acctttgagg tcaccatagg	2040
caactatggg aacgaagtgt atggcctgtc ccggccccag cggcctcggc cccggaagga	2100
gccgggggat gaggaagaag tagacctgat tcagaacgca agtgatgacg aggccggtga	2160
tgccggggac ctggcctcag tctcctccac tccaccaatg cggccccagg tcaccgacag	2220
gaactacttc catctgcctt acctggagcg aaagccctgc atctacatca agagctgggtg	2280
gccggaccag cgcgcgcgcc tctacaatgc caacatcatg gaccacattg ccgacaagct	2340
ggaagaaggc ctgaacgaca tacaggagat gatcaaaacg gagaagtcct acctgagcg	2400
tcgcctgcgg ggcgtcctgg aggagctgag ctgtggctgc tgccgcttcc tctccctcgc	2460
tgacaaggac cagggccact catcccgac caggcttgac cgggagcgcc tcaagtctg	2520
catgagggag ctgaaaaaca tggggcagca ggccaggatg ctgcgggcc aggtgaagcg	2580
gcacacgggt cgggacaagc tgaggctgtg ccagaacttc ctgcagaagc tgcgttctct	2640
ggcggacgag cccagcacca gcattcccga catcttcac tggtatgata gcaacaacaa	2700
gcgtgtcgcc tatgccgtg tgccctccaa ggacctgctc ttctccatcg tggaggagga	2760
gactggcaag gactgcgcca aggtcaagac gctcttctt aagctgccag ggaagcgggg	2820
cttcggctcg gcaggctgga cagtgcaggc caaggtggag ctgtacctgt ggctgggct	2880
cagcaaacag cgcaaggagt tcctgtgcgg cctgccctgt ggcttcagg aggtcaaggc	2940
agcccagggc ctgggcctgc atgccttccc acccgtcagc ctggtctaca ccaagaagca	3000
ggcgttccag ctccgagcgc acatgtacca ggcccgagc ctctttgccg ccgacagcag	3060
cggactctca gacctcttg cccgcgtctt ctcatcaat cagagtcagt gcacagaggt	3120
gctgaatgag acctgtgtc ccacctggga ccagatgctg gtgttcgaca acctggagct	3180
ctatggtgaa gctcatgagc tgagggacga tccgccctc attgtcattg aaatctatga	3240
ccaggattcc atgggcaaag ctgacttcat gggccggacc ttcgccaaac ccctggtgaa	3300
gatggcagac gaggcgctact gccacccccg ctcccccct cagctcagat actaccagat	3360
ctaccgtggc aacgccacag ctggagacct gctggcggcc ttcgagctgc tgcagattgg	3420
accagcaggg aaggctgacc tgccccccat caatggcccg gtggacgtgg accgaggtcc	3480
catcatgccc gtgcccatgg gcatccggcc cgtgctcagc aagtaccgag tggaggtgct	3540
gttctggggc ctacgggacc taaagcgggt gaacctggcc caggtggacc ggccacgggt	3600
ggacatcgag tgtgcaggga aggggggtgca gtcgtccctg atccacaatt ataagaagaa	3660
ccccaaactc aacaccctcg tcaagtgggt tgaagtggac ctcccagaga acgagctgct	3720

-continued

gcacccgccc	ttgaacatcc	gtgtggtgga	ctgccgggcc	tccggtcget	acacactggt	3780
gggtcccat	gccgtcagct	cctgcgacg	cttcattctac	cggccccag	accgctcggc	3840
ccccagctgg	aacaccacgg	tcaggcttct	cggcgctgc	cgtgtgctgt	gcaatggggg	3900
ctctctctct	cactccacag	gggaggttgt	ggtgactatg	gagccagagg	tacccatcaa	3960
gaaactggag	accatggtga	agctggacgc	gacttctgaa	gctgttgta	aggtggatgt	4020
ggctgaggag	gagaaggaga	agaagaagaa	gaagaagggc	actgcggagg	agccagagga	4080
ggaggagcca	gacgagagca	tgttggtgct	gtggtccaag	tactttgcct	ccattgacac	4140
catgaaggag	caacttcgac	aacaagagcc	ctctggaatt	gacttgagg	agaaggagga	4200
agtggacaat	accgagggcc	tgaaggggtc	aatgaagggc	aaggagaagg	caagggtgct	4260
caaagaggag	aagaagaaga	aaactcagag	ctctggctct	ggccaggggt	ccgaggcccc	4320
cgaagaaga	aaacccaaga	ttgatgagct	taagggtatac	cccaaagagc	tggagtcgga	4380
gtttgataac	tttgaggact	ggctgcacac	tttcaacttg	cttcggggca	agaccgggga	4440
tgatgaggat	ggctccaccg	aggaggagcg	cattgtggga	cgttcaagg	gctccctctg	4500
cgtgtacaaa	gtgccactcc	cagaggacgt	gtcccgggaa	gccggctacg	actccacct	4560
cggcatgttc	cagggcatcc	cagacaatga	ccccatcaat	gtgctggtcc	gagtctatgt	4620
ggtcggggcc	acggacctgc	accctgctga	catcaacggc	aaagctgacc	cctacatcgc	4680
catccggcta	ggcaagactg	acatccgcga	caaggagaac	tacatctcca	agcagctcaa	4740
ccctgtcttt	gggaagtctt	ttgacatcga	ggcctccttc	cccatggaat	ccatgctgac	4800
ggtggtctgt	tatgactggg	acctggtggg	cactgatgac	ctcattgggg	aaaccaagat	4860
cgaactggag	aaccgcttct	acagcaagca	cgcgcccacc	tgcggcatcg	cccagacct	4920
ctccacacat	ggctacaata	tctggcggga	ccccatgaag	cccagccaga	tctgacccg	4980
cctctgcaaa	gacggcaaa	tggaaggccc	ccactttggg	ccccctggga	gagtgaaggt	5040
ggccaaccgc	gtcttcaactg	ggccctctga	gattgaggac	gagaacggtc	agagggaagcc	5100
cacagacgag	catgtggcgc	tgttgccct	gaggcactgg	gaggacatcc	cccgcgcagg	5160
ctgcgcctg	gtgccagagc	atgtggagac	gaggccgctg	ctcaaccccg	acaagccggg	5220
catcgagcag	ggcgcctgg	agctgtgggt	ggacatgttc	cccatggaca	tgccagcccc	5280
tgggacgcct	ctggacatct	cacctcgga	gcccagaag	tacgagctgc	gggtcatcat	5340
ctggaacaca	gatgaggtgg	tcttgaggga	cgcgacttc	ttcacagggg	agaagtccag	5400
tgacatcttc	gtgagggggt	ggctgaagg	ccagcaggag	gacaagcagg	acacagacgt	5460
ccactaccac	tcctcactg	gcgagggcaa	cttcaactgg	cgtacactgt	tccccttcga	5520
ctacctggcg	gcgaggaga	agatcgat	ctccaagaag	gagtcacatg	tctcctggga	5580
cgaaccgag	tacaagatcc	cgcgcggct	cacctgcag	atctgggatg	cggaccactt	5640
ctccgctgac	gacttcttg	gggccatcga	gctggacctg	aaccggttc	cgcggggcgc	5700
aaagacagcc	aagcagtga	ccatggagat	ggccaccggg	gaggtggacg	tgccctcgt	5760
gtccatcttc	aagcaaaagc	gcgtcaaagg	ctggtggccc	ctcctggccc	gcaatgagaa	5820
cgatgagttt	gagctcacgg	gcaaggtgga	ggctgagctg	catttactga	cagcagagga	5880
ggcagagaag	aaccagtg	gcctggccc	caatgaacct	gacccctag	agaaacccaa	5940
cggcccgac	acgagcttc	tctggttcct	gaacctctc	aagtggctc	gctacttctt	6000
gtggcacacg	tatcgtggc	tgtcctcaa	actgttctg	ctcctgctgc	tgtcctcct	6060
cctcgccctg	tctcttact	ctgtgcctg	ctacctggtc	aagaaaatcc	tcggggcctg	6120

-continued

```

agcccagtgg cctcctggcc ggcccgcacac ggcccttcgtc tggttcctca accctctcaa 6180
gtccatcaag tacctcatct gcacccggta caagtggctc atcatcaaga tcgtgctggc 6240
gctgttgggg ctgctcatgt tggggctctt cctctacagc ctccctggct acatgggtcaa 6300
aaagctcctt ggggcatgaa ggccgccagc tcccgcagc cgtcccccag cctgcccga 6360
tttcttttca gtggcttgga ctcttttcca tctccctgg ggagcctgag gagccagcg 6420
tccactcttc atgccttggg ccgagcctgc ctctgcttg cggggggccgc ctgtcctcac 6480
tgccccaggc tgcggcttgc ccagtcgcc cctctgacc cctgcctgtg ggctggggag 6540
ccttggtatg ggtggggacc tggaatgggt ctctcttgcc ccacctggct gaggcgccac 6600
ccttcttcag gccagggctc cagaggaaga ctctgaaac cctccccagg tcttccaagt 6660
acaggattga agcttttagt aaattaacca aggaccatgg gtcagtggcc agggctttaa 6720
aaagaatgaa cgagcaaaag gtatccccgc cgtgaccct gcagatagca ccggtctttg 6780
atccgcagca ggggccagac cctgccaca agtcccagc cggctgttc tgccactgct 6840
gggtccact tggctctct cacttcccag ggggtgcct gtectgcctg tgggtttcca 6900
tggttcca gagctcctc tgcccagcc agcgcctcca gggcagctg aggagctgtg 6960
agaagcagca gaggggactc cccatcccgc gcacaccctg tcctcccacc cctgccccct 7020
tgcccttcca gccctttag ctgcagctgg gagctggccc gtcaagtgt gcccctgcct 7080
gtgtctgggt ttctgttggc tgttttctt ttcttgagt gtgattttt tctaaataa 7140
agaagtcaag cactgaaaaa aaaaaaaaaa a 7171

```

<210> SEQ ID NO 11

<211> LENGTH: 4969

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 11

```

ccgtgagttc tgcccaggcc ctgtgagctc accagagcca cagactcaca gccagaggt 60
ggcttcttcc ttcaggaact gaagaacccc catgaacacc aacatctcca ggttctgaga 120
acagaacctg ggaattgat gacttcttca tgatgaccga tactcaggat ggcctagcg 180
agagctccca gatcatgagg aagaaggcct gaacgacata caggagatga tcaaacgga 240
gaagtccac cctgagcgtc gctgcgggg cgtcctggag gagctgagct gtggtgctg 300
ccgttctctc tcctctgctg acaaggacca gggccactca tccgcacca ggttgaccg 360
ggagcgctc aagtctgca tgagggagct ggaaaacatg gggcagcagg ccaggatgct 420
gctggccag gtgaagcggc acacggtgcg ggacaagctg aggtgtgccc agaacttct 480
gcagaagctg cgttctctg cggacgagcc ccagcacagc attcccgaca tcttcatctg 540
gatgatgagc aacaacaagc gtgtcgcta tgcccggtg ccctccaagg acctgctctt 600
ctccatcgtg gaggaggaga ctggcaagga ctgcgccaa gtcaagacgc tcttcttaa 660
gtgcccaggg aagcggggct tcggctcggc aggtggaca gtgcaggcca aggtggagct 720
gtacctgtgg ctgggcctca gcaaacagcg caaggagtgc ctgtgcggcc tgccctgtgg 780
cttcaggag gtcaaggcag ccaggggcct gggcctgcat gccttcccac ccgtcagcct 840
ggtctacacc aagaagcagg cgttccagct ccgagcgcac atgtaccagg cccgcagcct 900
ctttgccgcc gacagcagcg gactctcaga cccctttgcc cgcgtcttct tcatcaatca 960
gagtcagtc acagaggtgc tgaatgagac cctgtgtccc acctgggacc agatgctggt 1020

```

-continued

gttcgacaac	ctggagctct	atggtgaagc	tcatgagctg	agggacgac	cgcccatcat	1080
tgtcattgaa	atctatgacc	aggattccat	gggcaaagct	gacttcatgg	gccggacctt	1140
cgccaaaccc	ctggtgaaga	tggcagacga	ggcgtactgc	ccaccccgct	tcccacctca	1200
gctcgagtac	taccagatct	accgtggcaa	cgccacagct	ggagacctgc	tggcggcctt	1260
cgagctgctg	cagattggac	cagcagggaa	ggctgacctg	cccccatca	atggcccggg	1320
ggacgtggac	cgaggtccca	tcatgcccgt	gcccattggc	atccggcccg	tgtcagcaa	1380
gtaccgagtg	gaggtgctgt	tctggggcct	acgggacctc	aagcgggtga	acctggccca	1440
ggtggaccgg	ccacgggtgg	acatcgagtg	tgcagggaa	gggtgcagt	cgtccctgat	1500
ccacaattat	aagaagaacc	ccaacttcaa	caccctcgtc	aagtggtttg	aagtggacct	1560
cccagagaac	gagctgctgc	accgcacctt	gaacatccgt	gtggtggact	gccgggcctt	1620
cggctcgctac	acactgtgtg	gctcccatgc	cgctcagctc	ctgcgacgct	tcatctaccg	1680
gccccagac	cgctcggccc	ccagctggaa	caccacgggg	gaggttgttg	tgactatgga	1740
gccagaggta	cccatcaaga	aactggagac	catggtgaag	ctggacgcga	cttctgaagc	1800
tgttgtcaag	gtggatgtgg	ctgaggagga	gaaggagaag	aagaagaaga	agaagggcac	1860
tgccgaggag	ccagaggagg	aggagccaga	cgagagcatg	ctggactggg	ggtccaagta	1920
ctttgctctc	attgacacca	tgaaggagca	acttcgacaa	caagagccct	ctggaattga	1980
cttgaggagg	aaggagggaag	tggacaatac	cgagggcctg	aaggggtcaa	tgaagggcaa	2040
ggagaaggca	agggtgtcca	aagaggagaa	gaagaagaaa	actcagagct	ctggctctgg	2100
ccaggggtcc	gaggcccccg	agaagaagaa	acccaagatt	gatgagctta	aggtataccc	2160
caaagagctg	gagtcagagt	ttgataactt	tgaggactgg	ctgcacactt	tcaacttgct	2220
tccggggcaag	accgggggatg	atgaggatgg	ctccaccgag	gaggagcgca	ttgtgggacg	2280
cttcaagggc	tccctctgcg	tgtacaaagt	gccactccca	gaggacgtgt	cccgggaagc	2340
cggctacgac	tccacctacg	gcatgttcca	gggcaccccg	agcaatgacc	ccatcaatgt	2400
gctggtccga	gtctatgtgg	tccggggccac	ggacctgcac	cctgctgaca	tcaacggcaa	2460
agctgacccc	tacatcgcca	tccggctagg	caagactgac	atccgcgaca	aggagaacta	2520
catctccaag	cagctcaacc	ctgtcttttg	gaagtccttt	gacatcgagg	cctccttccc	2580
catggaatcc	atgctgacgg	tggctgtgta	tgactgggac	ctggtgggca	ctgatgacct	2640
cattggggaa	accaagatcg	acctggagaa	ccgcttctac	agcaagcacc	gcgccacctg	2700
cggcatcgcc	cagacctact	ccacacatgg	ctacaataac	tggcgggacc	ccatgaagcc	2760
cagccagatc	ctgacccgcc	tctgcaaaga	cggcaaagtg	gacggccccc	actttggggc	2820
ccctgggaga	gtgaaggtgg	ccaaccgcgt	cttcaactggg	ccctctgaga	ttgaggacga	2880
gaacggctcg	aggaagccca	cagacgagca	tgtggcgctg	ttggccctga	ggcactggga	2940
ggacatcccc	cgcgagggct	gccgcctggg	gccagagcat	gtggagacga	ggccgctgct	3000
caaccccgac	aagccgggca	tgcagcaggg	ccgcctggag	ctgtgggtgg	acatgttccc	3060
catggacatg	ccagccctcg	ggacgcctct	ggacatctca	cctcggaagc	ccaagaagta	3120
cgagctgcgg	gtcatcatct	ggaacacaga	tgaggtggtc	ttggaggacg	acgacttctt	3180
cacaggggag	aagtccagtg	acatcttctg	gaggggggtg	ctgaagggcc	agcaggaggga	3240
caagcaggac	acagacgtcc	actaccactc	cctcaactggc	gagggcaact	tcaactggcg	3300
ctacctgttc	cccttcgact	acctggcggc	ggaggagaag	atcgatcatc	ccaagaagga	3360
gtccatgttc	tcctgggacg	agaccgagta	caagatcccc	gcgcggctca	ccctgcagat	3420

-continued

ctgggatgcg gaccacttct ccgctgacga cttcctgggg gccatcgagc tggacctgaa	3480
ccggttcccc cgggggcgcaa agacagccaa gcagtgcacc atggagatgg ccaccgggga	3540
ggtggacgtg cccctcgtgt ccatcttcaa gcaaaagcgc gtcaaaggct ggtggcccct	3600
cctggcccg c aatgagaacg atgagtttga gctcacgggc aaggtggagg ctgagctgca	3660
tttactgaca gcagaggagg cagagaagaa cccagtgggc ctggcccga atgaacctga	3720
ccccctagag aaacccaacc ggcccgacac gagcttcato tggttcctga accctctcaa	3780
gtcggtcgc tacttcttgt ggacacgta tcgtggctg ctctcaaac tgttgcgtgt	3840
cctgctgtg ctctctctcc tcgccctgtt cctctactct gtgcctggct acctggtcaa	3900
gaaaatcctc ggggcctgag cccagtggcc tcctggccgg ccgacacgg ccttcgtctg	3960
gttctcaac cctctcaagt ccataagta cctcatctgc acccggtaca agtggctcat	4020
catcaagatc gtgctggcgc tgttggggct gctcatgttg gggctcttcc tctacagcct	4080
ccctggctac atgggtcaaaa agctccttg ggcatgaagg ccgcccagctc ccgccagccg	4140
ctcccagcc ctgcgcatt tctttcagt ggcttggaact ctttcccatc tcccctgggg	4200
agcctgagga gccagcgtc cactcttcat gccttgggcc gagcctgcct cctgcttgcg	4260
ggggccgcct gtctcactg ccccaggctg cggcttgccc agtcccgcc ctctgacccc	4320
tgctgtggg ctggggagcc ttggatgggg tggggacctg gaatgggtct ctcttgccc	4380
acctggctga ggcgccacc ttcttcaggc ccaggtcca gaggaagact cctgaaaccc	4440
tcccaggtc ttccaagtac aggattgaag ctttagtgaa attaaccaag gacctgggt	4500
cagtgccag ggtcttataa agaataacg agcaaaagg atccccgcc tgacctctgc	4560
agatagcacc ggtctttgat ccgcagcagg gccagaccc tgcacacaag tcccagcgcg	4620
gctgcttctg ccaactgttg gctccacttg gctcctctca cttcccaggg ggtcgctgt	4680
cctgcctgtg ggtttccatg gcttccaga gctcctctg ccccagccag cgcctccagg	4740
cccagctgag gagctgtgag aagcagcaga ggggaactcc catcccgggc acacctgtc	4800
ctcccacccc tgcctccttg ccttccagc ccttccagct gcagctggga gctggcccg	4860
caagtgtgc cctgcctgt gtctgggtt ctgttggtg ttttctttt cttgagtgg	4920
gatttttctc taaataaaag aagtcaagca ctgaaaaaa aaaaaaaaa	4969

<210> SEQ ID NO 12

<211> LENGTH: 4771

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 12

ccgtgagttc tgcccaggcc ctgtgagctc accagagcca cagactcaca gccagagg	60
ggcttcttcc ttcaggaact gaagaacccc catgaacacc aacatctcca ggttctgaga	120
acagaacctg ggaaattgat gacttctca tgatgaccga tactcaggat gccctagcg	180
agagctccca gatcatgagg aagaaggcct gaacgacata caggagatga tcaaacgga	240
gaagtcttac cctgagcgtc gctgcgggg cgtcctggag gagctgagct gtggctgctg	300
ccgttctcct tccctcgtg acaaggacca gggccactca tcccgcacca ggcttgaccg	360
ggagcgctc aagtctgca tgagggagct ggaaaacatg gggcagcagg ccaggatgct	420
gcgggcccag gtgaagcggc acacgggtgc ggacaagctg aggctgtgcc agaacttct	480
gcagaagctg cgttctctgg cggacgagcc ccagcacagc attcccgaca tcttcatctg	540

-continued

gatgatgagc aacaacaagc gtgtgccta tgcccggtgtg cccccaagg acctgctctt	600
ctccatcgtg gaggaggaga ctggcaagga ctgcgccaag gtcaagacgc tcttccttaa	660
gctgccaggg aagcggggct tcggctcggc aggtggaca gtgcaggcca aggtggagct	720
gtacctgtgg ctgggctca gcaaacagcg caaggagttc ctgtgcgcc tgccctgtgg	780
cttcaggag gtcaaggcag cccagggcct gggcctgcat gccttccac ccgtcagcct	840
ggtctacacc aagaagcagg cgttcagct ccgagcgac atgtaccagg ccgcagcct	900
ctttgccgc gacagcagcg gactctcaga cccctttgcc cgcgtcttct tcatcaatca	960
gagtcagtgc acagaggtgc tgaatgagac cctgtgtccc acctgggacc agatgctggt	1020
gttcgacaac ctggagctct atggtgaagc tcatgagctg agggacgac cgcctcatcat	1080
tgtcattgaa atctatgacc aggattccat gggcaaagct gacttcatgg gccggacctt	1140
cgccaaacc ctggtgaaga tggcagacga ggcgtactgc ccaccccgct tcccacctca	1200
gctcgagtac taccagatct accgtggcaa cgcacagct ggagacctgc tggcgccctt	1260
cgagctgctg cagattggac cagcaggga ggcagacctg ccccatca atggcccggt	1320
ggacgtggac cgaggtccca tcatgcccgt gccatgggc atccggcccg tgctcagcaa	1380
gtaccgagtg gaggtgctgt tctggggcct acgggacctt aagcgggtga acctggcca	1440
ggtggaccgg ccacgggtgg acatcgagtg tgcagggaag ggggtgcagt cgtccctgat	1500
ccacaattat aagaagaacc ccaactcaa caccctcgtc aagtggtttg aagtggacct	1560
cccagagaac gagctgctgc acccgccctt gaacatccgt gtggtggact gccggccctt	1620
cggtcgctac aactggtgg gctcccatgc cgtcagctcc ctgcgacgt tcatctaccg	1680
gccccagac cgctcggccc ccagctggaa caccacgggg gaggttgttg tgactatgga	1740
gccagaggta cccatcaaga aactggagac catggtgaag ctggacgca cttctgaagc	1800
tgttgtcaag gtggtgtgg ctgaggagga gaaggagaag aagaagaaga agaaggcac	1860
tcgggaggag ccagaggagg aggagccaga cgagagcatg ctggactggt ggtccaagta	1920
ctttgcctcc attgacacca tgaaggagca acttcgaca caagagccct ctggaattga	1980
cttgaggag aaggaggaag tggacaatac cgagggcctg aaggggtcaa tgaaggcaa	2040
ggagaaggca agggctgcca aagaggagaa gaagaagaaa actcagagct ctggctctgg	2100
ccaggggtcc gagggccccc agaagaagaa acccaagatt gatgagctta aggtataccc	2160
caaagagctg ggtccgagt ttgataactt tgaggactgg ctgcacactt tcaacttgct	2220
tcggggcaag accggggtatg atgaggatgg ctccaccgag gaggagcga ttgtgggacg	2280
cttcaagggc tcctctcggt tgtacaaagt gccactccca gaggacgtgt cccgggaagc	2340
cggtacgac tccacctacg gcatgttcca gggcatcccg agcaatgacc ccatcaatgt	2400
gctggtccga gtctatgtgg tccgggccac ggacctgcac cctgctgaca tcaacggcaa	2460
agctgacccc tacatcgcca tccggctagg caagactgac atccgcgaca aggagaacta	2520
catctccaag cagctcaacc ctgtcttttg gaagtccttt gacatcgagg cctccttccc	2580
catggaatcc atgctgacgg tggtgtgta tgactgggac ctggtgggca ctgatgacct	2640
cattggggaa accaagatcg acctggagaa ccgcttctac agcaagcacc gcgccacctg	2700
cggcatcgcc cagacctact ccacacatgg ctacaatata tggcgggacc ccatgaagcc	2760
cagccagatc ctgacctgcc tctgcaaaga cggcaaagtg gacggccccc actttgggccc	2820
ccctgggaga gtgaaggtgg ccaaccgct cttcactggg ccctctgaga ttgaggacga	2880
gaacggtcag aggaagccca cagacgagca tgtggcgctg ttggccctga ggcactggga	2940

-continued

```

ggacatcccc cgcgagggct gccgcctggt gccagagcat gtggagacga ggccgctgct 3000
caaccccgac aagccgggca tcgagcaggg ccgcctggag ctgtgggtgg acatgttccc 3060
catggacatg ccagccctg ggacgcctct ggacatctca cctcggaagc ccaagaagta 3120
cgagctgcgg gtcacatctt ggaacacaga tgaggtggtc ttggaggacg acgacttctt 3180
cacaggggag aagtccagt acatcttcgt gaggggggtg ctgaagggcc agcaggagga 3240
caagcaggac acagacgtcc actaccactc cctcactggc gagggcaact tcaactggcg 3300
ctacctgttc cccttcgact acctggcggc ggaggagaag atcgatcatc ccaagaagga 3360
gtccatgttc tcctgggacg agaccgagta caagatcccc gcgcggctca ccctgcagat 3420
ctgggatgcg gaccacttct ccgctgacga ctctctgggg gccatcgagc tggacctgaa 3480
ccggttcccg cggggcgcaa agacagccaa gcagtgcacc atggagatgg ccaccgggga 3540
ggtggacgtg cccctcgtgt ccattctcaa gcaaaagcgc gtcaaaggct ggtggccctc 3600
cctggccccg aatgagaacg atgagtttga gctcacgggc aaggtggagg ctgagctgca 3660
tttactgaca gcagaggagg cagagaagaa cccagtgggc ctggcccgca atgaacctga 3720
ccccctagag aaacccaacc ggcccgacac ggccctcgtc tggttctca accctctcaa 3780
gtccatcaag tacctcatct gcacccggta caagtggctc atcatcaaga tcgtgctggc 3840
gtgttgggg ctgctcatgt tggggtctct cctctacagc ctccctggct acatggtcaa 3900
aaagctcctt ggggcatgaa ggccggcagc tcccggcagc cgctcccag ccctgcgcga 3960
tttcctttca gtggtctgga ctctttccca tctcccctgg ggagcctgag gagcccagcg 4020
tccactcttc atgccttggg ccgagcctgc ctctgcttg cggggggcgc ctgtctcac 4080
tgcccaggc tgccgcttgc ccagtccgc cctctgacc cctgcctgtg ggctggggag 4140
ccttgatagg ggtggggacc tggaatgggt ctctcttgcc ccacctggct gaggcgccac 4200
cttctctcag gccagggctc cagaggaaga ctctgaaac cctcccag tcttccaagt 4260
acaggattga agcttttagt aaattaacca aggaccatgg gtcagtgcgc agggctttaa 4320
aaagaatgaa cgagcaaaag gtatccccgc cgtgacccct gcagatagca ccggtctttg 4380
atccgcagca ggggccagac cctgccaca agtcccagc cggtctcttc tgccactgct 4440
gggtccact tggctcctct cacttcccag ggggtgcct gtctgcctg tgggtttcca 4500
tggttccca gagctccctc tgcccagcc agcgctcca ggcccagctg aggagctgtg 4560
agaagcagca gaggggactc cccatcccg gcacacctg tctcccacc cctgccccct 4620
tgccctcca gccctttcag ctgcagctgg gagctggccc gtcaagtgt gccctgcct 4680
gtgtctgggt ttctgttggc tgtttttctt ttcttgagt gtgatttttc tctaaataaa 4740
agaagtcaag cactgaaaaa aaaaaaaaaa a 4771

```

<210> SEQ ID NO 13

<211> LENGTH: 5123

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 13

```

ccgtgagttc tgcccaggcc ctgtgagctc accagagcca cagactcaca gcccagaggt 60
ggcttcttcc ttcaggaact gaagaacccc catgaacacc aacatctcca ggttctgaga 120
acagaacctg ggaattgat gacttcctca tgatgaccga tactcaggat ggccctagcg 180
agagctccca gatcatgagg tcctcactc ccctgatcaa cagggaggag gcatttgggg 240

```

-continued

aggctgggga ggcggggctg tgcccagca tccccacac tctgattca caggaagaag	300
gcctgaacga catacaggag atgatcaaaa cggagaagtc ctacctgag cgtcgctgc	360
ggggcgctct ggaggagctg agctgtggct gctgcgctt cctctccctc gctgacaagg	420
accagggcca ctcatccgc accaggcttg accgggagcg cctcaagtcc tgcagaggg	480
agctggaaaa catgggagcaggccagga tgctgcgggc ccagggtgaag cggcacacgg	540
tgccggacaa gctgaggtg tgccagaact tctgcagaa gctgcgcttc ctggcggacg	600
agccccagca cagcattccc gacatcttca tctggatgat gagcaacaac aagcgtgtcg	660
cctatgcccg tgtgccctcc aaggacctgc tcttctccat cgtggaggag gagactggca	720
aggactgcgc caaggtcaag acgctcttcc ttaagctgcc agggaagcgg ggttcggct	780
cggcaggctg gacagtgcag gccaaggtgg agctgtacct gtggtgggc ctcagcaaac	840
agcgcaagga gttcctgtgc ggctgcctt gtggttcca ggaggtcaag gcagcccagg	900
gcctgggcct gcctgccttc ccaccgtca gcctggctca caccaagaag caggcgttcc	960
agctccgagc gcacatgtac caggcccga gcctctttgc cgcgacagc agcggactct	1020
cagacccctt tgcccgcgtc ttcttcatca atcagagtca gtgcacagag gtgctgaatg	1080
agaccctgtg tcccactgg gaccagatgc tgggttccga caacctggag ctctatggtg	1140
aagctcatga gctgagggac gatccgccca tcattgtcat tgaaatctat gaccaggatt	1200
ccatgggcaa agctgacttc atgggcccga ccttcgccaa acccctggtg aagatggcag	1260
acgaggcgta ctgcccacc cgcttccac ctcagctcga gtactaccag atctaccgtg	1320
gcaacgccac agctggagac ctgctggcgg ccttcgagct gctgcagatt ggaccagcag	1380
ggaaggctga cctgcccccc atcaatggcc cggaggacgt ggaccgaggt cccatcatgc	1440
ccgtgcccat gggcatccgg ccggtgctca gcaagtacc agtggagggt ctgttctggg	1500
gcctacggga cctaaagcgg gtgaacctgg ccaggtgga ccggccacgg gtggacatcg	1560
agtgtgcagg gaaggggggt cagtcgtccc tgatccacaa ttataagaag aaccccaact	1620
tcaacaccct cgtcaagtgg tttgaagtgg acctccaga gaacgagctg ctgcaccgc	1680
ccttgaacat ccgtgtgggt gactgccggg ccttcggctg ctacacactg gtgggctccc	1740
atgccgtcag ctccctcgca cgcttcatct accggcccc agaccgctcg gccccagct	1800
ggaacaccac ggtcaggctt ctccggcgt gccgtgtgct gtgcaatggg ggtctctct	1860
ctactccac aggggaggtt gtggtgacta tggagccaga ggtaccatc aagaaactgg	1920
agaccatggt gaagctggac gcgacttctg aagctgtgt caagggtgat gtggctgagg	1980
aggagaagga gaagaagaag aagaagaagg gcactgcgga ggagccagag gaggaggagc	2040
cagacgagag catgctggac tgggtgtcca agtactttgc ctccattgac accatgaagg	2100
agcaacttcg acaacaagag ccctctggaa ttgacttga ggagaaggag gaagtggaca	2160
ataccgaggg cctgaagggg tcaatgaagg gcaaggagaa ggcaagggtg gccaaagagg	2220
agaagaagaa gaaaactcag agctctggct ctggccaggg gtccgaggcc cccgagaaga	2280
agaaacccaa gattgatgag cttaaggat accccaaaga gctggagtcc gagtttgata	2340
actttgagga ctggtgcac actttcaact tgcttcgggg caagaccggg gatgatgagg	2400
atggctccac cgaggaggag cgcattgtgg gacgcttcaa gggctccctc tgcgtgtaca	2460
aagtgccact ccagaggac gtgtccggg aagccggcta cgaactccac tacggcatgt	2520
tccagggcat ccgagcaat gaccccatca atgtgctggt ccgagtctat gtggtccggg	2580
ccacggacct gcacctgct gacatcaacg gcaaagctga ccctacatc gccatccggc	2640

-continued

taggcaagac	tgacatccgc	gacaaggaga	actacatctc	caagcagctc	aacctgtct	2700
ttgggaagtc	ctttgacatc	gaggctcct	tcccatgga	atccatgctg	acggtggtg	2760
tgtatgactg	ggacctggg	ggcactgatg	acctcattgg	ggaaccaag	atcgacctg	2820
agaaccgctt	ctacagcaag	caccgcgcca	cctgcggcat	cgccagacc	tactccacac	2880
atggctacaa	tatctggcgg	gacccatga	agccagcca	gatcctgacc	cgctctgca	2940
aagacggcaa	agtggacggc	ccccactttg	ggccccctgg	gagagtgaag	gtggccaacc	3000
gcgtcttcac	tgggccctct	gagattgagg	acgagaacgg	tcagagggaag	cccacagacg	3060
agcatgtggc	gctgttggcc	ctgaggcact	gggaggacat	cccccgcgca	ggctgcggcc	3120
tggtgccaga	gcattgtggg	acgaggccgc	tgctcaacct	cgacaagccg	ggcatcgagc	3180
agggccgcct	ggagctgtgg	gtggacatgt	tcccatgga	catgccagcc	cctgggacgc	3240
ctctggacat	ctcactcgg	aagccaaga	agtacgagct	gcgggtcatc	atctggaaca	3300
cagatgaggt	ggtcttgagg	gacgacgact	tcttcacagg	ggagaagtcc	agtgcacatc	3360
tcgtgagggg	gtggctgaag	ggccagcagg	aggacaagca	ggacacagac	gtccactacc	3420
actccctcac	tggcgagggc	aacttcaact	ggcgctacct	gttcccttc	gactacctgg	3480
cggcggagga	gaagatcgtc	atctccaaga	aggagtccat	gttctctgg	gacgagaccg	3540
agtacaagat	ccccgcggg	ctcaccctgc	agatctggga	tgccgaccac	ttctccgctg	3600
acgacttcct	gggggccatc	gagctggacc	tgaaccggtt	cccgcggggc	gcaaagacag	3660
ccaagcagtg	caccatggag	atggccaccg	gggaggtgga	cgtgccctc	gtgtccatct	3720
tcaagcaaaa	gcgcgtcaaa	ggctggtggc	ccctcctggc	ccgcaatgag	aacgatgagt	3780
ttgagctcac	gggcaagggt	gaggctgagc	tgcatctact	gacagcagag	gaggcagaga	3840
agaacccagt	gggcctggcc	cgcaatgaac	ctgacccct	agagaaacct	aaccggcccc	3900
acacgagctt	catctggttc	ctgaacctc	tcaagtcggc	tcgtacttc	ttgtggcaca	3960
cgtatcgctg	gctgtcctc	aaactgttg	tgctcctgct	gctgtcctc	ctcctcgccc	4020
tgttcctcta	ctctgtgct	ggctacctg	tcaagaaaat	cctcggggcc	tgagccagct	4080
ggcctcctgg	cggcccgac	acggccttcg	tctggttcct	caacctctc	aagtccatca	4140
agtacctcat	ctgcaccgg	tacaagtggc	tcattcatca	gatcgtgctg	gcgtgttgg	4200
ggctgctcat	gttggggctc	ttcctctaca	gcctccctgg	ctacatggtc	aaaaagctcc	4260
ttggggcatg	aaggccgcca	gctcccgcca	gcccctcccc	agccctgccg	catttccttt	4320
cagtggcttg	gactctttcc	catctccct	ggggagcctg	aggagcccag	cgccactct	4380
tcattgcctg	ggccgagcct	gcctcctgct	tgccggggcc	gcctgtcctc	actgccccag	4440
gctgcggctt	gccagtcctc	gccccctga	ccccctgctg	tgggctgggg	agccttgat	4500
ggggtgggga	cctggaatgg	gtctctcttg	ccccacctgg	ctgaggcgcc	acctttcttc	4560
agggccaggc	tccagaggaa	gactcctgaa	acctccccca	ggtcttccaa	gtacaggatt	4620
gaagctttag	tgaaattaac	caaggacat	gggtcagtcg	ccagggtttt	aaaaagaatg	4680
aacgagcaaa	aggtatcccc	gcctgacccc	ctgcagatag	caccggtctt	tgatccgcag	4740
cagggggccag	acctgcccc	caagtcccag	cgcggctgct	tctgccactg	ctgggctcca	4800
cttggtcctc	ctcaactccc	agggggtcgc	ctgtcctgcc	tgtgggtttc	catggcttcc	4860
cagagctccc	tctgccccag	ccagcgctc	caggccccag	tgaggagctg	tgagaagcag	4920
cagaggggac	tcccatccc	gggcacaccc	tgctctccca	ccccgcccc	cttgcccttc	4980

-continued

cagcccttcc agctgcagct gggagctggc ccgtcaagtg ctgccccctgc ctgtgtcttg	5040
gtttctgttg gctgttttcc ttttcttgag tgggtatttt tctctaaata aaagaagtca	5100
agcactgaaa aaaaaaaaaa aaa	5123

<210> SEQ ID NO 14
 <211> LENGTH: 6973
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 14

atcggagggg ggtcgggagg aggaggagga ggcagcggca gagaagagag aggcgtgtga	60
gccgtgctcc accggctagc tcttccccgc tgcctctgcc tggcagtgcc aggcagccca	120
caccagcatg gccttgctca tccacctcaa gacagtctcg gagctgcggg gcaggggcca	180
ccggatcgcc aaagtgaatt tccgagggca atcctttctac tctcgggtcc tggagaactg	240
tgaggatgtg gctgactttg atgagacatt tcggtggccg gtggccagca gcatcgacag	300
aaatgagatg ctggagattc aggttttcaa ctacagcaaa gtcttcagca acaagctcat	360
cgggaccttc cgcattgtgc tgcagaaggt ggtagaggag agccatgtgg aggtgactga	420
cacgtcgatt gatgacaaca atgctatcat caagaccagc ctgtgcgtgg aggtccgcta	480
tcaggccact gacggcacag tgggtctctg ggacgatggg gacttctctg gagatgagtc	540
tcttcaagag gaagagaagg acagccaaga gacggatgga ctgctccag gctccccgcc	600
cagctccccg cccccaggag agaagagctt ccggagagcc gggaggagcg tgttctccgc	660
catgaagctc ggcaaaaacc ggtctcaca ggaggagccc caaagaccag atgaaccggc	720
ggtgctggag atggaagacc ttgacctctt gccattccgg ctaggagatg gactggatcc	780
cgactcggtg tctctagcct cagtcacagc tctcaccact aatgtctcca acaagcgatc	840
taagccagac attaatgatg agccaagtgc tgggcggccc atggattacc aggtcagcat	900
cacggtgacg gaggccccgc agctggtggg cttgaacatg gaccctgtgg tgtgcgtgga	960
ggtgggtgac gacaagaagt acacatccat gaaggagtcc actaactgcc cctattacaa	1020
cgagtacttc gtcttcgact tccatgtctc tccggatgtc atgtttgaca agatcatcaa	1080
gatttcggtg attcactcca agaacctgct gcgcagtgcc accctggtgg gctccttcaa	1140
aatggacgtg ggaacctgtg actcgcagcc agagcaccag ttccatcaca agtgggccat	1200
cctgtctgac cccgatgaca tctcctcggg gctgaagggc tacgtgaagt gtgacgttgc	1260
cgtggtgggc aaaggggaca acatcaagac gcccacaag gccaatgaga ccgacgaaga	1320
tgacattgag gggaaactgc tgcctccccg gggggtgccc cccgaacgcc agtgggcccg	1380
gttctatgtg aaaatttacc gacgagagg gctgccccgt atgaacacaa gcctcatggc	1440
caatgtaaag aagcctttca tcggtgaaaa caaggacctc gtggaccctc acgtgcaagt	1500
cttctttgct ggccagaagg gcaagacttc agtgacagaag agcagctatg agccctgtg	1560
gaatgagcag gtcgtcttta cagacctctt cccccactc tgcaaacgca tgaaggtgca	1620
gatccgagac tcggacaagg tcaacgacgt ggccatcggc acccacttca ttgacctgcg	1680
caagatttct aatgacggag acaaaaggctt cctgcccaca ctgggcccag cctgggtgaa	1740
catgtacggc tccacacgta actacacgct gctggatgag catcaggacc tgaacgaggg	1800
cctgggggag ggtgtgtctc tccgggcccg gctcctgctg ggctggctg tggagatcgt	1860
agacacctcc aacctgagc tcaccagctc cacagagggt caggtggagc aggccacgcc	1920
catctcggag agctgtgcag gtaaaatgga agaattcttt ctctttggag ccttctctga	1980

-continued

ggcctcaatg atcgaccgga gaaacggaga caagcccatc acctttgagg tcaccatagg	2040
caactatggg aacgaagttg atggcctgtc ccggccccag cggcctcggc cccggaagga	2100
gccgggggat gaggaagaag tagacctgat tcagaacgca agtgatgacg aggccggtga	2160
tgccggggac ctggcctcag tctcctccac tccaccaatg cggccccagg tcaccgacag	2220
gaactacttc catctgccct acctggagcg aaagccctgc atctacatca agagctggtg	2280
gccggaccag cgccgcgcgc tctacaatgc caacatcatg gaccacattg ccgacaagct	2340
ggaagaaggc ctgaacgaca tacaggagat gatcaaaacg gagaagtctt acctgagcg	2400
tcgctgctgg ggctgctgg aggagctgag ctgtggctgc tgccgcttcc tctcctcgc	2460
tgacaaggac cagggccact catcccgac caggcttgac cgggagcgcc tcaagtctg	2520
catgagggag ctggaaaaca tggggcagca ggcaggatg ctgcgggccc aggtgaagcg	2580
gcacacgggtg cgggacaagc tgaggctgtg ccagaacttc ctgcagaagc tgcgcttct	2640
ggcggacgag ccccgacaca gcattcccga catcttcac tggtatgata gcaacaacaa	2700
gcgtgtgcgc tatgccgtg tgccctccaa ggacctgctc ttctccatcg tggaggagga	2760
gactggcaag gactgcgcca aggtcaagac gctcttctt aagctgccag ggaagcgggg	2820
cttcggctcg gcaggctgga cagtgcaggc caaggtggag ctgtacctgt ggctgggcct	2880
cagcaaacag cgcaaggagt tctgtgctg cctgcctgt ggcttccagg aggtcaaggc	2940
agcccagggc ctgggcctgc atgccttccc acccgtcagc ctggtctaca ccaagaagca	3000
ggcgttccag ctccgagcgc acatgtacca ggcccgacg ctctttgccg ccgacagcag	3060
cggactctca gacccctttg ccgcgctctt ctctcatcaat cagagtcagt gcacagaggt	3120
gctgaatgag acctgtgtc ccacctggga ccagatgctg gtgttcgaca acctggagct	3180
ctatggtgaa gctcatgagc tgagggacga tccgcccatc attgtcattg aaatctatga	3240
ccaggattcc atgggcaaag ctgacttcat gggccggacc ttccgcaaac cctggtgaa	3300
gatggcagac gaggcgtact gccaccccg ctcccccct cagctcgagt actaccagat	3360
ctaccgtggc aacccacag ctggagacct gctggcggcc ttcgagctgc tgcagattgg	3420
accagcaggg aaggtgacc tgcccccat caatggcccg gtggacgtgg accgaggtcc	3480
catcatgccc gtgcccattg gcacccggcc cgtgctcagc aagtaccgag tggaggtgct	3540
gttctggggc ctacgggacc taaagcgggt gaacctggcc caggtggacc ggccacgggt	3600
ggacatcgag tgtgcaggga aggggggtgca gtcgtccctg atccacaatt ataagaagaa	3660
ccccaaactc aacaccctcg tcaagtgggt tgaagtggac ctcccagaga acgagctgct	3720
gcaccgccc ttgaacatcc gtgtggtgga ctgccgggcc ttcggtcgct acacactggt	3780
gggtcccat gccgtcagct cctgcgacg ctctcatctac cggccccag accgctcggc	3840
ccccagctgg aacaccacgg tcaggcttct ccggcgctgc cgtgtgctgt gcaatggggg	3900
ctcctcctct cactccacag gggaggttgt ggtgactatg gagccagagg taccatcaa	3960
gaaactggag accatggtga agctggacgc gacttctgaa gctgtgtgca aggtggtgt	4020
ggctgaggag gagaaggaga agaagaagaa gaagaagggc actgcggagg agccagagga	4080
ggaggagcca gacgagagca tgctggactg gtgttccaag tactttgcct ccattgacac	4140
catgaaggag caacttcgac aacaagagcc ctctggaatt gacttgagg agaaggagga	4200
agtggacaat accgagggcc tgaaggggtc aatgaagggc aaggagaagg caaggctgc	4260
caaagaggag aagaagaaga aaactcagag ctctggctct ggccaggggt ccgaggcccc	4320

-continued

cgagaagaag aaacccaaga ttgatgagct taaggtatac cccaaagagc tggagtccga	4380
gtttgataac tttgaggact ggctgcacac tttcaacttg cttcggggca agaccgggga	4440
tgatgaggat ggctocaccg aggaggagcg cattgtggga cgcttcaagg gctccctctg	4500
cgtgtacaaa gtgccactcc cagaggacgt gtcccgggaa gccggctacg actccaccta	4560
cggcatgttc cagggcatcc cgagcaatga ccccatcaat gtgctgggtc gagtctatgt	4620
ggtcggggcc acggacctgc accctgctga catcaacggc aaagctgacc cctacatcgc	4680
catccggcta ggcaagactg acatccgca caaggagaac tacatctcca agcagctcaa	4740
ccctgtcttt gggaagtcc ttgacatga ggcctccttc cccatggaat ccatgctgac	4800
ggtggctgtg tatgactggg acctggtggg cactgatgac ctcatgggg aaaccaagat	4860
cgacctggag aaccgcttct acagcaagca ccgcgccacc tgcggcatcg cccagacct	4920
ctccacacat ggctacaata tctggcggga ccccatgaag cccagccaga tcttgacctg	4980
cctctgcaaa gacggcaaa tggaacggcc ccactttggg cccctggga gagtgaagg	5040
ggccaaccgc gtcttcaactg ggccctctga gattgaggac gagaacggc agaggaagcc	5100
cacagacgag catgtggcgc tgttggccct gaggcactgg gaggacatcc cccgcgcagg	5160
ctgccgcctg gtgccagagc atgtggagac gaggcgcctg ctcaaccocg acaagccggg	5220
catcgagcag ggccgcctgg agctgtgggt ggacatgttc cccatggaca tgccagcccc	5280
tgggacgcct ctggacatct cacctcgga gcccaagaag tacgagctgc gggatcatcat	5340
ctggaacaca gatgaggtg tcttgaggga cgacgacttc ttcacagggg agaagtccag	5400
tgacatcttc gtgagggggt ggctgaagg ccagcaggag gacaagcagg acacagacgt	5460
ccactaccac tccctcaactg gcgagggcaa cttcaactgg cgctacctgt tccccttcga	5520
ctacctggcg gcggaggaga agatcgatcat ctccaagaag gagtccatgt tctcctggga	5580
cgagaccgag tacaagatcc ccgcgcggct caccctgcag atctgggatg cggaccactt	5640
ctccgctgac gacttctctg gggccatcga gctggacctg aaccggttc ccgcgggcgc	5700
aaagacagcc aagcagtga ccatggagat ggccaccggg gaggtggacg tgccctctgt	5760
gtccatcttc aagcaaaagc gcgtcaaagg ctggtggccc ctctggccc gcaatgagaa	5820
cgatgagttt gagctcaccg gcaaggtgga ggctgagctg catttactga cagcagagga	5880
ggcagagaag aaccacgtgg gcctggccc caatgaacct gacccctag agaaacccaa	5940
ccggcccgc acggccttcg tctggttcct caaccctctc aagtccatca agtacctcat	6000
ctgcacccgg tacaagtggc tcatcatcaa gatcgtgctg gcgctgttg ggctgctcat	6060
gttggggctc ttcctctaca gctcctctgg ctacatggtc aaaaagctcc ttggggcatg	6120
aaggccgcca gctcccgcca gccgctcccc agccctgccg catttccttt cagtggcttg	6180
gactctttcc catctccctt ggggagcctg aggagcccag cgtccactct tcatgccttg	6240
ggccgagcct gctcctgct tgcggggggc gcctgtctc actgccccag gctgcggctt	6300
gcccagtcct gccctctga cccctgctg tgggctgggg agccttggat ggggtgggga	6360
cctggaatgg gtctctcttg cccacactgg ctgaggcgcc acccttcttc aggccaggc	6420
tccagaggaa gactcctgaa accctccca ggtcttccaa gtacaggatt gaagctttag	6480
tgaaattaac caaggacat gggtcagtgc ccagggtttt aaaaagaatg aacgagcaaa	6540
aggtatcccc gccgtgacce ctgcagatag caccggtctt tgatccgcag caggggccag	6600
accctgcca caagtccag cgcggctgct tctgccactg ctgggctcca cttggtcct	6660
ctcacttccc agggggtcgc ctgtcctgcc tgtgggtttc catggcttcc cagagctccc	6720

-continued

tctgccccag ccagcgccctc caggccccagc tgaggagctg tgagaagcag cagaggggac	6780
tccccatccc gggcacaccc tgtcctccca cccctgcccc cttgcccttc cagcccttcc	6840
agctgcagct gggagctggc ccgtcaagtg ctgccccctgc ctgtgtctgg gtttctgttg	6900
gctgtttttc tttcttgag tggtgatttt tctctaaata aaagaagtca agcactgaaa	6960
aaaaaaaaaaa aaa	6973

<210> SEQ ID NO 15

<211> LENGTH: 7125

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 15

ttggttgccct tggctctctgt gggcagcagc aggaggaggc ggcagcagcc agagaagagg	60
gaggcgtgtg agccacactc caccagcgag cttcttcccg ctgctctgga actgcccagg	120
ctctccccac cagcatggcc ctgattgttc acctcaagac tgtctcagag ctccgaggca	180
aaggtgaccg gattgccaaa gtcactttcc gagggcagtc tttctactcc cgggtcctgg	240
agaactgcga ggggtgtggct gactttgatg agacgttccg gtggccagtg gccagcagca	300
tcgaccggaa tgaagtgttg gagattcaga ttttcaacta cagcaaagtc ttcagcaaca	360
agctgatagg gaccttctgc atgggtgctgc agaaagtggg ggaggagaat cgggtagagg	420
tgaccgacac gctgatggat gacagcaatg ctatcatcaa gaccagcctg agcatggagg	480
tccggtatca ggccacagat ggcactgtgg gcccttgga tgatggagac ttctgggag	540
atgaatccct ccaggaggag aaggacagcc aggagacaga tgggctgcta cctgggttccc	600
gaccagcac ccgatatctt ggcgagaaga gctttcgcag agcgggaagg agtgtgttct	660
cggccatgaa actcggcaaa actcgggtccc acaaaggga gccccaaaga caagatgagc	720
cagcagtgtc ggagatggag gacctggacc acctagccat tcagctgggg gatgggctgg	780
atcctgactc cgtgtctcta gcctcggta ccgctctcac cagcaatgtc tccaacaaac	840
ggtctaagcc agatattaag atggagccca gtgctggaag gcccatggat taccaggcca	900
gcatcacagt gattgaggct cggcagctgg tgggcttgaa catggaccct gtgggtgtgtg	960
tggaggtggg tgatgacaag aaatacacgt caatgaagga gtccacaaac tgcccttact	1020
acaacgagta ctttgtcttc gacttccatg tctctcctga tgtcatgttt gacaagatca	1080
tcaagatctc gggttatccat tctaagaacc tgcttcggag cggcaccctg gtgggttcct	1140
tcaaaatgga tgtggggact gtgtattccc agcctgaaca ccagttccat cacaaatggg	1200
ccatcctgtc agaccccgat gacatctctg ctgggttgaa gggttatgta aagtgtgatg	1260
tcgctgtggg gggcaaggga gacaacatca agacacccca caaggccaac gagacggatg	1320
aggacgacat tgaagggaac ttgctgctcc ccgagggcgt gcccccgaa cggcagtggg	1380
cacggttcta tgtgaaaatt taccgagcag agggactgcc ccgatgaac acaagcctca	1440
tggccaacgt gaagaaggcg ttcacggtg agaacaagga cctcgtcgac cctatgtgc	1500
aagtcttctt tgctggacaa aagggaacaa catcagtcca gaagagcagc tatgagccgc	1560
tatggaatga gcaggctgctc ttcacagact tgttcccccc actctgcaaa cgcataaggg	1620
tgcagatccg ggactctgac aaggatcaat atgtggccat cggcaccac ttcacgacc	1680
tgcgcaagat ttccaacgat ggagacaaag gcttcctgcc taccctcggg ccagcctggg	1740
tgaacatgta cggtccacg cgcaactaca cactgctgga cgagcaccag gacttgaatg	1800

-continued

aaggcctggg	ggaggggtgtg	tccttccggg	cccgcccat	gttgggacta	gctgtggaga	1860
tcctggacac	ctccaacca	gagctcacca	gctccacgga	ggtgcaggtg	gagcaggcca	1920
cgctgtctc	ggagagctgc	acaggagaa	tggagaatt	ttttctattt	ggagccttct	1980
tggagcctc	aatgattgac	cggaaaaatg	gggacaagcc	aattaccttt	gaggtgacca	2040
taggaaacta	cggcaatgaa	gtcgtatgta	tgtcccgcc	cctgaggcct	cggccccgga	2100
aagagcctgg	ggatgaagaa	gaggtagacc	tgattcagaa	ctccagtgc	gatgaaggtg	2160
acgaagccgg	ggacctggcc	tcggtgtcct	ccccccacc	tatgcggccc	cagatcacgg	2220
acaggaacta	tttccacctg	cctacctgg	agcgcaagcc	ctgcatctat	atcaagagct	2280
ggtggcctga	ccagaggcgg	cgcctctaca	atgccaat	catggatcac	attgctgaca	2340
agctggaaga	aggcctgaat	gatgtacagg	agatgatcaa	aacggagaag	tcctaccogg	2400
agcgccgct	gcgggggtgtg	ctagaggaa	tcagctgtgg	ctgccaccgc	ttcctctccc	2460
tcctggacaa	ggaccagggc	cgtcgtccc	gcaccaggct	ggatcgagag	cgtcttaagt	2520
cctgtatgag	ggagttggag	agcatgggac	agcaggccaa	gagcctgagg	gctcagggtga	2580
agcgccacac	tgttcgggac	aagctgaggt	catgccagaa	ctttctgcag	aagctacgct	2640
tcctggcgga	tgagccccag	cacagcattc	ctgatgtgtt	catttggtatg	atgagcaaca	2700
acaaacgtat	cgcctatgcc	cgcgtgcctt	caaagacct	gctcttctcc	atcgtggagg	2760
aggaactggg	caaggactgc	gccaaagtca	agaccctctt	cctgaagctg	ccagggaaga	2820
ggggcttcgg	ctcggcaggc	tggacagtac	aggccaagct	ggagctctac	ctgtggctgg	2880
gcctcagcaa	gcagcgaaag	gacttcctgt	gtggtctgcc	ctgtggcttc	gaggagggtca	2940
aggcagccca	aggcctgggc	ctgcattcct	ttccgcccac	cagcctagtc	tacaccaaga	3000
agcaagcctt	ccagctccga	gcacacatgt	atcaggcccg	aagcctcttt	gctgctgaca	3060
gcagtgggct	ctctgatccc	tttgcccggtg	tcttcttcat	caaccagagc	caatgcactg	3120
aggttctaaa	cgagacactg	tgtcccacct	gggaccagat	gctggtat	gacaacctgg	3180
agctgtacgg	tgaagctcac	gagttacgag	atgatcccc	catcattgtc	attgaaatct	3240
acgaccagga	cagcatgggc	aaagccgact	tcattggccg	gaccttcgcc	aagccctggg	3300
tgaagatggc	agatgaagca	tactgcccac	ctcgcttccc	gccgcagctt	gagtactacc	3360
agatctaccg	aggcagtgcc	actgccggag	acctactggc	tgcttctgag	ctgctgcaga	3420
ttgggccatc	agggaaggct	gacctgccac	ccatcaatgg	cccagtgagc	atggacagag	3480
ggcccatcat	gcctgtgccc	gtgggaaatc	ggccagtgct	cagcaagtac	cgagtggagg	3540
tgctgttctg	gggctgagg	gacctaaaga	gggtgaacct	ggcccagggtg	gaccgaccac	3600
gggtggacat	cgagtgtgca	ggaaagggg	tacaatcctc	cctgattcac	aattataaga	3660
agaaccccaa	cttcaacacg	ctgggtcaagt	ggtttgaagt	ggacctcccc	gagaatgagc	3720
tcctgcaccc	acccttgaac	atccgagtg	tagattgccg	ggcctttgga	cgatacaccc	3780
tggtgggttc	ccacgcagtc	agctcactga	ggcgcttcat	ctaccgacct	ccagaccgct	3840
cagcccccaa	ctggaacacc	acagtcaggc	tgtccgggg	ctgccacagg	ctgcgcaatg	3900
ggggccccto	ttctcgcccc	acaggggagg	ttgtagtaag	catggagcct	gaggagccag	3960
ttaagaagct	ggagaccatg	gtgaaactgg	atgcgacttc	tgatgctgtg	gtcaagggtg	4020
atgtggctga	agatgagaag	gaaaggaaga	agaagaaaa	gaaaggcccg	tcagaggagc	4080
cagaggagga	agagcccgat	gagagcatgc	tggattgggtg	gtccaagtac	ttcgctcca	4140
tcgacacaat	gaaggagcaa	cttcgacaac	atgagacctc	tggaaactgac	ttggaagaga	4200

-continued

aggaagagat	ggaaagcgct	gagggcctga	agggaccaat	gaagagcaag	gagaagtcca	4260
gagctgcaaa	ggaggagaaa	aagaagaaaa	accagagccc	tggccctggc	cagggatcgg	4320
aggctcctga	gaagaagaaa	gccaagatcg	atgagcttaa	ggtgtacccc	aaggagctgg	4380
aatcggagtt	tgacagcttt	gaggactggc	tgacacacct	caacctgttg	aggggcaaga	4440
cgggagatga	tgaggatggc	tccacagagg	aggagcgcat	agtaggccga	ttcaagggtc	4500
ccctctgtgt	gtacaaagtg	ccactcccag	aagatgtatc	tcgagaagct	ggctatgatc	4560
ccacctatgg	aatgttccag	ggcatcccaa	gcaatgaccc	catcaatgtg	ctgggtccgaa	4620
tctatgtggt	cggggccaca	gacctgcacc	cggccgacat	caatggcaaa	gctgaccctc	4680
atattgccat	caagttaggg	aagaccgaca	tccgagacaa	ggagaactac	atctccaagc	4740
agctcaaccc	tgtgtttggg	aagtcttttg	acattgaggg	ctccttcccc	atggagtcca	4800
tgttgacagt	ggcctgttac	gactgggata	tgggtggcac	tgatgacctc	atcggagaaa	4860
ccaagattga	cctggaaaac	cgcttctaca	gcaagcatcg	cgccacctgc	ggcatcgcac	4920
agacctattc	catacatggc	tacaatatct	ggagggaccc	catgaagccc	agccagatcc	4980
tgacacgcct	ctgtaaagag	ggcaaagtgg	acggccccc	ctttggtccc	catgggagag	5040
tgagggttgc	caaccggtgc	ttcacggggc	cttcagaaat	agaggatgag	aatggtcaga	5100
ggaagcccac	agatgagcac	gtggcactgt	ctgctctgag	acactgggag	gacatcccc	5160
gggtgggctg	cgccttctgt	cgggaacacg	tggagaccag	gccgctgctc	aacctgaca	5220
agccaggcat	tgacagggc	cgcctggagc	tgtgggtgga	catgttcccc	atggacatgc	5280
cagccctcgg	gacacctctg	gatatatccc	ccaggaaaac	caagaagtac	gagctgcggg	5340
tcacgtgtgt	gaacacagac	gaggtgggtc	tggaagacga	tgatttcttc	acgggagaga	5400
agtccagtga	catttttctg	aggggggtgc	tgaaggcca	gcaggaggac	aaacaggaca	5460
cagatgtcca	ctatcactcc	ctcacggggg	agggcaactt	caactggaga	tacctcttcc	5520
ccttcgacta	cctagcggcc	gaagagaaga	tcgttatgtc	caaaaaggag	tctatgttct	5580
cctgggatga	gacggagtac	aagatccctg	cgcggctcac	cctgcagatc	tgggacgctg	5640
accacttctc	ggctgacgac	ttcctggggg	ctatcgagct	ggacctgaac	cggttcccg	5700
ggggcgctaa	gacagccaag	cagtgcacca	tggagatggc	caccggggag	gtggacgtac	5760
ccctggtttc	catctttaa	cagaaacgtg	tcaaaggctg	gtggccctc	ctggcccga	5820
atgagaatga	tgagtttgag	ctcacaggca	aagtggaggc	ggagctacac	ctactcacgg	5880
cagaggaggc	agagaagaac	cctgtggggc	tggctcgcaa	tgaacctgat	ccctagaaa	5940
aacccaatcg	gccggacaca	agcttcatct	ggttcttgaa	ccctctcaag	tctgcccgtc	6000
acttctctgt	gcatacctac	cgttggttac	tctctaaatt	cctgctgctc	ttcctcctgc	6060
tgtctctctt	cgcctgttt	ctctactctc	tgctgggcta	cctggccaag	aagatccttg	6120
gggctgagc	cctgcagtcg	cctaggcctg	cgggcctgac	acggcattcg	tctggttcct	6180
gaacccactc	aatctatca	agtacctcat	ctgcacccgg	tacaagtggc	tgatcatcaa	6240
gatcgtgctg	gcgctgctgg	ggctgctcat	gctggccctc	ttcctttaca	gcctcccagg	6300
ctacatggtc	aagaagctcc	taggggcctg	aagtgtgccc	cacccagccc	cgtccagca	6360
tccctccagg	ggctgctgcg	tattttgcct	tccctcacct	ggactctctc	ccaactccct	6420
gaggagccct	cccacgcctg	ccagccttga	gcaagacacc	tgttctgtgg	acttcatccc	6480
cacccacac	ccaaactgtt	gcttgctga	tcttgctcca	ggcctgcctg	gggtttgggg	6540

-continued

cacagttggc	ctccaaaacc	agataccctc	ttgtctaaag	taccaggttc	ctctgcccac	6600
ccccaaagat	ggtagtggcc	caaccctccc	tgtgctttcc	aaatcttgtc	ttaaggcacc	6660
agtgaatta	accaagaaac	gcggagcaat	gcccaggct	ctgatgagta	ggaacacgtg	6720
gaaagcacca	ggaatgccag	cagaggcgag	gcggcacacc	tctctgcaga	gcatccaggc	6780
cgagcggcgg	gcagcggcca	gctgcttctg	cgcattgctc	cctcttggtc	ctgcttcttt	6840
ctcacagtca	cagtcacttc	acagcttagc	cttgggcttc	ccatcacttc	caggggtgcc	6900
tctgccttgg	ccagtgtgtg	tcagctagta	cacaagctcc	aagtgtgaat	caggtgtact	6960
ggccgtcctg	aagactgact	gcctgtcctc	tcttgccgac	agccacaccc	gagtgtacac	7020
ttaaagcggg	gcccttctgc	ctctgtgggc	ctgtggctg	ctgttccttt	cttgagtgtg	7080
attttttttt	tctctccctc	aataaaataa	atcaaaactc	gagac		7125

<210> SEQ ID NO 16

<211> LENGTH: 7065

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 16

ttggttgcc	tggtctctgt	gggcagcagc	aggaggaggc	ggcagcagcc	agagaagagg	60
gaggcgtgtg	agccacactc	caccagcgag	cttcttcccg	ctgctctgga	actgcccagg	120
ctctccccac	cagcatggcc	ctgattgttc	acctcaagac	tgtctcagag	ctccagggca	180
aaggtgacgg	gattgccaaa	gtcactttcc	gagggcagtc	tttctactcc	cgggtcctgg	240
agaactgcga	gggtgtggct	gactttgatg	agacgttccg	gtggccagtg	gccagcagca	300
tcgaccggaa	tgaagtgttg	gagattcaga	ttttcaacta	cagcaaagtc	ttcagcaaca	360
agctgatagg	gaccttctgc	atgggtgctg	agaaagtggg	ggaggagaat	cgggtagagg	420
tgaccgacac	gctgatggat	gacagcaatg	ctatcatcaa	gaccagcctg	agcatggagg	480
tccggtatca	ggccacagat	ggcactgtgg	gcccctggga	tgatggagac	ttcctgggag	540
atgaatccct	ccaggaggag	aaggacagcc	aggagacaga	tgggctgcta	cctgggttcc	600
gaccacagac	ccggatatct	ggcgagaaga	gctttcgcag	agcgggaagg	agtgtgttct	660
cggccatgaa	actcggcaaa	actcggctcc	acaaagagga	gccccaaaga	caagatgagc	720
cagcagtgtc	ggagatggag	gacctggacc	acctagccat	tcagctgggg	gatgggctgg	780
atcctgactc	cgtgtctcta	gcctcggcca	ccgctctcac	cagcaatgtc	tccaacaaac	840
ggtctaagcc	agatattaag	atggagccca	gtgctggaag	gcccattggat	taccaggtca	900
gcatcacagt	gattgaggct	cggcagctgg	tgggcttgaa	catggaccct	gtgggtgtgtg	960
tggagggtggg	tgatgacaag	aaatacacgt	caatgaagga	gtccacaaac	tgcccttact	1020
acaacgagta	ctttgtcttc	gacttccatg	tctctcctga	tgtcatgttt	gacaagatca	1080
tcaagatctc	ggttatccat	tctaagaacc	tgcttcggag	cggcaccctg	gtgggttcct	1140
tcaaaatgga	tgtggggact	gtgtattccc	agcctgaaca	ccagttccat	cacaaatggg	1200
ccatcctgtc	agaccccgat	gacatctctg	ctgggttgaa	gggttatgta	aagtgtgatg	1260
tcgctgtggt	gggcaaggga	gacaacatca	agacacccca	caaggccaac	gagacggatg	1320
aggacgacat	tgaagggaac	ttgctgctcc	cggagggcgt	gcccccgaa	cggcagtggtg	1380
cacggttcta	tgtgaaaatt	taccgagcag	agggaactgcc	ccggatgaac	acaagcctca	1440
tggccaacgt	gaagaaggcg	ttcatcggtg	agaacaagga	cctcgctcgac	ccctatgtgc	1500
aagtcttctt	tgctggacaa	aagggaacaa	catcagtgtc	gaagagcagc	tatgagccgc	1560

-continued

tatggaatga	gcaggtcgtc	ttcacagact	tgttcccccc	actctgcaaa	cgcatgaagg	1620
tgcagatccg	ggactctgac	aaggtcaatg	atgtggccat	cggcacccac	ttcatcgacc	1680
tgcgcaagat	ttccaacgat	ggagacaaag	gcttcctgcc	taccctcggt	ccagcctggg	1740
tgaacatgta	cggtccacg	cgcaactaca	cactgctgga	cgagcaccag	gacttgaatg	1800
aaggcctggg	ggaggggtgtg	tccttcggg	cccgctcat	gttgggacta	gctgtggaga	1860
tcctggacac	ctccaacca	gagctcacca	gctccacgga	ggtgcagggtg	gagcaggcca	1920
cgctgtctc	ggagagctgc	acagggagaa	tggagaatt	ttttctat	ggagccttct	1980
tggagcctc	aatgattgac	cggaataatg	gggacaagcc	aattacctt	gaggtgacca	2040
taggaaacta	cggaatgaa	gtcgatggt	tgtcccgcc	cctgaggcct	cgccccgga	2100
aagagcctgg	ggatgaagaa	gaggtagacc	tgattcagaa	ctccagtgc	gatgaagggtg	2160
acgaagccgg	ggacctggcc	tcggtgtcct	ccacccacc	tatgcggccc	cagatcacgg	2220
acaggaacta	tttccacctg	ccctacctgg	agcgcaagcc	ctgcatctat	atcaagagct	2280
ggtggcctga	ccagaggcgg	cgctctaca	atgccaacat	catggatcac	attgctgaca	2340
agctggaaga	aggcctgaat	gatgtacagg	agatgatcaa	aacggagaag	tcctaccogg	2400
agcgccgct	cgggggtgtg	ctagaggaa	tcagctgtgg	ctgccaccgc	ttctctccc	2460
ttctggacaa	ggaccagggc	cgctctccc	gcaccaggct	ggatcgagag	cgctttaagt	2520
cctgtatgag	ggagttggag	agcatgggac	agcaggccaa	gagcctgagg	gctcagggtga	2580
agcggcacac	tgttcgggac	aagctgaggt	catgccagaa	ctttctgcag	aagctacgct	2640
tcctggcgga	tgagccccag	cacagcattc	ctgatgtgtt	catttgatg	atgagcaaca	2700
acaaacgtat	cgctatgcc	cgctgcctt	caaagacct	gctcttctcc	atcgtggagg	2760
aggaactggg	caaggactgc	gccaaagtca	agaccctctt	cctgaagctg	ccagggaaga	2820
ggggcttcgg	ctcggcaggc	tggacagtac	aggccaagct	ggagctctac	ctgtggctgg	2880
gcctcagcaa	gcagcgaaag	gacttcctgt	gtggtctgcc	ctgtggcttc	gaggagggtca	2940
aggcagccca	aggcctgggc	ctgcattcct	ttccgcccat	cagcctagtc	tacaccaaga	3000
agcaagcctt	ccagctccga	gcacacatgt	atcaggcccg	aagcctcttt	gctgctgaca	3060
gcagtgggct	ctctgatccc	tttgcccggt	tcttcttcat	caaccagagc	caatgcactg	3120
aggttctaaa	cgagacactg	tgtccacact	gggaccagat	gctggtat	gacaacctgg	3180
agctgtacgg	tgaagctcac	gagttacgag	atgatcccc	catcattgtc	attgaaatct	3240
acgaccagga	cagcatgggc	aaagccgact	tcattgggccc	gaccttcgcc	aagccctggg	3300
tgaagatggc	agatgaagca	tactgcccac	ctcgcttccc	gccgcagctt	gagtactacc	3360
agatctaccg	aggcagtgcc	actgccggag	acctactggc	tgcttctgag	ctgctgcaga	3420
ttgggccatc	agggaaggct	gacctgccac	ccatcaatgg	cccagtgagc	atggacagag	3480
ggcccatcat	gcctgtgccc	gtgggaatcc	ggccagtgct	cagcaagtac	cgagtggagg	3540
tgctgttctg	gggctgagg	gacctaaaga	gggtgaacct	ggcccagggtg	gaccgaccac	3600
gggtggacat	cgagtgtgca	ggaaaggggg	tacaatcctc	cctgattcac	aattataaga	3660
agaaccccaa	cttcaacacg	ctgggtcaagt	ggtttgaagt	ggacctcccc	gagaatgagc	3720
tcctgcaccc	acccttgaac	atccgagtg	tagattgccg	ggcctttgga	cgatacacc	3780
tggtgggttc	ccacgcagtc	agctcactga	ggcgcttcat	ctaccgacct	ccagaccgct	3840
cagcccccaa	ctggaacacc	acaggggagg	ttgtagtaag	catggagcct	gaggagccag	3900

-continued

ttaagaagct	ggagaccatg	gtgaaactgg	atgcgacttc	tgatgctgtg	gtcaagggtgg	3960
atgtggctga	agatgagaag	gaaaggaaga	agaagaaaaa	gaaaggcccg	tcagaggagc	4020
cagaggagga	agagcccgat	gagagcatgc	tggattgggtg	gtccaagtac	ttcgccctcca	4080
tcgacacaat	gaaggagcaa	cttcgacaaac	atgagacctc	tggaaactgac	ttggaagaga	4140
aggaagagat	ggaaagcgct	gagggcctga	agggaccaat	gaagagcaag	gagaagtcca	4200
gagctgcaaa	ggaggagaaa	aagaagaaaa	accagagccc	tggccctggc	cagggatcgg	4260
aggctcctga	gaagaagaaa	gccaagatcg	atgagcttaa	ggtgtacccc	aaggagctgg	4320
aatcggagtt	tgacagcttt	gaggactggc	tgacacacctt	caacctgttg	aggggcaaga	4380
cgggagatga	tgaggatggc	tcacagagg	aggagcgcat	agtaggcga	ttcaagggt	4440
ccctctgtgt	gtacaaagtg	ccactcccag	aagatgtatc	tcgagaagct	ggctatgatc	4500
ccacctatgg	aatgttcag	ggcatcccaa	gcaatgacct	catcaatgtg	ctggccgaa	4560
tctatgttgt	cggggccaca	gacctgcacc	cggccgacat	caatggcaaa	gctgaccct	4620
atattgccat	caagtttaggc	aagaccgaca	tccgagacaa	ggagaactac	atctccaagc	4680
agctcaacc	tgtgtttggg	aagtcctttg	acattgaggc	ctccttcccc	atggagtcca	4740
tgttgacagt	ggcgtgtac	gactgggac	tgggtggcac	tgatgacctc	atcgagaaa	4800
ccaagattga	cctggaaaac	cgcttctaca	gcaagcatcg	cgccacctgc	ggcatcgac	4860
agacctatc	catacatggc	tacaatatct	ggagggacct	catgaagccc	agccagatcc	4920
tgacacgcct	ctgtaaagag	ggcaaagtgg	acggccccc	ctttggtccc	catgggagag	4980
tgagggttgc	caacogtgc	ttcacggggc	cttcagaaat	agaggatgag	aatggtcaga	5040
ggaagccac	agatgagcac	gtggcactgt	ctgctctgag	acactgggag	gacatcccc	5100
gggtgggctg	cgccttgtg	cgggaacacg	tggagaccag	gccgctgctc	aacctgaca	5160
agccaggcat	tgagcagggc	cgcctggagc	tgtgggtgga	catgttcccc	atggacatgc	5220
cagccctgg	gacacctctg	gatatatccc	ccaggaaaac	caagaagtac	gagctgcggg	5280
tcacgtgtg	gaacacagac	gaggtgggtc	tggaaagcga	tgattttctc	acgggagaga	5340
agtccagtga	catttttgtg	aggggggtgg	tgaaggccca	gcaggaggac	aaacaggaca	5400
cagatgtcca	ctatcactcc	ctcacggggg	agggcaactt	caactggaga	tacctcttc	5460
ccttcgacta	cctagcggcc	gaagagaaga	tcgttatgtc	caaaaaggag	tctatgttct	5520
cctgggatga	gacggagtac	aagatccctg	cgcggctcac	cctgcagatc	tgggacgctg	5580
accacttctc	ggctgacgac	ttcctggggg	ctatcgagct	ggacctgaac	cggttcccga	5640
ggggcgctaa	gacagccaag	cagtgcacca	tggagatggc	caccggggag	gtggacgtac	5700
ccctggtttc	catctttaaa	cagaaacgtg	tcaaaggctg	gtggccctc	ctggcccgca	5760
atgagaatga	tgagtttgag	ctcacaggca	aagtggaggc	ggagctacac	ctactcacgg	5820
cagaggaggc	agagaagaac	cctgtggggc	tggctcgcaa	tgaacctgat	cccctagaaa	5880
aaccaaatcg	gccggacaca	agcttcactc	ggttcttgaa	ccctctcaag	tctgcccgt	5940
acttcctgtg	gcatacctac	cgttggtac	tcctcaaatt	cctgctgctc	ttcctcctgc	6000
tgctgctctt	cgcctgttt	ctctactctc	tgcttggtta	cctggccaag	aagatccttg	6060
gggctgagc	cctgcagtcg	cctaggcctg	cgggcctgac	acggcattcg	tctggttcct	6120
gaacccactc	aaatctatca	agtacctcat	ctgcacccgg	tacaagtggc	tgatcatcaa	6180
gatcgtgctg	gcgctgctgg	ggtgctcat	gctggccctc	ttcctttaca	gcctcccagg	6240
ctacatggtc	aagaagctcc	taggggcctg	aagtgtgccc	cacccagcc	cgctccagca	6300

-continued

tccctccagg ggetgctgcg tattttgcct tccctcacct ggactctctc ccaactccct	6360
gaggagccct cccacgcctg ccagccttga gcaagacacc tgcttgcctg acttcatccc	6420
caccccacac ccaaaactgtt gcttgccctga tcttgctcca ggctgcctg gggtttgggg	6480
cacagttggc ctccaaaacc agataccctc ttgtctaaag taccaggttc ctctgccaa	6540
ccccaaagag gtagtgggc caaccctccc tgtgctttcc aaatcttctc ttaaggcacc	6600
agtgaatta accaagaaac gcggagcaat gcccaaggct ctgatgagta ggaacacgtg	6660
gaaagcacca ggaatgccag cagaggcgag gcggcacacc tctctgcaga gcatccaggc	6720
cgagcggcgg gcagcggcca gctgcttctg cgcctgctct cctcttggtc ctgcttcttt	6780
ctcacagtca cagtcacttc acagcttagc cttgggcttc ccatcacttc caggggtgcc	6840
tctgccttgg ccagtgtgtg tcagctagta cacaagctcc aagtgtgaat cagggtgtact	6900
ggcgcctctg aagactgact gccctgtcct tcctgccgac agccacaccc gagtgtacac	6960
ttaaagcggg gcccttctgc ctctgtgggc ctgctggctg ctgttccctt cttgagtggtg	7020
atTTTTTTTT tctctccctc aataaaataa atcaaaactct gagac	7065

<210> SEQ ID NO 17

<211> LENGTH: 6907

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 17

ttggttgctt tggtctctgt gggcagcagc aggaggaggc ggcagcagcc agagaagagg	60
gaggcgtgtg agccacactc caccagcgag cttcttcccg ctgctctgga actgccagg	120
ctctcccccac cagcatggcc ctgattgttc acctcaagac tgtctcagag ctccgaggca	180
aaggtagaccg gattgccaaa gtcactttcc gagggcagtc tttctactcc cgggtcctgg	240
agaactgcga ggggttggtt gactttgatg agacgttccg gtggccagtg gccagcagca	300
tcgaccggaa tgaagtgttg gagattcaga ttttcaacta cagcaaagtc ttcagcaaca	360
agctgatagg gaccttctgc atgggtgctgc agaaagtggg ggaggagaat cgggtagagg	420
tgaccgcacac gctgatggat gacagcaatg ctatcatcaa gaccagcctg agcatggagg	480
tcgggtatca ggccacagat ggcactgtgg gcccttgga tgatggagac ttcctgggag	540
atgaatccct ccaggaggag aaggacagcc aggagacaga tgggctgcta cctggttccc	600
gaccagcac cggatatctt ggcgagaaga gctttcgag caaaggcaga gagaagacca	660
agggaggcag agatggcgag cacaaagcgg gaaggagtgt gttctcgcc atgaaactcg	720
gcaaaactcg gtcccacaaa gaggagcccc aaagacaaga tgagccagca gtgctggaga	780
tggaggacct ggaccaccta gccattcagc tgggggatgg gctggatcct gactccgtgt	840
ctctagcctc ggtcacgct ctcaccagca atgtctccaa caaacggtct aagccagata	900
ttaagatgga gccagtgct ggaaggccca tggattacca ggtcagatc acagtgattg	960
aggctcgga cctggtgggc ttgaacatgg accctgtggt gtgtgtggag gtgggtgatg	1020
acaagaaata cacgtcaatg aaggagtcca caaactgccc ttactacaac gagtactttg	1080
tcttcgactt ccatgtctct cctgatgtca tgtttgacaa gatcatcaag atctcggtta	1140
tccattctaa gaacctgctt cgagcggca ccctgggtggg ttccttcaaa atggatgtgg	1200
ggactgtgta ttcccagcct gaacaccagt tccatcacia atgggccatc ctgtcagacc	1260
ccgatgacat ctctgctggg ttgaagggtt atgtaaagtg tgatgtcgct gtgggtggga	1320

-continued

aggagagacaa catcaagaca ccccaacaagg ccaacgagac ggatgaggac gacattgaag	1380
ggaacttgct gctccccgag ggcgtgcccc cgaacggca gtgggcacgg ttctatgtga	1440
aaatttacgg agcagaggga ctgccccgga tgaacacaag cctcatggcc aacgtgaaga	1500
aggcgttcat cgggtgagaac aaggacctcg tcgacctcta tgtgcaagtc ttctttgctg	1560
gacaaaaggg caaacatca gtgcagaaga gcagctatga gccgctatgg aatgagcagg	1620
tcgtcttcac agacttgctc cccccactct gcaaacgcat gaagggtcag atccgggact	1680
ctgacaaggt caatgatgtg gccatcgcca cccacttcat cgacctgcgc aagatttcca	1740
acgatggaga caaaggcttc ctgcctaccc tcgggtccagc ctgggtgaac atgtacggct	1800
ccacgcgcaa ctacacactg ctggacgagc accaggactt gaatgaaggc ctgggggagg	1860
gtgtgtcctt ccggggccgc tctatgttgg gactagctgt ggagatcctg gacacctcca	1920
accagagct caccagctcc acggagggtgc aggtggagca ggccacgcct gtctcggaga	1980
gtgacacagg gagaatggaa gaattttttc tatttgagc cttcttgga gctcaatga	2040
ttgaccggaa aaatggggac aagccaatta cctttgaggt gaccatagga aactacggca	2100
atgaagtcga tggtatgtcc cggccctga gccctcgcc ccggaagag cctggggatg	2160
aagaagaggt agacctgatt cagaactcca gtgacgatga aggtgacgaa gccggggacc	2220
tggcctcggg gtctctcacc ccacctatgc ggccccagat cacggacagg aactatttcc	2280
acctgcccta cctggagcgc aagccctgca tctatatcaa gagctggtgg cctgaccaga	2340
ggcggcgct ctacaatgcc aacatcatgg atcacattgc tgacaagctg gaagaaggcc	2400
tgaatgatgt acaggagatg atcaaacgg agaagtccta ccggagcgc cgctgcggg	2460
gtgtgctaga ggaactcagc tgtggctgcc accgcttcc tccctctcg gacaaggacc	2520
agggccgctc gtccgcacc aggctggatc gagagcgtct taagtctgt atgagggagt	2580
tggagagcat gggacagcag gccaagagcc tgagggtca ggtgaagcgg cactgttc	2640
gggacaagct gaggtcatgc cagaacttcc tgcagaagct acgcttctg gcggatgagc	2700
cccagcacag cattcctgat gtgttcattt ggatgatgag caacaacaaa cgtatgcct	2760
atgcccgct gccttccaaa gacctgctct tctccatcgt ggaggaggaa ctgggcaagg	2820
actgcgcaa agtcaagacc ctcttctga agctgccagg gaagaggggc ttcggtcgg	2880
caggctggac agtacaggcc aagctggagc tctacctgtg gctgggcctc agcaagcagc	2940
gaaaggactt cctgtgtggt ctgccctgtg gcttcgagga ggtcaaggca gcccaaggcc	3000
tgggcctgca ttcctttccg cccatcagcc tagtctacac caagaagcaa gccttccagc	3060
tccgagcaca catgtatcag gcccaagcc tctttgtcgc tgacagcagt gggtctctg	3120
atccctttgc ccgtgtcttc ttcataacc agagccaatg cactgaggtt ctaaacgaga	3180
cactgtgtcc cacctgggac cagatgctgg tatttgacaa cctggagctg tacgggtgaag	3240
ctcacgagtt acgagatgat ccccccata ttgtcattga aatctacgac caggacagca	3300
tgggcaaagc cgacttcatg ggccggacct tcgccaagcc cctggtgaag atggcagatg	3360
aagcatactg cccacctcgc tccccgccgc agcttgagta ctaccagatc taccagggca	3420
gtgccactgc cggagaccta ctggctgcct tcgagctgct gcagattggg ccatcaggga	3480
aggctgacct gccaccatc aatggcccag tggacatgga cagagggccc atcatgcctg	3540
tgcccggtgg aatccggcca gtgctcagca agtacagat ggaggtgctg ttctggggcc	3600
tgagggacct aaagaggggtg aacctggccc aggtggaccg accacgggtg gacatcgagt	3660
gtgcaggaaa gggggtacaa tctccctga ttcacaatta taagaagaac cccaacttca	3720

-continued

acacgctggt caagtgggtt gaagtggacc tcccgagaa tgagctcctg caccaccct	3780
tgaacatccg agtggtagat tgcgggct ttggacgata caccctggtg ggttcccacg	3840
cagtacgctc actgaggcgc ttcactacc gacctccaga ccgctcagcc cccaactgga	3900
acaccacagg ggaggttgta gtaagcatgg agcctgagga gccagttaag aagctggaga	3960
ccatggtgaa actggatgcg acttctgatg ctgtggtaa ggtggatgtg gctgaagatg	4020
agaaggaaa gaagaagaag aaaaagaaag gcccgtcaga ggagccagag gaggaagagc	4080
ccgatgagag catgctggat tgggtgtcca agtacttcgc ctccatcgac acaatgaagg	4140
agcaacttcg acaacatgag acctctgga ctgacttga agagaaggaa gagatggaaa	4200
gcgctgaggg cctgaaggga ccaatgaaga gcaaggagaa gtccagagct gcaaaggagg	4260
agaaaaagaa gaaaaaccag agccctggcc ctggccaggg atcggaggct cctgagaaga	4320
agaaagcaa gatcgatgag ctaaggtgt accccaagga gctggaatcg gagtttgaca	4380
gcttgaggga ctggtgcac accttcaacc tgttgagggg caagacggga gatgatgagg	4440
atggtctcac agaggaggag cgcatagtag gccgattcaa gggctccctc tgtgtgtaca	4500
aagtgcact cccagaagat gtatctcgag aagctggcta tgatcccacc tatggaatgt	4560
tccagggcat ccaagcaat gaccccatca atgtgctggt ccgaatctat gtggtccggg	4620
ccacagacct gcacccggcc gacatcaatg gcaaagtga cccctatatt gccatcaagt	4680
taggcaagac cgacatccga gacaaggaga actacatctc caagcagctc aaccctgtgt	4740
ttgggaagtc ctttgacatt gaggcctcct tcccctgga gtccatgttg acagtggccg	4800
tgtagcactg ggatctggtg ggcactgatg acctcatcgg agaaaccaag attgacctgg	4860
aaaacgctt ctacagcaag catcgcgcca cctgcggcat cgcacagacc tattccatac	4920
atggctacaa tatctggagg gaccccatga agcccagcca gatcctgaca cgctctgta	4980
aagagggcaa agtggacggc ccccaacttg gtcccatgga gagagtgagg gttgccaaac	5040
gtgtcttcac ggggccttca gaaatagagg atgagaatgg tcagaggaag cccacagatg	5100
agcacgtggc actgtctgct ctgagacact gggaggacat ccccggtg ggctgccgcc	5160
ttgtgccgga acacgtggag accaggccgc tgcctcaacc tgacaagcca ggcattgagc	5220
agggccgct ggagctgtgg gtggacatgt tcccctgga catgccagcc cctgggacac	5280
ctctggatat atccccag aaaccaaga agtacgagct gcgggtcacc gtgtggaaca	5340
cagacgaggt ggtcctgga gacgatgatt tcttcacggg agagaagtcc agtgacattt	5400
ttgtgagggg gtggtgaag ggcagcagg aggacaaaca ggacacagat gtccactatc	5460
actccctcac gggggagggc aacttcaact ggagatacct cttccccttc gactacctag	5520
cggccgaaga gaagatcgtt atgtccaaa aggagtcctat gttctcctgg gatgagacgg	5580
agtacaagat cctgcgcgg ctcacctgc agatctggga cgctgaccac ttctcggctg	5640
acgacttcct gggggctatc gagctggacc tgaacgggtt cccgaggggc gctaagacag	5700
ccaagcagtg caccatggag atggccaccg gggaggtgga cgtaccctg gtttccatct	5760
ttaaacagaa acgtgtcaaa ggtggtggc cctcctggc ccgcaatgag aatgatgagt	5820
ttgagctcac aggcaaagtg gagggcggag tacacctact cacggcagag gaggcagaga	5880
agaacctgt gggcctggct cgcaatgaac ctgatccct agaaaaacc aaccggcctg	5940
acacggcatt cgtctggttc ctgaacccac tcaaatctat caagtacctc atctgaccc	6000
ggtacaagt gctgatcac aagatcgtgc tggcgctgct gggcgctgct atgctggccc	6060

-continued

tcttccttta cagcctccca ggctacatgg tcaagaagct cctagggggc tgaagtgtgc	6120
cccccccccag cccgctccag catccctcca ggggctgtctg cgtattttgc ctccctcac	6180
ctggactctc tcccaactcc ctgaggagcc ctcccacgcc tgcagcctt gagcaagaca	6240
cctgcttgct ggacttcac cccacccac acccaactg ttgcttgct gatctgtcc	6300
caggcctgcc tggggtttgg ggcacagtgt gctccaaaa ccagataccc tcttgtctaa	6360
agtaccaggt tcctctgcc aacccaaga gtggtagtgg cccaaccctc cctgtgcttt	6420
ccaaatcttg tcttaaggca ccagtgaat taaccaagaa acgcggagca atgcccaagg	6480
ctctgatgag taggaacacg tggaaagcac caggaatgcc agcagaggcg aggcggcaca	6540
cctctctgca gagcatccag gccgagcggc gggcagcggc cagctgcttc tgcgcatgct	6600
ctctcttggt ctctgcttct ttctcacagt cacagtcact tcacagctta gccttgggt	6660
tcccatcact tccaggggtg cctctgcctt ggcagtggtg tgtcagctag tacacaagct	6720
ccaagtgtga atcaggtgta ctggccgtcc tgaagactga ctgccctgct ctctctgccg	6780
acagccacac ccgagtgtac acttaaagcg gtgcccttct gcctctgtgg gcctgtgtgg	6840
tgctgttctt ttcttgatgt tgattttttt ttctctccc tcaataaaat aaatcaaact	6900
ctgagac	6907

<210> SEQ ID NO 18

<211> LENGTH: 6862

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 18

ttggttgctt tggtctctgt gggcagcagc aggaggaggc ggcagcagcc agagaagagg	60
gaggcgtgtg agccacactc caccagcgag cttcttcccg ctgctctgga actgccagg	120
ctctcccccac cagcatggcc ctgattgttc acctcaagac tgtctcagag ctccgaggca	180
aaggtagacc gattgccaaa gtcactttcc gagggcagtc tttctactcc cgggtcctgg	240
agaactgcga ggggttggtt gactttgatg agacgttccg gtggccagtg gccagcagca	300
tcgaccgaa tgaagtgttg gagattcaga ttttcaacta cagcaaagtc ttcagcaaca	360
agctgatagg gaccttctgc atggtgctgc agaaagtggg ggaggagaat cgggtagagg	420
tgaccgacac gctgatggat gacagcaatg ctatcatcaa gaccagcctg agcatggagg	480
tcoggtatca ggccacagat ggcactgttg gccctggga tgatggagac ttctgggag	540
atgaatccct ccaggaggag aaggacagcc aggagacaga tgggctgcta cctggttccc	600
gaccagcac ccgatatctt ggcgagaaga gctttcgag agcgggaagg agtgtgttct	660
cggccatgaa actcggcaaa actcgggtccc acaaagagga gcccacaaga caagatgagc	720
cagcagtgtc ggagatggag gacctggacc acctagccat tcagctgggg gatgggctgg	780
atctgactc cgtgtctcta gcctcgggtc ccgctctcac cagcaatgtc tccaacaaac	840
ggtctaagcc agatattaag atggagccca gtgctggaag gcccatggat taccaggtea	900
gcatcacagt gattgaggct cgcagctgg tgggcttgaa catggacct gtggtgtgtg	960
tggagggtgg tgatgacaag aaatacacgt caatgaagga gtccacaaac tgccttact	1020
acaacgagta ctttgtcttc gacttccatg tctctcctga tgtcatgttt gacaagatca	1080
tcaagatctc gggttatccat tctaagaacc tgcttcggag cggcaccctg gtgggttctt	1140
tcaaaatgga tgtggggact gtgtattccc agcctgaaca ccagttccat cacaatggg	1200
ccatcctgtc agaccccgat gacatctctg ctgggttgaa gggttatgta aagtgtgatg	1260

-continued

tcgctgtggt	gggcaaggga	gacaacatca	agacacccca	caaggccaac	gagacggatg	1320
aggacgacat	tgaagggaac	ttgctgctcc	cggagggcgt	gccccccgaa	cggcagtggg	1380
cacggttcta	tgtgaaaatt	taccgagcag	agggactgcc	cggatgaac	acaagcctca	1440
tgccaacgt	gaagaaggcg	ttcatcggtg	agaacaagga	cctcgtcgac	cctatgtgac	1500
aagtcttctt	tgctggacaa	aagggcaaaa	catcagtgc	gaagagcagc	tatgagccgc	1560
tatggaatga	gcaggctgtc	ttcacagact	tggtccccc	actctgcaa	cgcatgaagg	1620
tcgagatccg	ggactctgac	aaggtcaatg	atgtggccat	cggcacccac	ttcatcgacc	1680
tgcgcaagat	ttccaacgat	ggagacaaag	gcttcctgcc	taccctcggt	ccagcctggg	1740
tgaacatgta	cggctccacg	cgcaactaca	cactgctgga	cgagcaccag	gacttgaatg	1800
aaggcctggg	ggagggtgtg	tccttcggg	ccgcctcat	gttgggacta	gctgtggaga	1860
tcctggacac	ctccaacca	gagctcacca	gctccacgga	ggtgcagggtg	gagcaggcca	1920
cgctgtctc	ggagagctgc	acagggagaa	tgaagaatt	tttctatct	ggagccttct	1980
tgaagcctc	aatgattgac	cggaaaaatg	gggacaagcc	aattaccttt	gaggtagcca	2040
taggaaacta	cggcaatgaa	gtcgatggt	tggtccggcc	cctgaggcct	cgccccgga	2100
aagagcctgg	ggatgaagaa	gaggtagacc	tgattcagaa	ctccagtgc	gatgaagggtg	2160
acgaagccgg	ggacctggcc	tcggtgtcct	ccacccacc	tatgcggccc	cagatcacgg	2220
acaggaaacta	tttccacctg	ccctacctgg	agcgcaagcc	ctgcatctat	atcaagagct	2280
ggtggcctga	ccagaggcgg	cgctctaca	atgccaacat	catggatcac	attgctgaca	2340
agctggaaga	aggcctgaat	gatgtacagg	agatgatcaa	aacggagaag	tcctaccggg	2400
agcgccgct	gcggggtgtg	ctagaggaa	tcagctgtgg	ctgccaccgc	ttcctctccc	2460
tcctggacaa	ggaccagggc	cgctcgtccc	gcaccaggct	ggatcgagag	cgtcttaagt	2520
cctgtatgag	ggagttggag	agcatgggac	agcaggccaa	gagcctgagg	gctcagggtga	2580
agcggcacac	tggtcgggac	aagctgaggt	catgccagaa	ctttctgcag	aagctacgct	2640
tcctggcgga	tgagccccag	cacagcattc	ctgatgtgtt	catttgatg	atgagcaaca	2700
acaaacgtat	cgcctatgcc	cgcgtgcctt	ccaaagacct	gctcttctcc	atcgtggagg	2760
aggaactggg	caaggactgc	gccaagtca	agaccctctt	cctgaagctg	ccagggaaga	2820
ggggcttcgg	ctcggcaggc	tgacagctac	aggccaagct	ggagctctac	ctgtggctgg	2880
gcctcagcaa	gcagcgaaag	gacttcctgt	gtggtctgcc	ctgtggcttc	gaggagggtca	2940
aggcagccca	aggcctgggc	ctgcattcct	ttccgcccat	cagcctagtc	tacaccaaga	3000
agcaagcctt	ccagctccga	gcacacatgt	atcaggcccg	aagcctcttt	gctgctgaca	3060
gcagtgggct	ctctgatccc	tttgcccggtg	tctttctcat	caaccagagc	caatgcactg	3120
aggttctaaa	cgagacactg	tgccccacct	gggaccagat	gctgggtattt	gacaacctgg	3180
agctgtacgg	tgaagctcac	gagttacgag	atgatccccc	catcattgtc	attgaaatct	3240
acgaccagga	cagcatgggc	aaagccgact	tcattgggccc	gaccttcgcc	aagccccctgg	3300
tgaagatggc	agatgaagca	tactgcccac	ctcgcttccc	gccgcagctt	gagtactacc	3360
agatctaccg	aggcagtgcc	actgccggag	acctactggc	tgcttctgag	ctgctgcaga	3420
ttgggccatc	agggaaggct	gacctgccac	ccatcaatgg	cccagtggac	atggacagag	3480
ggcccatcat	gcctgtgccc	gtgggaatcc	ggccagtgtc	cagcaagtac	cgagtggagg	3540
tgctgttctg	gggcctgagg	gacctaaaga	gggtgaacct	ggcccagggtg	gaccgaccac	3600

-continued

gggtggacat	cgagtgtgca	ggaagggggg	tacaatcctc	cctgattcac	aattataaga	3660
agaaccccaa	cttcaacacg	ctgggtcaagt	ggtttgaagt	ggacctcccg	gagaatgagc	3720
tcctgcaccc	acccttgaac	atccgagtgg	tagattgccg	ggcctttgga	cgatacaccc	3780
tggtgggttc	ccacgcagtc	agctcactga	ggcgcttcat	ctaccgacct	ccagaccgct	3840
cagccccc	ctggaacacc	acaggggagg	ttgtagtaag	catggagcct	gaggagccag	3900
ttaagaagct	ggagaccatg	gtgaaactgg	atgcgacttc	tgatgctgtg	gtcaagggtg	3960
atgtggctga	agatgagaag	gaaaggaaga	agaagaaaaa	gaaaggcccg	tcagaggagc	4020
cagaggagga	agagcccgat	gagagcatgc	tggattgggtg	gtccaagtac	ttcgccctca	4080
tcgacacaat	gaaggagcaa	cttcgacaac	atgagacctc	tggaaactgac	ttggaagaga	4140
aggaagagat	ggaaagcgct	gagggcctga	agggaccaat	gaagagcaag	gagaagtcca	4200
gagctgcaaa	ggaggagaaa	aagaagaaaa	accagagccc	tggccctggc	cagggatcgg	4260
aggctcctga	gaagaagaaa	gccaaagatcg	atgagcttaa	gggtgtacccc	aaggagctgg	4320
aatcggagtt	tgacagcttt	gaggactggc	tgcacacctt	caacctgttg	aggggcaaga	4380
cgggagatga	tgaggatggc	tccacagagg	aggagcgcat	agtaggccga	ttcaagggct	4440
ccctctgtgt	gtacaaagtg	ccactcccag	aagatgtatc	tcgagaagct	ggctatgatc	4500
ccacctatgg	aatgttccag	ggcatcccaa	gcaatgacct	catcaatgtg	ctggccgaa	4560
tctatgtggt	cggggccaca	gacctgcacc	cggccgacat	caatggcaaa	gctgaccct	4620
atattgccat	caagtttagc	aagaccgaca	tccgagacaa	ggagaactac	atctccaagc	4680
agctcaaccc	tgtgtttggg	aagtcctttg	acattgaggc	ctccttcccc	atggagtcca	4740
tgttgacagt	ggcgtgtac	gactgggac	tggtgggcac	tgatgacctc	atcgagaaaa	4800
ccaagattga	cctggaaaaa	cgttcttaca	gcaagcatcg	gcgccacctg	ggcatcgac	4860
agacctatc	catacatggc	tacaatatct	ggagggaccc	catgaagccc	agccagatcc	4920
tgacacgcct	ctgtaaagag	ggcaaagtgg	acggccccc	ctttggctcc	catgggagag	4980
tgaggggttc	caaccgtgtc	ttcacggggc	cttcagaaat	agaggatgag	aatggtcaga	5040
ggaagccac	agatgagcac	gtggcactgt	ctgctctgag	acactgggag	gacatcccc	5100
gggtgggctg	cgccttgtg	cgggaacacg	tggagaccag	gccgctgctc	aaccctgaca	5160
agccaggcat	tgagcagggc	cgcctggagc	tgtgggtgga	catgttcccc	atggacatgc	5220
cagccctcgg	gacacctctg	gatatatccc	ccaggaaacc	caagaagtac	gagctgcggg	5280
tcatcgtgtg	gaacacagac	gaggtgggtc	tggaaagacga	tgatttcttc	acgggagaga	5340
agtccagtga	cattttttgtg	aggggggtggc	tgaaggggcca	gcaggaggac	aaacaggaca	5400
cagatgtcca	ctatcactcc	ctcacggggg	agggcaactt	caactggaga	tacctcttcc	5460
ccttcgacta	cctagcggcc	gaagagaaga	tcgttatgtc	caaaaaggag	tctatgttct	5520
cctgggatga	gacggagtac	aagatccctg	cgcggctcac	cctgcagatc	tgggacgctg	5580
accacttctc	ggctgacgac	ttcctggggg	ctatcgagct	ggacctgaac	cggttcccga	5640
ggggcgctaa	gacagccaag	cagtgcacca	tggagatggc	caccggggag	gtggacgtac	5700
ccctggtttc	catctttaa	cagaaacgtg	tcaaaggctg	gtggccctc	ctggcccgca	5760
atgagaatga	tgagtttgag	ctcacaggca	aagtggaggc	ggagctacac	ctactcacgg	5820
cagaggaggc	agagaagaac	cctgtggggc	tggtctgcaa	tgaacctgat	ccctagaaa	5880
aaccaaacg	gcctgacacg	gcattcgtct	ggttcctgaa	cccactcaaa	tctatcaagt	5940
acctcatctg	caccoggtac	aagtggctga	tcatcaagat	cgtgctggcg	ctgctggggc	6000

-continued

```

tgctcatgct ggccctcttc ctttacagcc tcccaggcta catggtcaag aagctcctag 6060
gggectgaag tgtgccccac cccagcccgc tccagcatcc ctccaggggc tgetgcgtat 6120
tttgcccttc ctcacctgga ctctctccca actccctgag gagccctccc acgcctgcca 6180
gccttgagca agacacctgc ttgctggact tcatccccc cccacaccca aactgttgct 6240
tgctgatct tgtcccaggc ctgctggggg ttgggggcac agttggcctc caaaaccaga 6300
taccctcttg tctaaagtac caggttcttc tgcccaacct caagagtggg agtggcccaa 6360
ccctccctgt gctttccaaa tctgtctta aggcaccagt gaaattaacc aagaaacgcg 6420
gagcaatgcc caaggctctg atgagtagga acacgtggaa agcaccagga atgccagcag 6480
aggcgaggcg gcacacctct ctgcagagca tccaggccga gcggcgggca gcggccagct 6540
gcttctgcgc atgctctctc cttgctctg cttctttctc acagtccag tcacttcaca 6600
gcttagcctt gggcttccca tcacttcag gggtgctct gccttgcca gtgtgtgtca 6660
gctagtacac aagctccaag tgtgaatcag gtgtactggc cgtcctgaag actgactgcc 6720
ctgtccttcc tgccgacagc cacaccgag tgtacactta aagcgggtgc cttctgctc 6780
tgtgggcctg ctggtgctg ttcctttctt gagtgtgatt tttttttct ctccctcaat 6840
aaaataaatc aaactctgag ac 6862

```

```

<210> SEQ ID NO 19
<211> LENGTH: 77
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 19

```

```

gggattttgc cgatttcggc ctattgggta aaaaatgagc tgatttaaca aaaatttaac 60
gcgaatttta acaaaat 77

```

```

<210> SEQ ID NO 20
<211> LENGTH: 82
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 20

```

```

gtaagtatca aggttacaag acagggttaa ggagaccaat agaaactggg cttgtcgaga 60
cagagaagac tcttgcgttt ct 82

```

```

<210> SEQ ID NO 21
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 21

```

```

gataggcacc tattggtctt actgacatcc actttgcctt tctctccaca g 51

```

```

<210> SEQ ID NO 22
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

-continued

<400> SEQUENCE: 22

gcctgcaaga actgggttcag cagcctgagc cacttcgtga tccacctg 48

<210> SEQ ID NO 23

<211> LENGTH: 548

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 23

gatccaatca acctctggat tacaaaattt gtgaaagatt gactgggtatt cttaactatg 60

ttgtctcttt tacgctatgt ggatacgtg ctttaatgcc tttgtatcat gctattgctt 120

cccgatggc tttcattttc tctccttgt ataaatcctg gttgctgtct ctttatgagg 180

agttgtggcc cggtgtcagg caacgtggcg tgggtgtgcac tgtgtttgct gacgcaaccc 240

ccactgggtg gggcattgcc accacctgtc agctccttcc cgggaacttcc gctttccccc 300

tccctattgc cacggcggaa ctcacgccc cctgccttgc ccgctgctgg acaggggctc 360

ggctgttggg cactgacaat tccgtgggtg tgcggggaa atcatcgtcc tttccttggc 420

tgctcgctg tggtgccacc tggattctgc gcgggacgtc cttctgctac gtcccttcgg 480

ccctcaatcc agcggacctt ccttcccgcg gcctgctgcc ggctctgcgg cctcttcgcg 540

gtcttcga 548

<210> SEQ ID NO 24

<211> LENGTH: 454

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 24

ctgcagctca gcctactact tgctttccag gctgttccca gttcccatgt cagctgcttg 60

tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc 120

ggtaaatgtg gcagcctgag cctcctttcc atctctgttg aggcagacat aggaccccca 180

acaaacagca tgcagggttg gagccagcca caggaccag gtaagggggc ctgggtcctt 240

aagcttctgc cactggctcc ggcattgcag agagaagaga aggggaggca gagctgaacc 300

ttagccttgc cttcctgggt acccttctga gcctcactgt cttctgtgag atgggcaaag 360

tgcggtgtg actccttggc aacggtgtta caccagggca ggtaaagtgt tagttatttg 420

tggggtacac caggactgtt aaagggtgaa ctat 454

<210> SEQ ID NO 25

<211> LENGTH: 1157

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 25

ggtctcacc agcattttca cttctaataa gttcaaatgt gatacggcac ctttctaaaa 60

attagttttc agggaaatag ggttcaaaac tggtagtggt aggggtccatt ctcacgaccc 120

ccaggcctgc taaccctgac caagctacct attacttacc ctcctcttcc tctcctcct 180

ctttctcct ctcctgttcc cctcttctct tctcctcccc tctctctccc tctcctccc 240

ccttggtgtg gatcagatcc agagcctgaa tgagcctcct gacccacac cccactagc 300

atgggcctgc aagtgccag aagtcctcc tgctcctaa actgcccagc cgatccatta 360

gctcttcctt cttccagtg aaagaagcag gcacagcctg tccctcccg tctacagaaa 420

-continued

```

ggaagctaca gcacagggag ggccaaaggc cttcctggga ctagacagtt gatcaacagc 480
aggactggag agctgggctc cttttttgtt ccttggtgcc ctgcccctcc ccatgacctg 540
cagagacatt cagcctgccca ggctttatga ggtgggagct gggctctccc tgatgtatta 600
ttcagctccc tggagtgggc cagctcctgt tacactggcc acagccctgg gcacccgctt 660
ctcacttcta gtttcccctc caaggtaatg tggtaggtca tgatcattct atcctggctt 720
cagggacctg actccacttt ggggccattc gaggggtcta gggtagatga tgtcccctg 780
tggggattaa tgtcctgctc tgtaaaactg agctagctga gatccaggag ggcttggcca 840
gagacagcaa gttgttgcca tggtgacttt aaagccaggt tgctgcccca gcacaggcct 900
cccagctctac cctcactaga aaacaacacc caggcacttt ccaccacctc tcaaaggatga 960
aaccacaggc tggctagag aatgaattat ggatcctcgc tgtccgtgcc acccagctag 1020
tcccagcggc tcagacactg aggagagact gtaggttcag ctacaagcaa aaagacctag 1080
ctggtctcca agcagtgtct ccaagtcctt gaacctgtga cacctgcccc aggcacatc 1140
aggcacagag ggccacc 1157

```

```

<210> SEQ ID NO 26
<211> LENGTH: 76
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

```

```

<400> SEQUENCE: 26

```

```

cccatgtcag ctgcttgtgc ttccagaga caaacagga ataatagatg tcattaaata 60
tacattgggc ccagg 76

```

```

<210> SEQ ID NO 27
<211> LENGTH: 95
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

```

```

<400> SEQUENCE: 27

```

```

agcctgagcc tcctttccat ctctgtggag gcagacatag gaccccaac aaacagcatg 60
caggttggga gccagccaca ggaccaggt aaggg 95

```

```

<210> SEQ ID NO 28
<211> LENGTH: 171
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 28

```

```

cccatgtcag ctgcttgtgc ttccagaga caaacagga ataatagatg tcattaaata 60
tacattgggc ccaggagacc tgagcctcct ttccatctct gtggaggcag acataggacc 120
cccaacaaac agcatgcagg ttgggagcca gccacaggac ccaggtaagg g 171

```

```

<210> SEQ ID NO 29
<211> LENGTH: 171
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

```

```

<400> SEQUENCE: 29

```

```

agcctgagcc tcctttccat ctctgtggag gcagacatag gaccccaac aaacagcatg 60

```

-continued

caggttggga gccagccaca ggaccaggt aagggccat gtcagctgct tgtgcttcc 120

agagacaaaa caggaataat agatgtcatt aaatatacat tgggcccag g 171

<210> SEQ ID NO 30

<211> LENGTH: 184

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 30

cccatgtcag ctgcttgtgc ttccagaga caaacagga ataatagatg tcattaaata 60

tacattgggc ccaggcggt caatgtggca gcctgagcct cctttccatc tctgtggagg 120

cagacatagg accccaaca aacagcatgc aggttgggag ccagccacag gaccaggtta 180

aggg 184

<210> SEQ ID NO 31

<211> LENGTH: 82

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 31

tgaggtggga gctgggctct cctgatgta ttattcagct ccctggagtt ggccagctcc 60

tgttacactg gccacagccc tg 82

<210> SEQ ID NO 32

<211> LENGTH: 106

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 32

cacaggcctc ccagtctacc ctactagaa aacaacaccc aggcacttcc caccacctct 60

caaaggtgaa acccaaggct ggtctagaga atgaattatg gatcct 106

<210> SEQ ID NO 33

<211> LENGTH: 188

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 33

tgaggtggga gctgggctct cctgatgta ttattcagct ccctggagtt ggccagctcc 60

tgttacactg gccacagccc tgcacaggcc tccagctcta ccctcactag aaaacaacac 120

ccaggcactt tccaccacct ctcaaagggtg aaaccaagg ctggtctaga gaatgaatta 180

tggatcct 188

<210> SEQ ID NO 34

<211> LENGTH: 188

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 34

cacaggcctc ccagtctacc ctactagaa aacaacaccc aggcacttcc caccacctct 60

caaaggtgaa acccaaggct ggtctagaga atgaattatg gatccttgag gtgggagctg 120

ggctctccct gatgtattat tcagctccct ggagttggcc agctcctgtt acactggcca 180

cagccctg 188

-continued

<210> SEQ ID NO 35
 <211> LENGTH: 430
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 35

tgaggtggga gctgggctct cctgatgta ttattcagct ccctggagtt ggccagctcc	60
tgttacactg gccacagccc tgggcatccg cttctcactt ctagtttccc ctccaaggta	120
atgtgggtggg tcatgatcat tctatcctgg cttcagggac ctgactccac tttggggcca	180
ttcgaggggt ctagggtaga tgatgtcccc ctgtggggat taatgtcctg ctctgtaaaa	240
ctgagctagc tgagatccag gagggcttgg ccagagacag caagtgttg ccatggtgac	300
tttaaagcca ggtgtctgcc ccagcacagg cctcccagtc taccctcact agaaaacaac	360
acccaggcac tttccaccac ctctcaaagg tgaaaccaa ggctggtcta gagaatgaat	420
tatggatcct	430

<210> SEQ ID NO 36
 <211> LENGTH: 1611
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 36

ctgcagctca gcctactact tgctttccag gctgttcccta gttcccatgt cagctgcttg	60
tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc	120
ggtcaatgtg gcagcctgag cctcctttcc atctctgttg aggcagacat aggaccccca	180
acaaacagca tgcagggttg gagccagcca caggaccagc gtaaggggcc ctgggtcctt	240
aagcttctgc cactggctcc ggcattgcag agagaagaga aggggcggca gagctgaacc	300
ttagccttgc cttcctgggt acccttctga gcctcactgt cttctgtgag atgggcaaag	360
tgcggtgtg actccttggc aacggtgtta caccagggca ggtaaaagtg tagttatttg	420
tggggtacac caggactgtt aaagggtgtaa ctatggtctc acccagcatt ttcaacttcta	480
ataagttcaa atgtgatagc gcacctttct aaaaattagt ttccagggaa atagggttca	540
aaactggtag tggtagggtc cattctcagc acccccaggc ctgctaacc tgaccaagct	600
acctattact taccctctct tttctctctc tctctttctt ccttctcctg ctteccctct	660
tccttctccc tcccttctct tccctctctc cctccttgg ctgtgatcag atccagagcc	720
tgaatgagcc tctgacccc acacccccac tagcatgggc ctgcaagtgc ccagaagtcc	780
ctcctgcctc ctaaaactgc cagccgatcc attagctctt ccttcttccc agtgaaagaa	840
gcaggcacag cctgtccctc ccgttctaca gaaaggaagc tacagcacag ggagggccaa	900
aggccttctt gggactagac agttgatcaa cagcaggact ggagagctgg gctccathtt	960
tgttccttgg tgccctgccc ctcccatga cctgcagaga cattcagcct gccaggcttt	1020
atgaggtggg agctgggctc tccctgatgt attattcagc tccctggagt tggccagctc	1080
ctgttacact ggccacagcc ctgggcatcc gcttctcact tctagtttcc cctccaagggt	1140
aatgtggtgg gtcgatgatca ttctatcctg gcttcaggga cctgactcca ctttggggcc	1200
attcgagggg tctagggtag atgatgtccc cctgtgggga ttaatgtcct gctctgtaaa	1260
actgagctag ctgagatcca ggagggcttg gccagagaca gcaagttggt gccatggtga	1320

-continued

ctttaagcc aggttgctgc cccagcacag gcctcccagt ctacctcac tagaaaacaa	1380
caccagga cttccacca cctctcaaag gtgaaacca aggttggtct agagaatgaa	1440
ttatggatcc tcgctgtccg tgcccccag ctagtcccag cggctcagac actgaggaga	1500
gactgtaggt tcagctacaa gcaaaaagac ctactgtgtc tccaagcagt gtctccaagt	1560
ccctgaacct gtgacacctg cccagggcat catcaggcac agagggccac c	1611

<210> SEQ ID NO 37
 <211> LENGTH: 1611
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 37

ggtctcacc agcattttca cttctaataa gttcaaagt gatacggcac ctttctaaaa	60
attagttttc agggaaatag ggttcaaac tggtagtggt agggtcatt ctacagacc	120
ccaggctgc taacctgac caagctacct attacttacc ctctctttc tctctctct	180
ctttctctt ctctgttc cctcttct tctctctcc tctctctcc tctctctct	240
ccttggtgt gatcagatc agagcctgaa tgagcctct gacccacac cccactagc	300
atgggctgc aagtgccag aagtcctcc tgcctctaa actgccagc cgatccatta	360
gctcttctt cttccagtg aaagaagcag gcacagcctg tccctccgt tctacagaaa	420
ggaagctaca gcacaggag ggccaaagg cttctggga ctacagagt gatcaacagc	480
aggactggag agctgggtc catttttgtt ccttggtgcc ctgcccctc ccatgacctg	540
cagagacatt cagcctgcca ggctttatga ggtgggagct gggctctcc tgatgtatta	600
ttcagctccc tggagtggc cagctcctgt tacactggc acagccctgg gcacccgtt	660
ctcacttcta gttcccctc caaggtaat tgggggtca tgatcattct atcctggctt	720
cagggaacct actccactt gggggcattc gaggggtcta gggtagatga tgtcccctg	780
tggggattaa tgtcctgtc tgtaaaact agctagtga gatccaggag ggcttgcca	840
gagacagcaa gttgttgcca tggtgacttt aaagccaggt tgctgcccc gcacaggcct	900
cccagtctac cctactaga aaacaacac caggcacttt ccaccacctc tcaaaggatga	960
aaccaaggc tggctagag aatgaattat ggatcctcgc tgtccgtgcc accagctag	1020
tcccagcggc tcagacactg aggagagact gtaggttcag ctacaagcaa aaagacctag	1080
ctggctcca agcagtgtc ccaagtcct gaacctgtga cacctgccc aggcacatc	1140
aggcacagag ggccacctg cagctcagc tactacttg tttccaggct gttcctagt	1200
cccatgtcag ctgctgtgc tttccagaga caaacagga ataatagatg tcattaaata	1260
tacattgggc cccaggcggc caatgtggc gcctgagcct cctttccatc tctgtggagg	1320
cagacatagg accccaaca aacagcatgc aggtggggag ccagccacag gaccaggt	1380
aggggccctg ggtccttaag cttctgccac tggctccgc attgcagaga gaagagaagg	1440
ggcggcagag ctgaacctta gccttgctt cctgggtacc cttctgagc tctgtgtct	1500
ctgtgagatg ggcaaagtgc ggtgtgact ccttggaac ggtgttacac cagggcaggt	1560
aaagtgtag ttatttgtg ggtacaccag gactgttaa ggtgtaacta t	1611

<210> SEQ ID NO 38
 <211> LENGTH: 965
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 38

```

ctgcagctca gctactact tgctttccag gctgttcta gttcccatgt cagctgcttg      60
tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc      120
ggccaatgtg gcagcctgag cctcctttcc atctctgttg aggcagacat aggacccccca      180
acaaacagca tgcagggttg gagccagcca caggacccag gtaagggggcc ctgggtcctt      240
aagctttctg cactggctcc ggcatgagc agagaagaga aggggcggca gactggagag      300
ctgggtccca tttttgttcc ttggtgccct gccctcccc atgacctgca gagacattca      360
gcctgccagg ctttatgagg tgggagctgg gctctccctg atgtattatt cagctccctg      420
gagttggcca gctcctgtta cactggccac agccctgggc atccgcttct cacttctagt      480
ttcccctcca aggtaatgtg gtgggtcatg atcattctat cctggcttca gggacctgac      540
tccacttttg ggccattcga ggggtctagg gtagatgatg tccccctgtg gggattaatg      600
tcctgctctg taaaactgag ctagctgaga tccaggaggg cttggccaga gacagcaagt      660
tgttgccatg gtgactttaa agccagggtg ctgccccagc acaggcctcc cagtctaccc      720
tcactagaaa acaacaccca ggcactttcc accacctctc aaagggtgaa cccaaggctg      780
gtctagagaa tgaattatgg atcctcgtg tccgtgccac ccagctagtc ccagcggtc      840
agacactgag gagagactgt aggttcagct acaagcaaaa agacctagct ggtctccaag      900
cagtgctctc aagtccctga acctgtgaca cctgccccag gcatcatcag gcacagaggg      960
ccacc                                              965

```

<210> SEQ ID NO 39

<211> LENGTH: 520

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 39

```

ctgcagctca gctactact tgctttccag gctgttcta gttcccatgt cagctgcttg      60
tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc      120
ggccaatgtg gcagcctgag cctcctttcc atctctgttg aggcagacat aggacccccca      180
acaaacagca tgcagggttg gagccagcca caggacccag gtaagggggcc ctgggtcctt      240
tttatgaggt gggagctggg ctctccctga tgtattatc agctccctgg agttggccag      300
ctcctgttac actggccaca gccctgggca tccgctgcca tggtgacttt aaagccagg      360
tgctgcccca gcacaggcct cccagtctac cctcactaga aaacaacacc caggcacttt      420
ccaccacctc tcaaagggtga aacccaaggc tgggtctagag aatgaattat ggatcctcgc      480
tgtccgtgcc acccagctag tcccagcggc tcagacactg      520

```

<210> SEQ ID NO 40

<211> LENGTH: 235

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 40

```

gtatgccttt tgagatggat gcagcagggt ctgtgaggct gccaggaggg gtagagtccc      60
cgggggcctc gggccccgct ggagtgtgga gcaggcccat gctcagctct ccaggctgtt      120

```

-continued

cgtggctccc ctgtcagctg ctcactcctt tccagagaca aaacaggaat aatagacatc	180
attaatatata catagggccc caggcggtcg gcgtggtggg ctgggcctcc cttcc	235

<210> SEQ ID NO 41
 <211> LENGTH: 688
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 41

tgccttgctt tctgagccgg cagcctggct cccacccca tgtattattc agctcctgag	60
agccagccag ctctgtgtac actgaccgca gccagcacc tgcctgccc attccctcc	120
tcccttgctt aggacctaga gggttcaaag ttctctcca agatgacttg gtgggctttg	180
gccatccac cctagggccc acttctggcc cagtgcaggt gtgctggtga ttagggcag	240
gtggcattcc atctctgtgg ctcaatgtct tctctgtga agccgaagtg acccaagggc	300
tcccttcagt ggggttagcc agctgtggcc caggaggggc ctaaccagga tgagcactga	360
tggtgccatg acgactccga ggccagaatg tctccccag cacaggcctc ataggcaggc	420
ttcccatcc tggtaaaca caccacaca ctttctacta ctgctctagg gtgaaacca	480
aggcgctcta gaggagatga attatggatc cgccctccc gaatcctggc tcggccctcc	540
ccacgccacc cagggccagt cgggtctgct cacagccga ggaggccgcg tgtccagccg	600
cgggcaagag acagagcagg tcctgtgtc tccaagtccc tgagcccgtg acaccggccc	660
caggccctgt agagagcagg cagccacc	688

<210> SEQ ID NO 42
 <211> LENGTH: 77
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

ccctgtcag ctgctcactc cttccagag acaaacagg aataatagac atcattaaat	60
atacataggg ccccagg	77

<210> SEQ ID NO 43
 <211> LENGTH: 82
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

tgagccggca gcctggctcc ccacccatg tattattcag ctctgagag ccagccagct	60
cctgttacac tgaccgcagc cc	82

<210> SEQ ID NO 44
 <211> LENGTH: 111
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

cacaggcctc ataggcaggc ttcccatcc tggtaaaca caccacaca ctttctacta	60
ctgctctagg gtgaaacca agcgctcta gaggagatga attatggatc c	111

<210> SEQ ID NO 45
 <211> LENGTH: 193
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

-continued

<400> SEQUENCE: 45

tgagccggca gcctggctcc ccaccccatg tattattcag ctctgagag ccagccagct	60
cctgttacac tgaccgcagc cccacaggcc tcataggcag gcttcccat cctggtaaac	120
aacaccaca cactttctac tactgctcta gggtgaaacc caaggcgctc tagaggagat	180
gaattatgga tcc	193

<210> SEQ ID NO 46

<211> LENGTH: 193

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 46

cacaggcctc ataggcaggc ttcccatcc tggtaacaa caccacaca ctttctacta	60
ctgctctagg gtgaaccca aggcgctcta gaggagatga attatggatc ctgagccggc	120
agcctggctc cccaccccat gtattattca gctcctgaga gccagccagc tctgtttaca	180
ctgaccgcag ccc	193

<210> SEQ ID NO 47

<211> LENGTH: 499

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 47

tgagccggca gcctggctcc ccaccccatg tattattcag ctctgagag ccagccagct	60
cctgttacac tgaccgcagc ccagcacctg ctctgccat tccctcctc ccttgccatg	120
gacctagagg gttcaaagt ctctccaag atgacttggg gggcttggc catccaccc	180
taggccccac ttctggccca gtgcagggtg gctgggtgatt tagggcaggg ggcattccat	240
ctctgtggct caatgtcttc ctctgtgaag ccgaagtgc ccaagggtc ccttcattggg	300
gttgagccag ctgtggccca gggagggcct aaccaggatg agcactgatg ttgcatgac	360
gactccgagg ccagaatgtc tccccagca caggcctcat aggcaggctt cccatcctg	420
gtaaacaca cccacacact ttctactact gctctagggt gaaaccaag gcgctctaga	480
ggagatgaat tatggatcc	499

<210> SEQ ID NO 48

<211> LENGTH: 1262

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 48

gtatgccttt tgagatggat gcagcagggt ctgtgaggct gccaggagg gtagagtcc	60
cgggggcctc gggcccgcgt ggagtgtgga gcaggcccat gctcagctct ccaggctgtt	120
cgtggctccc ctgtcagctg ctactcctt tccagagaca aaacaggaat aatagacatc	180
attaaatata catagggccc caggcggctg gcgtgggtggg ctgggcctcc cttcccata	240
acactgagct gctctgtggt gccaacctg ctctgggccc agccagagga ccccatgag	300
gcggcatgca ggcggggagc aggccacaga acgcaggtaa ggagacctta gcctagagtc	360
cttgggtct gtcactggcc accctcgcat cccaggctgc aggaactga ggccagaga	420

-continued

ggacaaggac tttcctggac ccacacagcc agtcagtgac agagcctagg gtctgagcca	480
ggcctgaccc aacctccatt tctgcctctc taccctcgcc cccgcccacacacacaca	540
cacacaagtg gagttccact gaaacgcccc tccttgccct gccttctgag cgggcagcct	600
ggctccccac cccatgtatt attcagctcc tgagagccag ccagctcctg ttacactgac	660
cgcagccccc cacctgctct gcccatctcc ctccctccct gcctaggacc tagaggggtc	720
aaagttctcc tccaagatga cttggtgggc ttggccatc ccacctagg cccacttct	780
ggcccagtg aggtgtgctg gtgatttagg gcaggtggca ttccatctct gtggctcaat	840
gtcttctct gtgaagccga agtgacccaa gggctccctt catggggttg agccagctgt	900
ggcccaggga gggcctaacc aggatgagca ctgatgttg catgacgact ccgaggccag	960
aatgtctccc ccagcacagg cctcataggc aggtctcccc atcctggtaa acaacacca	1020
cacactttct actactgctc tagggtgaaa cccaaggcgc tctagaggag atgaattatg	1080
gatccgcct cccggaatcc tggtcggcc ctccccacgc caccagggc cagtcgggtc	1140
tgtcacagc ccgaggaggc cgcgtgtcca gccgcgggca agagacagag caggtcctg	1200
tgtctccaag tccctgagcc cgtgacaccg gcccaggcc ctgtagagag caggcagcca	1260
cc	1262

<210> SEQ ID NO 49
 <211> LENGTH: 649
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 49

gcaggcccat gctcagctct ccaggtgtt cgtggctccc ctgtcagctg ctcaactctt	60
tccagagaca aaacaggaat aatagacatc attaaatata catagggcc caggcggtcg	120
gcgtgggtgg ctgggctcc ctccccata aactgagct gctctgctgg gccaaccgtg	180
ctctggggc agccagagga ccccatgag gcggcatgca ggccgggagc aggccacaga	240
acgcaggtaa ggagacctg cctctgagc cggcagcctg gctccccacc ccatgtatta	300
ttcagctcct gagagccagc cagctcctgt taaactgacc gcagcccagc acctgctctg	360
cccattcccc tctcccttg cctaggacct agagggttca aagttctct ccaagatgac	420
ttggtgggct ttggccatcg ggcctaacca ggatgagcac tgatgttgcc atgacgactc	480
cgaggccaga atgtctcccc cagcacaggc ctcataggca ggcttcccca tcttggtaaa	540
caacaccac acactttcta ctactgctct aggggtgaaac ccaaggcgt ctagaggaga	600
tgaattatgg atccgccctc ccggaatcct ggctcggccc tccccacgc	649

<210> SEQ ID NO 50
 <211> LENGTH: 240
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 50

ctgcagctca gcctactact tgctttccag gctgttcccta gttcccatgt cagctgcttg	60
tgctttccag agacaaaaca ggaataatag atgtcattaa atataattg ggccccaggc	120
ggtaaatgtg gcagcctgag cctcctttcc atctctgtgg aggcagacat aggaccccca	180
acaaacagca tgcagggttg gagccagcca caggaccag gtaagggggc ctgggtcctt	240

-continued

```

<210> SEQ ID NO 51
<211> LENGTH: 95
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 51

tttatgaggt gggagctggg ctctccctga tgtattattc agctccctgg agttggccag      60
ctcctgttac actggccaca gccctgggca tccgc                                     95

<210> SEQ ID NO 52
<211> LENGTH: 185
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 52

tgccatggtg actttaagc caggttgctg ccccgagcaca ggctcccag tctaccctca      60
ctagaaaaca acaccaggc actttccacc acctctcaaa ggtgaaaccc aaggctggtc      120
tagagaatga attatggatc ctgctgtgcc gtgccacca gctagtccca gcggctcaga      180
cactg                                             185

<210> SEQ ID NO 53
<211> LENGTH: 335
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 53

ctgcagctca gcctactact tgctttccag gctgttecta gttcccatgt cagctgcttg      60
tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc      120
ggtcaatgtg gcagcctgag cctcctttcc atctctgtgg aggcagacat aggaccccca      180
acaaacagca tgcaggttgg gagccagcca caggaccagc gtaagggggc ctgggtcctt      240
tttatgaggt gggagctggg ctctccctga tgtattattc agctccctgg agttggccag      300
ctcctgttac actggccaca gccctgggca tccgc                                     335

<210> SEQ ID NO 54
<211> LENGTH: 425
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 54

ctgcagctca gcctactact tgctttccag gctgttecta gttcccatgt cagctgcttg      60
tgctttccag agacaaaaca ggaataatag atgtcattaa atatacattg ggccccaggc      120
ggtcaatgtg gcagcctgag cctcctttcc atctctgtgg aggcagacat aggaccccca      180
acaaacagca tgcaggttgg gagccagcca caggaccagc gtaagggggc ctgggtcctt      240
tgccatggtg actttaagc caggttgctg ccccgagcaca ggctcccag tctaccctca      300
ctagaaaaca acaccaggc actttccacc acctctcaaa ggtgaaaccc aaggctggtc      360
tagagaatga attatggatc ctgctgtgcc gtgccacca gctagtccca gcggctcaga      420
cactg                                             425

<210> SEQ ID NO 55
<211> LENGTH: 280
<212> TYPE: DNA

```

-continued

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 55

tttatgaggt gggagctggg ctctccctga tgtattattc agctccctgg agttggccag	60
ctcctgttac actggccaca gccctgggca tccgctgcc tggtagcttt aaagccaggt	120
tgctgcccc gcacaggcct cccagtctac cctcactaga aaacaacacc caggcacttt	180
ccaccacctc tcaaagggtga aacccaaggc tggctctagag aatgaattat ggatccctcg	240
tgcccggtgc acccagctag tcccagcggc tcagacactg	280

<210> SEQ ID NO 56

<211> LENGTH: 520

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 56

tttatgaggt gggagctggg ctctccctga tgtattattc agctccctgg agttggccag	60
ctcctgttac actggccaca gccctgggca tccgctgca gctcagccta ctacttgctt	120
tccaggctgt tcttagttcc catgtcagct gcttgctgctt tccagagaca aaacaggaat	180
aatagatgtc attaaatata cattggggcc caggcgggtca atgtggcagc ctgagcctcc	240
tttccatctc tgtggaggca gacataggac ccccaacaaa cagcatgcag gttgggagcc	300
agccacagga cccaggtaag gggccctggg tcctttgcca tggtagcttt aaagccaggt	360
tgctgcccc gcacaggcct cccagtctac cctcactaga aaacaacacc caggcacttt	420
ccaccacctc tcaaagggtga aacccaaggc tggctctagag aatgaattat ggatccctcg	480
tgcccggtgc acccagctag tcccagcggc tcagacactg	520

<210> SEQ ID NO 57

<211> LENGTH: 520

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 57

tgccatggtg actttaagc cagggtgctg cccagcaca ggctcccag tctaccctca	60
ctagaaaaca acaccaggc actttccacc acctctcaaa ggtgaaacc aaggctggtc	120
tagagaatga attatggatc ctgctgtgcc gtgccacca gctagtccca gcggctcaga	180
cactgctgca gctcagccta ctacttgctt tccaggctgt tctagtctc catgtcagct	240
gcttgctgctt tccagagaca aaacaggaat aatagatgtc attaaatata cattggggcc	300
caggcgggtca atgtggcagc ctgagcctcc tttccatctc tgtggaggca gacataggac	360
cccaacaaa cagcatgcag gttgggagcc agccacagga cccaggtaag gggccctggg	420
tcctttttat gaggtgggag ctgggctctc cctgatgtat tttcagctc cctggagttg	480
gccagctcct gttacactgg ccacagccct gggcatccgc	520

<210> SEQ ID NO 58

<211> LENGTH: 520

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 58

-continued

tttatgaggt gggagctggg ctctccctga tgtattattc agtccctgg agttggccag	60
ctcctgttac actggccaca gccctgggca tccgctgcc tggtagcttt aaagccaggt	120
tgctgcccc gcacaggcct ccagctctac cctcactaga aaacaacacc caggcacttt	180
ccaccacctc tcaaagggtga aacccaaggc tggctctagag aatgaattat ggatcctcgc	240
tgtccgtgcc acccagctag tcccagcggc tcagacactg ctgcagctca gcctactact	300
tgctttccag gctgttccta gttcccatgt cagctgcttg tgctttccag agacaaaaca	360
ggaataatag atgtcattaa atatacattg gggcccaggc ggtcaatgtg gcagcctgag	420
cctcctttcc atctctgtgg aggcagacat aggaccccca acaaacagca tgcaggttgg	480
gagccagcca caggaccag gtaaggggcc ctgggtcctt	520

<210> SEQ ID NO 59
 <211> LENGTH: 964
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct
 <400> SEQUENCE: 59

tgcagctcag cctactactt gctttccagg ctgttcctag ttcccatgtc agctgcttgt	60
gctttccaga gacaaaacag gaataataga tgtcattaaa tatacattgg gcccaggcgc	120
gtcaatgtgg cagcctgagc ctctttcca tctctgtgga ggcagacata ggacccccaa	180
caaacagcat gcaggttggg agccagccac aggacccagg taagggggcc tgggtcctta	240
agcttctgcc actggctccg gcattgcaga gagaagagaa ggggcggcag actggagagc	300
tgggtccat tttgttctt tgggtccctg cccctcccca tgacctgcag agacattcag	360
cctgccaggc tttatgaggt gggagctggg ctctccctga tgtattattc agtccctgg	420
agttggccag ctcctgttac actggccaca gccctgggca tccgcttctc acttctagtt	480
tcccctccaa ggtaatgtgg tgggtcatga tcattctatc ctggcttcag ggacctgact	540
ccactttggg gccattcgag gggctctagg tagatgatgt cccctgtgg ggattaatgt	600
cctgctctgt aaaactgagc tagctgagat ccaggagggc ttggccagag acagcaagtt	660
gttgccatgg tgactttaaa gccaggttgc tgcccagca caggcctccc agtctaccct	720
cactagaaaa caacaccag gcactttcca ccacctctca aagtgaaac ccaaggctgg	780
tctagagaat gaattatgga tcctcgtgt cctgtccacc cagctagtc cagcggctca	840
gacactgagg agagactgta ggttcagcta caagcaaaaa gacctagctg gtctccaagc	900
agtgtctcca agtccctgaa cctgtgacac ctgcccagg catcatcagg cacagagggc	960
cacc	964

<210> SEQ ID NO 60
 <211> LENGTH: 519
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct
 <400> SEQUENCE: 60

tgcagctcag cctactactt gctttccagg ctgttcctag ttcccatgtc agctgcttgt	60
gctttccaga gacaaaacag gaataataga tgtcattaaa tatacattgg gcccaggcgc	120
gtcaatgtgg cagcctgagc ctctttcca tctctgtgga ggcagacata ggacccccaa	180

-continued

caaacagcat gcaggttggg agccagccac aggacccagg taaggggccc tgggtccttt	240
ttatgaggtg ggagctgggc tctccctgat gtattattca gctccctgga gttggccagc	300
tctgtttaca ctggccacag ccttgggcat ccgctgccat ggtgacttta aagccagggt	360
gctgccccag cacaggcctc ccagtctacc ctcactagaa aacaacaccc aggcactttc	420
caccacctct caaaggtgaa acccaaggct ggtctagaga atgaattatg gatccctcgt	480
gtccgtgccca ccagctagt ccagcggct cagacactg	519

<210> SEQ ID NO 61
 <211> LENGTH: 425
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 61

aatcaacctc tggattacaa aatttgtgaa agattgactg gtattcttaa ctatgttgct	60
ccttttacgc tatgtggata cgctgcttta atgcctttgt atcatgctat tgcttccgt	120
atggctttca tttctcctc ctgtataaa tcttggttag ttcttgccac ggcggaactc	180
atcgccgctc gccttgccg ctgctggaca ggggctcgcc tgttgggcaac tgacaattcc	240
gtggtgttta tttgtgaaat ttgtgatgct attgctttat ttgtaaccat ctagctttat	300
ttgtgaaatt tgtgatgcta ttgctttatt tgtaaccatt ataagctgca ataaacaagt	360
taacaacaac aattgcatc attttatgtt tcaggttcag ggggagatgt gggaggtttt	420
ttaaa	425

<210> SEQ ID NO 62
 <211> LENGTH: 865
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 62

ccccgggtgc gcggcgctcg tggtgccggc gggggggcgc aggtcgaggc cgggtgtaggg	60
ctccaggcag gcggcgaagg ccatgacgtg cgctatgaag gtctgctcct gcacgccgtg	120
aaccagggtc gcctgcgggc cgcgcgcgaa caccgccacg tcctcgctcg cgtgggtctc	180
ttcgtccagg ggcactgctg actgctgccg ataactgggg ctcccgctct cgtctctcgt	240
aacatccggc cgggcgcgct ccttgagcac atagcctgga ccgtttccgt ataggaggac	300
cgtgtaggcc ttctgtccc gggccttgcc agcggccagc ccgatgaagg agtcctctcg	360
cagggggtag cctccgaagg agaagacgtg ggagtggctg gcagtgcga ggctcagcgt	420
gtcctcctcg ctggtgagct ggcccgcct ctcaatggcg tcgtcgaaca tgatcgtctc	480
agtcagtgcc cggtaagccc tgccttcctg atgacctggc tcgatgcgac caccctccac	540
gaagaggaag aagccgcggg ggtgtctgct cagcaggcgc agggcagcct ctgtcatctc	600
catcaggagg gggctcagtg tggagtctcg gtggatctcg tatttcattg ctccaggctc	660
aaagagaccc atgagatggg tcacagacgg gtccaggga gcctgcatga gctcagtgcg	720
gttccacacg taccgggcac cctggcgctc gccagagccat tcctgcacca gattcttccc	780
gtccagcctg gtcccacett ggtgttagtc atctgggtac tcagggtctg gggttcccat	840
gcgaaacatg tactttcggc ctcca	865

<210> SEQ ID NO 63
 <211> LENGTH: 437

-continued

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 63

```

ccccgggtgc gcggcgtcgg tgggtgccgc ggggggcgcc aggtcgcagg cgggtgtagg 60
ctccaggcag gcggcgaagg ccatgacgtg cgctatgaag gtctgctcct gcacgccgtg 120
aaccagggtgc gcctgcgggc cgcgcgcgaa caccgccacg tcctgccttg cgtgggtctc 180
ttcgtccagg ggcaactgtg actgctgccg atactcgggg ctcccgtctt cgctctcggg 240
aacatccggc cgggcgcgtg ccttgagcac atagcctgga ccgtttccgt ataggaggac 300
cgtgtaggcc ttctgtccc gggccttgcc agcggccagc ccgatgaagg agctccctcg 360
cagggggtag cctccgaagg agaagacgtg ggagtggctg gcagtgcga ggctcagcgt 420
gtcctcctcg ctggtga 437

```

<210> SEQ ID NO 64

<211> LENGTH: 428

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 64

```

gctggccgc cctctcaatg gcgtcgtcga acatgatcgt ctacgtcagt gcccggttaag 60
ccctgcttcc atgatgacca tggtcgatgc gaccaccctc caccgaagg aagaagccgc 120
gggggtgtct gctcagcagg cgcagggcag cctctgtcat ctccatcagg gaggggtcca 180
gtgtggagtc tcggtggatc tcgtatttca tgtctccagg ctcaaagaga cccatgagat 240
gggtcacaga cgggtccagg gaagcctgca tgagctcagt gcggttcac acgtaccggg 300
caccctggcg ttcgcgcagc cattctgca ccagattctt cccgtccagc ctggtccac 360
cttggtgta gtcactggg tactcagggt ctgggggtcc catgcgaaac atgtactttc 420
ggcctcca 428

```

<210> SEQ ID NO 65

<211> LENGTH: 287

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 65

```

ccccgggtgc gcggcgtcgg tgggtgccgc ggggggcgcc aggtcgcagg cgggtgtagg 60
ctccaggcag gcggcgaagg ccatgacgtg cgctatgaag gtctgctcct gcacgccgtg 120
aaccagggtgc gcctgcgggc cgcgcgcgaa caccgccacg tcctgccttg cgtgggtctc 180
ttcgtccagg ggcaactgtg actgctgccg atactcgggg ctcccgtctt cgctctcggg 240
aacatccggc cgggcgcgtg ccttgagcac atagcctgga ccgtttc 287

```

<210> SEQ ID NO 66

<211> LENGTH: 290

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 66

```

cgataggag gaccgtgtg gccttctgt cccgggctt gccagcggc agcccgatga 60
aggagctccc tcgcagggg tagcctcca aggagaagac gtgggagtgg tcggcagtga 120
cgaggtcag cgtgtctctc tcgtggtga gctggccgc cctctcaatg gcgtcgtcga 180
acatgatcgt ctacgtcagt gcccggtta cctgcttcc atgatgacca tggtcgatgc 240

```

-continued

gaccaccctc cacgaagagg aagaagccgc gggggtgtct gctcagcagg 290

<210> SEQ ID NO 67
 <211> LENGTH: 288
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 67

cgcagggcag cctctgtcat ctccatcagg gaggggtcca gtgtggagtc tcggtggatc 60
 tcgtatttca tgtctccagg ctcaaagaga cccatgagat gggtcacaga cgggtccagg 120
 gaagcctgca tgagctcagt gcggttcac acgtaccggg caccctggcg ttgcgcgagc 180
 cattcctgca ccagattctt cccgtccagc ctgggtccac cttggctgta gtcattctggg 240
 tactcagggt ctggggttcc catgcgaaac atgtactttc ggcctcca 288

<210> SEQ ID NO 68
 <211> LENGTH: 84
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 68

gtaagtatca aggttacaag acaggttttaa ggagaccaat agaaactggg cttgtcgaga 60
 cagagaagac tcttgcgttt ctga 84

<210> SEQ ID NO 69
 <211> LENGTH: 49
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 69

taggcaccta ttggtcttac tgacatccac tttgccttcc tctccacag 49

40

The invention claimed is:

1. A dual vector system comprising:

- a first nucleic acid vector comprising a myosin 15 (Myo15) promoter operably linked to a first coding polynucleotide that encodes an N-terminal portion of an otoferlin (OTOF) protein; and
 a second nucleic acid vector comprising a second coding polynucleotide that encodes a C-terminal portion of an OTOF protein and a polyadenylation (poly(A)) sequence positioned at a 3' end of the second coding polynucleotide;

wherein the Myo15 promoter has at least 95% sequence identity to SEQ ID NO: 38, wherein neither the first nor second nucleic acid vector encodes a full-length OTOF protein, and, when introduced into a mammalian cell, the first and second nucleic acid vectors undergo homologous recombination or concatemerization to form a recombined nucleic acid that encodes a full-length OTOF protein.

2. The dual vector system of claim 1, wherein the first nucleic acid vector further comprises a splice donor signal sequence positioned at a 3' end of the first coding polynucleotide; and
 the second nucleic acid vector further comprises a splice acceptor signal sequence positioned 5' of the second coding polynucleotide;

wherein the first coding polynucleotide and the second coding polynucleotide do not overlap.

3. The dual vector system of claim 2, wherein the first nucleic acid vector further comprises a first recombinogenic region positioned 3' of the splice donor signal sequence; and

the second nucleic acid vector further comprises a second recombinogenic region positioned 5' of the splice acceptor signal sequence positioned.

4. The dual vector system of claim 3, wherein the first and second recombinogenic regions have the same nucleic acid sequence.

5. The dual vector system of claim 1, wherein the first and second nucleic acid vectors are adeno-associated virus (AAV) vectors.

6. The dual vector system of claim 1, wherein the first and second coding polynucleotides that encode the OTOF protein do not comprise introns.

7. The dual vector system of claim 1, wherein the Myo15 promoter has the sequence of SEQ ID NO: 38.

8. The dual vector system of claim 3, wherein each of the first and second recombinogenic regions is an AP gene fragment or an F1 phage AK gene.

9. The dual vector system of claim 1, wherein the OTOF protein is a human OTOF protein.

283

10. The dual vector system of claim 5, wherein the first and second nucleic acid vectors comprise an inverted terminal repeat (ITR) at each end of the nucleic acid sequence.

11. The dual vector system of claim 10, wherein the ITR is an AAV2 ITR.

12. The dual vector system of claim 1, wherein the poly(A) sequence is a bovine growth hormone (bGH) poly(A) signal sequence.

13. The dual vector system of claim 7, wherein the OTOF protein is a human OTOF protein.

14. The dual vector system of claim 13, wherein the first and second coding polynucleotides that encode the OTOF protein do not comprise introns.

15. The dual vector system of claim 14, wherein the first nucleic acid vector further comprises a splice donor signal sequence positioned at a 3' end of the first coding polynucleotide; and the second nucleic acid vector further comprises a splice acceptor signal sequence positioned 5' of the second coding polynucleotide;

284

wherein the first coding polynucleotide and the second coding polynucleotide do not overlap.

16. The dual vector system of claim 15, wherein the first nucleic acid vector further comprises a first recombinogenic region positioned 3' of the splice donor signal sequence; and

the second nucleic acid vector further comprises a second recombinogenic region positioned 5' of the splice acceptor signal sequence positioned.

17. The dual vector system of claim 16, wherein the first and second recombinogenic regions have the same nucleic acid sequence.

18. The dual vector system of claim 17, wherein each of the first and second recombinogenic regions is an AP gene fragment or an F1 phage AK gene.

19. The dual vector system of claim 16, wherein the first and second nucleic acid vectors are adeno-associated virus (AAV) vectors.

* * * * *